Hochschild Homology of the Cusp Calculus

Notes on Melrose–Nistor, *Homology of pseudodifferential operators I. Manifolds with boundary*

Expository notes on Melrose–Nistor

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Abstract

These notes are a reader's guide to the paper of R. Melrose and V. Nistor [15], which computes the continuous Hochschild homology of the algebra of *cusp* pseudodifferential operators of integral order with Laurent coefficients at the boundary of a compact manifold X, together with the ideals and quotients cut out by the two natural filtrations. The paper then interprets the index of a fully elliptic operator as a Hochschild 1-cocycle and writes it as a sum of a symbolic (Atiyah–Singer type) term and an eta invariant of the indicial family. We work the model cases first: closed manifolds, the half-line $[0, 1)$ with its suspended boundary algebra, the interval and the cylinder. After that we follow the sections of [15] in order and supply the arguments that the paper compresses: H-unitality, the weighted de Rham lemmas, the spectral sequences and their convergence, the analytic continuation behind the four trace functionals, the boundary maps, and the index formula. We also record a number of corrections. The most important are these. The homology statements in [15, (16)] tacitly assume that ∂X is connected and that $\dim X \geq 2$; for N boundary components the boundary map in degree two has rank $N - 1$, not 0. The signs of the symbolic terms in the index theorem [15, (93)] are inconsistent with the paper's own formula [15, (87)] and with the closed case [15, (3), (5)]. The regular value claimed in [15, Prop. 8] is wrong in general, and as a consequence the eta term of the index formula carries a local correction δ_Q, which vanishes in the cases we treat. We verify the corrected formula on explicit examples. Three appendices prove most of the background that [15] quote: the construction of the cusp calculus and its Fredholm theory (Appendix A); Wodzicki excision, Connes' theorem, the Künneth and Morita statements and the convergence of the spectral sequences (Appendix B); and the holomorphic families, residue traces and the suspended eta invariant (Appendix C). Section 2.5 lists what is still quoted. The main gap is the convergence of the order spectral sequences for the Laurent symbol algebras, where the argument used for the b-calculus fails.

1 Overview and roadmap

1.1 The algebras

Let X be a compact manifold with boundary, $\dim X = n$, and let $x \in C^\infty(X)$ be a boundary defining function. The function x fixes a *cusp structure* (Section 4.1), and with it the algebra $\Psi_c^{\mathbb{Z}}(X)$ of classical cusp pseudodifferential operators of integral order. Conjugation by powers of x preserves each $\Psi_c^m(X)$, so

$$x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) = \bigcup_{k\in\mathbb{Z}} \bigcup_{m\in\mathbb{Z}} x^k\Psi_c^m(X)$$

is an algebra. MN write $x^{-\mathbb{Z}}$ to stress that this second filtration, by powers of x, runs in the opposite direction to the order filtration. Each filtration has a residual part, and the doubly residual operators form the ideal

$$\mathcal{I} = x^\infty\Psi_c^{-\infty}(X) = \bigcap_{k,m} x^k\Psi_c^m(X).$$

Its elements are the operators with kernels in $\dot{C}^\infty(X^2)$, the smooth functions on X^2 vanishing to infinite order at $\partial(X^2)$. Following [15, (8)–(10)] we set

$$\begin{aligned} &\mathcal{A} = x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)/\mathcal{I}, && \mathcal{I}_\partial = x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)/\mathcal{I}, && \mathcal{I}_\sigma = x^\infty\Psi_c^{\mathbb{Z}}(X)/\mathcal{I}, \\ &\mathcal{A}_\sigma = \mathcal{A}/\mathcal{I}_\partial, && \mathcal{A}_\partial = \mathcal{A}/\mathcal{I}_\sigma, && \mathcal{A}_{\partial,\sigma} = \mathcal{A}/(\mathcal{I}_\sigma + \mathcal{I}_\partial). \end{aligned}$$

In words:

- $\mathcal{I}_\sigma$ consists of *complete symbols vanishing to infinite order at the boundary*;
- $\mathcal{A}_\sigma$ consists of *complete symbols with Laurent singularities at the boundary*;

- $\mathcal{I}_\partial$ consists of *Laurent series of smoothing indicial operators*;
- $\mathcal{A}_\partial$ consists of *Laurent series of indicial operators of finite order*;
- $\mathcal{A}_{\partial,\sigma}$ consists of *Laurent series in x of complete symbols at the boundary*.

The maps between them form the diagram of Figure 1, which reproduces the diagram on [15, p. 4].

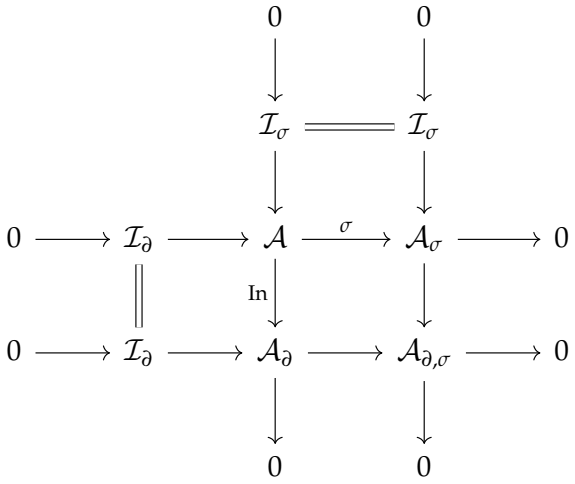


Figure 1: The extensions relating the algebras of [15]. Rows are the indicial (boundary) extensions and columns the symbol (interior) extensions. In addition there are the extensions $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) \to \mathcal{A} \to 0$, $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{-\infty}(X) \to \mathcal{I}_\partial \to 0$ and $0 \to \mathcal{I}_\sigma + \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_{\partial,\sigma} \to 0$. Since $\mathcal{I}_\sigma \cap \mathcal{I}_\partial = 0$ in $\mathcal{A}$, the central square is a pull-back.

The two quotient maps out of $\mathcal{A}$ are the *full symbol* σ, which records the complete symbol expansion as a Laurent series in x, and the *indicial map* In, which records the Taylor series at ∂X of the kernel at the cusp front face. After choices (a quantization, a boundary defining function and a normal fibration) there are isomorphisms of vector spaces [15, (13)–(15)]

$$\begin{gathered}\mathcal{I}_\sigma \cong \dot{C}^\infty({}^cS^*X)((q^{-1})), \qquad \mathcal{A}_\sigma \cong x^{-\mathbb{Z}}C^\infty({}^cS^*X)((q^{-1})),\\ \mathcal{I}_\partial \cong \Psi_{\mathrm{sus}}^{-\infty}(\partial X)((x)), \qquad \mathcal{A}_\partial \cong \Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x)), \qquad \mathcal{A}_{\partial,\sigma} \cong C^\infty({}^cS^*_{\partial X}X)((x,q^{-1})).\end{gathered} \tag{1}$$

Here $q = \sigma_1(Q)$ is the symbol of a positive elliptic $Q \in \Psi_c^1(X)$, and $((q^{-1}))$ denotes Laurent series in q^{-1}, with powers of q bounded above. Likewise $((x))$ denotes Laurent series in x, with powers of x bounded below; in $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x))$ the coefficients are moreover required to have bounded order (see (12)). (MN write these as $[q]]$ and $[[x]$; we use the standard notation for formal Laurent series.) Figure 2 places the two filtrations and their residual ideals side by side. The isomorphisms (1) are linear (Propositions A.22 and A.27); the products on the right-hand sides are the products of $\mathcal{A}$ and its quotients transported through them, that is, star products. In the indicial algebras the Laurent variable does not commute with the suspended operators:

$$[x, \partial_t] = x^2, \qquad t = 1/x, \tag{2}$$

where t is the linear variable of the suspension $\mathbb{R}_t \times \partial X$. The relation (2) is the *cusp twist*. We explain it in Section 4.5; it is responsible for most of the differences from the b-calculus (Section 18).

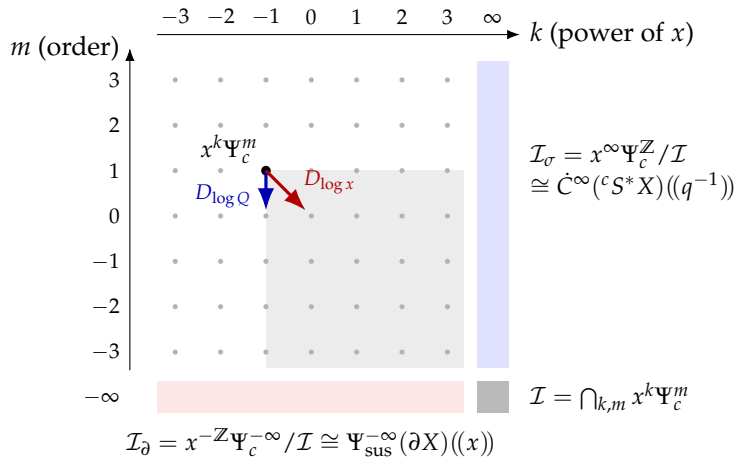


Figure 2: The two filtrations of $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$. The dot at (k,m) stands for $x^k\Psi_c^m(X)$, which contains $x^{k'}\Psi_c^{m'}(X)$ for $k' \geq k$ and $m' \leq m$ (shaded, for $(k,m) = (-1,1)$). Every element lies in such a quadrant: powers of x are bounded below, as in $((x))$, and orders are bounded above, as in $((q^{-1}))$. The residual edges are $k = \infty$, giving $\mathcal{I}_\sigma$ (blue), and $m = -\infty$, giving $\mathcal{I}_\partial$ (red); the corner $\mathcal{I}$ is divided out, so $\mathcal{I}_\sigma \cap \mathcal{I}_\partial = 0$ in $\mathcal{A}$. Then $\mathcal{A}_\sigma = \mathcal{A}/\mathcal{I}_\partial$ forgets the bottom edge, $\mathcal{A}_\partial = \mathcal{A}/\mathcal{I}_\sigma$ the right edge, and $\mathcal{A}_{\partial,\sigma}$ both. Arrows: on symbols $D_{\log Q}$ lowers the order by one, and $D_{\log x}$ lowers the order by one and raises the power of x by one (Section 11).

1.2 The main homological results

Write $Y = {}^cS^*X$ for the cosphere bundle of the cusp cotangent bundle; it is a compact manifold with boundary of dimension $2n-1$, and its boundary is $Z = {}^cS^*_{\partial X}X$, the part over ∂X. Let $\mathbb{S}_\sigma$ and $\mathbb{S}_\partial$ be circles. They record the homogeneity in the fibres and the Laurent variable x, and carry the classes $d\theta_\sigma = [d\log r]$ and $d\theta_\partial = [d\log x]$. All cohomology has complex coefficients, and Hochschild homology is the continuous homology of Section 5.

MN state their results for Hochschild *cohomology* [15, (16)]. In the form proved in these notes they read as follows.

Theorem 1.1 (MN (16), with the number of boundary components restored)**.** *Let X be connected, $n = \dim X \geq 2$, and let ∂X have $N \geq 1$ components $H_1, \dots, H_N$. Then Y is connected, and $Z = \bigsqcup_j Z_j$ with each $Z_j = {}^cS^*_{H_j}X$ connected. The continuous Hochschild cohomology groups are:*

$$
\begin{aligned}
\mathrm{HH}^k(\mathcal{I}) &\cong \begin{cases} \mathbb{C}, & k = 0, \\ 0, & k > 0, \end{cases} \\
\mathrm{HH}^k(\mathcal{I}_\sigma) &\cong H^k(Y \times \mathbb{S}_\sigma), \\
\mathrm{HH}^k(\mathcal{A}_\sigma) &\cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma), \\
\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) &\cong H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial), \\
\mathrm{HH}^k(\mathcal{I}_\partial) &\cong \begin{cases} \mathbb{C}^N, & k = 0, 1, \\ 0, & k \geq 2, \end{cases} \\
\mathrm{HH}^k(\mathcal{A}_\partial) &\cong \begin{cases} H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)/\mathbb{C}^N, & k = 1, 2, \\ H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial), & \textit{otherwise}, \end{cases}
\end{aligned}
$$

$$\mathrm{HH}^k(\mathcal{A}) \cong \begin{cases} \mathbb{C} \oplus H^1_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^1(Z \times \mathbb{S}_\sigma)/\mathbb{C}^N, & k = 1, \\ \big(H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^2(Z \times \mathbb{S}_\sigma)\big)/\mathbb{C}^{N-1}, & k = 2, \\ H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma), & \text{otherwise.} \end{cases}$$

The subspaces $\mathbb{C}^N$ are spanned by the classes $d\theta_\sigma \cdot 1_{Z_j}$ (for $k = 1$) and $d\theta_\partial \wedge d\theta_\sigma \cdot 1_{Z_j}$ (for $k = 2$), and $\mathbb{C}^{N-1}$ is the image of $\{\sum_j c_j\, d\theta_\sigma 1_{Z_j}\}$ under the de Rham connecting map $H^1(Z \times \mathbb{S}_\sigma) \to H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$; it is $(N-1)$-dimensional. Finally $\mathrm{HH}^k(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) \cong \mathrm{HH}^k(\mathcal{A})$ for $k \neq 1$, while $\mathrm{HH}^1(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X))$ is the quotient of $\mathrm{HH}^1(\mathcal{A})$ by the line spanned by the index cocycle.

For $N = 1$ these are exactly the formulas of [15, (16)]. The corresponding homology groups are obtained by Lefschetz duality on the $2n$-manifold $Y \times \mathbb{S}_\sigma$ and Poincaré duality on the closed manifolds $Z \times \mathbb{S}_\sigma$ and $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$. For example [15, Props. 2, 3]

$$\mathrm{HH}_k(\mathcal{I}_\sigma) \cong H^{2n-k}_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma), \qquad \mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y \times \mathbb{S}_\sigma) \oplus H^{2n-k-1}(Z \times \mathbb{S}_\sigma).$$

The precise sources of the hypotheses are discussed in Corrections 9.4 and 10.9. The proof rests on two quoted inputs, the boundary connecting map (B9) and the last step of the convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ (B10); everything else it uses is proved in the appendices (Section 2.5). The proof of Theorem 1.1 occupies Sections 6–12.

1.3 The index formula

A matrix A of elements of $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ is *fully elliptic* if its image in (a matrix algebra over) $\mathcal{A}$ is invertible, with the orders matched: for $A \in x^{-p}\Psi_c^m$ the inverse is the class of an element of $x^p\Psi_c^{-m}$ (Theorem A.34). For an operator of order m this means two things: A is elliptic as a cusp operator, and its indicial family $\mathrm{In}(A,\tau)$ is invertible for every real τ. Such an operator is Fredholm on $\dot{C}^\infty(X)$, and Fedosov's formula reads [15, (7)]

$$\mathrm{Ind}(A) = \dim\ker A - \dim\ker A^* = \mathrm{Tr}([A,B]) = \mathrm{IF}([A],[A]^{-1}), \tag{3}$$

where B is any inverse of A modulo $\mathcal{I}$. The *index functional* IF is the image of the trace on $\mathcal{I}$ under the connecting map of $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X) \to \mathcal{A} \to 0$. MN express it through the following data:

- the extensions $\widehat{\mathrm{Tr}}_\partial$ and $\widehat{\mathrm{Tr}}_\sigma$ of the traces on the two ideals $\mathcal{I}_\partial$ and $\mathcal{I}_\sigma$;
- the two exterior derivations $D_{\log x} = [\log x, \cdot\,]$ and $D_{\log Q} = [\log Q, \cdot\,]$.

In the form proved here (Proposition 15.3),

$$\mathrm{IF}(A,B) = \tfrac{1}{2}\big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big), \qquad \widetilde{D} = [\log x - \log Q, \cdot\,]. \tag{4}$$

This is [15, (87)]. In [15, (21)] the derivation is printed with the opposite sign (Correction 15.5). Inserting $B = A^{-1}$ splits the index into four terms. If A is translation invariant near the boundary with normal indicial family, two of them vanish, and one obtains the pseudodifferential APS formula (Theorem 15.13):

$$\mathrm{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\overline{\eta}(A), \qquad \mathrm{AS}(A) = \mathrm{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big), \qquad \overline{\eta}(A) = \eta(A_0) + \delta_Q(A). \tag{5}$$

Here $A_0 = \mathrm{In}(A)$, $t = 1/x$ is the suspension variable, $\eta(A_0) = 2\,\overline{\mathrm{Tr}}\big(A_0^{-1}[t, A_0]\big)$, $\overline{\mathrm{Tr}}$ is Melrose's regularized trace on the suspended algebra, and in terms of the indicial variable

$$\eta(A_0) = \frac{1}{\pi i}\,\overline{\int_{\mathbb{R}}} \mathrm{Tr}\Big(A_0(\tau)^{-1}\frac{dA_0(\tau)}{d\tau}\Big)\,d\tau.$$

The term $\delta_Q(A)$ is a local functional of the boundary symbols of A and Q. It comes from the corrected form of [15, Prop. 8] (item 9 of Section 2.6), and it vanishes in the cases treated in Section 16 (Proposition 15.8). For a Dirac operator in APS normal form $\eta(A_0)$ is the Atiyah–Patodi–Singer eta invariant of the boundary operator (Corollary 15.10). In MN, [15, (22)–(24)] and [15, (100)], the Atiyah–Singer term is written $\mathrm{Tr}_{\mathrm{R}}(A^{-1}[\log Q, A])$, which is $-\mathrm{AS}(A)$. The sign is wrong; see Correction 15.7 and the checks in Section 16.

1.4 Organization and how to read these notes

Section 2 fixes conventions and gives the dictionary to MN's notation. It also lists the background facts, with the appendix results that prove them and what remains quoted, and it collects the corrections. Section 3 treats the closed case (Brylinski–Getzler, Wodzicki), which is the template for everything else. Section 4 introduces the cusp calculus, starting from the half-line. Section 5 assembles the homological toolkit. Sections 6–12 follow [15, §§1–8] and prove Theorem 1.1. Sections 13–15 follow [15, §§9–12] and prove the index formula. Section 16 contains worked examples. Section 17 summarizes [15, §§13–15]. Section 18 compares the cusp and b-calculi and cites the companion notes [7]. Appendices A–C prove the background facts (B1)–(B8) of Section 2.5, apart from the inputs listed there as quoted. The main text can be read without them: each step that uses a background fact names the appendix result that proves it.

Figure 3 shows the logical dependencies.

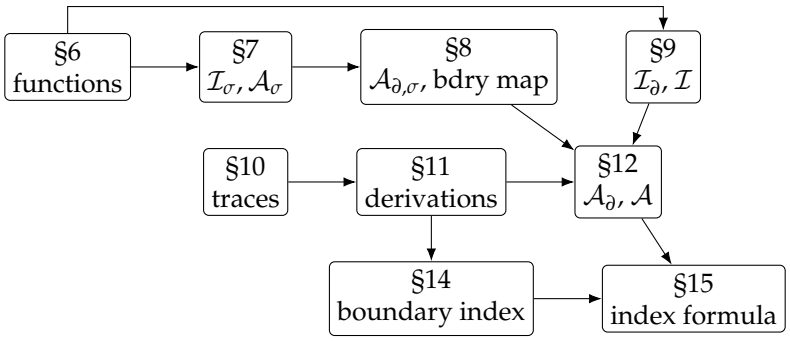


Figure 3: Logical structure of the main part of the notes.

2 Conventions, dictionary, background and corrections

2.1 Coordinates and Fourier conventions

Near a boundary point we use coordinates (x, y), $y \in \mathbb{R}^{n-1}$, where x is the fixed boundary defining function. The *cusp cotangent* coordinates (x, y, τ, η) are defined by writing a cusp covector as

$$\alpha = \tau \frac{dx}{x^2} + \eta \cdot dy, \tag{6}$$

as in [15, (139)]. Thus τ is the symbol of $x^2 D_x$, where $D = -i\partial$. The canonical symplectic form is $\omega = d\alpha = d\tau \wedge \frac{dx}{x^2} + \sum_j d\eta_j \wedge dy_j$. It agrees with the standard form $\sum d\xi \wedge dx$ in the interior, where $\tau = x^2\xi$. The Poisson bracket is

$$\{a, b\} = \partial_\tau a \, x^2 \partial_x b - x^2 \partial_x a \, \partial_\tau b + \partial_\eta a \cdot \partial_y b - \partial_y a \cdot \partial_\eta b, \tag{7}$$

and the principal symbol of a commutator is $\sigma([A,B]) = -i\{\sigma(A),\sigma(B)\}$. (In [15, (139)] the last term is misprinted as $\partial_y a \cdot \partial_\tau b$.) Formula (7) follows from the usual bracket $\partial_{\tilde\xi} a\,\partial_x b - \partial_x a\,\partial_{\tilde\xi} b$ by the change of variables $\tau = x^2\xi$. Indeed $\partial_{\tilde\xi}|_x = x^2\partial_\tau$ and $\partial_x|_{\tilde\xi} = \partial_x|_\tau + 2x^{-1}\tau\,\partial_\tau$, and the two terms involving $2x^{-1}\tau$ cancel.

Near the boundary we use the *suspension variable*

$$t = \frac{1}{x} \in (1,\infty), \qquad x^2\partial_x = -\partial_t, \qquad \frac{dx}{x^2} = -dt. \tag{8}$$

Fourier transformation is $\hat u(\xi) = \int e^{-it\xi} u(t)\,dt$, with inverse $u(t) = (2\pi)^{-1}\int e^{it\xi}\hat u(\xi)\,d\xi$, as in the companion notes [18, 7].

2.2 The indicial variable

MN define the indicial family by conjugation [15, Def. 4]:

$$\mathrm{In}(A,\tau) = \left(e^{i\tau/x} A e^{-i\tau/x}\right)\Big|_{\partial X}, \qquad \tau \in \mathbb{R}. \tag{9}$$

Then $\mathrm{In}(x^2D_x,\tau) = \tau$, so the indicial variable is the boundary value of the cusp covariable of (6). For a translation invariant operator on $\mathbb{R}_t \times \partial X$ with convolution kernel $\kappa(t-t',y,y')$, one finds

$$\mathrm{In}(A,\tau) = \int_{\mathbb{R}} e^{i\tau s}\kappa(s,\cdot,\cdot)\,ds = \hat\kappa(-\tau), \qquad \mathrm{In}(D_t,\tau) = -\tau, \qquad \mathrm{In}([t,A],\tau) = -i\,\partial_\tau \mathrm{In}(A,\tau). \tag{10}$$

The last identity holds because $[t,A]$ has kernel $(t-t')\kappa(t-t')$. Under the transcendental blow-up $x_b = e^{-1/x}$ of Section 4.6, the b-indicial family of the companion notes, $I(A,\lambda) = \int e^{-iz\lambda}\kappa_A(e^z)\,dz$ with $z = \log(x_b/x_b') = -(t-t')$, becomes (10) with $\tau = \lambda$. So *the cusp indicial variable τ of MN is the b-indicial variable λ of [18, 7].* In the b-calculus λ may be complex. In the cusp calculus only real τ occurs.

2.3 Densities and traces

Kernels are taken with respect to a cusp density ν_c in the right variable, equal to $x^{-2}dx\,\nu_\partial = dt\,\nu_\partial$ near the boundary, where ν_∂ is a smooth positive density on ∂X. This is the convention of Appendix A; MN normalize kernels as cusp half-densities [15, (118)], which gives the same operator spaces (Section 4.3). We use the following functionals; complex parameters are written s (for powers of Q) and z (for powers of x).

functional	definition	MN
Tr on $\mathcal{I}$	integral of the kernel on the diagonal	Tr
Tr_σ on $\mathcal{I}_\sigma$	$\mathrm{Res}_{s=0}\mathrm{Tr}(AQ^{-s})$ (Wodzicki residue)	(49)
Tr_∂ on $\mathcal{I}_\partial$	$\mathrm{Res}_{z=0}\mathrm{Tr}(Ax^z)$	Lemma 10
$\mathrm{Tr}_{\partial,\sigma}$ on $\mathcal{A}$	coefficient of $(sz)^{-1}$ in $Z(A;s,z)$	(44)
$\widehat{\mathrm{Tr}}_\sigma$, $\widehat{\mathrm{Tr}}_\partial$	the mixed coefficients of $Z(A;s,z)$	(44)
$\overline{\mathrm{Tr}}$ on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(\partial X)$	$(2\pi)^{-1}\overline{\int}\,\mathrm{Tr}\,\mathrm{In}(B,\tau)\,d\tau$	(57)–(59)

Here $Z(A;s,z) = \frac12\big(\mathrm{Tr}(Ax^zQ^{-s}) + \mathrm{Tr}(AQ^{-s}x^z)\big)$. **MN write τ for our s** in [15, (43)–(62)], and also use τ for the cusp covariable in [15, (139)]. We keep τ for the covariable throughout.

2.4 Hochschild conventions

The Hochschild boundary is $b = \sum_{i=0}^{k} (-1)^i d_i$, where d_i multiplies a_i and a_{i+1} for $i < k$ and $d_k(a_0 \otimes \cdots \otimes a_k) = a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}$. The bar differential is $b' = \sum_{i<k} (-1)^i d_i$. For a derivation D,

$$i_D(a_0 \otimes a_1 \otimes \cdots \otimes a_k) = a_0 D(a_1) \otimes a_2 \otimes \cdots \otimes a_k \qquad \text{[15, §6]},$$

and on cochains $i_D \varphi = \varphi \circ i_D$. For a 0-cochain φ this reads

$$(i_D \varphi)(A, B) = \varphi\big(AD(B)\big).$$

2.5 Background facts: where they are proved

The paper [15] quotes the analytic foundations of the cusp calculus and the homological tools. Appendices A–C prove most of them. For each item we state the fact as it is used, say which appendix result proves it, and record what remains quoted. Every step in the main text that uses one of these facts says so.

(B1) *Cusp calculus.* The cusp operators form a doubly filtered $*$-algebra, and conjugation by x^z preserves both filtrations. There are the symbol sequence [15, (11)], the indicial sequence [15, (12)], a global quantization map [15, Prop. 24], which gives the *linear* isomorphisms [15, (13)], and the Taylor series isomorphism at the front face [15, Prop. 25], which is an algebra isomorphism for the twisted product. *Proved*: Theorem A.20, Propositions A.21, A.25, A.22, A.27, A.26; agreement with [15, Def. 2] is Proposition A.17, and the lift of b-operators [15, Prop. 26] is Proposition A.28. *Quoted*: the standard uniform pseudodifferential calculus on the cylinder $\mathbb{R} \times \partial X$ (Background fact A.4), and the density half of [15, Prop. 26] (Remark A.29), which is not used.

(B2) *Fredholm theory.* A fully elliptic element of $x^{-p}\Psi_c^m(X; E)$ has an inverse modulo $\mathcal{I}$ in $x^p \Psi_c^{-m}(X; E)$. It is Fredholm on $\dot{C}^\infty(X; E)$ and on the cusp Sobolev spaces, its kernel and cokernel lie in $\dot{C}^\infty$, and it has a generalized inverse in the calculus [15, p. 4, §12]. *Proved*: Theorems A.34 and A.35.

(B3) *Holomorphic families and meromorphy.* [15, Lemma 4, App. B] use the complex powers Q^{-s} of a positive elliptic $Q \in \Psi_c^1(X)$, and the meromorphy of $\operatorname{Tr}(A(s, z) x^z Q^{-s})$, with at most simple poles on the lines $s = m + n - k$ and $z = -p + 1 - j$. *Proved* in the form (B3′) of Appendix C: any admissible family $Q(s)$ of order $-s$ with $Q(0) = \mathrm{Id}$ may replace Q^{-s} (Proposition C.12 constructs one explicitly); Theorem C.14 proves the two-variable meromorphy and identifies the residues, and Proposition C.15 shows which results of Sections 10–15 hold for such families. In the main text Q^{-s} may be read as any admissible family, with the single change noted after Proposition 15.3. *Quoted (sketched)*: the cusp calculus with holomorphic parameters (Lemmas C.5(1) and C.11).

(B4) *Wodzicki excision.* A short exact sequence of topological algebras whose kernel is topologically H-unital gives a long exact sequence in continuous Hochschild homology [23]. *Proved*: Theorem B.7, with Lemma B.1 for the exactness of the chain spaces. *Quoted*: the facts (F1)–(F3) about nuclear Fréchet spaces [20].

(B5) *Connes' theorem.* The Hochschild–Kostant–Rosenberg map gives $\mathrm{HH}_k(C^\infty(M)) \cong C^\infty(M; \Lambda^k)$ for compact manifolds, possibly with boundary or corners, and similarly for the formal, Laurent, flat and Schwartz coefficient algebras [5]. *Proved*: Theorems B.17 and B.18, and Corollary B.21 for homogeneous functions.

(B6) *Künneth and Morita.* The Künneth formula in the form used, and Morita invariance for matrices, smoothing-kernel algebras and full corners $eM_N(\mathfrak{A})e$ [5, 9]. *Proved*: Proposition B.20 and Corollary B.21 (Künneth); Lemma B.22, Proposition B.24 and Remark B.25 (Morita). *Quoted*: the Eilenberg–Zilber theorem (Background fact B.19); and Morita invariance for the H-unital ideals with bundle coefficients, which is only sketched in Remark B.25. The topological Künneth formula of [8] is not needed.

(B7) *The suspended algebra* [12]. The regularized trace $\overline{\mathrm{Tr}}$ is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}$; its regularization by a holomorphic family carries a local correction (this corrects [15, Prop. 8]); the eta invariant $\eta(A) = (\pi i)^{-1}\overline{\int}\,\mathrm{Tr}(A^{-1}\partial_\tau A)\,d\tau$ is the APS eta invariant for Dirac operators; and $\overline{\eta} = \eta(A_0) + \delta_Q$. *Proved*: Theorems C.23, C.24, C.29 and Proposition C.30(i). *Quoted*: in Theorem C.29, the small-time heat expansion and the regularity of $\eta_B(s)$ at $s = 0$ from [18]; in Proposition C.30(ii), the fact that $(D_t^2 + B^2)^{-s/2}$ is an entire suspended family (Seeley's complex powers with a parameter [19]); the parity argument of Remark C.27 (sketch); and, in Remark C.31, the local cancellation of Bismut–Freed [2].

(B8) *Wodzicki residue* [21, 6]: it is a trace, given by the local formula $(2\pi)^{-n}\int_{S^*M} a_{-n}\,\nu$, independent of Q, and for $n \geq 2$ unique among traces vanishing on smoothing operators. *Proved*: Theorem C.7 and Proposition C.9. *Quoted, not used*: Wodzicki's stronger uniqueness, that every trace on $\Psi^{\mathbb{Z}}$ vanishes on $\Psi^{-\infty}$.

(B9) *The boundary connecting map* [15, Prop. 5]: part (3) of Proposition 8.2, i.e. that the connecting map $\mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}_{k-1}(\mathcal{I}_\sigma)$ is a non-zero universal multiple of the de Rham connecting map on the summand without $d\theta_\partial$ and vanishes on the other summand. MN's proof uses a multiplicative splitting s_0 that we could not reconstruct (Remark 8.3). *Quoted*.

(B10) *Convergence for the Laurent symbol algebras.* The order spectral sequences of $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ converge, so that HH_k is isomorphic to $E^2_{k-n,k}$ by the leading symbol. *Proved*: the algebraic convergence statement (Proposition B.26), completeness through mixed complexes (Lemma B.27), convergence for the closed symbol algebra and for $\mathcal{I}_\sigma$ (Theorem B.30), and the partial result Proposition B.33 for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$. The stage argument used for the b-calculus in [7] fails here, because the cusp Lie derivative $\mathcal{L}_{x^2\partial_x}$ is nilpotent on leading coefficients (Example B.31). *Quoted*: the remaining step, from [15, p. 12] (Remark B.34). (The x-adic spectral sequence of $\mathcal{I}_\partial$ raises a similar but milder question, discussed in the proof of Proposition 9.2.)

In summary, the following remain quoted or only sketched:

- (B9), part (3) of [15, Prop. 5];
- (B10), the last step of the convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ (Remark B.34);
- the uniform calculus on cylinders (Background fact A.4);
- the nuclear-space facts (F1)–(F3) of Appendix B;
- the Eilenberg–Zilber theorem (Background fact B.19);
- Morita invariance for the H-unital ideals with bundle coefficients (sketched in Remark B.25);
- the calculus with holomorphic parameters (Lemmas C.5(1) and C.11, sketched);

- the inputs of the eta identification: the heat expansion and the regularity of η_B in Theorem C.29 [18], the entire suspended family in Proposition C.30(ii) [19], the parity sketch of Remark C.27, and Bismut–Freed in Remark C.31 [2];
- Wodzicki's stronger uniqueness of the residue (not used);
- the density half of [15, Prop. 26] (Remark A.29; not used).

2.6 Corrections and unstated hypotheses

For convenience we list here the points where we disagree with the text of [15]. Each is proved in the place indicated.

1. *Connected boundary.* [15, (16)], Proposition 6, Lemma 12 and Proposition 12 treat $\mathcal{I}_\partial$ as if ∂X were connected. For N components, $\mathrm{HH}^0(\mathcal{I}_\partial) \cong \mathrm{HH}^1(\mathcal{I}_\partial) \cong \mathbb{C}^N$, the double residue splits into N traces, and the boundary map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ has rank $N-1$ (Correction 9.4).
2. *Dimension one.* For $n = 1$ the boundary cosphere bundle Z_j consists of two points. Then the residue traces Tr_σ and $\mathrm{Tr}_{\partial,\sigma}$ split into two (Correction 10.9); the boundary trace Tr_∂ on $\mathcal{I}_\partial$ does not, and the claims "HH^0 is one-dimensional" fail.
3. *The cusp de Rham complex.* The remark after [15, (33)] says that the cusp de Rham complex has b-cohomology. In fact it has an extra summand $H^{k-1}(\partial M)$ (Correction 6.8). This is not used elsewhere.
4. *The cut-off in Lemma 1.* The function ϕ in [15, (26)] must be supported in $(-1, 1)$ (Remark 6.3).
5. *The derivation $\widetilde{D}$.* [15, (21)] and [15, (83)] have opposite signs; (4) is correct (Correction 15.5).
6. *Signs of $\overline{\mathrm{AS}}$ and β in [15, (93)].* With the definitions [15, (90), (91)] the index theorem reads $\mathrm{Ind} = -\overline{\mathrm{AS}} - \frac{1}{2}\overline{\eta} - \beta_Q + \gamma$. The symbolic term in [15, (23), (100)] has the wrong sign (Correction 15.7).
7. *The eta term.* The last line of the proof of [15, Prop. 17] should read $\overline{\eta}(A) = 2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$. The proposition needs Q to be asymptotically translation invariant (Correction 15.9).
8. *The residue on the suspended algebra.* The identification of the residue with $\overline{\mathrm{Tr}}([t, B])$ in [15, Prop. 8] fails, for instance when ∂X is a point; the same parity argument applies to Q_0^{-n} on any boundary (Remark 10.13). We use the residue itself.
9. *The regular value in MN Proposition 8.* [15, Prop. 8] also asserts that the regular value at $s = 0$ of $\overline{\mathrm{Tr}}(BQ_0^{-s})$ is $\overline{\mathrm{Tr}}(B)$. This is false in general: for ∂X a point and $B = \langle D_t \rangle$ the two differ by $-1/2\pi$ (Example C.25). The regular value is $\overline{\mathrm{Tr}}(B) + \gamma'(0)/2\pi$, with a local correction $\gamma'(0)$ (Theorem C.24, Correction C.26). The error propagates to [15, Cor. 2], which acquires the correction term (Corollary 10.14(2)), and to [15, Prop. 17], which becomes $\overline{\eta}(A) = \eta(A_0) + \delta_Q(A)$ with δ_Q local (Proposition 15.8). The correction δ_Q vanishes in the cases used in Section 16, including the interval example (Example 16.2). It is compensated by β_Q in general, since the index does not depend on Q.
10. *The compactification in Appendix A.* The map [15, (130)] gives $[-1, 1]$ the wrong smooth structure at the endpoints for the cusp calculus. One must compactify by $1/z$ (Remark 4.2).

11. *The cusp metric in Appendix B.* The metric $dx^2/x^4 + h/x^2$ of [15, App. B] is a scattering metric, and its Laplacian is not cusp-elliptic. An exact cusp metric is $dx^2/x^4 + h$ (Remark 4.6).

12. *Minor misprints.* We note these as they occur. Examples: $i_{\log r}$ for $i_{\log x}$ in Lemma 13, and A^z for a^z in the proof of Lemma 9. Also the factor 2 in [15, (80)]: in fact $C(0) = [\log x, \log Q]$.

To avoid a clash with the eta invariant, we write β_Q for the functional MN call β in [15, (91)]. Two further points are not corrections to [15] but concern our use of the appendices. For a general admissible family $Q(s)$ in place of Q^{-s}, Proposition 15.3 acquires a coboundary term, which does not affect the index (Remark 15.4). And the chain model of Section 5.1 is not complete for the order filtration, so the convergence arguments must be run in the mixed complexes of Appendix B (Remark 5.7). Figure 4 records, for the main results, which background facts they use and which of the items above affect them.

	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	items
$\mathcal{I}_\sigma$, $\mathcal{A}_\sigma$ (Props. 7.1, 7.2)	●				●	●				●	4
$\mathcal{A}_{\partial,\sigma}$, boundary sequence (Props. 8.1, 8.2)	●			●	●				●	●	–
$\mathcal{I}_\partial$, $\mathcal{I}$ (Props. 9.3, 9.5)	●		●	●	●	●					1
residue functionals (§10)			●				●	●			2, 8, 9, 11
$\mathcal{A}_\partial$, $\mathcal{A}$, Thm. 1.1 (Props. 12.3, 12.5)			●	●					●	●	1, 2
index formula (Lemma 15.2, Thm. 15.6)		●	●			●					5, 6
eta term, reduced formula (Prop. 15.8, Thm. 15.13)			●				●				6, 7, 9

Figure 4: Which main results use which background facts of Section 2.5 (dots), and which items of the list in Section 2.6 concern them (last column). Blue dots: facts proved in Appendices A–C; on an orange column the proof uses a step that is sketched or quoted ((B3): the calculus with holomorphic parameters; (B6): Eilenberg–Zilber and Morita for H-unital ideals with bundles; (B7): the inputs of the eta identification). Red columns are quoted: (B9), part (3) of MN Proposition 5 (Remark 8.3), and (B10), the last step of the convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ (Remark B.34). Everything built on the homology of $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$, or on the boundary map of $0 \to \mathcal{I}_\sigma \to \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma} \to 0$, including the corrected groups of $\mathcal{A}$, rests on them. (The index formula uses the trace computations of Section 12, not the homology groups.) Each row also uses the rows above it, as in Figure 3. Items 3 (not used elsewhere), 10 (the foundations behind (B1)) and 12 (misprints) are not shown.

3 The closed case: Brylinski–Getzler and Wodzicki

Everything in [15] is modelled on the case $\partial X = \varnothing$, which MN recall in their introduction [15, (1)–(5)]. We treat it first, with proofs of the formulas that will be generalized later.

3.1 The symbol algebra and its homology

Let M be a closed manifold of dimension n, let $\Psi^{\mathbb{Z}}(M)$ be the algebra of classical pseudodifferential operators of integral order, and let $\mathcal{A}_M = \Psi^{\mathbb{Z}}(M)/\Psi^{-\infty}(M)$. A quantization identifies $\mathcal{A}_M$ with

formal sums $\sum_{j\le m} a_j$ of smooth functions a_j on $T^*M\setminus 0$, homogeneous of degree j, with a star product

$$a\star b=\sum_{l\ge 0} P_l(a,b),\qquad P_0(a,b)=ab,\qquad P_1(a,b)-P_1(b,a)=-i\{a,b\},$$

where P_l is bidifferential and lowers the total degree by l.

Theorem 3.1 (Brylinski–Getzler, Wodzicki [4, 22])**.** $\mathrm{HH}_k(\mathcal{A}_M)\cong H^{2n-k}(S^*M\times \mathbb{S}_\sigma)$, *and dually* $\mathrm{HH}^k(\mathcal{A}_M)\cong H^k(S^*M\times\mathbb{S}_\sigma)$.

The proof is the model for Sections 7–8. We give it in general form in Section 5; see also [14, Ch. 10], whose argument we follow. In outline:

1. filter Hochschild chains by total order;
2. the associated graded algebra is commutative, so E^1 is given by homogeneous differential forms (HKR and Connes, (B5));
3. the first differential is the Poisson differential $-i\delta$, and the symplectic star operator turns it into the de Rham differential;
4. homogeneous forms of non-zero degree are exact, and forms of degree 0 on $T^*M\setminus 0\cong S^*M\times(0,\infty)$ give $H^*(S^*M)\oplus H^{*-1}(S^*M)\,dr/r$, which is $H^*(S^*M\times\mathbb{S}_\sigma)$;
5. the E^2 term lies on one line, so the spectral sequence degenerates and converges.

Question 3.2. *Where does the circle* $\mathbb{S}_\sigma$ *come from, and why must one take operators of* all *integral orders?*

Answer. The fibres of $T^*M\setminus 0$ are cones, $S^{n-1}\times(0,\infty)_r$. Homogeneous forms see the radial factor only through dr/r, which is invariant under dilation. So the homogeneous de Rham complex is that of $S^*M\times\mathbb{S}_\sigma$, where $\mathbb{S}_\sigma=(0,\infty)/e^{\mathbb{Z}}$ is the quotient of the radial half-line by integral dilations [15, §1]. The class $d\theta_\sigma=[dr/r]$ corresponds to $i_{\log Q}$ (Proposition 11.3). If only operators of order ≤ 0 were allowed, the chains would not be closed under the shifts of order that produce dr/r. The homology is then infinite dimensional; see [15, §15] and Section 17.3. MN make the same point on [15, p. 3]. □

3.2 The residue trace, the exterior derivation and the index

Let $Q\in\Psi^1(M)$ be positive, elliptic and invertible. For $A\in\Psi^m(M)$ the function $\mathrm{Tr}(AQ^{-s})$ is holomorphic for $\mathrm{Re}\,s>m+n$ and extends meromorphically with simple poles. Wodzicki's residue is $\mathrm{Tr}_{\mathrm{R}}(A)=\mathrm{Res}_{s=0}\mathrm{Tr}(AQ^{-s})$ [15, (2)]. It is a trace, independent of Q, vanishes on $\Psi^{-\infty}$, and is given locally by (B8) (Theorem C.7):

$$\mathrm{Tr}_{\mathrm{R}}(A)=(2\pi)^{-n}\int_{S^*M} a_{-n}\,\nu,\qquad \nu=\iota_R\,\frac{\omega^n}{n!}\ \text{up to the orientation convention,}\tag{11}$$

where R is the radial vector field. The *regularized trace* $\widehat{\mathrm{Tr}}(A)$ is the constant term of $\mathrm{Tr}(AQ^{-s})$ at $s=0$. The formula

$$D_{\log Q}(A)=[\log Q,A]=\frac{d}{ds}\Big(Q^sAQ^{-s}\Big)\Big|_{s=0}$$

defines a derivation of $\Psi^{\mathbb{Z}}(M)$, lowering order by one, which is not inner [15, §6].

Lemma 3.3 (Trace defect)**.** *For $A, B \in \Psi^{\mathbb{Z}}(M)$,*

$$\widehat{\mathrm{Tr}}([A,B]) = \mathrm{Tr_R}(A[\log Q, B]) = -\,\mathrm{Tr_R}([\log Q, A]B), \qquad \mathrm{Tr_R}([\log Q, C]) = 0.$$

Proof. For $\operatorname{Re} s$ large, the trace property gives $\mathrm{Tr}([A,B]Q^{-s}) = \mathrm{Tr}(A[B,Q^{-s}])$. Write

$$[B, Q^{-s}] = BQ^{-s} - Q^{-s}B = s\,M(s)Q^{-s}, \qquad M(s) = \frac{B - Q^{-s}BQ^{s}}{s}.$$

Then M is an entire family of fixed order with $M(0) = \frac{d}{ds}(-Q^{-s}BQ^{s})|_{s=0} = [\log Q, B]$. Hence $\mathrm{Tr}([A,B]Q^{-s}) = s\,\mathrm{Tr}(AM(s)Q^{-s})$. The function $\mathrm{Tr}(AM(s)Q^{-s})$ has at most a simple pole at $s=0$, with residue $\mathrm{Tr_R}(AM(0))$. So the left side is regular at $s=0$, with value $\mathrm{Tr_R}(A[\log Q, B])$. For the last identity take $A = \mathrm{Id}$: the function $\mathrm{Tr}([\mathrm{Id}, B]Q^{-s})$ vanishes identically, so $\mathrm{Tr_R}([\log Q, B]) = 0$. Applying this to $C = AB$ gives $\mathrm{Tr_R}([\log Q, A]B) + \mathrm{Tr_R}(A[\log Q, B]) = 0$. □

Proposition 3.4 (The index functional, MN (3)–(5))**.** *The connecting map of $0 \to \Psi^{-\infty} \to \Psi^{\mathbb{Z}} \to \mathcal{A}_M \to 0$ sends the trace to the Hochschild 1-cocycle $\mathrm{IF}(A,B) = \mathrm{Tr_R}(A[\log Q, B])$ on $\mathcal{A}_M$. If $A \in \Psi^m(M)$ is elliptic with parametrix B, then*

$$\mathrm{Ind}(A) = \mathrm{Tr}([A,B]) = \mathrm{Tr_R}\big(A[\log Q, A^{-1}]\big).$$

Proof. The trace on $\Psi^{-\infty}$ extends to the functional $\widehat{\mathrm{Tr}}$ on $\Psi^{\mathbb{Z}}$. Its Hochschild coboundary $(b\widehat{\mathrm{Tr}})(A,B) = \widehat{\mathrm{Tr}}([A,B])$ vanishes if A or B is smoothing, because $\mathrm{Tr_R}$ vanishes on $\Psi^{-\infty}$. By Lemma 3.3 it equals $\mathrm{Tr_R}(A[\log Q, B])$. Lemma 5.2 below then identifies it with the image of $[\mathrm{Tr}]$ under the connecting map. For the index: $1 - BA$ and $1 - AB$ are smoothing, $\mathrm{Ind}\,A = \mathrm{Tr}(1-BA) - \mathrm{Tr}(1-AB) = \mathrm{Tr}([A,B])$ (Fedosov), and $\mathrm{Tr}([A,B]) = \widehat{\mathrm{Tr}}([A,B])$ because $[A,B]$ is smoothing. □

Since $A[\log Q, A^{-1}] = -[\log Q, A]A^{-1}$, we also have

$$\mathrm{Ind}(A) = -\,\mathrm{Tr_R}\big([\log Q, A]A^{-1}\big) = -\,\mathrm{Tr_R}\big(A^{-1}[\log Q, A]\big).$$

Keep this sign in mind; it is the source of Correction 15.7.

3.3 Example: the circle

Let $M = S^1$, with modes $e_k = e^{ik\theta}$, and let Q act by $Qe_k = \langle k\rangle e_k$, where $\langle k\rangle = (1+k^2)^{1/2}$.

Homology. $S^*S^1 = S^1 \times \{\pm 1\}$, so $S^*S^1 \times \mathbb{S}_\sigma$ is two tori, and Theorem 3.1 gives

$$\dim \mathrm{HH}_k(\mathcal{A}_{S^1}) = 2,\ 4,\ 2 \qquad (k = 0,1,2).$$

In degree 0 there are *two* independent residue traces,

$$\mathrm{Tr}^{\pm}_{\mathrm{R}}(A) = \frac{1}{2\pi}\int_0^{2\pi} a_{-1}(\theta, \pm 1)\, d\theta,$$

one for each direction of ξ. Their sum is $\mathrm{Tr_R}$. This is the $n = 1$ phenomenon that will reappear at the boundary (Correction 10.9).

Index. Let Π be the orthogonal projection onto $\operatorname{span}\{e_k : k \geq 0\}$; it is a classical operator of order 0. Put

$$A = e^{i\theta}\Pi + (1 - \Pi), \qquad B = e^{-i\theta}\Pi + (1 - \Pi).$$

Then $Ae_k = e_{k+1}$ for $k \geq 0$ and $Ae_k = e_k$ for $k < 0$. So A is injective and its range misses exactly e_0, whence $\operatorname{Ind} A = -1$. A direct check gives $BA = 1$ and $1 - AB = R$, where R is the rank one operator $e_0 \mapsto e_0 - e_{-1}$. Therefore $\operatorname{Tr}([A, B]) = -\operatorname{Tr} R = -1$, as Fedosov's formula requires. Now compute the residue formula. For $k \geq 1$,

$$A[\log Q, B]e_k = A\big(\log\langle k-1\rangle - \log\langle k\rangle\big)e_{k-1} = \big(\log\langle k-1\rangle - \log\langle k\rangle\big)e_k = \Big(-\frac{1}{k} + O(k^{-2})\Big)e_k.$$

For $k \leq -1$ the operator $A[\log Q, B]$ vanishes on e_k, and e_0 contributes a finite rank term. Hence

$$\operatorname{Tr}_{\mathrm{R}}\big(A[\log Q, B]\big) = \operatorname{Res}_{s=0} \sum_{k\geq 1}\Big(-\frac{1}{k} + O(k^{-2})\Big)\langle k\rangle^{-s} = -1 = \operatorname{Ind} A.$$

In symbol terms, $A[\log Q, B]$ has $a_{-1}(\theta, \xi) = -\xi^{-1}$ for $\xi > 0$ and 0 for $\xi < 0$. Only $\operatorname{Tr}_{\mathrm{R}}^{+}$ contributes. Figure 5 shows both computations.

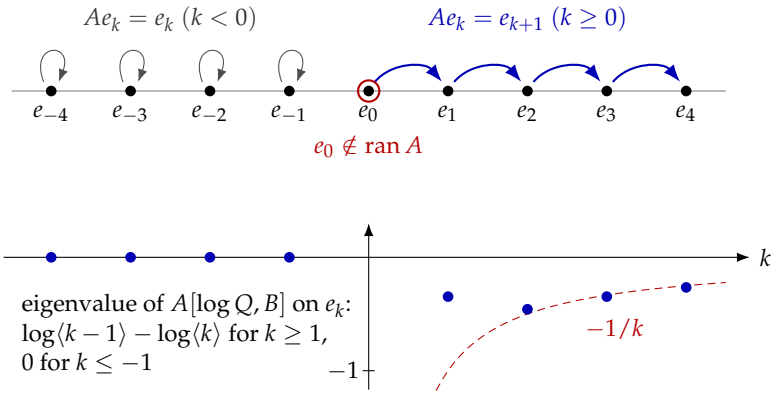


Figure 5: The example of Section 3.3, $A = e^{i\theta}\Pi + (1 - \Pi)$. Top: A shifts the modes e_k, $k \geq 0$, up by one and fixes the others, so A is injective, its range misses e_0, and $\operatorname{Ind} A = -1$. Bottom: $A[\log Q, B]$ is diagonal on e_k, $k \neq 0$, with eigenvalue $-1/k + O(k^{-2})$ for $k \geq 1$ and 0 for $k \leq -1$ (e_0 gives a finite rank term). The tail $-1/k$ on one side only gives $\operatorname{Tr}_{\mathrm{R}}(A[\log Q, B]) = -1 = \operatorname{Ind} A$: only $\operatorname{Tr}_{\mathrm{R}}^{+}$ contributes, and the opposite order $\operatorname{Tr}_{\mathrm{R}}(A^{-1}[\log Q, A])$ gives $+1$.

Discussion. The example fixes the sign convention for everything that follows:

$$\operatorname{Ind} A = \operatorname{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big), \qquad \textit{not} \qquad \operatorname{Tr}_{\mathrm{R}}\big(A^{-1}[\log Q, A]\big).$$

The second expression equals $+1$ here.

4 The cusp calculus

This section collects the facts about cusp operators that [15] uses. It follows [15, App. A–C], [10] and [13]. It is an informal tour: the proofs are in Appendix A, which constructs the calculus directly on the cylinder $\mathbb{R}_t \times \partial X$ and does not use the triple space. We begin with the half-line, where everything is explicit.

4.1 Cusp structures and cusp vector fields

A *cusp structure* on X is a choice of boundary defining function x, modulo two kinds of change: multiplication by a positive constant on each boundary component, and addition of a term in $x^2C^\infty(X)$ [15, App. A]. Equivalently, it is a trivialization of the normal bundle of ∂X, up to constant multiples on each component. The *cusp vector fields* are [15, (6), (114)]

$$\mathcal{V}_c(X) = \{V \in C^\infty(X; TX) : \ Vx \in x^2C^\infty(X)\}.$$

Lemma 4.1 (MN Lemma 20 and Corollary 4)**.** *In coordinates (x, y) near a boundary point, $V \in \mathcal{V}_c(X)$ if and only if $V = a\,x^2\partial_x + \sum_j b_j\partial_{y_j}$ with a, b_j smooth. If x' is another boundary defining function, then $\mathcal{V}_c$ is the same for x and x' exactly when x' defines the same cusp structure.*

Proof. Write $V = \alpha\partial_x + \sum \beta_j\partial_{y_j}$. Then $Vx = \alpha$, so $V \in \mathcal{V}_c$ if and only if $\alpha = ax^2$. For the second statement write $x' = e^f x$, with f smooth. Then

$$(x^2\partial_x)x' = x^2e^f(1 + x\partial_x f), \qquad \partial_{y_j}x' = x\,e^f\partial_{y_j}f.$$

The first expression lies in $x'^2C^\infty = x^2C^\infty$. The second lies in x^2C^∞ if and only if $\partial_y f = O(x)$ at ∂X, that is, if and only if $f|_{\partial X}$ is locally constant. In that case $x' = cx + O(x^2)$ on each component, which is the stated equivalence. □

The cusp vector fields form a Lie algebra and a $C^\infty(X)$-module. Near the boundary they are locally free on $x^2\partial_x, \partial_{y_j}$. Hence there is a vector bundle ${}^cTX$ with $\mathcal{V}_c = C^\infty(X; {}^cTX)$. Its dual is ${}^cT^*X$, with local frame $dx/x^2, dy_j$. Over the interior ${}^cTX = TX$, and globally ${}^cTX \cong TX$ as abstract bundles. So the cosphere bundle $Y = {}^cS^*X = ({}^cT^*X \setminus 0)/\mathbb{R}^+$ is diffeomorphic to S^*X. Its boundary is $Z = {}^cS^*_{\partial X}X$, the sphere bundle of ${}^cT^*_{\partial X}X = T^*\partial X \times \mathbb{R}_\tau$. If $n \geq 2$, each component Z_j is an S^{n-1}-bundle over H_j, hence connected. If $n = 1$, each Z_j is the pair of points $\tau = \pm 1$.

MN also show that any two cusp structures are related by a diffeomorphism isotopic to the identity [15, Lemma 19]. So the algebras for different structures are isomorphic. The proof integrates a vector field transverse to the boundary that interpolates between the two defining functions.

4.2 The model: the half-line

Let $X = [0, 1)_x$. In the variable $t = 1/x \in (1, \infty)$, the cusp vector field is $x^2\partial_x = -\partial_t$ by (8). So the cusp structure makes the end $x \to 0$ into a *cylindrical end* $t \to \infty$ of a complete manifold. The dictionary:

on $[0, 1)_x$	on $(1, \infty)_t$
$C^\infty([0, 1))$	smooth functions with expansions in powers of $1/t$
$\dot{C}^\infty([0, 1))$	functions rapidly decreasing as $t \to \infty$
$x^{-\mathbb{Z}}C^\infty([0, 1))$	functions with expansions $\sum_{j \geq -p} c_j t^{-j}$
cusp density dx/x^2	dt
exact cusp metric dx^2/x^4	dt^2
$\sum_j a_j(x)(x^2D_x)^j$	$\sum_j a_j(1/t)(-D_t)^j$
$\mathrm{In}(P, \tau)$	$\sum_j a_j(0)\tau^j$

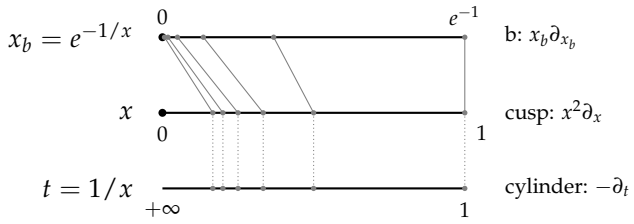


Figure 6: The half-line in three coordinates. The cusp vector field $x^2\partial_x$ is $-\partial_t$ in $t = 1/x$, the generator of translations of a cylindrical end, and it is the b-vector field $x_b\partial_{x_b}$ in the transcendental coordinate $x_b = e^{-1/x}$ (Section 4.6). The gray marks are the points $t = 1, 2, \dots, 6$: equally spaced in t, at $x = 1/t$, and at $x_b = e^{-t}$, where they crowd exponentially towards 0. All powers of x_b vanish to infinite order in x.

A cusp differential operator is thus a differential operator on the half-line $(1,\infty)_t$ whose coefficients have expansions in $1/t$. Its indicial family is its "value at $t = \infty$", Fourier transformed in t. The pseudodifferential operators near the boundary are the operators

$$Au(t) = \frac{1}{2\pi}\iint e^{i(t-t')\xi}\, a\big(1/t, \xi\big)\, u(t')\, dt'\, d\xi,$$

where $a(x,\xi)$ is a classical symbol in ξ, smooth in x up to $x = 0$. The kernel is smooth off the diagonal and rapidly decreasing in $t - t'$ [15, Props. 22, 23]. The translation invariant operators (with $a = a(\xi)$) form the suspended algebra $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(\mathrm{pt})$, the convolution operators by kernels that are classical conormal at 0 and Schwartz at infinity. It is *commutative*.

Remark 4.2 (The compactification in [15], (130)). MN identify $\mathbb{R}$ with $(-1,1)$ via $z \mapsto z/(1+z^2)^{1/2}$. Near $z = +\infty$ this gives $1 - z/\langle z\rangle = \frac{1}{2}z^{-2} + O(z^{-4})$, so the smooth structure it induces on $[-1,1]$ is that of ρ^2 with $\rho = 1/z$. In that structure ∂_z is not a cusp vector field, and translations do not act smoothly. The identification [15, Prop. 22] and the description of Ψ_{sus} on [15, p. 44] need the compactification by $\rho = 1/|z|$. For example one can use $z \mapsto \frac{2}{\pi}\arctan z$, for which $1 - \frac{2}{\pi}\arctan z = \frac{2}{\pi}\arctan(1/z)$ is a smooth defining function in ρ. Then $\partial_z = -\rho^2\partial_\rho$ is a cusp vector field, and the translation $z \mapsto z + c$ acts by $\rho \mapsto \rho/(1 + c\rho)$, which is smooth. With this change the statements of [15, App. A] hold as written.

Example 4.3. The operator $P = x^2 D_x + c$ on $[0,1)$ is $-D_t + c$, and $\mathrm{In}(P,\tau) = \tau + c$. Its inverse on $\mathbb{R}_t$, when $c \notin \mathbb{R}$, is convolution with an exponentially decreasing kernel. The indicial family is invertible for all real τ precisely when $c \notin \mathbb{R}$. This is the cusp version of "no indicial roots on the real line". There is no condition off the real axis; compare Section 18.

4.3 Kernels and the cusp double space

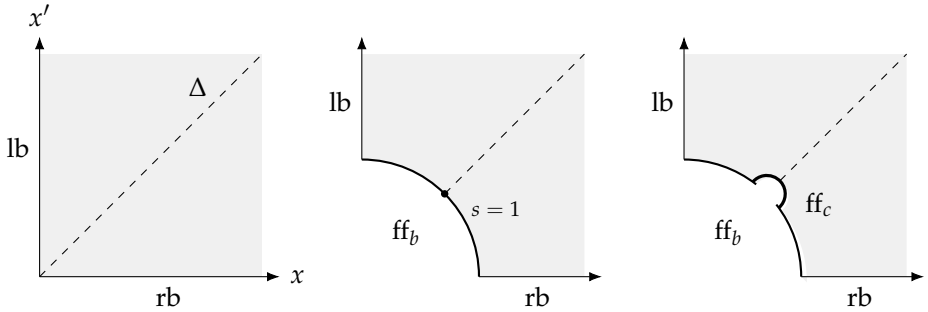
Globally, kernels are described on a blown-up double space [15, App. A]. First blow up the corners $H \times H$ of X^2 to obtain the b-double space X^2_b, whose front face has the coordinate $s = x/x'$. The cusp structure picks out the point $s = 1$ on each fibre of the b-front face, more precisely the fibre diagonal $\Phi_2(H) = \{s = 1\} \times H^2$. Blowing these up gives

$$X^2_c = [X^2_b;\, \{\Phi_2(H)\}_H], \qquad \mu = \frac{1}{x} - \frac{1}{x'} = t - t' \quad \text{on the new front face } \mathrm{ff}_c.$$

The interior of ff_c is $\mathbb{R}_\mu \times H^2$; see Figure 7. The lifted diagonal meets the boundary of X^2_c only in ff_c, at $\mu = 0$ [15, Lemma 22]. Then $\Psi^m_c(X)$ consists of the kernels on X^2_c that are classical conormal of

order m at the lifted diagonal, smooth up to ff_c, and rapidly vanishing at all other boundary faces [15, Def. 2]. In particular their restriction to ff_c is rapidly decreasing as $|\mu| \to \infty$. The kernels in $\Psi_c^{-\infty}$ that vanish to infinite order also at ff_c are those of $\mathcal{I}$ [15, Def. 1]. Proofs: Lemma A.6 (coordinates on X_c^2), Proposition A.17 (agreement of this description with the construction of Appendix A) and Lemma A.19 (the ideal $\mathcal{I}$). There kernels are normalized by the right cusp density ν_c, equal to $x^{-2}dx\,\nu_\partial$ near ∂X (Section A.1); MN use half-densities [15, (118)]. Passing from one to the other multiplies the kernel by $(\nu_c(z')/\nu_c(z))^{1/2}$. Near ff_c this is $1 - x\mu$ times a smooth positive function of (y, y'), and at the other faces it grows at most polynomially, where the kernels vanish to infinite order. So the two conventions define the same spaces of operators.

Cut-offs and pseudolocality. We often multiply operators by cut-off functions. Pseudolocality at the boundary, in the form $\phi A\psi \in \mathcal{I}$ when $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \varnothing$, holds only for cut-offs that are functions of x alone (Lemma A.18(1)). If ϕ and ψ depend on y, then $\phi A\psi$ is smoothing but restricts to ff_c as $\phi|_{\partial X} N(A) \psi|_{\partial X}$, which need not vanish, so it is in $\Psi_c^{-\infty}$ but not in $\mathcal{I}$. All the cut-offs used in the main text to localize near or away from ∂X are functions of x.



(a) X^2 (b) X_b^2 (c) X_c^2

Figure 7: Near a corner $H \times H$ of X^2 (suppressing H^2). (b) The b-blow-up replaces the corner by the front face ff_b with coordinate $s = x/x'$; the diagonal meets it at $s = 1$. (c) The cusp double space blows up the fibre diagonal $\{s = 1\}$ once more. The new face ff_c carries the coordinate $\mu = 1/x - 1/x' = t - t' \in [-\infty, \infty]$, the difference of the suspension variables, and cusp kernels restrict to it as convolution kernels in μ, rapidly decreasing at $\mu = \pm\infty$. b-kernels are instead smooth on ff_b and convolution kernels in $\log s$.

4.4 Symbols, quantization and the star product

The principal symbol gives exact sequences [15, (11)]

$$0 \to \Psi_c^{m-1}(X) \to \Psi_c^m(X) \xrightarrow{\sigma_m} \mathcal{G}^m({}^cT^*X) \to 0,$$

where $\mathcal{G}^m({}^cT^*X)$ is the space of smooth functions on ${}^cT^*X \setminus 0$ that are homogeneous of degree m. The symbol of a product is the product of the symbols, and $\sigma_{m+m'-1}([A, B]) = -i\{\sigma_m(A), \sigma_{m'}(B)\}$ with the bracket (7) (Proposition A.21). A global quantization $x^{-\mathbb{Z}} S^{\mathbb{Z}}({}^cT^*X) \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ exists [15, Prop. 24] (Proposition A.22). It is linear, not multiplicative. It induces the isomorphisms (1) for $\mathcal{I}_\sigma$ and $\mathcal{A}_\sigma$ [15, (13)], which are isomorphisms of vector spaces, and transporting the product of $\mathcal{A}_\sigma$ through them gives a star product

$$a \star b = \sum_{j \geq 0} P_j(a, b), \qquad P_0(a, b) = ab, \qquad P_1(a, b) - P_1(b, a) = -i\{a, b\} \qquad \text{[15, (137)–(138)]},$$

with P_j bidifferential of total homogeneity $-j$. In local coordinates with the left quantization, $P_j(a,b) = \sum_{|\alpha|=j} (\alpha!)^{-1} \partial_\zeta^\alpha a \, {}^cD^\alpha b$, where $\zeta = (\tau, \eta)$ and ${}^cD = (x^2 D_x, D_y)$ up to lower order terms in the x-derivatives. Only P_0 and the antisymmetric part of P_1 will matter.

4.5 The indicial family, the suspended algebra and the twist

For a compact manifold H without boundary, $\Psi^m_{\mathrm{sus}}(H)$ is the algebra of pseudodifferential operators on $\mathbb{R}_t \times H$ that are invariant under translation in t and whose convolution kernels decrease rapidly as $|t| \to \infty$ [12]. Through (10) it is the same as the algebra of families $\mathbb{R} \ni \tau \mapsto B(\tau) \in \Psi^m(H)$ that are classical symbols jointly in τ and in the cotangent variables of H. The smoothing part is $\Psi^{-\infty}_{\mathrm{sus}}(H) \cong \mathcal{S}(\mathbb{R}) \widehat{\otimes} \Psi^{-\infty}(H)$. The indicial map gives [15, (12)]

$$0 \to x\Psi^m_c(X) \to \Psi^m_c(X) \xrightarrow{\mathrm{In}} \Psi^m_{\mathrm{sus}}(\partial X) \to 0.$$

More precisely, a choice of normal fibration identifies a collar with $[0,\epsilon) \times \partial X$. The Taylor series T of the kernel at ff_c then gives [15, Prop. 25]

$$\mathsf{T}: \; x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_c(X)/x^\infty\Psi^{\mathbb{Z}}_c(X) \;\cong\; \Psi^{\mathbb{Z}}_{\mathrm{sus}}(\partial X)((x)), \qquad A \sim \sum_{j \geq -p} x^j A_{(j)}. \tag{12}$$

Here $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(\partial X)((x)) = \bigcup_m \Psi^m_{\mathrm{sus}}(\partial X)((x))$ consists of the Laurent series *whose coefficients have bounded order*, the powers of x are written *on the left*, and MN write A_{-j} for our $A_{(j)}$ [15, (53)]. The product on the right is not the naive one, because x does not commute with Ψ_{sus}; with the twisted product of Proposition A.27, T is an algebra isomorphism (Proposition A.27; the exact indicial sequence is Proposition A.25).

Lemma 4.4 (The cusp twist)**.** *Let $g \in \Psi^m_{\mathrm{sus}}(\partial X)$, acting on functions of (t,y) with $t = 1/x$ near the boundary. Write g' for the element with $\mathrm{In}(g',\tau) = \partial_\tau \mathrm{In}(g,\tau)$. Then:*

1. *$[t, g] = -i\, g'$;*
2. *$[x, g] = i\, xg'x = i\, x^2 g' + O(x^3)$, and in particular $[x, \partial_t] = x^2$;*
3. *$[\log x, g] = -\sum_{k \geq 1} \frac{1}{k}\, x^k \, \mathrm{ad}_t^k(g)$, where $\mathrm{ad}_t^k(g)$ has indicial family $(-i\partial_\tau)^k \mathrm{In}(g)$.*

Parts (1) and (2) are also proved in Lemma A.5 and Proposition A.26. The associated graded algebra of $\Psi_{\mathrm{sus}}(\partial X)((x))$ for the x-filtration is commutative modulo $\Psi(\partial X)$. Its Poisson bracket is $\{x, \tau\} = -x^2$, that is, $\{t, \tau\} = 1$.

Proof. (1) follows from (10), since $[t, g]$ has kernel $(t - t')\kappa(t - t')$. For (2),

$$[t^{-1}, g] = t^{-1}g - gt^{-1} = -t^{-1}(tg - gt)t^{-1} = -t^{-1}[t, g]t^{-1} = i\, t^{-1} g' t^{-1}.$$

Moving the right-hand factor x past g' costs a further power of x, by the same identity. For $g = \partial_t$ we have $\mathrm{In}(\partial_t, \tau) = -i\tau$, so $g' = -i$ and $[x, \partial_t] = x^2$. For (3), the kernel of $[\log x, g] = -[\log t, g]$ is $-(\log t - \log t')\kappa(t - t')$. With $s = t - t'$,

$$\log t - \log t' = -\log\Big(1 - \frac{s}{t}\Big) = \sum_{k \geq 1} \frac{1}{k}\, \frac{s^k}{t^k},$$

and $t^{-k} = x^k$ acts on the left. The kernel $s^k \kappa(s)$ is that of $\mathrm{ad}_t^k(g)$, so its indicial family is $(-i\partial_\tau)^k \mathrm{In}(g)$ by (10). The bracket of the leading symbols follows from (2): $\sigma([x, \tau]) = -i\{x, \tau\}$ and $[x, \tau]$ has leading term ix^2, so $\{x, \tau\} = -x^2$. Then $\{t, \tau\} = -x^{-2}\{x, \tau\} = 1$. □

Discussion. **What the twist means.** In $\mathcal{I}_\partial$ and $\mathcal{A}_\partial$ the "Laurent variable" x is the function $1/t$ on the suspension line, and the coefficients are functions of D_t. So these algebras are quantizations of functions of (t, τ) near $t = +\infty$, and t and τ are canonically conjugate. In a Laurent series in x, a derivative in τ costs one power of x. This is why the associated graded algebra is commutative and why the first differential of the x-adic spectral sequence is a Poisson differential, exactly as for the order filtration (Section 9). In the b-calculus the corresponding relation is $x^{-1}Ax = \alpha(A)$, where α is the complex shift $\lambda \mapsto \lambda - i$. This automorphism does *not* become trivial on the associated graded algebra; see Section 18. Figure 8 contrasts the two.

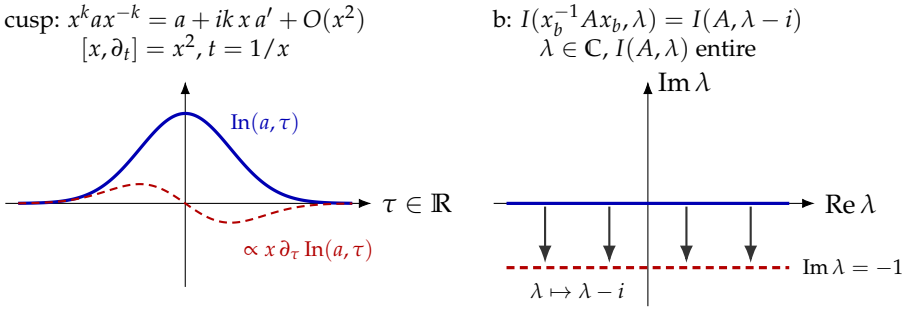


Figure 8: Conjugation by powers of x. Left, the cusp case (Lemmas 4.4 and 9.1): for a of x-degree zero, $x^k a x^{-k} = a + ik\,x\,a' + O(x^2)$, where a' has indicial family $\partial_\tau \mathrm{In}(a, \tau)$. The correction (dashed) is a derivative in τ and costs one power of x, so it disappears in the associated graded algebra, which is commutative with Poisson bracket $\{t, \tau\} = 1$. Right, the b-case: conjugation by x_b evaluates the indicial family on the line $\mathrm{Im}\,\lambda = -1$. This is a finite shift, an automorphism that survives in the associated graded algebra (Section 18).

4.6 The transcendental blow-up and the b-calculus

The b-vector fields are spanned by $x_b\partial_{x_b}$ and ∂_y. Put [15, (143)]

$$x_b = e^{-1/x}, \qquad \text{so that} \qquad \frac{dx_b}{dx} = \frac{x_b}{x^2}, \qquad x_b\partial_{x_b} = x^2\partial_x.$$

Thus b-differential operators in (x_b, y) pull back to cusp differential operators in (x, y). A change $x_b' = a(y)x_b + O(x_b^2)$ of the b-defining function changes x to $x' = x/(1 - x\log a) + O(x^\infty) = x + x^2\log a + O(x^3)$. If a is constant on each component, this is the same cusp structure. In general the two structures are related by Lemma 4.1 up to the isotopy of [15, Lemma 19]. MN quote that b-pseudodifferential operators lift to cusp operators and that the lifts are dense [15, Prop. 26]. The first statement is Proposition A.28; the density is not proved here and not used (Remark A.29).

Two features of the lift explain why it is useful.

1. The b-indicial family $I(A, \lambda)$ of [18] becomes the cusp indicial family with $\tau = \lambda \in \mathbb{R}$ (Section 2.2).

2. A b-operator is $A_0 + x_b A_1 + \ldots$, with A_0 dilation invariant. After the lift, A_0 is translation invariant in t, and $x_b = e^{-1/x}$ vanishes to infinite order in x. So the Laurent series (12) of a lifted b-operator is just $\mathrm{In}(A)$, with no higher terms.

A b-operator whose indicial family is invertible on $\mathbb{R}$ (that is, Fredholm on L_b^2) therefore lifts to a fully elliptic cusp operator. The cusp index formula applies to it, and its eta term is the b-eta invariant of

the indicial family, up to the local correction δ_Q of Proposition 15.8. The b-Laurent algebra $x_b^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}$ would require invertibility for *all* $\lambda \in \mathbb{C}$ (Section 18). This is the "problem with completeness" mentioned in [15, p. 3]. Note also that Laurent series in x_b and Laurent series in $x = 1/\log(1/x_b)$ are different algebras, so the lift does not identify the two Hochschild computations [7].

Question 4.5. *Why does the cusp calculus avoid the completeness problem, if the two calculi have isomorphic norm closures [15, p. 3]?*

Answer. The norm closures see only the boundary behaviour on the real line. The C^∞ algebras used for Hochschild homology see more. In the small b-calculus, indicial families are *entire* functions of $\lambda \in \mathbb{C}$. Inverting an operator modulo $x_b^\infty\Psi_b^{-\infty}$ means inverting its Taylor series at ff_b, so the inverse of the indicial family must again be entire. Hence $I(A,\lambda)$ must be invertible for *every* $\lambda \in \mathbb{C}$ [7, Lemma "invertible"]. Dirac operators never satisfy this. In the cusp calculus conjugation by x acts through the twist of Lemma 4.4, which only adds terms of higher order in x. So only invertibility on $\mathbb{R}$ is needed. □

4.7 Ellipticity, cusp metrics and complex powers

An element of $x^{-p}\Psi_c^m(X)$ is *elliptic* if its principal symbol (after removing x^{-p}) is invertible on ${}^cT^*X \setminus 0$. It is *fully elliptic* if, in addition, its leading indicial family is invertible for all real τ. In the Laurent setting this is invertibility in $\mathcal{A}$ with matched orders: $A \in x^{-p}\Psi_c^m(X)$ is fully elliptic if and only if there is $B \in x^p\Psi_c^{-m}(X)$ with $AB - 1, BA - 1 \in \mathcal{I}$ (Theorem A.34, (B2)).

Remark 4.6 (Exact cusp metrics). [15, App. B] writes an exact cusp metric as $dx^2/x^4 + h/x^2$. The orthonormal frames of that metric are spanned by $x^2\partial_x$ and $x\partial_y$, which are the *scattering* vector fields. Its Laplacian is a cusp differential operator, but its cusp symbol $\tau^2 + x^2|\eta|^2$ degenerates at $x = 0$. So it is not cusp-elliptic, and $(\Delta + 1)^{-s/2}$ is not a cusp family of order $-s$. The metric compatible with $\mathcal{V}_c$ is

$$g = \frac{dx^2}{x^4} + h = dt^2 + h \quad \text{near } \partial X,$$

a metric with cylindrical ends. Its Laplacian has cusp symbol $\tau^2 + |\eta|_h^2$. This is the metric to use in [15, §5].

For a positive invertible elliptic $Q \in \Psi_c^1(X)$, such as $(\Delta_g + 1)^{1/2}$ for the metric of Remark 4.6, the complex powers Q^{-s} form an entire family of order $-s$ (B3). Appendix C avoids the spectral theory of Q: any admissible family $Q(s)$ of order $-s$ with $Q(0) = \mathrm{Id}$ can be used (Proposition C.12, Definition C.10). The formal logarithm

$$\log Q = \frac{d}{ds} Q^s \Big|_{s=0}$$

is not in the calculus, but commutators with it are. If Q is asymptotically translation invariant, $\mathrm{In}(Q) = Q_0 = (D_t^2 + \Delta_{\partial X} + 1)^{1/2}$, and this Q_0 is the regularizer used in [15, (60)].

5 The Hochschild homology toolkit

This section collects the homological algebra used in [15], in the form in which we use it. Several statements are adapted from the companion notes [7] and from [14, Ch. 10]; we say so in each case.

5.1 Chains of topological algebras

For a Fréchet algebra $\mathfrak{A}$, the space of k-chains is the completed projective tensor product $C_k(\mathfrak{A}) = \mathfrak{A}^{\widehat{\otimes}(k+1)}$ [15, (36)]. Most of our algebras are increasing unions $\mathfrak{A} = \bigcup_N \mathfrak{A}_N$ of nuclear Fréchet spaces with $\mathfrak{A}_N\mathfrak{A}_M \subset \mathfrak{A}_{N+M}$; for example $\mathfrak{A}_N = x^{-N}\Psi_c^N(X)$. For these we take

$$C_k(\mathfrak{A}) = \varinjlim_N \mathfrak{A}_N^{\widehat{\otimes}(k+1)}.$$

Ideals and quotients carry the induced filtrations [7, §"The chain model"]. The basic examples are

$$C^\infty(M)^{\widehat{\otimes}(k+1)} = C^\infty(M^{k+1}), \qquad \dot{C}^\infty(M)^{\widehat{\otimes}(k+1)} = \dot{C}^\infty(M^{k+1})$$

for compact manifolds with boundary [15, proof of Lemma 1]. Here $\dot{C}^\infty(M^{k+1})$ denotes functions vanishing to infinite order at all boundary faces of the manifold with corners M^{k+1}. Since nuclear completed tensor products preserve short exact sequences of Fréchet spaces [20], for a closed ideal $J \subset \mathfrak{A}$ the sequence

$$0 \to \mathcal{K}_\bullet \to C_\bullet(\mathfrak{A}) \to C_\bullet(\mathfrak{A}/J) \to 0 \tag{13}$$

is exact. Here $\mathcal{K}_k$ is the closed span of the chains with at least one entry in J [7, Lemma "Exactness of chain spaces"]; see Lemma B.1 for a proof that needs no continuous section of $\mathfrak{A} \to \mathfrak{A}/J$.

5.2 H-unitality and excision

An algebra J is *H-unital* if its bar complex $(C_\bullet(J), b')$ is acyclic [23]. It is *topologically* H-unital if there is a continuous contraction E with $b'E + Eb' = 1$. A unital algebra is H-unital, with $E(a_0 \otimes \cdots \otimes a_k) = 1 \otimes a_0 \otimes \cdots \otimes a_k$.

Background fact 5.1 (Wodzicki's excision theorem, (B4); proved in Theorem B.7)**.** If $J \subset \mathfrak{A}$ is a closed ideal that is topologically H-unital, then $C_\bullet(J) \to \mathcal{K}_\bullet$ is a quasi-isomorphism. Hence (13) gives a long exact sequence

$$\cdots \to \mathrm{HH}_k(J) \to \mathrm{HH}_k(\mathfrak{A}) \to \mathrm{HH}_k(\mathfrak{A}/J) \xrightarrow{\partial} \mathrm{HH}_{k-1}(J) \to \cdots.$$

To compute $\partial[c]$, lift the cycle c to a chain $\tilde{c}$ of $\mathfrak{A}$. Then $b\tilde{c} \in \mathcal{K}$, and $\partial[c]$ is its class.

The proof, by filtering $\mathcal{K}$ by the position of the entries from J and contracting the graded pieces with the bar contraction of J, is Theorem B.7; Example B.3 shows what goes wrong without H-unitality. The topological version needs the contraction to be continuous, and this is what Lemma 5.4 provides.

Lemma 5.2 (Connecting maps on cocycles)**.** *Under the hypotheses of 5.1, let φ be a $(k-1)$-cocycle on J. Let $\hat{\varphi}$ be a continuous $(k-1)$-cochain on $\mathfrak{A}$ that extends φ and such that $b\hat{\varphi}$ vanishes whenever one of its arguments lies in J. Then $b\hat{\varphi}$ descends to a k-cocycle on $\mathfrak{A}/J$ that represents $\partial^*[\varphi]$; that is, $\langle \varphi, \partial[c]\rangle = \langle b\hat{\varphi}, c\rangle$.*

Proof. This is [7, Lemma "Connecting maps on cocycles"]. The cochain $\hat{\varphi}$ restricted to $\mathcal{K}$ is a cocycle there, because $b\hat{\varphi}$ vanishes on $\mathcal{K}$. It restricts to φ on $C(J)$, so it represents $[\varphi]$ under the quasi-isomorphism. Hence $\langle \varphi, \partial[c]\rangle = \langle \hat{\varphi}, b\tilde{c}\rangle = \langle b\hat{\varphi}, \tilde{c}\rangle$, and the last pairing depends only on the image c of $\tilde{c}$. □

This lemma is how every boundary map in [15] is computed: extend a trace from an ideal by analytic continuation, and compute the coboundary of the extension. MN's formula (17) is the case of the traces on $\mathcal{I}_\sigma$ and $\mathcal{I}_\partial$.

5.3 Derivations

Lemma 5.3 (MN §6). *Let D be a derivation of $\mathfrak{A}$ and $i_D(a_0 \otimes a_1 \otimes \cdots \otimes a_k) = a_0 D(a_1) \otimes a_2 \otimes \cdots \otimes a_k$. Then $bi_D + i_D b = 0$. If $D = [h, \cdot\,]$ with $h \in \mathfrak{A}$ ($\mathfrak{A}$ unital), then*

$$i_D = bR_h - R_h b, \qquad R_h(a_0 \otimes \cdots \otimes a_k) = a_0 h \otimes a_1 \otimes \cdots \otimes a_k.$$

So i_D induces a map $\mathrm{HH}_k \to \mathrm{HH}_{k-1}$ that vanishes for inner derivations. Dually, $b(i_D\varphi) = -i_D(b\varphi)$ for cochains.

Proof. Both identities are checked on elementary tensors. We do $k = 2$; the general case is the same bookkeeping. On the one hand,

$$i_D b(a_0 \otimes a_1 \otimes a_2) = a_0 a_1 D(a_2) - a_0 D(a_1 a_2) + a_2 a_0 D(a_1) = -a_0 D(a_1) a_2 + a_2 a_0 D(a_1),$$

while $bi_D(a_0 \otimes a_1 \otimes a_2) = b(a_0 D(a_1) \otimes a_2) = a_0 D(a_1) a_2 - a_2 a_0 D(a_1)$. The sum vanishes. For the inner case,

$$bR_h(a_0 \otimes a_1 \otimes a_2) - R_h b(a_0 \otimes a_1 \otimes a_2) = a_0 h a_1 \otimes a_2 - a_0 a_1 h \otimes a_2 = a_0 [h, a_1] \otimes a_2,$$

since the terms $a_0 h \otimes a_1 a_2$ and $a_2 a_0 h \otimes a_1$ occur in both bR_h and $R_h b$ and cancel. □

In [15] the derivations $D_{\log Q}$ and $D_{\log x}$ are commutators with elements *not* in the algebra [15, (63)–(64)]. Changing Q or x changes them by commutators with $\log(Q'/Q) \in \Psi_c^0(X)$ or $\log(x'/x) \in C^\infty(X)$, which are inner. So the maps $i_{\log Q}$ and $i_{\log x}$ on homology do not depend on the choices. The ideals $\mathcal{I}_\sigma$, $\mathcal{I}_\partial$, $\mathcal{I}$ are preserved by both derivations, and R_h maps chains of an ideal to chains of the same ideal. So the same independence holds for the ideals.

5.4 Explicit H-unitality contractions

Lemma 5.4. *Let V, W be nuclear Fréchet spaces with a continuous pairing $\langle \cdot, \cdot \rangle : W \times V \to \mathbb{C}$. Let $\mathcal{K} = V \widehat{\otimes} W$ be the algebra with product $(v \otimes w)(v' \otimes w') = \langle w, v' \rangle v \otimes w'$, and fix $\psi \in W$, $\tilde{\psi} \in V$ with $\langle \psi, \tilde{\psi} \rangle = 1$. Then*

$$E\big((v \otimes w) \otimes c\big) = (v \otimes \psi) \otimes (\tilde{\psi} \otimes w) \otimes c$$

is a continuous contraction of the bar complex. Hence the algebras

$$\Psi^{-\infty}(H) = C^\infty(H) \widehat{\otimes} C^\infty(H), \qquad \mathcal{I} = \dot{C}^\infty(X) \widehat{\otimes} \dot{C}^\infty(X)$$

are topologically H-unital. Moreover $\mathrm{HH}_0(\mathcal{K}) \cong \mathbb{C}$, spanned by the trace $v \otimes w \mapsto \langle w, v \rangle$, and $\mathrm{HH}_k(\mathcal{K}) = 0$ for $k > 0$.

Proof. The contraction is [7, Lemma "Explicit contractions"]: $d_0 E = 1$ because $(v \otimes \psi)(\tilde{\psi} \otimes w) = v \otimes w$, and $d_{i+1} E = E d_i$ for $i \geq 0$, so $b'E + Eb' = 1$. The homology statement is Morita invariance for the smoothing kernels (Lemma B.22, adapted from [7, Lemma "Morita invariance for smoothing kernels"]), applied with $\mathfrak{A} = \mathbb{C}$. □

5.5 The Hochschild–Kostant–Rosenberg map and Connes' theorem

For a commutative algebra the map [15, after (26)]

$$\chi(f_0 \otimes f_1 \otimes \cdots \otimes f_k) = \frac{1}{k!}\, f_0\, df_1 \wedge \cdots \wedge df_k$$

satisfies $\chi \circ b = 0$. Conversely the antisymmetrization $\varphi(f_0; f_1, \dots, f_k) = \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma) f_0 \otimes f_{\sigma(1)} \otimes \cdots \otimes f_{\sigma(k)}$ is a cycle with $\chi\varphi(f_0; f_1, \dots, f_k) = f_0\, df_1 \wedge \cdots \wedge df_k$.

Background fact 5.5 (Connes' theorem, (B5); proved in Theorems B.17 and B.18)**.** For a compact manifold M, possibly with boundary or corners, χ induces isomorphisms $\mathrm{HH}_k(C^\infty(M)) \cong C^\infty(M; \Lambda^k)$ and $\mathrm{HH}_k(\dot{C}^\infty(M)) \cong \dot{C}^\infty(M; \Lambda^k)$. The same holds, fibrewise, for smooth functions on a manifold with values in a commutative nuclear algebra of Laurent or formal series.

The proof for compact manifolds with corners, adapted from [14, Ch. 10], is Theorem B.17. The formal, Laurent, flat and Schwartz versions are Theorem B.18, and the version for homogeneous functions on a cone bundle, which is what the symbol algebras need, is Corollary B.21.

5.6 Filtered complexes and the convergence of spectral sequences

Let $C_k = \{\sum_{p \le P} c_p : c_p \in C_{p,k}\}$ be a complex of formal sums, bounded above in p, with differential $b = \sum_{l \ge 0} b_l$, where $b_l : C_{p,k} \to C_{p-l,k-1}$. The filtration $F_p = \{c : c_q = 0 \text{ for } q > p\}$ is exhaustive and separated, and we assume it is complete. (This assumption is not automatic: the chain model of Section 5.1 is *not* complete for the order filtration, and completeness is restored by passing to the mixed complexes of Lemma B.27; see Remark 5.7.) Put $E^1_{p,k} = H_k(C_{p,\bullet}, b_0)$. Then b_1 induces $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$, and we write $E^2 = H(E^1, d^1)$.

Proposition 5.6 (Degeneration and convergence [14, Prop. "convergence"])**.** *Suppose there is an integer n such that $E^2_{p,k} = 0$ whenever $p \neq k - n$. Then every class in $H_k(C, b)$ has a representative in $F_{k-n}C_k$, and*

$$\Theta : H_k(C, b) \to E^2_{k-n,k}, \qquad [c] \mapsto [c_{k-n}],$$

is an isomorphism. All differentials d^r, $r \ge 2$, vanish.

Proof. This is Proposition B.26, where the proof is given in full. The key step lowers the order of a cycle whose leading term has vanishing E^2-class by subtracting $b(x + y)$, with x a b_0-cycle one order up and y a correction at the same order; completeness is used to sum the resulting series of corrections, and the higher differentials vanish because they leave the line $p = k - n$. □

Remark 5.7 (Completeness and pole orders). Proposition 5.6 needs a complete filtration, and the chain model of Section 5.1 is not complete: a series such as $\sum_j r^j \otimes r^{-2j}$ has total orders tending to $-\infty$ but lies in no single $\mathfrak{A}_N^{\widehat{\otimes}2}$ (Appendix B, after Proposition B.26). The remedy, taken from [7, §"Convergence and degeneration"], is to compute with the mixed complexes $C(\mathfrak{B}, \mathfrak{A})$ of Lemma B.27, whose stages are complete and which compute $\mathrm{HH}(\mathfrak{A})$. For the closed symbol algebra and for $\mathcal{I}_\sigma$ this gives convergence (Theorem B.30).

For the algebras with Laurent singularities at the boundary ($\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$) one also needs the corrections in the proof to stay in a fixed pole stage. MN say convergence "follows directly, as in the boundaryless case" [15, p. 12]. In the b-calculus this is proved in [7, §"Convergence and degeneration"], using the invertibility of $\mathcal{L}_{x\partial_x}$ on leading coefficients. That argument does *not* apply with the cusp Poisson structure: the cusp Lie derivative $\mathcal{L}_{x^2\partial_x}$ raises the power of x, so it is nilpotent

on leading coefficients, and a cycle such as $x^{-L}|\tau|^p$ has no primitive in its own stage although it is exact in the direct limit (Example B.31). What survives is Proposition B.33: representatives on the critical line exist, Θ is well defined, and the corrections can be found if the pole bound is allowed to grow by two at each order. The remaining step, a uniform pole bound, is quoted from [15, p. 12] (Remark B.34); it is item (B10) of Section 2.5. In what follows *convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ is to be read as resting on (B10).* For $\mathcal{I}_\sigma$ (rapid vanishing at the boundary) and for the closed case no issue arises.

5.7 The Poisson differential and the symplectic star

Let $V \to M$ be a real vector bundle with a symplectic structure on $V \setminus 0$ that is homogeneous of degree 1 under dilations. The examples are T^*M and ${}^cT^*X$ with the forms of Section 2.1. Let Π be the Poisson bivector and ι_Π contraction with it. The Koszul–Brylinski differential is $\delta = \iota_\Pi d - d\iota_\Pi$, which maps k-forms of homogeneity p to $(k-1)$-forms of homogeneity $p-1$. On functions,

$$\delta(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_j (-1)^{j+1}\{a_0, a_j\}\, da_1 \wedge \ldots \widehat{da_j} \cdots \wedge da_k + \sum_{j<l} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge \cdots .$$

Proposition 5.8 (First differential [14, Prop. "d1"])**.** *For a star product with $P_0(a,b) = ab$ and $P_1(a,b) - P_1(b,a) = -i\{a,b\}$, the differential d^1 of the order spectral sequence is $-i\delta$ under the HKR identification of E^1 with homogeneous forms.*

The point of the proof is that the symmetric part of P_1 cancels in pairs on antisymmetrized chains, and the antisymmetric part produces δ term by term; see [14, Ch. 10].

Proposition 5.9 (Brylinski [3])**.** *Let $*$ be the symplectic star operator, defined by $\alpha \wedge *\beta = \Lambda(\alpha,\beta)\,\omega^n/n!$. Then $\delta = (-1)^{k-1} *^{-1} d *$ on k-forms, and $*$ maps forms of degree k and homogeneity p to forms of degree $2n-k$ and homogeneity $p+n-k$. Moreover $\iota_{H_f}(*\eta) = \pm * (df \wedge \eta)$.*

For ${}^cT^*X$ the volume form is $\omega^n/n! = x^{-2}\times$(a smooth volume). So on smooth forms $*$ introduces a factor $x^{\pm 2}$ [15, p. 12]. In the Laurent algebras this is harmless. In $\mathcal{I}_\sigma$, where everything vanishes to infinite order at the boundary, it is harmless as well.

Proposition 5.10 (Homogeneous de Rham cohomology [15, Lemma 3])**.** *Let $\Omega^\bullet_{(c)}$ be the forms on $V \setminus 0$ of homogeneity c. If $c \neq 0$ the complex $(\Omega^\bullet_{(c)}, d)$ is acyclic. For $c = 0$, writing forms as $\alpha' + \alpha'' \wedge dr/r$ with r a positive homogeneous function of degree one,*

$$H^j(\Omega^\bullet_{(0)}, d) \cong H^j(SV) \oplus H^{j-1}(SV)\,\frac{dr}{r} \cong H^j(SV \times \mathbb{S}_\sigma).$$

The same holds for Laurent forms, flat forms, and formal Laurent forms over manifolds with boundary, with the corresponding de Rham cohomology of SV (Section 6).

Proof. Let R be the radial vector field. It commutes with dilations, so ι_R preserves homogeneity, and $\mathcal{L}_R = c$ on $\Omega_{(c)}$. If $c \neq 0$ and $d\alpha = 0$, Cartan's formula gives $\alpha = d(\iota_R\alpha/c)$. If $c = 0$, the forms α' and α'' are pulled back from SV, and $d(\alpha' + \alpha'' \wedge dr/r) = d\alpha' + d\alpha'' \wedge dr/r$. The operator ι_R preserves Laurent, flat or formal behaviour in x, so the same proof applies to those classes of forms. □

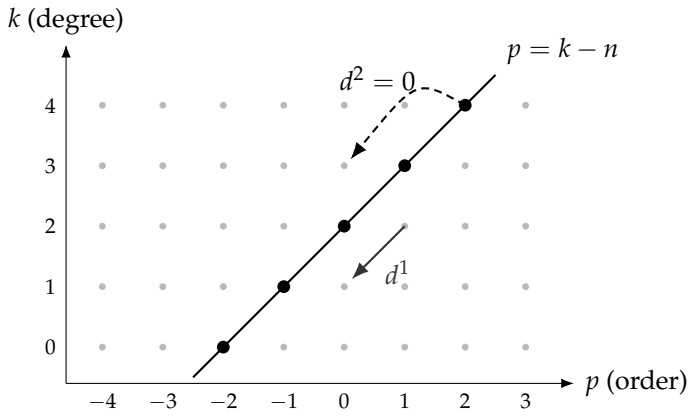


Figure 9: The spectral sequence of the order filtration for a symbol algebra, drawn for $n = 2$. Gray dots mark the positions (p, k) where $E^1_{p,k}$ (homogeneous k-forms of degree p) can be non-zero. The differential d^1 moves along lines parallel to $p = k - n$. After the symplectic star and Proposition 5.10, E^2 lives only on the line $p = k - n$ (black dots), and the higher differentials leave that line, so they vanish. Adapted from [14, Ch. 10].

5.8 Morita invariance

If $e \in M_N(\mathfrak{A})$ is an idempotent not contained in any proper two-sided ideal, then $eM_N(\mathfrak{A})e$ is Morita equivalent to $\mathfrak{A}$ and has the same Hochschild homology (B6) [15, Prop. 13], [9]. For matrices and for algebras $\mathcal{K}\widehat{\otimes}\mathfrak{A}$ as in Lemma 5.4, with $\mathfrak{A}$ unital or only H-unital, the generalized trace Tr_* is a quasi-isomorphism (Lemma B.22, adapted from [7, Lemma "Morita invariance for smoothing kernels"]). Full corners are Proposition B.24, and the application to bundles is Remark B.25. We use the smoothing-kernel case with $\mathcal{K} = \Psi^{-\infty}(\partial X)$ to remove the boundary smoothing operators in Section 9.

6 Functions and weighted de Rham cohomology (MN §1)

Question 6.1. *Why do MN allow arbitrary negative powers of x, that is, Laurent coefficients, instead of working with $\Psi_c^{\mathbb{Z}}(X)$ itself?*

Answer. For the same reason that one allows operators of all integral orders. The order filtration and the x-filtration are treated symmetrically: in $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ both Q and x are invertible modulo the residual ideals. The homology is then finite dimensional, and each filtration contributes a circle, $\mathbb{S}_\sigma$ for the order and $\mathbb{S}_\partial$ for the powers of x, with classes $d\log r$ and $d\log x$. Without negative powers, closed forms such as $\beta \wedge dx/x^2$ would no longer be exact (Correction 6.8), and the homology would acquire infinite-dimensional pieces, as it does for order zero (Section 17.3). On the analytic side, $x^{-p}A$ has the same index as A when both are fully elliptic [15, p. 31], so nothing is lost. □

Throughout this section M is a compact manifold with boundary, and x is a boundary defining function with a fixed collar $[0, \epsilon)_x \times \partial M$. We consider the three algebras of the Taylor sequence [15, (25)],

$$0 \to \dot{C}^\infty(M) \to x^{-\mathbb{Z}}C^\infty(M) \to C^\infty(\partial M)((x)) \to 0, \tag{14}$$

where the last map takes the Taylor series at ∂M in the collar and $C^\infty(\partial M)((x))$ is the algebra of formal Laurent series $\sum_{k \geq -p} f_k x^k$.

6.1 H-unitality of the flat functions

Lemma 6.2 (MN Lemma 1)**.** *$\dot{C}^\infty(M)$ is topologically H-unital.*

Proof. The k-chains are $\dot{C}^\infty(M^{k+1})$. Choose $\phi \in C_c^\infty((-1,1))$ with $\phi(0) = 1$, and for $f \in \dot{C}^\infty(M^{k+1})$ define

$$(Ef)(z^{(0)}, z^{(1)}, \dots, z^{(k+1)}) = \phi\Big(\frac{x^{(0)} - x^{(1)}}{x^{(0)} + x^{(1)}}\Big) f(z^{(1)}, \dots, z^{(k+1)}), \tag{15}$$

where $x^{(j)}$ is x on the j-th factor [15, (26)].

Ef is flat. Put $w = (x^{(0)} - x^{(1)})/(x^{(0)} + x^{(1)}) \in [-1, 1]$. On the support of $\phi(w)$ we have $|w| \le 1 - \delta$ for some $\delta > 0$. Hence $x^{(0)}$ and $x^{(1)}$ are comparable there: $c\, x^{(1)} \le x^{(0)} \le C\, x^{(1)}$. The function $\phi(w)$ is smooth away from $x^{(0)} = x^{(1)} = 0$. Each derivative of $\phi(w)$ is bounded by a power of $(x^{(0)} + x^{(1)})^{-1}$, which is bounded by a power of $1/x^{(1)}$ on the support. Since f vanishes to infinite order as $x^{(1)} \to 0$, every derivative of Ef is $O((x^{(1)})^\infty) = O((x^{(0)})^\infty)$ there. At the faces $x^{(j)} = 0$, $j \ge 1$, flatness is inherited from f. So $E : \dot{C}^\infty(M^{k+1}) \to \dot{C}^\infty(M^{k+2})$ continuously.

E is a contraction. The face d_0 restricts to $z^{(0)} = z^{(1)}$, where $w = 0$, so $d_0 E f = f$. For $i \ge 1$, the face d_i identifies $z^{(i)}$ and $z^{(i+1)}$, and does not touch the factor $\phi(w)$ unless $i = 1$. When $i = 1$ it replaces $z^{(1)}$ by itself, so in all cases $d_i E = E d_{i-1}$. Therefore

$$b' E f = d_0 E f + \sum_{i \ge 1} (-1)^i d_i E f = f - E\Big(\sum_{i \ge 0} (-1)^i d_i f\Big) = f - E b' f. \qquad \square$$

Remark 6.3 (The support of ϕ). MN only ask that $\phi \in C_c^\infty(\mathbb{R})$ with $\phi(0) = 1$. This is not enough. If $\phi(-1) \ne 0$, then at a point with $x^{(0)} = 0 < x^{(1)}$ we have $w = -1$, and $Ef = \phi(-1) f(z^{(1)}, \dots)$ need not vanish, so $Ef \notin \dot{C}^\infty(M^{k+2})$. The support condition $\operatorname{supp}\phi \subset (-1, 1)$ is what makes $x^{(0)}$ and $x^{(1)}$ comparable.

The algebra $x^{-\mathbb{Z}} C^\infty(M)$ is unital, and so is $C^\infty(\partial M)((x))$. By excision (B4, Theorem B.7), (14) gives a long exact sequence in Hochschild homology.

6.2 Hochschild homology of the three function algebras

The HKR map χ is defined on the chains of all three algebras [15, (27)]:

$$\begin{aligned}
&\chi : \dot{C}^\infty(M^{k+1}) \to \dot{C}^\infty(M; \Lambda^k),\\
&\chi : x_0^{-\mathbb{Z}} \cdots x_k^{-\mathbb{Z}} C^\infty(M^{k+1}) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k),\\
&\chi : C^\infty((\partial M)^{k+1})((x_0, \dots, x_k)) \to C^\infty(\partial M; \Lambda^k)((x)) \oplus C^\infty(\partial M; \Lambda^{k-1})((x))\, \frac{dx}{x^2}.
\end{aligned}$$

(MN print M for ∂M in the last target.) In the last line a k-form on the formal neighbourhood of ∂M is split into its tangential part and its dx-part. Any of dx, dx/x, dx/x^2 may be used as the normal basis element, since the coefficients are Laurent series. MN use dx/x^2 to match the cusp geometry. We use dx/x when computing cohomology.

Lemma 6.4 (MN Lemma 2)**.** *The three HKR maps are isomorphisms on Hochschild homology. The long exact sequence of* (14) *splits into the short exact sequences [15, (28)]*

$$0 \to \dot{C}^\infty(M; \Lambda^k) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k) \to C^\infty(\partial M; \Lambda^k)((x)) \oplus C^\infty(\partial M; \Lambda^{k-1})((x))\, \frac{dx}{x} \to 0.$$

Proof. The HKR maps for the three algebras are isomorphisms by Theorem B.18 (the flat, Laurent and formal Laurent cases of (B5)); for $\mathbb{C}((x))$ this is the Koszul computation $\mathrm{HH}_0 = \mathbb{C}((x))$, $\mathrm{HH}_1 = \mathbb{C}((x))\,dx$, $\mathrm{HH}_k = 0$ for $k \geq 2$. No Künneth formula for formal Laurent series is needed. The HKR maps commute with the maps of (14), since restriction to Taylor series commutes with d (Theorem B.18). Finally, the right-hand map at the level of forms is surjective by Borel's lemma (every formal Laurent form is the Taylor series of a Laurent form). So the connecting maps vanish, and the long exact sequence breaks into the displayed short exact sequences. □

6.3 Weighted de Rham cohomology

The de Rham differential acts on all three spaces of forms in Lemma 6.4.

Lemma 6.5 (Formal Laurent forms)**.** *The cohomology of the de Rham complex of formal Laurent forms $\sum_{j\geq -p} x^j(\alpha_j + \beta_j \wedge dx/x)$, with α_j, β_j forms on ∂M, is $H^k(\partial M) \oplus H^{k-1}(\partial M)\,dx/x$. It is spanned by the classes with $j = 0$.*

Proof. For a k-form α on ∂M, $d(x^j\alpha) = x^j\big(d\alpha + (-1)^k j\,\alpha \wedge dx/x\big)$, and $d(x^j\beta \wedge dx/x) = x^j\,d\beta \wedge dx/x$. So the complex splits into the subcomplexes of fixed x-degree j. On degree j the differential is $(\alpha, \beta) \mapsto (d\alpha,\ d\beta + (-1)^k j\alpha)$. This is the mapping cone of multiplication by j on the de Rham complex of ∂M. For $j \neq 0$ that multiplication is invertible, so the cone is acyclic; explicitly $h(\alpha, \beta) = ((-1)^{k-1}j^{-1}\beta, 0)$ is a contracting homotopy. For $j = 0$ the complex is the direct sum of the de Rham complex of ∂M and its shift. □

Proposition 6.6 (MN Proposition 1)**.** *The de Rham cohomology of the three complexes in Lemma 6.4 is*

$$H^k_{\mathrm{rel}}(M), \qquad H^k(M) \oplus H^{k-1}(\partial M), \qquad H^k(\partial M) \oplus H^{k-1}(\partial M),$$

and the long exact sequence is [15, (29)]

$$\cdots \to H^k_{\mathrm{rel}}(M) \to H^k(M) \oplus H^{k-1}(\partial M) \xrightarrow{\iota^* \oplus \mathrm{Id}} H^k(\partial M) \oplus H^{k-1}(\partial M) \to H^{k+1}_{\mathrm{rel}}(M) \to \cdots,$$

the direct sum of the long exact sequence of the pair $(M, \partial M)$ and the identity on $H^{k-1}(\partial M)$.

Proof. Flat forms. The complex $\dot{C}^\infty(M; \Lambda^\bullet)$ contains the compactly supported forms on the interior, and the inclusion is a quasi-isomorphism [15, p. 9]. In the collar a flat form is $\alpha + \beta \wedge dx$ with α, β flat at $x = 0$. The homotopy $h(\alpha + \beta \wedge dx) = \pm \int_0^x \beta$, integrated from the boundary and cut off away from the collar, preserves flatness. It shows that a closed flat form is cohomologous to one that vanishes near ∂M. The cohomology is therefore $H^k_c(M^\circ) = H^k_{\mathrm{rel}}(M)$.

Formal Laurent forms. This is Lemma 6.5.

Laurent forms. Let $\chi_0 \in C^\infty(M)$ be a cut-off equal to 1 near ∂M and supported in the collar. Consider

$$\Psi : C^\infty(M; \Lambda^k) \oplus C^\infty(\partial M; \Lambda^{k-1}) \to x^{-\mathbb{Z}} C^\infty(M; \Lambda^k), \qquad (\gamma, \beta) \mapsto \gamma + \chi_0\,\beta \wedge \frac{dx}{x}.$$

It is a chain map for the differential $(\gamma, \beta) \mapsto (d\gamma, d\beta)$, because $d\chi_0 \wedge \beta \wedge dx = 0$ ($d\chi_0$ is a multiple of dx in the collar). It fits into a map of short exact sequences: from [15, (30)],

$$0 \to \dot{C}^\infty(M; \Lambda^\bullet) \to C^\infty(M; \Lambda^\bullet) \oplus C^\infty(\partial M; \Lambda^{\bullet-1}) \to C^\infty(\partial M; \Lambda^\bullet \oplus \Lambda^{\bullet-1}) \to 0,$$

where the second map is restriction to ∂M plus the identity, to the sequence of Lemma 6.4. On the outer terms the induced maps in cohomology are the identity and the inclusion of the $j = 0$

part of Lemma 6.5. Both are isomorphisms, so Ψ is a quasi-isomorphism by the five lemma. The cohomology of the source is $H^k(M) \oplus H^{k-1}(\partial M)$, by the de Rham theorem for manifolds with boundary. The long exact sequence of the source sequence is the sequence of the pair together with the identity on $H^{k-1}(\partial M)$. □

Example 6.7. For $M = [0,1)$ (non-compact, but the computation is local), the Laurent de Rham cohomology is $H^0 = \mathbb{C}$ and $H^1 = \mathbb{C}\, dx/x$. For the compact interval $M = [0,1]$ with $x = s(1-s)$, it is $H^0 = \mathbb{C}$ and $H^1 = \mathbb{C}^2$, spanned by dx/x localized near either end. The relative cohomology is $H^1_{\mathrm{rel}}([0,1]) = \mathbb{C}$, and the map $H^1_{\mathrm{rel}} \to H^1 \oplus H^0(\partial M)$ is zero, as it must be.

6.4 b- and cusp de Rham complexes

The Laurent forms $x^{-\mathbb{Z}} C^\infty(M; \Lambda^k)$ coincide with the Laurent b-forms and the Laurent cusp forms. The smooth b-forms $C^\infty(M; {}^b\Lambda^k)$, locally generated by dx/x and dy, and the smooth cusp forms $C^\infty(M; {}^c\Lambda^k)$, generated by dx/x^2 and dy, are subcomplexes [15, (32)–(33)]. By the proof of Proposition 6.6 the b-complex has the cohomology of the Laurent complex,

$$ {}^bH^k(M) \cong H^k(M) \oplus H^{k-1}(\partial M)$$

(Nest–Tsygan [16], quoted in [15]).

Correction to MN 6.8 (The cusp de Rham complex). After [15, (33)] it is stated that the cusp de Rham complex also has cohomology ${}^bH^k(M)$. This is not so: its cohomology is

$$ {}^cH^k(M) \cong H^k(M) \oplus H^{k-1}(\partial M) \oplus H^{k-1}(\partial M).$$

The extra classes are $\beta \wedge dx/x^2$, with β closed on ∂M. They are exact in the Laurent complex ($\beta \wedge dx/x^2 = \pm d(\beta/x)$) but not in the smooth cusp complex.

Proof. Expand a smooth cusp form near ∂M as $\sum_{j\geq 0} x^j(\alpha_j + \beta_j \wedge dx/x^2)$. From $d(x^j\alpha) = x^j\, d\alpha + (-1)^k j\, x^{j+1}\alpha \wedge dx/x^2$ we see that the differential pairs α_j with β_{j+1}, with coefficient j. For $j \geq 1$ this pairing is invertible, and those pairs cancel as in Lemma 6.5. What survives is α_0, β_0 and β_1. Since α_{-1} is not available, β_0 is not hit. This gives $H^k(\partial M) \oplus H^{k-1}(\partial M) \oplus H^{k-1}(\partial M)$ for the formal boundary complex. A form $\chi_0\beta \wedge dx/x^2$ with $d\beta = 0$ is closed on M, so the connecting map vanishes on the new classes, and the five-lemma argument of Proposition 6.6 gives the result. On $[0,1)$: dx/x^2 and dx/x are closed smooth cusp 1-forms, and neither is df for a smooth f. □

This plays no role in [15], which only uses the Laurent complex.

6.5 Homogeneous Laurent forms

Let $V \to M$ be a real vector bundle and r a positive function on $V \setminus 0$, homogeneous of degree one. Combining Proposition 5.10 with Proposition 6.6 gives [15, (35)]: the de Rham cohomology of formal sums of homogeneous Laurent forms on $V \setminus 0$, with degrees bounded above, is

$$ {}^bH^k(SV \times \mathbb{S}_\sigma) = {}^bH^k(SV) \oplus {}^bH^{k-1}(SV)\, \frac{dr}{r} \cong H^k(SV \times \mathbb{S}_\sigma) \oplus H^{k-1}(\partial SV \times \mathbb{S}_\sigma)\, \frac{dx}{x}. \qquad (16)$$

The same holds with flat forms (relative cohomology) and with formal Laurent forms at the boundary (cohomology of $\partial SV \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$).

7 The symbol algebras $\mathcal{A}_\sigma$ and $\mathcal{I}_\sigma$ (MN §2)

Proposition 7.1 (MN Proposition 2). $\mathrm{HH}_k(\mathcal{A}_\sigma) \cong {}^bH^{2n-k}(Y \times \mathbb{S}_\sigma)$, *that is,*

$$\mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y) \oplus H^{2n-k-1}(Y)\, d\theta_\sigma \oplus H^{2n-k-1}(Z)\, \frac{dx}{x} \oplus H^{2n-k-2}(Z)\, d\theta_\sigma \wedge \frac{dx}{x}.$$

Equivalently, $\mathrm{HH}_k(\mathcal{A}_\sigma) \cong H^{2n-k}(Y \times \mathbb{S}_\sigma) \oplus H^{2n-k-1}(Z \times \mathbb{S}_\sigma)$, *and by duality* $\mathrm{HH}^k(\mathcal{A}_\sigma) \cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma)$.

Proof. Chains and filtration. By the quantization (B1, Proposition A.22), as vector spaces,

$$\mathcal{A}_\sigma \cong \varinjlim_{m,N} x^{-N} \prod_{j \le m} \mathcal{G}^j({}^cT^*X),$$

with the star product [15, (37)]. Completing tensor products gives the chains [15, (38)]: formal sums $\sum_{p \le P} c_p$, where c_p is a function on $({}^cT^*X \setminus 0)^{k+1}$ with Laurent singularities in each x_j, homogeneous in each factor, of total homogeneity p. The order filtration is independent of the quantization, and the Hochschild differential decomposes as $b = \sum_l b_l$ with b_l coming from P_l. So we are in the setting of Section 5.6.

E^1. The differential b_0 is the Hochschild differential of the commutative algebra $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X)$. By Corollary B.21 (Connes' theorem (B5) for Laurent functions on the sphere bundle, with a radial variable), $E^1_{p,k} \cong x^{-\mathbb{Z}}\mathcal{G}^p({}^cT^*X; \Lambda^k)$, the homogeneous Laurent k-forms of degree p; in the mixed complexes of Remark 5.7 this is Lemma B.28.

d^1. By Proposition 5.8, $d^1 = -i\delta$, where δ is the Koszul–Brylinski differential of the cusp Poisson structure (7). This is the extension by continuity of the Poisson differential of the interior [15, p. 12].

E^2. The cusp symplectic form is $\omega = d\tau \wedge dx/x^2 + d\eta \wedge dy$, homogeneous of degree one, with only a finite order singularity at $x = 0$. The symplectic star operator therefore maps $x^{-\mathbb{Z}}\mathcal{G}^p({}^cT^*X; \Lambda^k)$ isomorphically onto $x^{-\mathbb{Z}}\mathcal{G}^{p+n-k}({}^cT^*X; \Lambda^{2n-k})$ and intertwines δ with $\pm d$ (Proposition 5.9). In local coordinates, $*(f\, dx^I\, d\zeta^J) = \pm x^{\pm 2} f\, dx^{J^{\complement}} d\zeta^{I^{\complement}}$, which explains the factor x^2 in [15, p. 12]. Hence

$$E^2_{p,k} \cong H^{2n-k}\big(x^{-\mathbb{Z}}\mathcal{G}^{p+n-k}({}^cT^*X; \Lambda^\bullet), d\big),$$

which vanishes for $p \neq k - n$ by Proposition 5.10. For $p = k - n$ it equals ${}^bH^{2n-k}(Y \times \mathbb{S}_\sigma)$ by (16).

Degeneration and convergence. E^2 is concentrated on the line $p = k - n$. Proposition B.33 then shows that every class has a representative on that line and that the leading-symbol map Θ is well defined. That Θ is an isomorphism is the quoted step (B10) (Remark 5.7, Remark B.34).

Duality. $Y \times \mathbb{S}_\sigma$ is a compact oriented $2n$-manifold with boundary $Z \times \mathbb{S}_\sigma$. It is oriented by the symplectic structure and the orientation of the circle. Lefschetz duality gives $H^{2n-k}(Y \times \mathbb{S}_\sigma)^* \cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$, and Poincaré duality on the closed $(2n-1)$-manifold $Z \times \mathbb{S}_\sigma$ gives $H^{2n-k-1}(Z \times \mathbb{S}_\sigma)^* \cong H^k(Z \times \mathbb{S}_\sigma)$. □

Discussion. **The singular symplectic form.** Over the interior, ${}^cT^*X = T^*X$ and ω is the usual symplectic form. At the boundary, $\omega = d\tau \wedge dx/x^2 + \cdots$ blows up to second order. Its inverse, the Poisson bivector $x^2\partial_\tau \wedge \partial_x + \cdots$, *vanishes* to second order. This is why δ extends smoothly to the boundary and why the commutator of two cusp operators is again a cusp operator. On Laurent forms the factor $x^{\pm 2}$ is invisible, and the computation is word for word the closed one. What changes is the de Rham cohomology of the Laurent forms, which picks up the boundary term $H^{*-1}(Z)\, dx/x$ (Proposition 6.6).

Proposition 7.2 (MN Proposition 3). *$\mathcal{I}_\sigma$ is H-unital,* $\mathrm{HH}_k(\mathcal{I}_\sigma) \cong H^{2n-k}_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$, *and* $\mathrm{HH}^k(\mathcal{I}_\sigma) \cong H^k(Y \times \mathbb{S}_\sigma)$.

Proof. The computation is that of Proposition 7.1 with flat forms in place of Laurent forms. By Proposition 6.6 these give relative cohomology. Here no pole orders occur, and convergence is proved in Theorem B.30, using the mixed complexes of Lemma B.27. For H-unitality, one can filter the bar complex by order. Its E^1 term is the bar homology of the commutative algebra $\dot{C}^\infty$-valued in homogeneous functions. This vanishes by Lemma 6.2, whose contraction (15) only involves the base variables and preserves homogeneity. Concluding that the bar complex of $\mathcal{I}_\sigma$ is acyclic needs completeness, which the chain model lacks (Remark 5.7), so we prefer the direct argument: use the contraction (15) directly with a quantization in which Id has full symbol 1 and which is a left quantization in the collar. Then the cut-off factor in (15) is independent of the fibre variable in slot 0, and $P_j(\phi(w), \cdot) = 0$ for $j \geq 1$ [15, p. 13]. □

Remark 7.3. One can also obtain Proposition 7.2 from the closed case: the operators with kernels supported in the interior are dense in $\mathcal{I}_\sigma$, and $\mathcal{I}_\sigma$ is an ideal in $\Psi^{\mathbb{Z}}(2X)/\Psi^{-\infty}(2X)$ for the double $2X$ [15, (47)]. In Section 10 this identification shows that Tr_σ is Wodzicki's residue.

8 The boundary symbol algebra and the boundary sequence (MN §3)

Proposition 8.1 (MN Proposition 4). $\mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \cong H^{2n-k}(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)$ *and* $\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) \cong H^k(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial)$.

Proof. $\mathcal{A}_{\partial,\sigma} \cong C^\infty(Z)((x, q^{-1}))$ (as vector spaces, with the transported star product) consists of formal Laurent series in x of complete symbols over ∂X. The proof of Proposition 7.1, including its reliance on (B10) for convergence, applies with formal Laurent forms on the formal neighbourhood of ${}^cT^*_{\partial X}X \setminus 0$. The homogeneous de Rham cohomology is generated over $H^*(Z)$ by 1, $d\log r$, $d\log x$ and $d\log x \wedge d\log r$ [15, p. 13] (Lemma 6.5 and Proposition 5.10). The total dimension of $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ is $2n$. Since $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ is closed, Poincaré duality gives the cohomology statement. □

Proposition 8.2 (MN Proposition 5; part (3) quoted as (B9)). *The ideal $\mathcal{I}_\sigma$ is H-unital. Under the isomorphisms of Propositions 7.1, 7.2 and 8.1, the long exact sequence of $0 \to \mathcal{I}_\sigma \to \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma} \to 0$ is the direct sum of the long exact sequence of the pair $(Y \times \mathbb{S}_\sigma, Z \times \mathbb{S}_\sigma)$ (in degrees $2n-k$) and the identity on $H^{2n-k-1}(Z \times \mathbb{S}_\sigma)\, dx/x$:*

1. $\mathrm{HH}_k(\mathcal{I}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_\sigma)$ *is $\iota^* \oplus 0$, relative to absolute cohomology;*
2. $\mathrm{HH}_k(\mathcal{A}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma})$ *is restriction to $Z \times \mathbb{S}_\sigma$ on the first summand and the identity on the summand with dx/x;*
3. *the connecting map $\partial : \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}_{k-1}(\mathcal{I}_\sigma)$ is a non-zero universal multiple of the de Rham connecting map $H^{2n-k}(Z \times \mathbb{S}_\sigma) \to H^{2n-k+1}_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$ on the summand without $d\theta_\partial$, and vanishes on the summand $H^{2n-k-1}(Z \times \mathbb{S}_\sigma)\, d\theta_\partial$.*

Proof. Excision applies because $\mathcal{I}_\sigma$ is H-unital (Proposition 7.2). The maps in (1) and (2) come from algebra morphisms that preserve the order filtration. They therefore induce maps of spectral sequences, and on E^2 they are the maps induced on homogeneous forms by inclusion and restriction. This gives (1) and (2) [15, p. 14].

For (3) we follow MN. By Wodzicki's theorem the connecting map is that of $0 \to \mathcal{K} \to C(\mathcal{A}_\sigma) \to C(\mathcal{A}_{\partial,\sigma}) \to 0$. First let $X = H \times [0,1]$ be a product, with cusp structure given by $x(1-x)$. Then $\mathcal{A}_{\partial,\sigma}$

splits as a sum of the two boundary algebras, $\partial = \partial_0 \oplus \partial_1$, and by symmetry $\partial_1 = -\partial_0$. MN use the subalgebra of operators that commute with $x^2(1-x)^2\partial_x$ to construct a splitting $s_0 : \mathcal{A}^{(0)}_{\partial,\sigma} \to \mathcal{A}_\sigma$ that is an algebra morphism; this is special to the product situation [15, p. 14]. Let $\varphi \in C^\infty([0,1])$ be 1 near 0 and 0 near 1. Define

$$s(f_0 \otimes \cdots \otimes f_k) = s_0(f_0)\varphi \otimes s_0(f_1) \otimes \cdots \otimes s_0(f_k),$$
$$\sigma(f_0 \otimes \cdots \otimes f_k) = s_0(f_0)[\varphi, s_0(f_1)] \otimes s_0(f_2) \otimes \cdots \otimes s_0(f_k).$$

Since s_0 is multiplicative, $bs - sb$ can be computed term by term, and the only defect comes from moving φ past $s_0(f_1)$. Up to the convention for the cyclic face this gives $\sigma = [b, s]$. The chain $\sigma(z)$ lies in $\mathcal{K}$, because $[\varphi, \cdot\,]$ is supported where $d\varphi \neq 0$, in the interior. (Here φ is a function of x alone, so pseudolocality applies: $[\varphi, P]$ is a compactly supported interior operator modulo $\mathcal{I}$, by Lemma A.18(1),(2).) So σ_* is the connecting map. It lowers order by one, and on leading symbols

$$\chi\big(\sigma(z)\big) = -i\,\iota_{H_\varphi}\chi(z) \qquad \text{for antisymmetric } z,$$

because $\sigma([\varphi, f]) = -i\{\varphi, f\} = -iH_\varphi f$. By Proposition 5.9, $\iota_{H_\varphi}(*\eta) = \pm * (d\varphi \wedge \eta)$. So if $\chi(z) = *\eta$ with $d\eta = 0$, the image class is $\pm i[d\varphi \wedge \eta] = \pm i[d(\varphi\eta)]$, where η is extended to the collar as a product. This is the de Rham connecting map applied to $[\eta]$. If $\eta = \eta' \wedge dx/x$, then $d\varphi \wedge \eta = 0$, because $d\varphi$ is a multiple of dx. This gives the vanishing on the $d\theta_\partial$ summand.

For general X, operators supported in a collar $V \cong \partial X \times [0,1)$ embed in the algebras of the product $\partial X \times [0,1]$. The result follows from the naturality of the exact sequences and of the isomorphisms of Propositions 7.1, 7.2, 8.1 [15, p. 15]. □

Remark 8.3 (On the splitting s_0). The existence of a multiplicative splitting s_0 is asserted in [15] without details. We were unable to reconstruct it for formal Laurent series in x, since an element of $\mathcal{A}_\sigma$ has only finitely many poles and cannot carry an infinite formal series. What the proof uses is the leading-symbol computation of σ_* on E^2. For that it suffices to have a splitting that is multiplicative modulo terms of lower order and modulo $\mathcal{I}_\sigma$. The constant $\pm i$ in (3) (MN write ι) is irrelevant for everything that follows, which uses only kernels and images. We have not supplied such a splitting or an alternative argument, so the proof of (3) above is a sketch of MN's argument and not a complete proof. We treat (3) as the quoted input (B9). Every result below that uses the boundary map of $0 \to \mathcal{I}_\sigma \to \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma} \to 0$ depends on (B9), in particular Lemma 12.4's identification and the corrected homology tables in Section 12 and Examples 12.7 and 12.8.

9 The indicial algebra $\mathcal{I}_\partial$ (MN §4)

9.1 The model: a boundary point

Let ∂X be a point; for instance $X = [0,1)$. Then $\Psi^{-\infty}_{\mathrm{sus}}(\mathrm{pt})$ is the convolution algebra $\mathcal{S}(\mathbb{R}_t)$. By (10) it is isomorphic to $\mathcal{S}(\mathbb{R}_\tau)$ with pointwise multiplication, and

$$\mathcal{I}_\partial \cong \mathcal{S}(\mathbb{R}_\tau)((x)), \qquad A \sim \sum_{j \geq -p} x^j a_j(\tau),$$

with the product determined by the twist (Lemma 4.4). Let us make the product explicit to second order. For $a, b \in \mathcal{S}(\mathbb{R}_\tau)$ (elements of x-degree zero),

$$x^k a = a x^k + ik\, x^{k+1} a' + O(x^{k+2}),$$

since $[x^k, a] = \sum_j x^j [x, a] x^{k-1-j}$ and $[x, a] = i x^2 a' + O(x^3)$.

Lemma 9.1 (Commutators in $\mathcal{I}_\partial$)**.** *For $f, g \in \mathcal{S}(\mathbb{R}_\tau)$ and $k \in \mathbb{Z}$:*

1. $[f, g] = 0$ *when both have x-degree zero;*
2. $[x^k f,\, g] = ik\, x^{k+1} g' f + O(x^{k+2})$;
3. $[x^k f,\, x^{-k} g] = ik\, x\, (fg)' + O(x^2)$.

Proof. (1) Translation invariant operators commute. (2) $[x^k f, g] = x^k[f, g] + [x^k, g]f = ikx^{k+1}g'f + O(x^{k+2})$. (3) Write $x^k f x^{-k} = f + ik\, xf' + O(x^2)$ and $x^{-k} g x^k = g - ik\, xg' + O(x^2)$. Then $x^k f x^{-k} g - x^{-k} g x^k f = fg + ikxf'g - gf + ikxg'f + O(x^2)$. □

Proposition 9.2 ($\mathcal{I}_\partial$ for a boundary point)**.** $\mathrm{HH}_0(\mathcal{I}_\partial) \cong \mathbb{C}$, *detected by*

$$\mathrm{Tr}_\partial(A) = \overline{\mathrm{Tr}}(a_1) = \frac{1}{2\pi} \int_{\mathbb{R}} a_1(\tau)\, d\tau,$$

where a_1 is the coefficient of x^1. $\mathrm{HH}_1(\mathcal{I}_\partial) \cong \mathbb{C}$, *generated by the cycle $a \otimes b$ with a, b of x-degree zero and $\int ab'\, d\tau \neq 0$, and detected by the cocycle*

$$(i_{\log x}\, \mathrm{Tr}_\partial)(A, B) = \mathrm{Tr}_\partial(A[\log x, B]), \qquad (i_{\log x}\, \mathrm{Tr}_\partial)(a, b) = \frac{i}{2\pi} \int_{\mathbb{R}} a(\tau) b'(\tau)\, d\tau.$$

$\mathrm{HH}_k(\mathcal{I}_\partial) = 0$ *for $k \geq 2$, and $\mathcal{I}_\partial$ is H-unital.*

Proof. Spectral sequence. Filter by x-degree. The chains are formal sums in increasing powers of x, complete in that direction, and b raises x-degree. The associated graded algebra is the commutative algebra $\mathcal{S}(\mathbb{R}_\tau)[x, x^{-1}]$ (a graded piece for each power of x). By HKR (B5) for Schwartz and formal Laurent coefficients (Theorem B.18) and Lemma 6.4,

$$E^1 \cong \big\{\text{forms on } \mathbb{R}_\tau \times (\text{punctured formal disc}),\ \text{Schwartz in } \tau,\ \text{Laurent in } x\big\} \qquad [15,\ (42)].$$

By Lemma 4.4, the first differential is $-i\delta$ for the Poisson structure $\{x, \tau\} = -x^2$, whose symplectic form is $\omega = d\tau \wedge dx/x^2$ [15, p. 16]. The symplectic star operator in two dimensions is $*1 = \omega$, $*\omega = 1$ and, on 1-forms, $*\, d\tau = d\tau$, $*\, dx = dx$. Indeed $dx \wedge *d\tau = \{x, \tau\}\omega = -x^2\, d\tau \wedge dx/x^2 = dx \wedge d\tau$. So

$$E_k^2 \cong H_{\mathrm{dR}}^{2-k}\big(\text{Schwartz-in-}\tau,\ \text{Laurent-in-}x\ \text{forms}\big) \cong H_c^{2-k}(\mathbb{R}_\tau \times \mathbb{S}_\partial) \cong H^{1-k}(\mathbb{S}_\partial),$$

where the Laurent de Rham cohomology in x is spanned by 1 and dx/x (Lemma 6.5) and the Schwartz de Rham cohomology in τ is $H_c^1(\mathbb{R}) = \mathbb{C}$, detected by $\int d\tau$. Concretely:

- in degree $k = 0$, a function $\sum x^j f_j$ goes to $\sum x^j f_j\, d\tau \wedge dx/x^2$, whose class is $\int f_1\, d\tau$ times $[d\tau \wedge dx/x]$;
- in degree $k = 1$, a form $\sum x^j (g_j\, d\tau + h_j\, dx)$ goes to a class determined by $\int g_0\, d\tau$.

The E^2 term sits in x-degree 1 for $k = 0$ and in x-degree 0 for $k = 1$. Higher differentials raise x-degree by $r \geq 2$ and lower k by one, so they vanish. Convergence holds by completeness in the direction of increasing powers of x. Unlike the order filtration of Remark 5.7, the filtration here is by the powers of x themselves. In a fixed pole stage (each tensor factor in $x^{-L}\mathcal{S}(\mathbb{R})[[x]]$) only finitely many multidegrees have a given total x-degree, so each stage is complete. As in [15], we take for granted that the corrections can be kept in a fixed stage. The obstruction of Example B.31

concerns the order filtration and does not apply directly here, but Appendix B does not treat this case.

The generators. Lemma 9.1(3) shows directly that the x^1-coefficient of a commutator is a derivative, so its integral vanishes. Hence Tr_∂ is a trace. Lemma 9.1(2) shows that $x^{k+1}h$ is a commutator modulo higher order for $k \neq 0$ and h a sum of products $g'f$. Such sums exhaust $\mathcal{S}(\mathbb{R})$. Indeed, if $\int h = 0$ then $h = G'$ with $G \in \mathcal{S}$; writing $G = uv$ with $u, v \in \mathcal{S}$ (every Schwartz function is such a product) gives $h = u'v + uv'$. And $\psi'\psi'$ has positive integral for any non-zero $\psi \in \mathcal{S}$. For degree one: $a \otimes b$ is a cycle by Lemma 9.1(1). By Lemma 4.4(3), $[\log x, b] = -x[t, b] + O(x^2)$ and $\mathrm{In}([t, b]) = -ib'$. Hence

$$\mathrm{Tr}_\partial(a[\log x, b]) = \overline{\mathrm{Tr}}\big(-a[t, b]\big) = -\frac{1}{2\pi}\int a(-ib')\,d\tau = \frac{i}{2\pi}\int ab'\,d\tau.$$

This is non-zero for suitable a, b, so $i_{\log x}\,\mathrm{Tr}_\partial$ is a non-zero class (it is a cocycle by Lemma 5.3).

H-unitality. The same filtration applied to the bar complex has E^1 equal to the bar homology of the commutative graded algebra. That algebra is H-unital: it is $\mathcal{S}(\mathbb{R}) \otimes \mathbb{C}[x, x^{-1}]$, and $\mathcal{S}(\mathbb{R}) = \mathcal{S}(\mathbb{R}) \cdot \mathcal{S}(\mathbb{R})$ has the contraction $(Ef)(\tau', \tau_0, \dots) = \phi(\tau' - \tau_0) f(\tau_0, \dots)$ with $\phi \in \mathcal{S}(\mathbb{R})$ equal to 1 near 0. Hence $E^1 = 0$, and the bar complex of $\mathcal{I}_\partial$ is acyclic by completeness. With $\phi \equiv 1$ near 0, the same E is in fact a contraction for the twisted product: all derivatives of $\phi(\tau' - \tau_0)$ vanish on the diagonal, so the twist corrections of Lemma 4.4 do not contribute to $d_0 E$. □

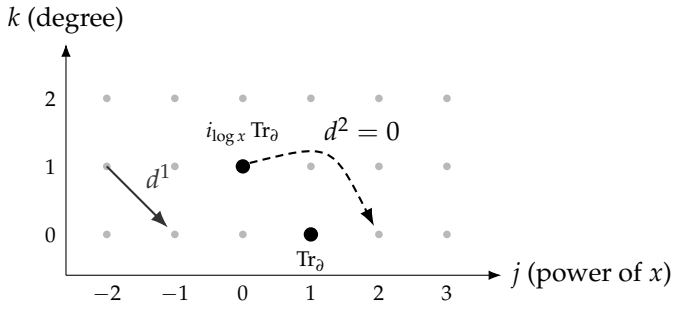


Figure 10: The x-adic spectral sequence of Proposition 9.2 (one boundary point). Gray dots: E^1, forms on $\mathbb{R}_\tau \times \mathrm{S}_\partial$ of fixed x-degree j. The first differential $d^1 = -i\delta$ raises j by one and lowers k by one. E^2 survives only at $(j, k) = (1, 0)$, dual to Tr_∂, and at $(0, 1)$, dual to $i_{\log x}\,\mathrm{Tr}_\partial$. A differential d^r goes from (j, k) to $(j + r, k - 1)$, so for $r \geq 2$ it misses both positions.

9.2 General boundary

In a product collar, the variable $x = 1/t$ commutes with operators acting only in y. Hence, componentwise,

$$\mathcal{I}_\partial \cong \bigoplus_{j=1}^{N} \Psi^{-\infty}(H_j) \widehat{\otimes} \big(\mathcal{S}(\mathbb{R}_\tau)((x))\big) \qquad \text{[15, p. 16].}$$

Since $\Psi^{-\infty}(H) = C^\infty(H) \widehat{\otimes} C^\infty(H)$ is of the form in Lemma 5.4, Morita invariance for smoothing kernels (Lemma B.22) removes it. MN use the Künneth formula [8] for the same purpose.

Proposition 9.3 (MN Proposition 6, for N components)**.** *$\mathcal{I}_\partial$ is H-unital, and*

$$\mathrm{HH}_k(\mathcal{I}_\partial) \cong \bigoplus_{j=1}^{N} H^{1-k}(\mathrm{S}_\partial).$$

That is, $\mathrm{HH}_0(\mathcal{I}_\partial) \cong \mathrm{HH}_1(\mathcal{I}_\partial) \cong \mathbb{C}^N$*, and* $\mathrm{HH}_k(\mathcal{I}_\partial) = 0$ *for* $k \geq 2$*. The dual bases are* $\{\mathrm{Tr}_{\partial,j}\}$ *and* $\{i_{\log x}\,\mathrm{Tr}_{\partial,j}\}$*, where* $\mathrm{Tr}_{\partial,j}(A) = \overline{\mathrm{Tr}}(A_{(1)}|_{H_j})$ *[15, Lemma 10].*

Correction to MN 9.4 (Several boundary components). [15, (16), Prop. 6, Lemma 12] assert $\mathrm{HH}^k(\mathcal{I}_\partial) \cong \mathbb{C}$ for $k = 0, 1$. This holds only if ∂X is connected, which is never stated, although MN allow disconnected boundary in Appendix A and use two boundary components in the proof of their Proposition 5. In general the answer is $\mathbb{C}^N$ (Proposition 9.3). The consequences for $\mathcal{A}_\partial$ and $\mathcal{A}$ are worked out in Section 12. The algebras $\mathcal{A}_\partial$, $\mathcal{A}_{\partial,\sigma}$ and $\mathcal{I}_\partial$ are direct sums over the components, so for them nothing else changes. For $\mathcal{A}$, the boundary map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ has rank $N-1$, not 0 (Proposition 12.5).

9.3 The residual ideal and the Laurent smoothing operators

Proposition 9.5. *$\mathcal{I}$ is H-unital,* $\mathrm{HH}_0(\mathcal{I}) \cong \mathbb{C}$ *via the trace and* $\mathrm{HH}_k(\mathcal{I}) = 0$ *for* $k > 0$ *[15, end of §4]. Moreover, for* $x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)$*,*

$$\mathrm{HH}_0 \cong \mathbb{C}^N, \qquad \mathrm{HH}_1 \cong \mathbb{C}^{N-1}, \qquad \mathrm{HH}_k = 0 \ (k \geq 2),$$

and the connecting map $\mathrm{HH}_1(\mathcal{I}_\partial) \to \mathrm{HH}_0(\mathcal{I})$ *is dual to* $\mathrm{Tr} \mapsto -\sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j}$*.*

Proof. $\mathcal{I} = \dot{C}^\infty(X)\widehat{\otimes}\dot{C}^\infty(X)$, so the first part is Lemma 5.4. For the second part, apply Lemma 5.2 to $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{-\infty} \to \mathcal{I}_\partial \to 0$. Extend the trace by the *cusp trace* $\mathrm{Tr}_c(A) = \mathrm{FP}_{z=0}\,\mathrm{Tr}(Ax^z)$, the constant term at $z = 0$. For $\mathrm{Re}\,z$ large,

$$\mathrm{Tr}([A,B]x^z) = \mathrm{Tr}(A[B,x^z]) = -z\ \mathrm{Tr}\big(AL(z)x^z\big), \qquad L(z) = \frac{x^zBx^{-z} - B}{z}, \quad L(0) = [\log x, B].$$

Since $\mathrm{Tr}(AL(z)x^z)$ has at most a simple pole at $z = 0$ with residue $\mathrm{Tr}_\partial(AL(0))$, we get $\mathrm{Tr}_c([A,B]) = -\,\mathrm{Tr}_\partial(A[\log x, B])$. This vanishes if A or B is in $\mathcal{I}$. So the image of $[\mathrm{Tr}]$ is $-\sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j}$, which is non-zero on every component by Proposition 9.2. The connecting map $\mathrm{HH}_1(\mathcal{I}_\partial) = \mathbb{C}^N \to \mathrm{HH}_0(\mathcal{I}) = \mathbb{C}$ is therefore surjective. The long exact sequence gives the result. □

This is the cusp counterpart of the b-trace defect formula and of part (iii) of the main theorem of [7]. MN do not state it; we include it because it is the simplest place where the rank $N-1$ appears.

10 The residue functionals (MN §5)

Fix x and a positive, elliptic, invertible $Q \in \Psi_c^1(X)$ (Remark 4.6). All the trace functionals of [15] come from one meromorphic function of two variables.

10.1 The double zeta function

Definition 10.1 (MN (43))**.** For $A \in x^p\Psi_c^m(X)$ put

$$Z(A;s,z) = Z_{x,Q}(A;s,z) = \tfrac{1}{2}\big(\mathrm{Tr}(Ax^zQ^{-s}) + \mathrm{Tr}(AQ^{-s}x^z)\big), \qquad Z^L(A;s,z) = \mathrm{Tr}(Ax^zQ^{-s}).$$

By (B3) (Theorem C.14), Z is holomorphic for $\operatorname{Re} z > -p+1$ and $\operatorname{Re} s > m+n$. It extends meromorphically, with at most simple poles on the lines $z = -p+1-j$ and $s = m+n-k$, $j,k \in \mathbb{N}_0$ [15, Lemma 4]. The reason for the shift $-p+1$ in z is that the cusp density is $x^{-2}dx\,\nu_\partial$: the integral $\int_0 x^{z+p}x^{-2}dx$ converges for $\operatorname{Re} z > 1-p$. Near the origin $sz\,Z(A;s,z)$ is holomorphic, and we write [15, (44)]

$$sz\,Z(A;s,z) = \operatorname{Tr}_{\partial,\sigma}(A) + s\,\widehat{\operatorname{Tr}}_\partial(A) + z\,\widehat{\operatorname{Tr}}_\sigma(A) + O\big(|s|^2+|z|^2\big). \tag{17}$$

Thus:

- $\operatorname{Tr}_{\partial,\sigma}(A)$ is the coefficient of $(sz)^{-1}$, the *double residue*;
- $\widehat{\operatorname{Tr}}_\partial(A)$ is the residue in z of the regular part in s; it regularizes the boundary trace by Q;
- $\widehat{\operatorname{Tr}}_\sigma(A)$ is the residue in s of the regular part in z; it is the Wodzicki residue regularized at the boundary by x.

Question 10.2. *Why are two complex parameters needed?*

Answer. An operator in $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ fails to be trace class for two independent reasons: its order may be too large, which is a divergence of the kernel at the diagonal, and its kernel may not decay at the boundary, which is a divergence of the integral over the cusp end $\int^\infty dt$. The factor Q^{-s} cures the first and x^z the second. The two filtrations have their own residual ideals $\mathcal{I}_\sigma$ and $\mathcal{I}_\partial$, and their own residues Tr_σ and $\operatorname{Tr}_\partial$. The double residue $\operatorname{Tr}_{\partial,\sigma}$ is the residue of a residue, and it lives on the doubly quotient algebra $\mathcal{A}_{\partial,\sigma}$. It is the trace dual to the top class of $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$: it picks the coefficient of x^1 (the residue in x, because $x \cdot dx/x^2 = dx/x$) and the homogeneity $-n$ (the residue in the fibre). □

On the diagonal $s = z$ we get $\widetilde{Z}(A;z) = Z(A;z,z)$, with

$$z^2\widetilde{Z}(A;z) = \operatorname{Tr}_{\partial,\sigma}(A) + z\big(\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma\big)(A) + O(z^2).$$

Figure 11 shows the picture.

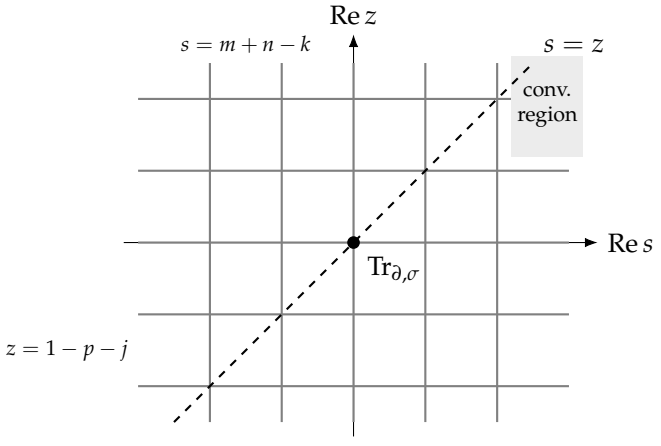


Figure 11: The poles of $Z(A;s,z)$ lie on the gray lines (drawn schematically through the origin; after shifting A by powers of x and Q, lines pass through $(0,0)$). The crossing at the origin produces the double residue $\operatorname{Tr}_{\partial,\sigma}$. The residue along the line $z=0$ and the residue along $s=0$, each regularized in the other variable, give $\widehat{\operatorname{Tr}}_\partial$ and $\widehat{\operatorname{Tr}}_\sigma$. Restricting to the diagonal $s=z$ gives a double pole with leading coefficient $\operatorname{Tr}_{\partial,\sigma}$.

Lemma 10.3 (MN Lemma 5)**.** *$Z^L(A;s,z) - Z(A;s,z)$ is holomorphic near $(0,0)$. Hence the three functionals in* (17) *are the same for Z, Z^L and $Z^R = \operatorname{Tr}(AQ^{-s}x^z)$.*

Proof. By the trace property for Re large, $Z^R - Z^L = \operatorname{Tr}(A[Q^{-s}, x^z])$, and

$$[Q^{-s}, x^z] = \big(Q^{-s}x^zQ^sx^{-z} - 1\big)x^zQ^{-s}.$$

The family $Q^{-s}x^zQ^sx^{-z} - 1$ is entire of order 0 and vanishes on both axes $s = 0$ and $z = 0$. So it equals $sz\,M(s,z)$ with M entire of order 0. Then $Z^R - Z^L = sz\operatorname{Tr}(AMx^zQ^{-s})$, which has no pole at the origin, since $\operatorname{Tr}(AMx^zQ^{-s})$ has at most simple poles in each variable. Finally $Z = \frac{1}{2}(Z^L + Z^R)$. □

Lemma 10.4 (MN Lemma 6)**.** $\operatorname{Tr}_{\partial,\sigma}$ *is a trace on $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$. It does not depend on x or Q, and it vanishes on $\mathcal{I}_\sigma + \mathcal{I}_\partial$. Hence it defines a trace on $\mathcal{A}$, $\mathcal{A}_\sigma$, $\mathcal{A}_\partial$ and $\mathcal{A}_{\partial,\sigma}$.*

Proof. Trace. By Lemma 10.3 we may use Z^L. For Re large,

$$Z^L([A,B]) = \operatorname{Tr}(A[B, x^zQ^{-s}]) = \operatorname{Tr}\big(AK(s,z)x^zQ^{-s}\big), \qquad K = B - x^zQ^{-s}BQ^sx^{-z}.$$

The family K is entire, of fixed order, and vanishes at $(0,0)$. Hence $K = sK_1 + zK_2$ with K_1, K_2 entire. This is Hadamard's lemma in two variables, $K_1 = \int_0^1 \partial_s K(\theta s, \theta z)\,d\theta$, and similarly for K_2. Then

$$sz\,Z^L([A,B]) = s \cdot sz\operatorname{Tr}(AK_1x^zQ^{-s}) + z \cdot sz\operatorname{Tr}(AK_2x^zQ^{-s}),$$

and both brackets are holomorphic near 0. So the value at $(0,0)$ is 0.

Independence. If $x' = ax$, then $x'^z = x^z(1 + zB(z))$ with $B(z) = (a^z - 1)/z$ entire. So $sz(Z^L_{x'} - Z^L_x) = z \cdot sz\operatorname{Tr}(AB(z)x^zQ^{-s})$ vanishes at the origin. The case of Q is the same, with $Q'^{-s} = (1 + sC(s))Q^{-s}$.

Vanishing. If $A \in x^\infty\Psi_c^{\mathbb{Z}}$, then Ax^z is entire in z with values in a fixed space of flat operators. So Z has no pole in z, and $\operatorname{Tr}_{\partial,\sigma}(A) = 0$. If $A \in x^{-\mathbb{Z}}\Psi_c^{-\infty}$, then Z has no pole in s. □

10.2 The interior residue Tr_σ

If $A \in \mathcal{I}_\sigma$, then Z is entire in z. Its residue in s is $\operatorname{Tr}_\sigma(A) = \operatorname{Res}_{s=0}\operatorname{Tr}(AQ^{-s})$, and $\widehat{\operatorname{Tr}}_\sigma$ restricts to Tr_σ on $\mathcal{I}_\sigma$.

Lemma 10.5 (MN Lemma 7 and Proposition 7)**.** Tr_σ *is the extension by continuity of Wodzicki's residue on the double $2X$. If $\sum_j a_j$ is the full symbol of $A \in \mathcal{I}_\sigma$, then*

$$\operatorname{Tr}_\sigma(A) = (2\pi)^{-n}\int_{{}^cS^*X} a_{-n}\,{}^c\nu,$$

where ${}^c\nu$ is the contraction of ω^n with the radial vector field. Since ${}^c\nu = x^{-2}\nu'$ with ν' smooth, the same formula defines Tr_σ on $x^p\Psi_c^{\mathbb{Z}}$ for $p > 1$.

Proof. For A with kernel supported in $(X^\circ)^2$, choose $Q' \in \Psi^1(2X)$ positive and elliptic with $Q'A - QA$ smoothing. Then $\operatorname{Tr}(AQ^{-s}) - \operatorname{Tr}(AQ'^{-s})$ is regular at 0, so $\operatorname{Tr}_\sigma(A) = \operatorname{Tr}_R(A)$. Such operators are dense in $\mathcal{I}_\sigma$, and both sides are continuous. The formula is the local formula (B8) (Theorems C.7 and C.14(1)). The singular density x^{-2} comes from ω^n, which contains dx/x^2. □

10.3 The double residue

For general $A \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ the function $\operatorname{Tr}_\sigma(x^z A)$ is defined for $\operatorname{Re} z$ large and is meromorphic in z [15, (52)].

Lemma 10.6 (MN Lemma 8)**.** $\operatorname{Tr}_{\partial,\sigma}(A) = \operatorname{Res}_{z=0}\operatorname{Tr}_\sigma(x^z A)$. *Explicitly,*

$$\operatorname{Tr}_{\partial,\sigma}(A) = (2\pi)^{-n}\int_{{}^cS^*_{\partial X}X} a_{-n,(1)}\,{}^c\nu_\partial,$$

where $a_{-n,(1)}$ is the term of homogeneity $-n$ in the symbol of the coefficient $A_{(1)}$ of x^1 in (12)*, and ${}^c\nu_\partial$ is the density on ${}^cS^*_{\partial X}X$ obtained by contracting $\omega_\partial^{n-1}d\tau$ with the radial field on ${}^cT^*_{\partial X}X = T^*\partial X \times \mathbb{R}_\tau$.*

Proof. By the definitions, $\operatorname{Tr}_{\partial,\sigma}(A)$ is the z-residue of the s-residue, which is $\operatorname{Res}_z\operatorname{Tr}_\sigma(x^z A)$. Insert the formula of Lemma 10.5: $\int_0 x^z\, x^j\, x^{-2}dx$ has a pole at $z=0$ only for $j=1$, with residue 1. □

Example 10.7 (A boundary point)**.** Let $n=1$ and let $A = x\,c(D_t)$ near the boundary point, with c a classical symbol. The kernel of $xc(D_t)Q^{-s}$ on the diagonal is $x\cdot(2\pi)^{-1}\int c(\tau)\langle\tau\rangle^{-s}d\tau$, and the cusp density is $dt = dx/x^2$. So

$$\operatorname{Res}_z\int_0^1 x^{1+z}\frac{dx}{x^2} = 1, \qquad \operatorname{Tr}_{\partial,\sigma}(A) = \frac{1}{2\pi}\big(c_{-1}(1)+c_{-1}(-1)\big),$$

where $c \sim c_{-1}(\pm1)|\tau|^{-1}$ as $\tau\to\pm\infty$. The two points of $Z=\{\tau=\pm1\}$ contribute separately, *with positive weights*. The density ${}^c\nu_\partial$ is a density, not a form.

Example 10.8 (The four functionals on the half-line)**.** Let $n=1$, and near the boundary point let $A = \varphi(x)\,x^k\,c(D_t)$, where φ is a cut-off equal to 1 near $x=0$, c is a classical symbol of order m, and Q is translation invariant near the boundary with $q=\langle\tau\rangle$. The diagonal value of the kernel of $x^k c(D_t)Q^{-s}$ is $x^k(2\pi)^{-1}\int c(\tau)\langle\tau\rangle^{-s}d\tau$. Integrating against $x^z\,dx/x^2$ gives

$$Z(A;s,z) \approx \Big(\int_0^1 \varphi\, x^{k+z-2}\,dx\Big)\cdot\frac{1}{2\pi}\int_{\mathbb{R}} c(\tau)\langle\tau\rangle^{-s}\,d\tau,$$

up to terms that are holomorphic near the origin. The first factor has a pole at $z=0$ only if $k=1$, with residue 1. The second factor has a pole at $s=0$ with residue $(2\pi)^{-1}(c_{-1}(1)+c_{-1}(-1))$, and regular value $\overline{\operatorname{Tr}}(c)+\gamma'(0)/2\pi$ (Proposition 10.12). Writing $c\sim\sum_j c_j^\pm|\tau|^j$ as $\tau\to\pm\infty$ and expanding $\langle\tau\rangle^{-s} = |\tau|^{-s}\sum_l\binom{-s/2}{l}\tau^{-2l}$, one finds $\gamma'(0) = \sum_{l\ge1}\frac{(-1)^l}{2l}(c^+_{2l-1}+c^-_{2l-1})$. So, for $k=1$,

$$\operatorname{Tr}_{\partial,\sigma}(A) = \frac{c_{-1}(1)+c_{-1}(-1)}{2\pi}, \qquad \widehat{\operatorname{Tr}}_\partial(A) = \overline{\operatorname{Tr}}(c)+\frac{\gamma'(0)}{2\pi},$$
$$\widehat{\operatorname{Tr}}_\sigma(A) = \operatorname{FP}_{z=0}\int_0^1 \varphi\, x^{z-1}dx\cdot\operatorname{Tr}_{\partial,\sigma}(A),$$

and for $k\neq1$, $\operatorname{Tr}_{\partial,\sigma}(A) = \widehat{\operatorname{Tr}}_\partial(A) = 0$. The correction vanishes if c has order ≤ 0; for $c=\langle\tau\rangle$ it is $-1/2\pi$ (Example C.25). This illustrates Lemma 10.6 and Corollary 10.14. The functional $\widehat{\operatorname{Tr}}_\sigma$ is a residue in the fibre variable with a Hadamard finite part in x, and $\widehat{\operatorname{Tr}}_\partial$ is a regularized trace in the fibre variable with a residue in x.

Correction to MN 10.9 (Dimension one). For $n=1$ the set Z_j consists of two points, $\tau=\pm1$, over each boundary point. Then $\operatorname{Tr}_{\partial,\sigma}$ splits as $\operatorname{Tr}^+_{\partial,\sigma}+\operatorname{Tr}^-_{\partial,\sigma}$, and each summand is a trace (the proof of Lemma 10.4 applies with a cut-off in τ). Hence $\mathrm{HH}^0(\mathcal{A}_\sigma)\cong\mathrm{HH}^0(\mathcal{A}_{\partial,\sigma})\cong\mathbb{C}^{2N}$, and similarly $\mathrm{HH}^0(\mathcal{I}_\sigma)\cong H^0(Y\times\mathbb{S}_\sigma)\cong\mathbb{C}^2$ for connected X. The statements "HH^0 is one-dimensional" after [15, (16)] and [15, Lemma 12] require $n\ge2$, exactly as Wodzicki's uniqueness of the residue does. The cohomological formulas of Theorem 1.1 for $\mathcal{I}_\sigma$, $\mathcal{A}_\sigma$, $\mathcal{A}_{\partial,\sigma}$ remain true for $n=1$, with Z_j disconnected.

10.4 The regularized interior residue $\widehat{\mathrm{Tr}}_\sigma$

By Lemma 10.6, $\widehat{\mathrm{Tr}}_\sigma(A)$ is the regular value at $z=0$ of $\mathrm{Tr}_\sigma(x^z A)$. Equivalently, it is the Hadamard finite part [15, (55)]:

$$\widehat{\mathrm{Tr}}_\sigma(A) = \lim_{\epsilon\downarrow 0}\Big((2\pi)^{-n}\int_{{}^cS^*X\cap\{x>\epsilon\}} a_{-n}\,{}^c\nu + (\log\epsilon)\,\mathrm{Tr}_{\partial,\sigma}(A) + \sum_{l>0}\gamma_l\epsilon^{-l}\Big). \tag{18}$$

The sign of the logarithm is right, since $\int_\epsilon^1 x^{-1}dx = -\log\epsilon$.

Lemma 10.10 (MN Lemma 9)**.** $\widehat{\mathrm{Tr}}_\sigma$ *vanishes on $\mathcal{I}_\partial$, does not depend on Q, and for $x' = ax$*

$$\widehat{\mathrm{Tr}}_\sigma(A;x') - \widehat{\mathrm{Tr}}_\sigma(A;x) = \mathrm{Tr}_{\partial,\sigma}\big(A\log a\big).$$

Proof. The residue Tr_σ vanishes on smoothing operators and does not depend on Q. For the change of x, $\mathrm{Tr}_\sigma((x'^z - x^z)A) = \mathrm{Tr}_\sigma(zx^zB(z))$ with $B(z) = z^{-1}(a^z-1)A$ entire (MN print A^z for a^z). Its value at $z=0$ is $\mathrm{Res}_z\,\mathrm{Tr}_\sigma(x^zB(z)) = \mathrm{Tr}_{\partial,\sigma}(B(0)) = \mathrm{Tr}_{\partial,\sigma}(A\log a)$. □

10.5 The boundary trace and the regularized suspended trace

Lemma 10.11 (MN Lemma 10)**.** *On $\mathcal{I}_\partial$, $\mathrm{Tr}_\partial(A) = \mathrm{Res}_{z=0}\mathrm{Tr}(Ax^z) = \overline{\mathrm{Tr}}(A_{(1)})$, where $\overline{\mathrm{Tr}}(B) = (2\pi)^{-1}\int \mathrm{Tr}\,\mathrm{In}(B,\tau)\,d\tau$ [15, (57)]. It is a trace and does not depend on x or Q.*

Proof. On the diagonal of X_c^2 the kernel of A is $B\nu$ with $\nu = x^{-2}\nu'$, and $B\in x^{-\mathbb{Z}}C^\infty(X)$. Write $B\sim\sum_k x^kB_{(k)}$. The residue at $z=0$ of $\int x^zB\,x^{-2}dx\,\nu_\partial$ comes from $k=1$ and equals $\int_{\partial X}B_{(1)}\nu_\partial$. This is the kernel of $A_{(1)}$ at $\{0\}\times\Delta_{\partial X}$ integrated over the boundary, which is $\overline{\mathrm{Tr}}(A_{(1)})$ by Fourier inversion (Theorem C.14(2)). The trace property was proved in Proposition 9.2. Independence is as in Lemma 10.4. □

Melrose's regularization of $\overline{\mathrm{Tr}}$ to all of $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(H)$ [12], [15, (58)–(59)], takes p large, integrates $h_p(\tau) = \mathrm{Tr}(\partial_\tau^pB(\tau))$ p times from 0, and takes $(2\pi)^{-1}$ times the constant term of $\int_{-T}^T$ as $T\to\infty$ (Definition C.22). It does not depend on p, it vanishes on odd families, and it is a trace on $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(H)$ (Theorem C.23).

Proposition 10.12 (MN Proposition 8, corrected)**.** *Let $Q_0 = (D_t^2+\Delta_H+1)^{1/2}$, or more generally let $Q_0(s)$ be a holomorphic suspended family of order $-s$ with $Q_0(0)=\mathrm{Id}$. For $B\in\Psi_{\mathrm{sus}}^m(H)$, the function $\overline{\mathrm{Tr}}(BQ_0^{-s})$ (defined by the integral for* Re s *large) is meromorphic, and*

$$s\,\overline{\mathrm{Tr}}(BQ_0^{-s}) = \widetilde{\mathrm{Tr}}(B) + s\Big(\overline{\mathrm{Tr}}(B) + \frac{\gamma_B'(0)}{2\pi}\Big) + O(s^2),$$

where $\widetilde{\mathrm{Tr}}(B) = (2\pi)^{-n}\int b_{-n}$ is the Wodzicki residue of B regarded as an operator on $\mathbb{R}\times H$ ($n=\dim H+1$), and $\gamma_B(s)$ is the sum over $\tau\to\pm\infty$ of the coefficients of $|\tau|^{-n-s}$ (after integration over the fibres of T^*H) in the symbol of BQ_0^{-s}. The correction $\gamma_B'(0)$ is local, and it is not zero in general.

Proof. This is Theorem C.24 with $C(s) = BQ_0^{-s}$. The residue is $\gamma_B(0)/2\pi = \widetilde{\mathrm{Tr}}(B)$. For the regular value one compares the p-fold integration by parts in the definition of $\overline{\mathrm{Tr}}$ with the continuation in s; the boundary terms produce $\gamma_B'(0)$. [15, Prop. 8] omits this term. For H a point and $B=\langle D_t\rangle$, $\gamma_B(s) = 1-s$, so the regular value is $\overline{\mathrm{Tr}}(B) - 1/2\pi$ (Example C.25). □

Remark 10.13 (The functional in MN Proposition 8). MN write the residue as $\widetilde{\mathrm{Tr}}(B) = \overline{\mathrm{Tr}}([t,B])$. This cannot be right as it stands. Take $H = \mathrm{pt}$ and $B = \langle D_t\rangle^{-1}$. The residue is $(2\pi)^{-1}(1+1) = 1/\pi$, whereas $\mathrm{In}([t,B]) = -i\partial_\tau\langle\tau\rangle^{-1}$ is odd, so $\overline{\mathrm{Tr}}([t,B]) = 0$. (In general $\overline{\mathrm{Tr}}([t,B])$ picks out the difference of the $|\tau|^0$-coefficients at $\pm\infty$, not the $|\tau|^{-1}$ coefficients.) Only the residue description is used later, through Corollary 10.14, so we take it as the definition. The other claim of [15, Prop. 8], about the regular value, is corrected in Proposition 10.12 (item 9 of Section 2.6).

Corollary 10.14 (MN Corollaries 1 and 2; (2) corrected)**.** *For $A \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$:*

1. $\mathrm{Tr}_{\partial,\sigma}(A) = \widetilde{\mathrm{Tr}}(A_{(1)})$*, the residue of* $\mathrm{Tr}_\partial(AQ^{-s})$ *at* $s = 0$ *[15, (61)];*
2. *if Q is translation invariant to all orders at ∂X, with Taylor series Q_0^{-s} at the front face, then*

$$\widehat{\mathrm{Tr}}_\partial(A) = \overline{\mathrm{Tr}}(A_{(1)}) + \frac{\gamma'_{A_{(1)}}(0)}{2\pi},$$

with the local correction $\gamma_{A_{(1)}}$ of Proposition 10.12, computed from $A_{(1)}Q_0^{-s}$. [15] state this without the correction term.

Proof. If Q is translation invariant to all orders, then $(AQ^{-s})_{(1)} = A_{(1)}Q_0^{-s}$, and by Lemma 10.3 and Theorem C.14(2), $\widehat{\mathrm{Tr}}_\partial(A)$ is the regular value at $s = 0$ of $\overline{\mathrm{Tr}}(A_{(1)}Q_0^{-s})$. Proposition 10.12 applies (Correction C.26). In general the residue does not depend on Q. □

Lemma 10.15 (MN Lemma 11)**.** *For two regularizers Q, Q',*

$$\widehat{\mathrm{Tr}}_\partial(A;Q') - \widehat{\mathrm{Tr}}_\partial(A;Q) = -\mathrm{Tr}_{\partial,\sigma}\big(A\log(Q'/Q)\big), \qquad \log(Q'/Q) = \frac{d}{ds}Q'^sQ^{-s}\Big|_{s=0} \in \Psi_c^0(X).$$

Proof. Write $Q'^{-s} = (1+sB(s))Q^{-s}$ with $B(s) = s^{-1}(Q'^{-s}Q^s - 1)$, so that $B(0) = -\log(Q'/Q)$. Then $\mathrm{Tr}_\partial(AQ'^{-s}) - \mathrm{Tr}_\partial(AQ^{-s}) = s\,\mathrm{Tr}_\partial(AB(s)Q^{-s})$. Its value at $s = 0$ is $\mathrm{Res}_s\,\mathrm{Tr}_\partial(AB(0)Q^{-s}) = \mathrm{Tr}_{\partial,\sigma}(AB(0))$. □

Thus $\widehat{\mathrm{Tr}}_\partial$ does not depend on x but does depend on Q, while $\widehat{\mathrm{Tr}}_\sigma$ does not depend on Q but does depend on x.

11 Exterior derivations and explicit cocycles (MN §§6–7)

11.1 The two derivations

The derivations are [15, (63)–(64)]

$$D_{\log Q}A = [\log Q, A] = \frac{d}{ds}Q^sAQ^{-s}\Big|_{s=0}, \qquad D_{\log x}A = [\log x, A] = \frac{d}{dz}x^zAx^{-z}\Big|_{z=0}.$$

On symbols, $\sigma([\log Q, A]) = -i\{\log q, a\}$ lowers the order by one. Similarly $\sigma([\log x, A]) = -i\{\log x, a\} = i\,x\,\partial_\tau a$, which lowers the order by one and raises the power of x by one, by (7). Both derivations preserve $\mathcal{I}$, $\mathcal{I}_\sigma$, $\mathcal{I}_\partial$. Changing Q or x changes them by inner derivations, so the induced maps $i_{\log Q}$, $i_{\log x}$ are canonical (Lemma 5.3).

Lemma 11.1. $\mathrm{Tr}_{\partial,\sigma}([\log Q, B]) = \mathrm{Tr}_{\partial,\sigma}([\log x, B]) = 0$ *for all B [15, (84)].*

Proof. Put $Q^u B Q^{-u} = B + uM(u)$, with M entire and $M(0) = [\log Q, B]$. By the trace property, for Re large,

$$\operatorname{Tr}\big(Q^u B Q^{-u} x^z Q^{-s}\big) = \operatorname{Tr}\big(B x^z (x^{-z} Q^{-u} x^z Q^u) Q^{-s}\big).$$

The family $x^{-z}Q^{-u}x^zQ^u - 1$ vanishes for $u = 0$ and for $z = 0$, so it equals $uz\,N(u,z)$ with N entire. Comparing the two sides, $u\,Z^L(M(u);s,z) = uz\operatorname{Tr}(Bx^zNQ^{-s})$, so $Z^L(M(0);s,z) = z\operatorname{Tr}(Bx^zN(0,z)Q^{-s})$. Multiplying by sz and evaluating at the origin gives $\operatorname{Tr}_{\partial,\sigma}([\log Q, B]) = 0$. For $\log x$, put $x^w B x^{-w} = B + wL(w)$ and use $\operatorname{Tr}(x^w B x^{-w} x^z Q^{-s}) = \operatorname{Tr}\big(Bx^z(x^{-w}Q^{-s}x^wQ^s)Q^{-s}\big)$, where now $x^{-w}Q^{-s}x^wQ^s - 1 = ws\,P(w,s)$. The same argument gives $\operatorname{Tr}_{\partial,\sigma}([\log x, B]) = 0$. This is [15, (84)], where only the combination $[\log x - \log Q, B]$ is treated. □

11.2 Traces generate degree zero

Lemma 11.2 (MN Lemma 12, for N components, $n \geq 2$)**.** $\mathrm{HH}^0(\mathcal{I}_\sigma) = \mathbb{C}\ \operatorname{Tr}_\sigma$. *Moreover* $\mathrm{HH}^0(\mathcal{A}_\sigma)$ *and* $\mathrm{HH}^0(\mathcal{A}_{\partial,\sigma})$ *are spanned by the double residues* $\operatorname{Tr}_{\partial,\sigma,j}$ *of the components (the restriction of* $\operatorname{Tr}_{\partial,\sigma}$ *to* Z_j*), and* $\mathrm{HH}^0(\mathcal{I}_\partial)$ *is spanned by the* $\operatorname{Tr}_{\partial,j}$*.*

Proof. By Theorem 1.1 and Propositions 7.1–9.3, the dimensions are $1, N, N, N$. The functionals listed are traces (Lemmas 10.4, 10.5, 10.11, applied componentwise), and they are linearly independent: by Example 10.7 and Proposition 9.2 each one is non-zero on an element supported near a single component. □

11.3 The isomorphisms Φ_j and the action of the derivations

Using Lefschetz and Poincaré duality as in Proposition 7.1, the homology computations give isomorphisms [15, (65)–(68)]

$$\Phi_1 : H^*(Z \times \mathbb{S}_\sigma) \oplus H^*_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \to \mathrm{HH}^*(\mathcal{A}_\sigma), \qquad \Phi_2 : H^*(Y \times \mathbb{S}_\sigma) \to \mathrm{HH}^*(\mathcal{I}_\sigma),$$
$$\Phi_3 : H^*(Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial) \to \mathrm{HH}^*(\mathcal{A}_{\partial,\sigma}), \qquad \Phi_4 : H^*(\mathbb{S}_\partial)^{\oplus N} \to \mathrm{HH}^*(\mathcal{I}_\partial).$$

(MN print S^*M for Y in (65).) The class $\Phi(\eta)$ is characterized by its values on antisymmetric chains [15, (69)]:

$$\Phi([\eta])\big(\mathsf{A}(f_0 \otimes \cdots \otimes f_p)\big) = \int * \big(\chi_0(f_0 \otimes \cdots \otimes f_p)\big) \wedge \eta, \tag{19}$$

where χ_0 is the HKR map on principal symbols and A is antisymmetrization. MN normalize the circle classes as $d\theta_\sigma = [i\,dr/r]$ and $d\theta_\partial = [i\,dx/x]$. The factor i absorbs the $-i$ in $\sigma([\cdot,\cdot]) = -i\{\cdot,\cdot\}$.

Proposition 11.3 (MN Proposition 9)**.** $i_{\log Q}\Phi_j(\xi) = \Phi_j(d\theta_\sigma \wedge \xi)$ *for* $j = 1,2,3$*, and* $i_{\log x}\Phi_j(\xi) = \Phi_j(d\theta_\partial \wedge \xi)$ *for* $j = 3,4$*.*

Proof. Choose Q with principal symbol r. On leading symbols, $i_{\log Q}$ replaces $f_0 \otimes f_1 \otimes \cdots$ by $f_0\sigma([\log Q, f_1]) \otimes \cdots = -if_0H_{\log r}f_1 \otimes \cdots$. After antisymmetrization this gives

$$\chi_0 \circ i_{\log Q} \circ \mathsf{A} = -i\,\iota_{H_{\log r}} \circ \chi_0 \circ \mathsf{A}.$$

By Proposition 5.9, $\iota_{H_f} * \eta' = \pm * (df \wedge \eta')$. Inserting this into (19) moves $d\log r$ onto η, which is wedge with $d\theta_\sigma$ up to MN's normalization. The same argument with $\log x$ gives $d\theta_\partial$ on the boundary algebras, where x is a variable of the symplectic space. On $\mathcal{A}_\sigma$ and $\mathcal{I}_\sigma$ the derivation $i_{\log x}$ is not of this form, because $\log x$ is singular at the boundary. □

11.4 Explicit cocycles

For $\mathcal{I}_\partial$ the degree one generator is $i_{\log x}\operatorname{Tr}_\partial$. Its value on translation invariant elements was computed in Proposition 9.2. In general, by Lemma 4.4(3),

$$(i_{\log x}\operatorname{Tr}_\partial)(B,C) = -\sum_{k\geq 1}\frac{1}{k}\,\operatorname{Tr}_\partial\big(B\,x^k\,\mathrm{ad}_t^k(C)\big),$$

and only finitely many terms contribute to the coefficient of x^1. [15, (70)] gives a kernel formula for this cocycle. We were not able to match its index conventions with [15, Lemma 10]: with $A\sim\sum x^{-j}A_j$ the products $B_{-j}C_{j-k}$ in (70) have total power x^k, not x^1. We therefore use the formula above.

For the symbol algebras, MN describe cocycles $\tau_b(g_0,\dots,g_p)=\operatorname{Tr}_\sigma(f_0g_0[f_1,g_1]\cdots[f_p,g_p])$ built from chains $b=f_0\otimes\cdots\otimes f_p$. When $\chi_0(b)$ represents η, the class of τ_b is $\Phi_2([\eta])$. Composing with $i_{\log Q}$ and $i_{\log x}$ gives the classes with $d\theta_\sigma$ and $d\theta_\partial$ [15, p. 24]. We do not need these beyond degree two.

12 The homology of $\mathcal{A}_\partial$ and $\mathcal{A}$ (MN §8)

12.1 The boundary maps of the traces: MN (17)

Lemma 12.1 (MN Lemma 13)**.** *For $A,B\in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$,*

$$\widehat{\operatorname{Tr}}_\partial([A,B]) = \operatorname{Tr}_{\partial,\sigma}(A[\log Q,B]) = (i_{\log Q}\operatorname{Tr}_{\partial,\sigma})(A,B),$$
$$\widehat{\operatorname{Tr}}_\sigma([A,B]) = -\operatorname{Tr}_{\partial,\sigma}(A[\log x,B]) = -(i_{\log x}\operatorname{Tr}_{\partial,\sigma})(A,B).$$

(MN print $i_{\log r}$ in the second line.)

Proof. First line. $\widehat{\operatorname{Tr}}_\partial(C)$ is the regular value at $s=0$ of $\operatorname{Tr}_\partial(CQ^{-s})$, by Lemma 10.3. For $C=[A,B]$,

$$\operatorname{Tr}_\partial([A,B]Q^{-s}) = \operatorname{Tr}_\partial(A[B,Q^{-s}]) = s\,\operatorname{Tr}_\partial\big(AM(s)Q^{-s}\big),\qquad M(s)=\frac{B-Q^{-s}BQ^s}{s},$$

using that $\operatorname{Tr}_\partial=\operatorname{Res}_z\operatorname{Tr}(\cdot\,x^z)$ is a trace on operators of order below $-n$. Since $M(0)=[\log Q,B]$, the regular value at $s=0$ is $\operatorname{Res}_s\operatorname{Tr}_\partial(A[\log Q,B]Q^{-s})=\operatorname{Tr}_{\partial,\sigma}(A[\log Q,B])$ by Corollary 10.14.

Second line. $\widehat{\operatorname{Tr}}_\sigma(C)$ is the regular value at $z=0$ of $\operatorname{Tr}_\sigma(Cx^z)$. Since Tr_σ is a trace,

$$\operatorname{Tr}_\sigma([A,B]x^z) = \operatorname{Tr}_\sigma(A[B,x^z]) = -z\,\operatorname{Tr}_\sigma\big(AL(z)x^z\big),\qquad L(z)=\frac{x^zBx^{-z}-B}{z},$$

with $L(0)=[\log x,B]$. The regular value is $-\operatorname{Res}_z\operatorname{Tr}_\sigma(A[\log x,B]x^z)=-\operatorname{Tr}_{\partial,\sigma}(A[\log x,B])$ by Lemma 10.6. □

Both right-hand sides vanish if A or B lies in $\mathcal{I}_\sigma+\mathcal{I}_\partial$. By Lemma 5.2, Lemma 12.1 computes the connecting maps of $0\to\mathcal{I}_\partial\to\mathcal{A}_\partial\to\mathcal{A}_{\partial,\sigma}\to 0$ and $0\to\mathcal{I}_\sigma\to\mathcal{A}_\sigma\to\mathcal{A}_{\partial,\sigma}\to 0$ on the traces [15, (17)–(19)]. The corresponding statements hold for each boundary component separately. Figure 12 puts Lemma 12.1 next to the other trace defects of these notes.

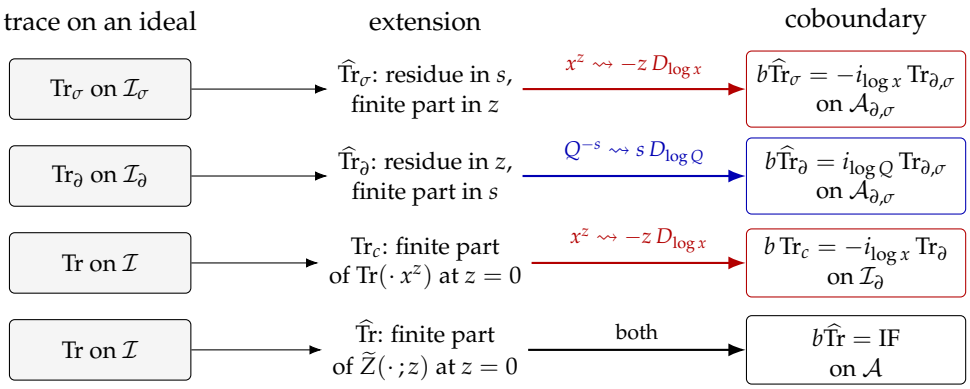


Figure 12: Every boundary map in [15] is computed by Lemma 5.2: extend a trace from an ideal by analytic continuation, then take the Hochschild coboundary $(b\varphi)(A,B) = \varphi([A,B])$. Commuting past the regularizer produces a derivation: $[B, Q^{-s}] = s\,M(s)Q^{-s}$ with $M(0) = [\log Q, B]$, and $[B, x^z] = -z\,L(z)x^z$ with $L(0) = [\log x, B]$. The factor s or z cancels the pole, leaving a residue of $A\,D(B)$. Rows 1–2: Lemma 12.1 [15, (17)]; their sum is $\mathrm{Bf} = (i_{\log Q} - i_{\log x})\,\mathrm{Tr}_{\partial,\sigma}$ (Proposition 14.2). Row 3: Proposition 9.5. Row 4: Lemma 15.2 and Proposition 15.3, where both derivations occur through $\widetilde{D} = [\log x - \log Q, \cdot\,]$.

12.2 The algebra $\mathcal{A}_\partial$

Since $\mathcal{I}_\partial$ is H-unital, $0 \to \mathcal{I}_\partial \to \mathcal{A}_\partial \to \mathcal{A}_{\partial,\sigma} \to 0$ gives a long exact sequence. By Proposition 9.3 it reduces in cohomology to [15, (71)–(72)]

$$0 \to \mathrm{HH}^0(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^0(\mathcal{A}_\partial) \to \mathrm{HH}^0(\mathcal{I}_\partial) \xrightarrow{\partial} \mathrm{HH}^1(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^1(\mathcal{A}_\partial) \to \cdots$$
$$\cdots \to \mathrm{HH}^1(\mathcal{I}_\partial) \xrightarrow{\partial} \mathrm{HH}^2(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}^2(\mathcal{A}_\partial) \to 0,$$

together with $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})$ for $k \geq 3$.

Proposition 12.2 (MN Proposition 10)**.** *For each component H_j,*

$$\partial[\mathrm{Tr}_{\partial,j}] = [i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma,j}], \qquad \partial[i_{\log x}\,\mathrm{Tr}_{\partial,j}] = -[i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma,j}],$$

and both maps ∂ are injective.

Proof. The first formula is Lemma 12.1 with Lemma 5.2, applied to the extension $\widehat{\mathrm{Tr}}_\partial$ of Tr_∂. For the second, the cochain $i_{\log x}\widehat{\mathrm{Tr}}_\partial$ extends $i_{\log x}\,\mathrm{Tr}_\partial$, and by Lemma 5.3

$$b(i_{\log x}\widehat{\mathrm{Tr}}_\partial) = -i_{\log x}(b\widehat{\mathrm{Tr}}_\partial) = -i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}\,.$$

This vanishes when one argument is in $\mathcal{I}_\partial$, because $\mathrm{Tr}_{\partial,\sigma}$ vanishes on $\mathcal{I}_\partial$ and $\mathcal{I}_\partial$ is an ideal preserved by both derivations. By Proposition 11.3, under Φ_3 the two images are $d\theta_\sigma \cdot 1_{Z_j}$ and $-d\theta_\partial \wedge d\theta_\sigma \cdot 1_{Z_j}$. These are linearly independent over j, and non-zero because each Z_j is non-empty. □

Proposition 12.3 (MN Proposition 11)**.** $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})/\mathbb{C}^N$ *for $k = 1, 2$, and* $\mathrm{HH}^k(\mathcal{A}_\partial) \cong \mathrm{HH}^k(\mathcal{A}_{\partial,\sigma})$ *otherwise. Dually,* $\mathrm{HH}_k(\mathcal{A}_\partial)$ *is the kernel of* $\partial : \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}) \to \mathrm{HH}_{k-1}(\mathcal{I}_\partial)$ *for $k = 1, 2$ [15, (75)].*

Proof. Insert the injectivity of Proposition 12.2 into the exact sequence. □

12.3 The algebra $\mathcal{A}$

Now consider $0 \to \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_\sigma \to 0$ [15, (77)]. The quotient maps give a morphism of extensions to $0 \to \mathcal{I}_\partial \to \mathcal{A}_\partial \to \mathcal{A}_{\partial,\sigma} \to 0$. By naturality of the connecting maps,

$$\partial_{\mathcal{A}} = p^* \circ \partial_{\mathcal{A}_\partial} : \mathrm{HH}^i(\mathcal{I}_\partial) \to \mathrm{HH}^{i+1}(\mathcal{A}_\sigma), \qquad p : \mathcal{A}_\sigma \to \mathcal{A}_{\partial,\sigma}. \tag{20}$$

Lemma 12.4 (The map p^*)**.** *Write* $\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) \cong H^k(Z \times \mathbb{S}_\sigma) \oplus H^{k-1}(Z \times \mathbb{S}_\sigma)\, d\theta_\partial$ *and* $\mathrm{HH}^k(\mathcal{A}_\sigma) \cong H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^k(Z \times \mathbb{S}_\sigma)$*. Then p^* is the identity from the first summand to the second, and $\pm\delta$ from the $d\theta_\partial$-summand to H^k_{rel}. Here $\delta : H^{k-1}(Z \times \mathbb{S}_\sigma) \to H^k_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$ is the de Rham connecting map.*

Proof. p^* is dual to $p_* : \mathrm{HH}_k(\mathcal{A}_\sigma) \to \mathrm{HH}_k(\mathcal{A}_{\partial,\sigma})$, which is restriction plus the identity on the $d\theta_\partial$-part (Proposition 8.2). Under Poincaré duality on $Z \times \mathbb{S}_\sigma \times \mathbb{S}_\partial$ the summand without $d\theta_\partial$ in homology pairs with the summand with $d\theta_\partial$ in cohomology, and conversely. The dual of restriction $H^{2n-k}(Y \times \mathbb{S}_\sigma) \to H^{2n-k}(Z \times \mathbb{S}_\sigma)$ under Lefschetz and Poincaré duality is $\pm\delta$, by Stokes' theorem: $\int_Z a|_Z \wedge b = \pm \int_Y a \wedge d(\chi_0 \tilde{b})$ for closed a and b. This is the content of the computation in the proof of [15, Prop. 12], where it is written $i \oplus (-1)^q \partial$. □

Proposition 12.5 (MN Proposition 12, corrected)**.** *Let X be connected and $n \geq 2$. Then:*

1. $\partial_{\mathcal{A}}[\mathrm{Tr}_{\partial,j}] = [i_{\log Q}\, \mathrm{Tr}_{\partial,\sigma,j}] \leftrightarrow d\theta_\sigma \cdot 1_{Z_j} \in H^1(Z \times \mathbb{S}_\sigma)$*, and $\partial_{\mathcal{A}}$ is injective on* $\mathrm{HH}^0(\mathcal{I}_\partial)$*;*
2. $\partial_{\mathcal{A}}[i_{\log x}\, \mathrm{Tr}_{\partial,j}] = \pm\Phi_1\big(\delta(d\theta_\sigma \cdot 1_{Z_j})\big)$*, and $\partial_{\mathcal{A}}$ has rank $N-1$ on* $\mathrm{HH}^1(\mathcal{I}_\partial)$*, with kernel spanned by* $\sum_j i_{\log x}\, \mathrm{Tr}_{\partial,j} = i_{\log x}\, \mathrm{Tr}_\partial$*.*

Consequently $\mathrm{HH}^k(\mathcal{A})$ *is as stated in Theorem 1.1.*

Proof. (1) By (20), Proposition 12.2 and Lemma 12.4, the image is the class $d\theta_\sigma \cdot 1_{Z_j}$ in the summand $H^1(Z \times \mathbb{S}_\sigma)$ of $\mathrm{HH}^1(\mathcal{A}_\sigma)$. These are independent.

(2) Similarly the image is $\pm\delta(d\theta_\sigma \cdot 1_{Z_j})$. By exactness of $H^1(Y \times \mathbb{S}_\sigma) \to H^1(Z \times \mathbb{S}_\sigma) \xrightarrow{\delta} H^2_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma)$, a combination $\sum_j c_j d\theta_\sigma 1_{Z_j}$ lies in $\ker\delta$ if and only if it is the restriction of a class on $Y \times \mathbb{S}_\sigma$. By the Künneth decomposition $H^1(Y \times \mathbb{S}_\sigma) = H^1(Y) \oplus H^0(Y)\, d\theta_\sigma$, and restriction preserves it. Since Y is connected ($n \geq 2$), $H^0(Y) = \mathbb{C}$, and the restriction of $d\theta_\sigma$ is $\sum_j d\theta_\sigma 1_{Z_j}$. So $\ker\delta \cap \mathrm{span}\{d\theta_\sigma 1_{Z_j}\}$ is the line $c_1 = \cdots = c_N$, and the rank is $N-1$.

For the groups, insert (1) and (2) into the exact sequence of $0 \to \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_\sigma \to 0$:

$$\begin{aligned}
&\mathrm{HH}^0(\mathcal{A}) \cong \mathrm{HH}^0(\mathcal{A}_\sigma) \cong \mathbb{C}^N,\\
&\mathrm{HH}^1(\mathcal{A}) \cong \mathrm{HH}^1(\mathcal{A}_\sigma)/\mathbb{C}^N \oplus \ker\big(\partial_{\mathcal{A}}|_{\mathrm{HH}^1(\mathcal{I}_\partial)}\big) \cong H^1_{\mathrm{rel}}(Y \times \mathbb{S}_\sigma) \oplus H^1(Z \times \mathbb{S}_\sigma)/\mathbb{C}^N \oplus \mathbb{C},\\
&\mathrm{HH}^2(\mathcal{A}) \cong \mathrm{HH}^2(\mathcal{A}_\sigma)/\mathbb{C}^{N-1}, \qquad \mathrm{HH}^k(\mathcal{A}) \cong \mathrm{HH}^k(\mathcal{A}_\sigma),\ k \geq 3.
\end{aligned}$$

The splitting in degree one depends on a choice of extension of $i_{\log x}\, \mathrm{Tr}_\partial$ to $\mathcal{A}$. One choice is $-\mathrm{IF}$ (Section 15). □

Correction to MN 12.6 (MN Proposition 12). MN find $\partial[i_{\log x}\, \mathrm{Tr}_\partial] = 0$, because "the form $d\theta_\sigma$ has an extension to the whole manifold" [15, p. 27]. That argument is correct for the *sum* $\sum_j d\theta_\sigma 1_{Z_j}$, which is all there is when $N = 1$. For $N \geq 2$ the individual classes $d\theta_\sigma 1_{Z_j}$ do not extend, and $\mathrm{HH}^2(\mathcal{A})$ is smaller than [15, (16)] claims by $N-1$ dimensions. The same rank appears in the b-calculus [7, Remark "Several boundary components"]. If X has c components, each with non-empty boundary, the rank is $N - c$. Figure 13 illustrates the count.

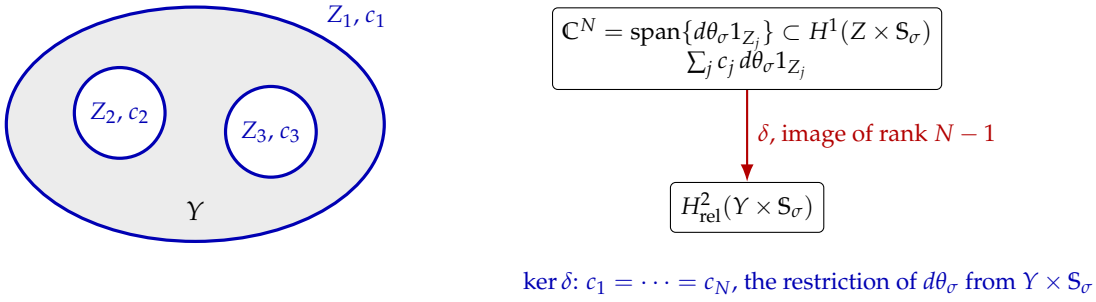


Figure 13: Why $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ has rank $N-1$ (Proposition 12.5, Correction 12.6), drawn schematically for $N=3$, with the connected manifold Y shown as a planar domain whose boundary Z has components Z_1, Z_2, Z_3. The class $i_{\log x}\,\mathrm{Tr}_{\partial,j}$ goes to $\pm\Phi_1(\delta(d\theta_\sigma 1_{Z_j}))$, where δ is the de Rham connecting map. A combination $\sum_j c_j d\theta_\sigma 1_{Z_j}$ lies in $\ker\delta$ exactly when it is the restriction of a class on $Y \times \mathbb{S}_\sigma$; since Y is connected ($n \geq 2$) this means $c_1 = \cdots = c_N$. Dually, by Stokes' theorem, $\sum_j \int_{Z_j} \beta = 0$ for every closed β on Y. The kernel is spanned by $i_{\log x}\,\mathrm{Tr}_\partial = \sum_j i_{\log x}\,\mathrm{Tr}_{\partial,j}$; MN's argument sees only this sum.

Discussion. **How the index formula comes out of the boundary maps.** The index functional is a class in $\mathrm{HH}^1(\mathcal{A})$ whose restrictions to the two ideals are known: $i_{\log Q}\,\mathrm{Tr}_\sigma$ on $\mathcal{I}_\sigma$ (Atiyah–Singer) and $-i_{\log x}\,\mathrm{Tr}_\partial$ on $\mathcal{I}_\partial$ (Toeplitz). A pair of cocycles on the two ideals extends to $\mathcal{A}$ only if their images in $\mathrm{HH}^2(\mathcal{A}_{\partial,\sigma})$ agree. By Lemma 12.1 and Proposition 12.2, both images equal $[i_{\log x} i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}]$. The extension is unique up to the image of $\mathrm{HH}^1(\mathcal{A}_\sigma)$ and $\mathrm{HH}^1(\mathcal{A}_\partial)$, and analytic continuation picks one: the regularized value of $\widetilde{Z}([A,B];z)$. Evaluating it on $[A] \otimes [A^{-1}]$ splits the index into an interior symbolic term and a boundary indicial term (Section 15).

12.4 The full operator algebra

The ideal $\mathcal{I}$ is H-unital with $\mathrm{HH}_*(\mathcal{I}) = \mathbb{C}$ in degree 0 (Proposition 9.5). The connecting map $\mathrm{HH}_1(\mathcal{A}) \to \mathrm{HH}_0(\mathcal{I}) = \mathbb{C}$ is the index functional, which is non-zero: there are fully elliptic operators of non-zero index (Example 16.2). Hence $\mathrm{HH}_k(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) \cong \mathrm{HH}_k(\mathcal{A})$ for $k \neq 1$, and $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)) = \ker \mathrm{IF} \subset \mathrm{HH}_1(\mathcal{A})$. This completes the proof of Theorem 1.1.

12.5 Examples

Example 12.7 (The cylinder)**.** Let $X = [0,1] \times S^1$, so $n=2$ and $N=2$. The cusp cotangent bundle is trivial, so $Y \cong [0,1] \times T^2$ and $Z = \{0,1\} \times T^2$. The cohomology of tori and $H^k_{\mathrm{rel}}([0,1] \times M) \cong H^{k-1}(M)$ give the following dimensions, for $k = 0,1,2,3,4$:

	$k=0$	1	2	3	4
$\mathrm{HH}^k(\mathcal{I}_\sigma) = H^k(T^3)$	1	3	3	1	0
$\mathrm{HH}^k(\mathcal{A}_\sigma)$	2	7	9	5	1
$\mathrm{HH}^k(\mathcal{A}_{\partial,\sigma}) = 2H^k(T^4)$	2	8	12	8	2
$\mathrm{HH}^k(\mathcal{I}_\partial)$	2	2	0	0	0
$\mathrm{HH}^k(\mathcal{A}_\partial)$	2	6	10	8	2
$\mathrm{HH}^k(\mathcal{A})$, Theorem 1.1	2	6	8	5	1
$\mathrm{HH}^k(\mathcal{A})$, [15, (16)] read literally	2	7	9	5	1

In degree one the correct quotient is by $\mathbb{C}^N$, not $\mathbb{C}$, and in degree two by the rank $N-1=1$ image of Correction 12.6. In degree zero the formula of [15, (16)] already gives 2, although the text asserts that HH^0 is one-dimensional. Note that MN's own product computation in the proof of [15, Prop. 5] (with $\partial_1 = -\partial_0$) is the cylinder case, and is consistent with these numbers.

Example 12.8 (The interval)**.** Let $X=[0,1]$, so $n=1$ and $N=2$. Then $Y=[0,1]\times\{\pm\}$, two intervals, and Z consists of four points. The same bookkeeping (Correction 10.9) gives

$$\dim \mathrm{HH}^k(\mathcal{A}_\sigma) = 4,6,2, \qquad \dim \mathrm{HH}^k(\mathcal{A}_\partial) = 4,6,2, \qquad \dim \mathrm{HH}^k(\mathcal{A}) = 4,5,1 \qquad (k=0,1,2).$$

Here $\partial_{\mathcal{A}}$ on $\mathrm{HH}^1(\mathcal{I}_\partial)=\mathbb{C}^2$ has rank one: the two classes $\delta(d\theta_\sigma(1_{j,+}+1_{j,-}))$, $j=0,1$, are opposite, since $1_{0,\pm}+1_{1,\pm}$ is the restriction of $1_{Y_\pm}$.

13 Morita invariance and bundles (MN §9)

To have operators of non-zero index one needs bundles. Embed $E\subset X\times\mathbb{C}^N$, and let $e\in M_N(C^\infty(X))$ be the orthogonal projection onto E. Then $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E) = eM_N(x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X))e$, and similarly for all the quotient algebras and ideals. The idempotent e lies in no proper two-sided ideal, so each corner algebra is Morita equivalent to the trivial-bundle algebra, and the Hochschild groups are the same (B6) [15, Prop. 13] (Proposition B.24 and Remark B.25; for the H-unital ideals the bundle version is sketched there). In the symbol algebras this replaces $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X)$ by $x^{-\mathbb{Z}}\mathcal{G}({}^cT^*X;\hom E)$, and the HKR map uses the fibre trace. All trace functionals extend by composing with the fibre trace tr. From now on all algebras act on sections of E.

14 The boundary index (MN §10)

Definition 14.1 (MN (78))**.** For $A,B\in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$ put $\mathrm{Bf}(A,B)=\lim_{z\to0} z\,\widetilde{Z}([A,B];z)$.

The limit exists because $\mathrm{Tr}_{\partial,\sigma}([A,B])=0$ (Lemma 10.4), so the double pole of $\widetilde{Z}([A,B];z)$ is absent.

Proposition 14.2 (MN Proposition 14)**.** *For all A,B,*

$$\mathrm{Bf}(A,B) = \mathrm{Tr}_{\partial,\sigma}\big(B[\log x-\log Q,A]\big) = \big(-i_{\log x}\,\mathrm{Tr}_{\partial,\sigma} + i_{\log Q}\,\mathrm{Tr}_{\partial,\sigma}\big)(A,B) = \big(\widehat{\mathrm{Tr}}_\partial+\widehat{\mathrm{Tr}}_\sigma\big)([A,B]).$$

Bf *vanishes if A or B lies in $\mathcal{I}_\sigma$ or $\mathcal{I}_\partial$. It is a Hochschild* 1*-cocycle on $\mathcal{A}_{\partial,\sigma}$, and under Φ_3 it corresponds to $-d\theta_\partial+d\theta_\sigma$ on each component Z_j.*

Proof. By the trace property, $\mathrm{Tr}([A,B]Q^{-z}x^z) = \mathrm{Tr}\big([A,B](1+z^2C(z))x^zQ^{-z}\big)$, where

$$C(z) = z^{-2}\big(Q^{-z}x^zQ^zx^{-z}-1\big)$$

is entire because the bracket vanishes on both axes. Hence [15, (79)]

$$\widetilde{Z}([A,B];z) = \mathrm{Tr}([A,B]x^zQ^{-z}) + \tfrac12 z^2\,\mathrm{Tr}\big([A,B]C(z)x^zQ^{-z}\big).$$

The second term has at most a simple pole, so it does not contribute to Bf. By the Baker–Campbell–Hausdorff formula, $C(0) = -[\log Q,\log x] = [\log x,\log Q]$. MN write $2[\log Q,\log x]$ in [15, (80)], but only the fact that C is entire matters. Next,

$$\mathrm{Tr}([A,B]x^zQ^{-z}) = \mathrm{Tr}\big(B[x^zQ^{-z},A]\big),$$

$$[x^z Q^{-z}, A] = (x^z A x^{-z} - A) x^z Q^{-z} + x^z (Q^{-z} A Q^z - A) Q^{-z}.$$

Write $zL_1(z) = x^z A x^{-z} - A$ and $zL_2(z) = A - Q^z A Q^{-z}$, so that $L_1(0) = [\log x, A]$ and $L_2(0) = -[\log Q, A]$. Since $x^z(Q^{-z}AQ^z - A)Q^{-z} = x^z Q^{-z} \cdot zL_2(z)$, the trace property gives [15, (81)]

$$z \operatorname{Tr}([A,B] x^z Q^{-z}) = z^2 \operatorname{Tr}\big(B L_1(z) x^z Q^{-z}\big) + z^2 \operatorname{Tr}\big(L_2(z) B x^z Q^{-z}\big).$$

At $z = 0$ the right side is $\operatorname{Tr}_{\partial,\sigma}(BL_1(0) + L_2(0)B) = \operatorname{Tr}_{\partial,\sigma}(B[\log x, A] - [\log Q, A]B)$. This equals $\operatorname{Tr}_{\partial,\sigma}(B[\log x - \log Q, A])$, since $\operatorname{Tr}_{\partial,\sigma}$ is a trace. The second expression follows from $\operatorname{Tr}_{\partial,\sigma}(A[D,B]) = -\operatorname{Tr}_{\partial,\sigma}([D,A]B) = -\operatorname{Tr}_{\partial,\sigma}(B[D,A])$ (Lemma 11.1). The third is Lemma 12.1. Vanishing on $\mathcal{I}_\sigma + \mathcal{I}_\partial$ holds because $\operatorname{Tr}_{\partial,\sigma}$ vanishes there and the ideals are preserved by the derivations. The cohomology class is given by Proposition 11.3. □

Since $\mathcal{I}_\sigma \cdot \mathcal{I}_\partial \subset \mathcal{I}_\sigma \cap \mathcal{I}_\partial = 0$ in $\mathcal{A}$, the ideal $\mathcal{I}_\sigma + \mathcal{I}_\partial = \mathcal{I}_\sigma \oplus \mathcal{I}_\partial$ is H-unital. The last expression in Proposition 14.2 shows, by Lemma 5.2, that Bf is the image of the trace $\operatorname{Tr}_\partial + \operatorname{Tr}_\sigma$ under the connecting map of $0 \to \mathcal{I}_\sigma + \mathcal{I}_\partial \to \mathcal{A} \to \mathcal{A}_{\partial,\sigma} \to 0$ [15, p. 28].

Proposition 14.3 (MN Proposition 15)**.** *For invertible $A \in \mathcal{A}_{\partial,\sigma}$ put* c-ind$(A) = \operatorname{Bf}(A, A^{-1})$*. This is constant along smooth paths of invertible elements, and it vanishes if A is the image of an invertible element of $\mathcal{A}$.*

Proof. Let $\widetilde{D} = [\log x - \log Q, \cdot\,]$ [15, (83)]. By Proposition 14.2, $\operatorname{Bf}(A,B) = \operatorname{Tr}_{\partial,\sigma}(B\widetilde{D}(A))$. For a smooth path A_t,

$$\frac{d}{dt} \operatorname{Bf}(A_t, A_t^{-1}) = \operatorname{Tr}_{\partial,\sigma}\big(A^{-1}\widetilde{D}(\dot A) - A^{-1}\dot A A^{-1}\widetilde{D}(A)\big) = \operatorname{Tr}_{\partial,\sigma}\big(\widetilde{D}(A^{-1}\dot A)\big),$$

because $\widetilde{D}(A^{-1}\dot A) = -A^{-1}\widetilde{D}(A)A^{-1}\dot A + A^{-1}\widetilde{D}(\dot A)$ and $\operatorname{Tr}_{\partial,\sigma}$ is a trace. The right side vanishes by Lemma 11.1. If A lifts to $A' \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}$ with an inverse B' modulo $\mathcal{I}$, then $[A', B'] \in \mathcal{I}$. So $\widetilde{Z}([A', B']; z)$ is entire and Bf $= 0$. □

Invertibility in $\mathcal{A}_{\partial,\sigma}$ means ellipticity of the full symbol near the boundary, in the Laurent sense. Liftability to an invertible element of $\mathcal{A}$ means that, in addition, a representative is elliptic in the interior and has invertible indicial family. So c-ind is an obstruction to completing boundary symbol data to a fully elliptic operator [15, Prop. 15].

Example 14.4 (A boundary index in dimension one)**.** Let $n = 1$ with one boundary point. The algebra $\mathcal{A}_{\partial,\sigma} = C^\infty(\{\pm\})((x, q^{-1}))$ splits as $\mathcal{A}^+_{\partial,\sigma} \oplus \mathcal{A}^-_{\partial,\sigma}$ by the central idempotents $1_\pm$, the symbols that are 1 for $\pm\tau > 0$. For $k \in \mathbb{Z}$ let

$$A = x^k 1_+ + 1_-, \qquad A^{-1} = x^{-k} 1_+ + 1_-.$$

We compute $\operatorname{Bf}(A, A^{-1}) = \operatorname{Tr}_{\partial,\sigma}(A^{-1}[\log x - \log Q, A])$ with Q translation invariant near the boundary and $q = |\tau|$ there. First, $[\log x, A] = 0$ in $\mathcal{A}_{\partial,\sigma}$: $\log x$ commutes with x^k, and $[\log x, 1_\pm]$ has symbol $ix\partial_\tau 1_\pm = 0$. Second, the star product expansion gives

$$[\log Q, x^k] = -i\{\log|\tau|, x^k\} + (\text{order} \le -2) = -ik\,\frac{x^{k+1}}{\tau} + (\text{order} \le -2),$$

by (7). Hence the coefficient of x^1 in $-A^{-1}[\log Q, A]$ has homogeneous part of degree -1 equal to $ik\tau^{-1}1_+$. By Example 10.7,

$$\text{c-ind}(A) = \frac{1}{2\pi}\Big(\frac{ik}{1}\Big) = \frac{ik}{2\pi}.$$

For $A = x^k$ (that is, $x^k 1_+ + x^k 1_-$) the two points contribute $ik/2\pi$ and $ik/(2\pi\cdot(-1))$, which cancel, as they must: x^k is invertible in $\mathcal{A}$. For $k \neq 0$, $x^k 1_+ + 1_-$ cannot be lifted to a fully elliptic operator. Indeed any lift has leading indicial coefficient vanishing on a half-line of τ.

15 The index cocycle and the index formula (MN §§11–12)

15.1 The index cocycle

Definition 15.1 (MN (85))**.** For $A, B \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$ let $\mathrm{IF}(A,B)$ be the regular value at $z=0$ of $\widetilde{Z}([A,B];z)$, that is, $\mathrm{IF}(A,B) = \lim_{z\to 0}\big(\widetilde{Z}([A,B];z) - z^{-1}\,\mathrm{Bf}(A,B)\big)$.

Lemma 15.2 (MN Lemma 14; Fedosov's formula)**.** *If A is fully elliptic and $B \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$ satisfies $[B] = [A]^{-1}$ in $\mathcal{A}$, then A is Fredholm on $\dot{C}^\infty(X;E)$ and* $\mathrm{Ind}(A) = \mathrm{IF}(A,B)$.

Proof. By (B2) (Theorem A.35(3)), A has a generalized inverse B_0 with $AB_0 = 1-\Pi_1$ and $B_0A = 1-\Pi_0$, where Π_0 projects onto $\ker A$ and Π_1 onto a complement of the range. Both are finite rank and lie in $\mathcal{I}$. Then

$$\operatorname{Ind} A = \operatorname{Tr}\Pi_0 - \operatorname{Tr}\Pi_1 = \operatorname{Tr}(AB_0 - B_0A) = \operatorname{Tr}([A,B_0]).$$

Since $[A,B_0] \in \mathcal{I}$, the function $\widetilde{Z}([A,B_0];z)$ is entire with value $\operatorname{Tr}([A,B_0])$ at 0. So $\mathrm{Bf}(A,B_0) = 0$ and $\mathrm{IF}(A,B_0) = \operatorname{Ind} A$. Finally IF vanishes when either argument lies in $\mathcal{I}$, and $B - B_0 \in \mathcal{I}$. □

Lemma 15.2 says that IF represents the image of $[\mathrm{Tr}]$ under the connecting map of $0 \to \mathcal{I} \to x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}} \to \mathcal{A} \to 0$. This is Lemma 5.2 with the extension $\widehat{\mathrm{Tr}}(C) =$ the regular value of $\widetilde{Z}(C;z)$. Nistor's compatibility theorem [17] identifies this connecting map with the index map in K-theory [15, p. 29].

Proposition 15.3 (MN Proposition 16)**.** IF *descends to a* 1*-cocycle on $\mathcal{A}$, and*

$$\mathrm{IF}(A,B) = \tfrac{1}{2}\big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big), \qquad \widetilde{D} = [\log x - \log Q, \cdot\,].$$

Proof. By definition $\mathrm{IF}(A,B) = \frac{d}{dz}\big(z\widetilde{Z}([A,B];z)\big)|_{z=0}$. Symmetrizing the computation in the proof of Proposition 14.2 gives [15, (88)]

$$2\widetilde{Z}([A,B];z) = z\operatorname{Tr}\big((BL_1 + L_2B)x^zQ^{-z}\big) + z\operatorname{Tr}\big((BL_3 + L_4B)Q^{-z}x^z\big),$$

with $zL_3(z) = Q^{-z}AQ^z - A$ and $zL_4(z) = A - x^{-z}Ax^z$. Hence $L_1(0) = L_4(0) = [\log x, A]$ and $L_2(0) = L_3(0) = -[\log Q, A]$. Expanding $x^{\pm z}Ax^{\mp z}$ to second order shows $L_1'(0) + L_4'(0) = 0$, and similarly $L_2'(0) + L_3'(0) = 0$. By (17) and Lemma 10.3, both $z^2\operatorname{Tr}(Cx^zQ^{-z})$ and $z^2\operatorname{Tr}(CQ^{-z}x^z)$ equal $\mathrm{Tr}_{\partial,\sigma}(C) + z(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(C) + O(z^2)$, also for holomorphic families $C(z)$. Therefore

$$\begin{aligned} 2\,\mathrm{IF}(A,B) = {}& \big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(BL_1(0) + L_2(0)B + BL_3(0) + L_4(0)B\big) \\ & + \mathrm{Tr}_{\partial,\sigma}\big(B(L_1' + L_3')(0) + (L_2' + L_4')(0)B\big). \end{aligned}$$

Since $\mathrm{Tr}_{\partial,\sigma}$ is a trace, the last term equals $\mathrm{Tr}_{\partial,\sigma}\big(B(L_1' + L_4' + L_2' + L_3')(0)\big) = 0$. The first term is $(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(B\widetilde{D}(A) + \widetilde{D}(A)B)$. Finally IF vanishes if an argument lies in $\mathcal{I}$. □

Remark 15.4 (General regularizing families). The proof uses that Q^{-s} is a one-parameter group, through $L_2'(0) + L_3'(0) = 0$. If Q^{-s} is replaced by an arbitrary admissible family $Q(s)$ (Appendix C), then $Q(s)Q(u) - Q(s+u) = su\,W$, and Proposition C.15(2) shows that

$$\mathrm{IF}_Q(A,B) = \tfrac{1}{2}\big(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big) + \tfrac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}\big([A,B]\kappa_Q\big), \qquad \kappa_Q = Q''(0) - Q'(0)^2 \in \Psi_c^0.$$

The extra term is the coboundary of $C \mapsto \frac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}(C\kappa_Q)$. It vanishes when $[A,B] \in \mathcal{I}$ (indeed when $[A,B] \in \mathcal{I}_\sigma + \mathcal{I}_\partial$), in particular when $[B] = [A]^{-1}$ in $\mathcal{A}$, and it vanishes identically for genuine complex powers. So Theorem 15.6 and everything after it hold verbatim for any admissible family.

Correction to MN 15.5 (The sign of $\widetilde{D}$). [15, (21)] defines $\widetilde{D}(A) = [\log Q - \log x, A]$, while [15, (83)] and the proof of [15, Prop. 16] use $[\log x - \log Q, A]$. The computation above, which follows (88) of [15], gives $[\log x - \log Q, A]$. The examples in Section 16 confirm this sign independently.

15.2 Restrictions to the ideals

On $\mathcal{I}_\partial$ the Q-regularization is unnecessary. By Proposition 9.5, $\mathrm{IF}|_{\mathcal{I}_\partial} = -i_{\log x}\operatorname{Tr}_\partial$, the *Toeplitz cocycle*. On $\mathcal{I}_\sigma$ the x-regularization is unnecessary, and the closed-manifold computation (Proposition 3.4) gives $\mathrm{IF}|_{\mathcal{I}_\sigma} = i_{\log Q}\operatorname{Tr}_\sigma$, the *Atiyah–Singer cocycle*. So IF extends $-i_{\log x}\operatorname{Tr}_\partial$ from $\mathcal{I}_\partial$ to $\mathcal{A}$. This is why $\partial[i_{\log x}\operatorname{Tr}_\partial] = 0$ (Proposition 12.5). The compatibility $\partial[i_{\log Q}\operatorname{Tr}_\sigma] = \partial[i_{\log x}\operatorname{Tr}_\partial]$ in $\mathrm{HH}^2(\mathcal{A}_{\partial,\sigma})$ is the statement that both equal $[i_{\log x} i_{\log Q}\operatorname{Tr}_{\partial,\sigma}]$ (Proposition 12.2) [15, p. 30].

15.3 The four terms

For invertible $A \in \mathcal{A}$ MN define [15, (89)–(92)]

$$\overline{\eta}(A) = -\widehat{\operatorname{Tr}}_\partial\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big), \quad \overline{\mathrm{AS}}(A) = \tfrac{1}{2}\widehat{\operatorname{Tr}}_\sigma\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big),$$
$$\beta_Q(A) = \tfrac{1}{2}\widehat{\operatorname{Tr}}_\partial\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big), \quad \gamma(A) = \tfrac{1}{2}\widehat{\operatorname{Tr}}_\sigma\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big).$$

(MN call β_Q simply β.) Inserting $B = A^{-1}$ in Proposition 15.3 and using $\widetilde{D} = D_{\log x} - D_{\log Q}$ gives the following.

Theorem 15.6 (Index theorem, MN (93), corrected)**.** *For every fully elliptic $A \in x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$,*

$$\operatorname{Ind}(A) = -\overline{\mathrm{AS}}(A) - \tfrac{1}{2}\overline{\eta}(A) - \beta_Q(A) + \gamma(A).$$

Equivalently, with $\mathrm{AS}(A) = -\overline{\mathrm{AS}}(A) = \tfrac{1}{2}\widehat{\operatorname{Tr}}_\sigma\big(A[\log Q, A^{-1}] + [\log Q, A^{-1}]A\big)$ *and* $\beta(A) = -\beta_Q(A)$*,*

$$\operatorname{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\overline{\eta}(A) + \beta(A) + \gamma(A).$$

Proof. By Proposition 15.3 with $B = A^{-1}$,

$$\begin{aligned} 2\,\mathrm{IF}(A, A^{-1}) &= (\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma)\big(A^{-1}[\log x, A] + [\log x, A]A^{-1}\big) \\ &\quad - (\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma)\big(A^{-1}[\log Q, A] + [\log Q, A]A^{-1}\big). \end{aligned}$$

In order, the four pieces are $-\overline{\eta}$, 2γ, $-2\beta_Q$ and $-2\overline{\mathrm{AS}}$. For the second form note that

$$A[\log Q, A^{-1}] = -[\log Q, A]A^{-1}, \qquad [\log Q, A^{-1}]A = -A^{-1}[\log Q, A]. \qquad \square$$

Correction to MN 15.7 (Signs in MN (93), (22)–(23), (98)–(100)). MN state $\operatorname{Ind}(A) = \overline{\mathrm{AS}}(A) - \tfrac{1}{2}\overline{\eta}(A) + \beta(A) + \gamma(A)$ with $\overline{\mathrm{AS}}$, β as in [15, (90), (91)]. With those definitions the signs of $\overline{\mathrm{AS}}$ and β are wrong. The following checks support this. The second and third are not independent of each other, since the circle is a closed manifold, and the sign of β rests on the derivation alone:

1. The derivation from [15, (87)] (Proposition 15.3) gives Theorem 15.6.
2. On a closed manifold $\overline{\eta} = \beta = \gamma = 0$, and MN's own [15, (3), (5)] give (Proposition 3.4)
$$\operatorname{Ind} A = \operatorname{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big) = -\overline{\mathrm{AS}}(A).$$
3. On the circle, Section 3.3 gives $\operatorname{Ind} A = -1$ and $\overline{\mathrm{AS}}(A) = +1$ for the Toeplitz-type operator there.

Consequently the symbolic term in [15, (23)] and [15, (100)] should be $\operatorname{Tr}_{\mathrm{R}}(A[\log Q, A^{-1}]) = -\operatorname{Tr}_{\mathrm{R}}(A^{-1}[\log Q, A])$. The sign of the eta term is correct; see Example 16.2. Identifying $\overline{\eta}$ with the eta invariant of [12] assumes that its parametrization of the suspended variable agrees with MN Definition 4; reversing that parametrization reverses the sign. (Here $\overline{\eta}$ is identified with $\eta(A_0)$ up to the local term δ_Q of Proposition 15.8.) The interval example and the Atiyah–Patodi–Singer convention support the choice made here.

Figure 14 shows how the four terms arise and what becomes of them under the hypotheses of Theorem 15.13 below.

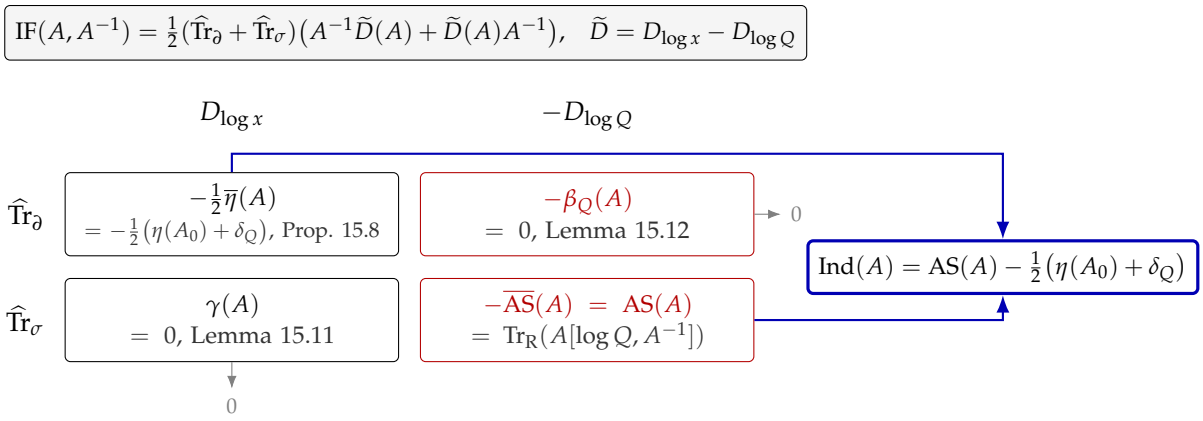


Figure 14: The four terms of Theorem 15.6. Inserting $B = A^{-1}$ in Proposition 15.3 and splitting $\widetilde{D} = D_{\log x} - D_{\log Q}$ gives one term for each pair of a regularized trace and a derivation: $\mathrm{Ind}(A) = -\overline{\mathrm{AS}}(A) - \frac{1}{2}\overline{\eta}(A) - \beta_Q(A) + \gamma(A)$. The terms in red carry the signs corrected from [15, (93)] (Correction 15.7). For A fully elliptic and translation invariant near the boundary with normal indicial family, and Q as in Theorem 15.13, the grey annotations apply: γ and β_Q vanish, $\overline{\eta}$ is the eta invariant of A_0 plus the local correction δ_Q coming from [15, Prop. 8] (item 9 of Section 2.6), and the result is the reduced formula. In the examples of Section 16, $\delta_Q = 0$.

15.4 The eta term

Proposition 15.8 (MN Proposition 17, corrected)**.** *Let $A \in \Psi_c^m(X;E)$ be fully elliptic, and let Q be translation invariant to all orders at ∂X, with Taylor series Q_0^{-s} at the front face. Let $A_0 = \mathrm{In}(A) \in \Psi_{\mathrm{sus}}^m(\partial X;E)$ and $t = 1/x$. Then*

$$\overline{\eta}(A) = \eta(A_0) + \delta_Q(A), \qquad \eta(A_0) = 2\,\overline{\mathrm{Tr}}\big(A_0^{-1}[t,A_0]\big) = \frac{1}{\pi i}\overline{\int_{\mathbb{R}} \mathrm{Tr}\big(A_0(\tau)^{-1}\partial_\tau A_0(\tau)\big)d\tau},$$

where $\eta(A_0)$ is Melrose's eta invariant of [12], (B7), and $\delta_Q(A) = (2\pi)^{-1}\gamma'(0)$, with γ the correction of Proposition 10.12 computed from $A_0^{-1}[t,A_0]Q_0^{-s}$ and $[t,A_0]A_0^{-1}Q_0^{-s}$. The term $\delta_Q(A)$ is a local functional of the boundary symbols of A and Q. It vanishes:

1. *when the terms of homogeneity $-n-s$ of these two symbols are odd in τ; in particular when ∂X is a point and Q_0^{-s} is even in τ (Example 16.2);*
2. *when n is odd and A is a differential operator (Remark C.27, a sketch);*
3. *when $A_0 = \sigma(i\tau + B_\partial)$ is in APS normal form and $Q_0^{-s} = (D_t^2 + B_\partial^2)^{-s/2}$ (Proposition C.30(ii));*
4. *when A is a Dirac operator compatible with a Clifford structure and $Q = (A^*A+1)^{1/2}$ (Remark C.31, quoting Bismut–Freed [2]).*

Proof. This is Proposition C.30(i). In outline: a parametrix $B \in \Psi_c^{-m}$ has $\mathrm{In}(B) = A_0^{-1}$. By Lemma 4.4(3), $[\log x, A] \in x\Psi_c^{m-1}$ with $\mathrm{In}(x^{-1}[\log x, A]) = -[t,A_0]$ [15, (95)]. Since x commutes with everything to leading order, the coefficient of x^1 in $B[\log x, A]$ is $-C_1$, $C_1 = A_0^{-1}[t,A_0]$, and in $[\log x, A]B$ it is $-C_2$, $C_2 = [t,A_0]A_0^{-1}$. By the corrected Corollary 10.14(2),

$$\overline{\eta}(A) = \overline{\mathrm{Tr}}(C_1) + \overline{\mathrm{Tr}}(C_2) + \frac{\gamma'_{C_1}(0) + \gamma'_{C_2}(0)}{2\pi},$$

and $\overline{\mathrm{Tr}}(C_2) = \overline{\mathrm{Tr}}(C_1)$ because $\overline{\mathrm{Tr}}$ is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}$ (Theorem C.23). By (10), $\mathrm{In}([t, A_0]) = -i\partial_\tau A_0$, so $2\overline{\mathrm{Tr}}(C_1) = 2(2\pi)^{-1}\overline{\int}\,\mathrm{Tr}(A_0^{-1}(-i)\partial_\tau A_0)\,d\tau$, which is the second expression for $\eta(A_0)$. For (1): $\gamma(s)$ is the sum of the values at $\tau = +1$ and $\tau = -1$ of the fibre integrals of the terms of homogeneity $-n-s$, which cancel for odd terms. If ∂X is a point, that term of $C_j Q_0^{-s}$ is the product of the principal symbols, $-im\tau^{-1}$ times that of Q_0^{-s}, since $C_j = -iA_0^{-1}\partial_\tau A_0$ up to order; it is odd when Q_0^{-s} is even. For (2)–(4) see the cited results. □

Correction to MN 15.9 (The proof of MN Proposition 17). The proof in [15] correctly obtains $\widehat{\mathrm{Tr}}_\partial(B[\log x, A]) = -\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0]) = \widehat{\mathrm{Tr}}_\partial([\log x, A]B)$, up to the local correction of Corollary 10.14(2), but then concludes $\overline{\eta}(A) = -\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0])$. With definition [15, (89)] the conclusion is $2\overline{\mathrm{Tr}}(A_0^{-1}[t, A_0]) + \delta_Q(A)$, in agreement with [15, (24)] when $\delta_Q = 0$; there $[\log x, A_0]$ should read $[t, A_0]$. Also, the proof cites [15, (54)]; what is needed is [15, Cor. 2], in its corrected form, which assumes that Q is translation invariant to all orders at the boundary. For general Q, Lemma 10.15 gives $\overline{\eta}(A; Q) = \eta(A_0) + \delta_{Q_{\mathrm{ti}}}(A) + \mathrm{Tr}_{\partial,\sigma}\big((A^{-1}[\log x, A] + [\log x, A]A^{-1})\log(Q/Q_{\mathrm{ti}})\big)$, with Q_{ti} translation invariant to all orders. The corrections are compensated by β_Q, since the index does not depend on Q.

Corollary 15.10 (MN Corollary 3)**.** *Suppose that near the boundary $A = \sigma(-\partial_t + B_\partial)$, with σ a unitary bundle map and B_∂ a self-adjoint first-order elliptic differential operator on ∂X with $0 \notin \operatorname{spec} B_\partial$. This is the APS normal form $\sigma(\partial_u + B_\partial)$ in the inward normal variable $u = -t$. Then $\eta(A_0) = \eta_{\mathrm{APS}}(B_\partial) = \sum_\lambda \operatorname{sgn}\lambda$ (regularized). If Q is as in Proposition 15.8(3) or (4), then also $\overline{\eta}(A) = \eta_{\mathrm{APS}}(B_\partial)$.*

Proof. $\mathrm{In}(A, \tau) = e^{i\tau t}Ae^{-i\tau t} = \sigma(i\tau + B_\partial)$. Formally, on the λ-eigenspace $\mathrm{Tr}(A_0^{-1}\partial_\tau A_0) = i/(\lambda + i\tau) = (i\lambda + \tau)/(\lambda^2 + \tau^2)$; the odd part integrates to zero over $[-T, T]$, and $(\pi i)^{-1}\int i\lambda/(\lambda^2 + \tau^2)\,d\tau = \operatorname{sgn}\lambda$. The proof that the regularized sum is $\eta_{B_\partial}(0)$ is Theorem C.29. The last statement is Proposition 15.8(3),(4). □

15.5 Simplifications

Lemma 15.11 (MN Lemma 15)**.** *Let $A \in \Psi_c^m(X; E)$ be fully elliptic and translation invariant in a collar $\{x' < \epsilon\}$ with respect to a normal fibration. Choose $x = f(x')$ with $f(x') = x'$ near 0 and $x \equiv 1$ outside $\{x' < \epsilon\}$. Then $\gamma(A) = 0$.*

Proof. In the collar write $\log x = G(t')$ with $t' = 1/x'$. So $G(t') = -\log t'$ for large t', and G is constant ($= 0$) near $t' = 1/\epsilon$. For translation invariant A, Taylor expansion of $G(t) - G(t')$ about the diagonal gives

$$[\log x, A] \sim \sum_{j\geq 1} G^{(j)}(t')R_j,$$

with R_j translation invariant of order $m - j$ [15, p. 32]. The same holds for $B[\log x, A]$ and $[\log x, A]B$, with other translation invariant coefficients R_j', since a parametrix B can also be taken translation invariant there. The residue density of $G^{(j)}(t')R_j'$ is $G^{(j)}(t')$ times the constant residue density of R_j'. By (18), $\widehat{\mathrm{Tr}}_\sigma$ integrates it against $dt'\nu_\partial$ and takes the Hadamard finite part as $x \to 0$. For $j \geq 2$, $\int G^{(j)}dt' = [G^{(j-1)}]$ vanishes at both ends, since $G^{(j-1)} = O(t'^{1-j})$ at infinity and G is constant at the inner end. For $j = 1$, $\int_{1/\epsilon}^{T} G'\,dt' = G(T) - G(1/\epsilon) = \log x|_{x=1/T} - 0$, which is purely logarithmic and is removed by the finite part. Outside the collar $d\log x = 0$, so there is no further contribution. □

Lemma 15.12 (MN Lemma 16)**.** *Suppose that $A \in \Psi_c^m(X;E)$, $m > 0$, is fully elliptic and normal to first order at the boundary: $[A^*, A] \in x^2\Psi_c^{2m-1}(X;E)$. Let $Q = (A^*A+1)^{1/2m}$. Then $[\log Q, A] \in x^2\Psi_c^{m-1}$, so $\beta_Q(A) = 0$, and the integral defining $\overline{\mathrm{AS}}(A)$ converges without regularization.*

Proof. $[A, A^*A] = [A, A^*]A \in x^2\Psi_c^{3m-1}$, so by the functional calculus A commutes with functions of A^*A modulo x^2. Hence $[\log Q, A] \in x^2\Psi_c^{m-1}$. The functional $\widehat{\mathrm{Tr}}_\partial$ only sees the coefficient of x^1: for $C \in x^2\Psi_c$ the function $Z(C;s,z)$ has no pole at $z = 0$, since its poles lie at $z = -1-j$. So $\beta_Q(A) = 0$. Similarly $\mathrm{Tr}_\sigma(x^z C)$ is regular at $z = 0$ for $C \in x^2\Psi_c$, so no finite part is needed in (18). For operators of order ≤ 0, replace 1 by an operator that is positive in the interior and vanishes near the boundary [15, p. 33]. □

Theorem 15.13 (Reduced index theorem, MN (99)–(100), corrected)**.** *Let $A \in \Psi_c^m(X;E)$ be fully elliptic and translation invariant near the boundary, with normal indicial family A_0. Let Q be positive, translation invariant near the boundary and commuting with A there. Then*

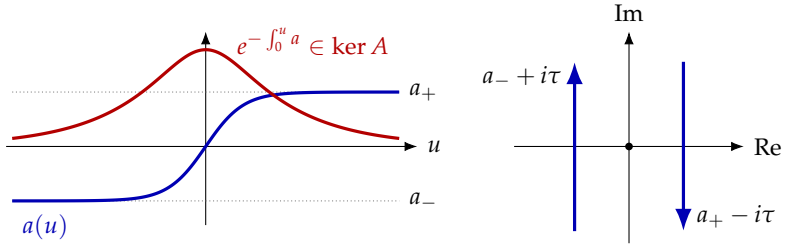
$$\mathrm{Ind}(A) = \mathrm{AS}(A) - \tfrac{1}{2}\big(\eta(A_0) + \delta_Q(A)\big),$$
$$\mathrm{AS}(A) = \mathrm{Tr}_{\mathrm{R}}\big(A[\log Q, A^{-1}]\big) = -\mathrm{Tr}_{\mathrm{R}}\big(A^{-1}[\log Q, A]\big),$$

where the residue integral converges, and $\eta(A_0)$ and the local correction $\delta_Q(A)$ are as in Proposition 15.8. In particular $\mathrm{Ind}(A) = \mathrm{AS}(A) - \frac{1}{2}\eta(A_0)$ whenever $\delta_Q(A) = 0$, for instance in the cases (1)–(4) of Proposition 15.8. [15, (99)–(100)] state the formula without δ_Q.

Proof. By Lemmas 15.11 and 15.12, $\gamma = \beta_Q = 0$, and by Proposition 15.8, $\overline{\eta}(A) = \eta(A_0) + \delta_Q(A)$. The two terms of AS agree, because $\widehat{\mathrm{Tr}}_\sigma([C_1, C_2]) = -\mathrm{Tr}_{\partial,\sigma}(C_1[\log x, C_2])$ vanishes for $C_2 = [\log Q, A] \in x^2\Psi_c$ (Lemma 12.1). □

16 Worked examples of the index formula

Example 16.1 (The closed circle)**.** This is Section 3.3. With $A = e^{i\theta}\Pi + (1-\Pi)$, $\mathrm{Ind}\, A = -1 = \mathrm{AS}(A)$ and $\overline{\mathrm{AS}}(A) = +1$. This fixes the sign of the symbolic term.



$\mathrm{sgn}\, a_+$	$\mathrm{sgn}\, a_-$	$\mathrm{Ind}\, A$	$-\frac{1}{2}\eta(A_0)$
+	−	1	1
−	+	−1	−1
+	+	0	0
−	−	0	0

Figure 15: Example 16.2, $A = \partial_u + a(u)$ with a real and $a = a_\pm \neq 0$ near $u = \pm\infty$. Left: for $a_+ > 0 > a_-$ the solution $e^{-\int_0^u a}$ of $Au = 0$ decays at both ends, so $\ker A \neq 0$. Middle: the indicial families $a_+ - i\tau$ (right end) and $a_- + i\tau$ (left end) trace vertical lines, which miss 0 because $a_\pm$ are real and non-zero. Right: since $\mathrm{AS}(A) = 0$, $\delta_Q = 0$ and the ends contribute $\eta = -\mathrm{sgn}\, a_+$ and $\eta = \mathrm{sgn}\, a_-$, the formula $\mathrm{Ind}\, A = \mathrm{AS}(A) - \frac{1}{2}\eta(A_0)$ gives $\frac{1}{2}(\mathrm{sgn}\, a_+ - \mathrm{sgn}\, a_-)$ in all four cases.

Example 16.2 (First order operators on an interval)**.** Let X be the compactification of $\mathbb{R}_u$ at both ends by $1/|u|$. So $X \cong [-1,1]$, $n = 1$, $N = 2$, and near the ends $x = 1/|u|$ and $t = |u|$. Let $a \in C^\infty(\mathbb{R};\mathbb{R})$ be real-valued, with $a(u) = a_\pm$ for $\pm u > R$ and $a_\pm \neq 0$, and consider

$$A = \partial_u + a(u) \in \Psi_c^1(X).$$

Full ellipticity. The principal symbol is $i\xi \neq 0$. At the right end $t = u$, so $A = \partial_t + a_+$ and $\mathrm{In}(A,\tau) = e^{i\tau t}(\partial_t + a_+)e^{-i\tau t} = a_+ - i\tau$. At the left end $t = -u$, so $A = -\partial_t + a_-$ and $\mathrm{In}(A,\tau) = a_- + i\tau$. Since $a_\pm$ are real and non-zero, both are invertible for all real τ. (For complex $a_\pm$ this fails: $a_+ = i$ gives $a_+ - i\tau = 0$ at $\tau = 1$. Reality of a is also used below for the formal adjoint $-\partial_u + a$.)

Index directly. A solution of $Au = 0$ is a multiple of $\exp(-\int a)$. It lies in $\dot{C}^\infty(X)$ (rapid decay as $|u| \to \infty$) if and only if $a_+ > 0 > a_-$. For the formal adjoint $-\partial_u + a$, the condition is $a_+ < 0 < a_-$. Hence

$$\operatorname{Ind} A = \tfrac{1}{2}\big(\operatorname{sgn} a_+ - \operatorname{sgn} a_-\big).$$

Figure 15 shows the case $a_+ > 0 > a_-$ and the four sign cases. *The formula.* Take $Q = (A^*A + 1)^{1/2}$. Near the ends $A^*A = -\partial_u^2 + a_\pm^2$ commutes with A, so A is normal there and Theorem 15.13 applies. The symbol of $[\log Q, A]$ is $-i\{\log\langle\xi\rangle + c_{-1} + \cdots,\ i\xi + a\}$, of order -1, because the principal symbol $i\xi$ of A does not depend on u. So $A^{-1}[\log Q, A]$ has order -2, and its residue density (homogeneity $-n = -1$) vanishes: $\mathrm{AS}(A) = 0$.

The correction δ_Q vanishes. Here ∂X consists of two points, so $d = 0$, and near each end $Q_0^{-s} = (\tau^2 + a_\pm^2 + 1)^{-s/2}$ is even in τ. At the right end $C_1 = C_2 = A_0^{-1}[t, A_0] = -(a_+ - i\tau)^{-1} = -(a_+ + i\tau)/(a_+^2 + \tau^2)$, whose expansion begins with $-i\tau^{-1}$. The term of homogeneity $-1 - s$ of $C_1 Q_0^{-s}$ is therefore $-i\tau^{-1}|\tau|^{-s}$, which is odd in τ: its values at $\tau = \pm 1$ cancel, so $\gamma(s) \equiv 0$ in Proposition 10.12 (Theorem C.24). The left end is the same, with $C_1 = (a_- + i\tau)^{-1}$. Hence $\delta_Q(A) = 0$ exactly, by case (1) of Proposition 15.8 (this is also an instance of the parity statement (2), with $n = 1$, but the computation here is complete), and $\overline{\eta}(A) = \eta(A_0)$. By Proposition 15.8, at the right end

$$\eta = \frac{1}{\pi i}\int \frac{-i}{a_+ - i\tau}\,d\tau = -\frac{1}{\pi}\int \frac{a_+ + i\tau}{a_+^2 + \tau^2}\,d\tau = -\operatorname{sgn} a_+,$$

and at the left end $\eta = \frac{1}{\pi i}\int \frac{i}{a_- + i\tau} d\tau = \operatorname{sgn} a_-$. So

$$\mathrm{AS}(A) - \tfrac{1}{2}\eta(A_0) = 0 - \tfrac{1}{2}\big(-\operatorname{sgn} a_+ + \operatorname{sgn} a_-\big) = \tfrac{1}{2}\big(\operatorname{sgn} a_+ - \operatorname{sgn} a_-\big) = \operatorname{Ind} A.$$

This confirms the sign of the eta term in [15, (93)] and the sign of $\widetilde{D}$ in Proposition 15.3. If instead $\widetilde{D}$ had the sign of [15, (21)], the eta term would enter with $+\frac{1}{2}$ and the formula would give $-\operatorname{Ind} A$. At the right end the boundary operator in APS form is $B_\partial = -a_+$ (Corollary 15.10), and $\eta_{\mathrm{APS}}(B_\partial) = -\operatorname{sgn} a_+$, as found.

Example 16.3 (Dirac operators on manifolds with cylindrical ends)**.** Let $\eth$ be a Dirac operator for an exact cusp metric (Remark 4.6) and a product structure near the boundary, $\eth = \sigma(-\partial_t + \eth_\partial)$. If $\eth_\partial$ is invertible, $\eth$ is fully elliptic. In the $\mathbb{Z}_2$-graded setting of Section 17.2, Theorem 15.13 and Corollary 15.10 give

$$\operatorname{Ind} \eth^+ = \mathrm{AS}(\eth) + \nu - \tfrac{1}{2}\eta(\eth_\partial),$$

where ν is the interior correction of [15, (110)]. Here we take $Q = (\eth^*\eth + 1)^{1/2}$; the correction δ_Q of Proposition 15.8 vanishes by Remark C.31, which quotes the local cancellation of Bismut–Freed [2]. For a general first-order $\eth_\partial$ a local term $-\frac{1}{2}\delta_Q$ would remain. This is the APS theorem in the invertible case. The term AS is the residue form of the Atiyah–Singer integrand, by the local index theorem. MN do not evaluate it further.

17 Cyclic homology, the graded case and other algebras (MN §§13–15)

We summarize the remaining sections of [15], with comments.

17.1 Cyclic homology

For a unital algebra the Connes–Loday–Quillen sequence $\cdots \to \mathrm{HH}_n \xrightarrow{I} \mathrm{HC}_n \xrightarrow{S} \mathrm{HC}_{n-2} \xrightarrow{B} \mathrm{HH}_{n-1} \to \cdots$ relates cyclic and Hochschild homology [15, (101)–(103)]. The key point is [15, Lemma 17]: $B = 0$ for $\mathcal{A}_\sigma, \mathcal{I}_\sigma, \mathcal{A}_{\partial,\sigma}, \mathcal{I}_\partial$. The reason is an order count. By the spectral sequences, HH_q is generated by cycles of order $q - n$ (x-degree for $\mathcal{I}_\partial$), and all cycles of lower order are boundaries. Inductively $\mathrm{HC}_{m-1} \cong \bigoplus_k \mathrm{HH}_{m-1-2k}$ is generated by elements of order at most $m - 1 - n$. The operator B does not raise order, so it maps them into cycles of order less than $m - n$, which are boundaries. Hence [15, Prop. 18]

$$\mathrm{HC}_m(\mathfrak{A}) \cong \bigoplus_{k \geq 0} \mathrm{HH}_{m-2k}(\mathfrak{A}), \qquad \mathfrak{A} = \mathcal{A}_\sigma, \mathcal{I}_\sigma, \mathcal{A}_{\partial,\sigma}, \mathcal{I}_\partial, \mathcal{A}_\partial,$$

with S the projection, and periodic cyclic homology obtained by dropping the restriction $k \geq 0$. For $\mathcal{A}_\partial$ MN write $\mathrm{HC}_i(\mathcal{A}_\partial) \cong \mathrm{HC}_i(\mathcal{A}_{\partial,\sigma})/\mathbb{C}$ for $i > 0$. With Proposition 12.3, the codimension is N in general, and the statement is about dimensions: in homology $\mathrm{HH}_{1,2}(\mathcal{A}_\partial)$ are subspaces of $\mathrm{HH}_{1,2}(\mathcal{A}_{\partial,\sigma})$, not quotients.

17.2 Bundles and grading

Dirac operators map between different bundles $E^\pm$. MN therefore pass to $\mathbb{Z}_2$-graded algebras [15, §14]. A graded embedding $E \subset X \times (\mathbb{C}^N \oplus \mathbb{C}^N)$ gives a graded HKR map with the supertrace and the Grassmann connection [15, (105)]. The Hochschild groups have the same dimensions as in the ungraded case and do not depend on the embedding [15, Lemma 18, Prop. 19]. If $E^+ = E^-$ near the boundary, the index of an odd fully elliptic A is [15, Thm. 1, (111)]

$$\mathrm{Ind}(A) = \mathrm{IF}([A], [A]^{-1}) + \nu, \qquad \nu = \mathrm{Tr}\big(e^+ Q_N^{-s}\big) - \mathrm{Tr}\big(e^- Q_N^{-s}\big)\big|_{s=0},$$

a local interior correction coming from the embedding. Proposition 15.3 holds verbatim, and so do our sign corrections.

17.3 Other algebras

The method applies to $\Psi_c^{\mathbb{Z}}(X)$, $\Psi_c^0(X)$, complete symbol algebras and the b-calculus, but the groups can be infinite dimensional [15, §15]. For $\mathcal{A}_6 = x^{-\mathbb{Z}}\Psi_c^0(X)/x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)$ the E^1 term agrees with that of $\mathcal{A}_{\partial,\sigma}$ for $p \leq 0$ and vanishes for $p > 0$. The E^2 term is unchanged for $p < 0$, while $E^2_{0,q}$ consists of spaces of closed forms [15, Prop. 20]. This is the precise sense in which restricting to order zero destroys finiteness. (MN state the E^1 comparison with $\mathcal{A}_{\partial,\sigma}$. Since $\mathcal{A}_6$ is the order-≤ 0 part of $\mathcal{A}_\sigma$, and the stated $E^2_{0,q}$ involves forms on Y as well as on Z, this is presumably a misprint for $\mathcal{A}_\sigma$.)

18 Comparison with the b-calculus

The companion notes [7] carry out the b-calculus analogue of [15], as MN suggest [15, p. 3]. Table 1 summarizes the differences.

	cusp calculus [15]	b-calculus [7]
vector fields	$x^2\partial_x,\ \partial_y$	$x\partial_x,\ \partial_y$
boundary variable	$t = 1/x$, translations	$\log x$, dilations
indicial family	$\mathrm{In}(A,\tau)$, $\tau \in \mathbb{R}$, symbol class	$I(A,\lambda)$, entire in $\lambda \in \mathbb{C}$
conjugation by x	$x^{-1}Ax = A + O(x)$: a *derivation* at leading order, $[x, \partial_t] = x^2$	$I(x^{-1}Ax,\lambda) = I(A,\lambda - i)$: an *automorphism*
graded algebra of $\mathcal{I}_\partial$	commutative, Poisson	skew Laurent series
method for $\mathcal{I}_\partial$	x-adic spectral sequence	weight decomposition, crossed products
invertibility in $\mathcal{A}$	$\mathrm{In}(A,\tau)$ invertible for $\tau \in \mathbb{R}$	$I(A,\lambda)$ invertible for all $\lambda \in \mathbb{C}$
$\mathrm{HH}^{0,1}(\mathcal{I}_\partial)$	$\mathbb{C}^N, \mathbb{C}^N$	$\mathbb{C}^N, \mathbb{C}^N$
rank of $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$	$N-1$ (MN: 0)	$N-1$
trace defect on $\mathcal{I}_\partial$	$b\,\mathrm{Tr}_c = -i_{\log x}\,\mathrm{Tr}_\partial$	b-trace defect formula

Table 1: Cusp versus b-calculus.

18.1 Derivation twist versus automorphism twist

In the cusp calculus, Lemma 4.4 shows that commuting x past an indicial operator produces terms of *higher* order in x, governed by the derivation ∂_τ. The associated graded algebra is therefore commutative, with Poisson bracket $\{t, \tau\} = 1$. The x-adic spectral sequence is then a Brylinski–Getzler computation on $\mathbb{R}_\tau \times \mathbb{S}_\partial$ (Proposition 9.2). In the b-calculus, $x^{-1}Ax$ has indicial family $I(A, \lambda - i)$. The complex shift by $-i$ is an automorphism that is *not* close to the identity, so the associated graded algebra is a skew Laurent series ring. The computation proceeds by a weight decomposition and the homology of crossed products by $\mathbb{Z}$ [7, §"The boundary ideal"]. The answers agree, $\mathrm{HH}_{0,1}(\mathcal{I}_\partial) \cong \mathbb{C}^N$. The generators are also closely related: Tr_∂ and $i_{\log x}\,\mathrm{Tr}_\partial$ in the cusp case, and the boundary trace Tr_∂ and the cocycle φ of [7] in the b-case.

18.2 Several boundary components

In both calculi the boundary algebras are direct sums over components, and the only global interaction is through the interior symbol algebra. The map $\mathrm{HH}^1(\mathcal{I}_\partial) \to \mathrm{HH}^2(\mathcal{A}_\sigma)$ sends $i_{\log x}\,\mathrm{Tr}_{\partial,j}$ to the de Rham coboundary of $d\theta_\sigma 1_{Z_j}$ (Proposition 12.5). Its kernel is spanned by the sum over j, because only $\sum_j 1_{Z_j} = 1_Z$ extends over the connected Y. Equivalently, by Stokes' theorem, $\sum_j \int_{Z_j} \beta = 0$ for every closed β on Y. This is the rank $N-1$ phenomenon of [7, Remark "Several boundary components"]. In the cusp case MN's argument ("$d\theta_\sigma$ extends") covers only the sum, and their statements (16) and Proposition 12 are correct exactly when $N = 1$. Proposition 9.5 shows the same rank in the simplest setting: $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi_c^{-\infty}(X)) \cong \mathbb{C}^{N-1}$, the cusp analogue of [7, Thm., part (iii)].

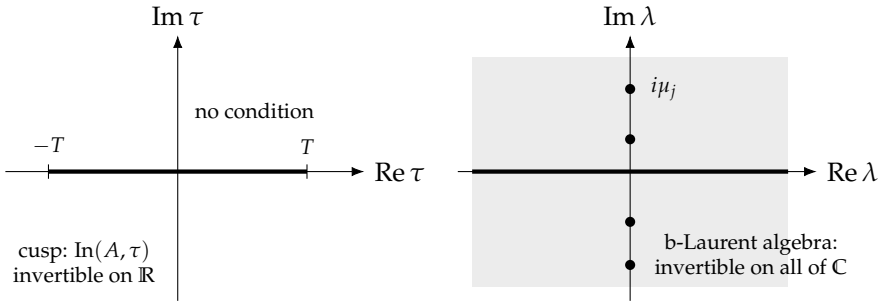


Figure 16: Left: a cusp operator is fully elliptic when $\mathrm{In}(A,\tau)$ is invertible for real τ (thick line). The eta term is the regularized integral of $\mathrm{Tr}(A_0^{-1}\partial_\tau A_0)$ over $[-T,T]$, with $T\to\infty$ by Hadamard's finite part. Right: in the b-calculus $I(A,\lambda)$ is entire, and invertibility in the b-Laurent algebra requires $I(A,\lambda)$ to be invertible on all of $\mathbb{C}$ (shaded). A Dirac operator $x\partial_x + B_\partial$ has indicial roots $i\mu_j$, $\mu_j \in \operatorname{spec} B_\partial$ (dots), so it is never invertible there. Fredholmness on L^2_b needs only the real axis, which is exactly the cusp condition after the transcendental blow-up.

18.3 Invertibility, completeness and the trace defects

In the b-calculus, invertibility in $\mathcal{A}_b$ forces the indicial family to be invertible on all of $\mathbb{C}$ (Section 4.6). Dirac operators never satisfy this, so the b-Laurent algebra cannot see APS-type indices. The cusp calculus needs invertibility only on $\mathbb{R}$, which is the APS condition $\ker B_\partial = 0$. Under the transcendental blow-up $x_b = e^{-1/x}$ the b-indicial family becomes the cusp indicial family with $\tau = \lambda$. A b-operator with indicial family invertible on $\mathbb{R}$ lifts to a fully elliptic cusp operator, and its index is given by Theorem 15.6. The trace defects also match. For translation invariant A, B, Proposition 9.2 and Proposition 9.5 give

$$\mathrm{Tr}_c([A,B]) = -\,\mathrm{Tr}_\partial(A[\log x, B]) = -\frac{i}{2\pi}\int_{\mathbb{R}} \mathrm{Tr}\big(\mathrm{In}(A,\tau)\,\partial_\tau\,\mathrm{In}(B,\tau)\big)\,d\tau.$$

The b-trace satisfies the same identity in the variable λ: ${}^b\mathrm{Tr}([A,B]) = -\operatorname{Res}_z \mathrm{Tr}(A[\log x_b, B]x_b^z)$, and $I([\log x_b, B]) = i\partial_\lambda I(B)$ [7, Lemma "Conjugation by powers of x"]. This is Melrose's b-trace defect formula. The two regularizations differ (x^z versus $x_b^z = e^{-z/x}$), but on the leading terms they agree.

A Construction of the cusp calculus

This appendix proves the background facts (B1) and (B2) of Section 2.5, as they are stated there. We construct the cusp calculus, prove that it is a doubly filtered algebra, and derive the symbol sequence [15, (11)], the indicial sequence [15, (12)], the global quantization [15, Prop. 24], the Taylor series isomorphism [15, Prop. 25] and the invariance of orders under conjugation by x^z. We then prove the Fredholm theorem (B2).

What is proved and what is quoted. We quote one thing: the standard calculus of pseudodifferential operators on the cylinder $\mathbb{R}\times H$, H closed, with symbol estimates uniform in the cylinder variable (Background fact A.4). This is a boundaryless calculus. It is the calculus of [14, Ch. 2] on $\mathbb{R}^n$, transplanted to $\mathbb{R}\times H$ by a partition of unity on H, and it contains nothing about cusps. Every statement that is specific to the cusp calculus is proved here, except the density statement in [15, Prop. 26] (Remark A.29), which the notes do not use.

Strategy. The usual route to composition is the triple space [10]. We avoid it. Near the boundary a cusp operator is an operator on the cylinder $\mathbb{R}_t \times \partial X$, $t = 1/x$, whose kernel is a smooth function of $x = 1/t$ with values in suspended kernels. We show (Lemma A.9) that this smoothness is equivalent to stability of the operator under the derivation $\delta = t^2[\partial_t, \cdot\,]$. The derivation obeys a Leibniz rule (Lemma A.12). Composition is then reduced to the uniform calculus, with no new estimates.

A.1 Conventions

X is a compact manifold with boundary, of dimension n, with a cusp structure (Section 4.1) represented by the boundary defining function x. We fix a collar $[0,\epsilon)_x \times \partial X \hookrightarrow X$ whose first coordinate is x (a normal fibration). On the collar $t = 1/x$, so the collar is the end $\{t > 1/\epsilon\}$ of the cylinder $\mathbb{R}_t \times \partial X$. We fix a smooth positive density ν_∂ on ∂X and a cusp density ν_c on X, smooth and positive in the interior and equal to $x^{-2}dx\,\nu_\partial = dt\,\nu_\partial$ on the collar. Operators are identified with their kernels with respect to ν_c in the second variable:

$$Au(z) = \int_X K_A(z,z')u(z')\,\nu_c(z').$$

$\dot{C}^\infty(X)$ is the space of smooth functions vanishing to infinite order at ∂X, and $\dot{C}^\infty(X^2)$ the space of smooth functions on X^2 vanishing to infinite order at $\partial(X^2)$. Since x'^{-2} is a smooth weight, a kernel lies in $\dot{C}^\infty(X^2)$ with respect to ν_c if and only if it does with respect to a smooth density.

The cusp cotangent coordinates are (x, y, τ, η) as in (6). So τ is the symbol of $x^2D_x = -D_t$. If σ denotes the covariable dual to t, then $\tau = -\sigma$. Orders are integers, as in [15]. Nothing changes for real orders.

We write the proofs for scalar operators and connected ∂X. Several boundary components are handled one at a time. For bundles E, we trivialize E over the collar as the pull-back of $E|_{\partial X}$; kernels then take values in $\mathrm{Hom}(E)$, and every argument below applies entrywise.

A.2 Background: conormal kernels and the uniform calculus on a cylinder

Let H be a closed manifold of dimension $n-1$ with a smooth positive density ν_H. In the application $H = \partial X$.

Definition A.1 (Suspended kernels)**.** For $m \in \mathbb{Z}$, $\mathfrak{K}^m(H)$ is the space of distributions κ on $\mathbb{R}_\mu \times H \times H$ such that

1. away from $\{0\} \times \Delta_H$, κ is smooth, and for every N and every derivative ∂^α in (μ, y, y'), $\langle\mu\rangle^N \partial^\alpha \kappa$ is bounded on $\{|\mu| \geq 1\}$;
2. near each point of $\{0\} \times \Delta_H$, in local coordinates y on H and modulo a smooth function,

$$\kappa(\mu, y, y') = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{-i\mu\tau + i(y-y')\cdot\eta}\, a(y,\tau,\eta)\, d\tau\, d\eta, \tag{21}$$

with a a classical symbol of order m in $\zeta = (\tau, \eta)$.

The leading part of a is a well-defined function $\sigma_m(\kappa)$ on $(T^*H \times \mathbb{R}_\tau) \setminus 0$, homogeneous of degree m: the principal symbol. We give $\mathfrak{K}^m(H)$ its usual Fréchet topology and put $\mathfrak{K}^{-\infty}(H) = \bigcap_m \mathfrak{K}^m(H)$.

Definition A.2 (The suspended algebra)**.** $\Psi^m_{\mathrm{sus}}(H)$ is the space of operators on $\mathbb{R}_t \times H$ given by convolution in t with some $\kappa \in \mathfrak{K}^m(H)$:

$$(\kappa * u)(t,y) = \int_{\mathbb{R}} \int_H \kappa(t-t',y,y')u(t',y')\,\nu_H(y')\,dt'.$$

Its *indicial family* is the Fourier transform

$$\mathrm{In}(\kappa,\tau) = \int_{\mathbb{R}} e^{i\tau\mu}\kappa(\mu,\cdot,\cdot)\,d\mu, \qquad \tau \in \mathbb{R}, \tag{22}$$

a family of pseudodifferential operators on H. In local coordinates its full symbol is $a(y,\tau,\eta)$, by (21). This is (10).

The sign in the exponent of (21) makes τ the cusp covariable. For example $-D_t = x^2 D_x$ has convolution kernel $i\delta'(\mu) = (2\pi)^{-1}\int \tau e^{-i\mu\tau}d\tau$, so $\mathrm{In}(x^2D_x,\tau) = \tau$.

Definition A.3 (Uniform operators)**.** $\Psi^m_{\mathrm{u}}(\mathbb{R}\times H)$ is the space of operators on $\mathbb{R}_t \times H$ whose kernel, with respect to $dt'\,\nu_H(y')$, has the form

$$K(t,t',y,y') = \widetilde{K}(t,t-t',y,y'), \qquad \widetilde{K} \in C^\infty_{\mathrm{b}}\big(\mathbb{R}_t;\mathfrak{K}^m(H)\big),$$

where $C^\infty_{\mathrm{b}}(\mathbb{R};F)$ is the space of smooth maps into the Fréchet space F all of whose derivatives are bounded in every seminorm. The principal symbol is $\sigma_m(A)(t,\cdot) = \sigma_m(\widetilde{K}(t))$. We write $\mu = t - t'$ throughout.

Translation invariant elements of Ψ^m_{u} are exactly the elements of $\Psi^m_{\mathrm{sus}}(H)$: there $\widetilde{K}(t,\mu) = \kappa(\mu)$.

Background fact A.4 (Uniform calculus on a cylinder; quoted)**.** The following are standard [14, Ch. 2, Ch. 4].

(K1) *Conormal kernels.* $\mathfrak{K}^m \subset \mathfrak{K}^{m+1}$ and $\mathfrak{K}^{-\infty}(H) = \mathcal{S}(\mathbb{R}_\mu; C^\infty(H\times H))$. The map σ_m is continuous, and its kernel is $\mathfrak{K}^{m-1}$, with the topology of $\mathfrak{K}^{m-1}$ equal to the one induced from $\mathfrak{K}^m$. There are continuous linear maps $\mathrm{Op} : S^m_{\mathrm{cl}} \to \mathfrak{K}^m$ (S^m_{cl} the classical symbols of order m on $T^*H\times\mathbb{R}_\tau$) and $\Sigma : \mathfrak{K}^m \to S^m_{\mathrm{cl}}$ such that $\sigma_m(\mathrm{Op}(a))$ is the leading part of a, $\mathrm{Op}(S^{-\infty}_{\mathrm{cl}}) \subset \mathfrak{K}^{-\infty}$, and $\kappa - \mathrm{Op}(\Sigma\kappa) \in \mathfrak{K}^{-\infty}$ depends continuously on κ. Multiplication by μ maps $\mathfrak{K}^m$ to $\mathfrak{K}^{m-1}$, with $\sigma_{m-1}(\mu\kappa) = -i\partial_\tau\sigma_m(\kappa)$. Multiplication by a smooth function $g(\mu)$ with $|\partial^b_\mu g| \le C_b\langle\mu\rangle^N$ for all b is continuous on $\mathfrak{K}^m$, jointly continuous in (g,κ) for g in such a bounded set, and $\sigma_m(g\kappa) = g(0)\sigma_m(\kappa)$.

(K2) *Fourier transform.* For $\kappa \in \mathfrak{K}^m(H)$, $\mathrm{In}(\kappa,\tau)$ is a smooth family of operators $H^s(H) \to H^{s-m}(H)$, and the norms of $\partial^j_\tau \mathrm{In}(\kappa,\tau)$ are bounded by $C\langle\tau\rangle^{|m|}$, where C is a seminorm of κ. The map $\kappa \mapsto \mathrm{In}(\kappa,\cdot)$ is an isomorphism from $\mathfrak{K}^{-\infty}(H)$ onto $\mathcal{S}(\mathbb{R}_\tau; C^\infty(H\times H))$.

(K3) *Composition.* If $A \in \Psi^m_{\mathrm{u}}$ and $B \in \Psi^{m'}_{\mathrm{u}}$, then $AB \in \Psi^{m+m'}_{\mathrm{u}}$, $\mathrm{supp}\,K_{AB} \subset \mathrm{supp}\,K_A \circ \mathrm{supp}\,K_B$ and $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$. The formal adjoint A^* and the transpose A^t (with respect to $dt\,\nu_H$) lie in Ψ^m_{u}, and $\sigma_m(A^*) = \overline{\sigma_m(A)}$. These operations are continuous in the seminorms of $\widetilde{K}$, and so preserve smooth and holomorphic dependence on parameters.

(K4) *Mapping.* $A \in \Psi^m_{\mathrm{u}}$ maps $\mathcal{S}(\mathbb{R}\times H)$ to itself and $H^s(\mathbb{R}\times H)$ to $H^{s-m}(\mathbb{R}\times H)$ continuously, with norms bounded by seminorms of $\widetilde{K}$.

Lemma A.5. *$\Psi^{\mathbb{Z}}_{\mathrm{sus}}(H)$ is a filtered algebra, and* $\mathrm{In}(\kappa_1 * \kappa_2,\tau) = \mathrm{In}(\kappa_1,\tau)\,\mathrm{In}(\kappa_2,\tau)$. *If $\kappa \in \Psi^m_{\mathrm{sus}}(H)$, then* $\mathrm{ad}_t\,\kappa = [t,\kappa]$ *has kernel $\mu\kappa \in \mathfrak{K}^{m-1}(H)$, and* $\mathrm{In}([t,\kappa],\tau) = -i\partial_\tau\,\mathrm{In}(\kappa,\tau)$.

Proof. By (K3) the composite of two translation invariant elements of Ψ_{u} lies in Ψ_{u}, and it is translation invariant. The Fourier transform of a convolution is the product of the Fourier transforms. The kernel of $t\kappa - \kappa t$ is $(t-t')\kappa(t-t') = \mu\kappa$, and $\int e^{i\tau\mu}\mu\kappa\,d\mu = -i\partial_\tau\,\mathrm{In}(\kappa,\tau)$. □

A.3 The b- and cusp double spaces

We use the blow-up formalism of Paul's notes [18]. Near the corner $\partial X \times \partial X$ of X^2 we use the coordinates (x, y, x', y'). The b-double space is $X_b^2 = [X^2; \partial X \times \partial X]$. Away from the left boundary face lb it has projective coordinates

$$(x,\ s = x'/x,\ y,\ y'), \qquad \mathrm{ff}_b = \{x = 0\}, \quad \mathrm{rb} = \{s = 0\},$$

and the lifted diagonal is $\{s = 1,\ y = y'\}$. On the interior of ff_b the coordinate $s \in (0, \infty)$ is the ratio of the two defining functions. A change $x \mapsto ax + O(x^2)$ with $a > 0$ constant does not change $s|_{\mathrm{ff}_b}$. A change with $a = a(y)$ replaces it by $a(y')s/a(y)$. So the *fibre diagonal*

$$\Phi = \mathrm{ff}_b \cap \{s = 1\} = \{x = 0,\ s = 1\} \cong \partial X \times \partial X$$

is fixed by the cusp structure, as in [15, (116)]. It is a p-submanifold: a point of the half-plane $\{x \geq 0\}$ times ∂X^2. The *cusp double space* is $X_c^2 = [X_b^2; \Phi]$, with new boundary hypersurface ff_c.

Lemma A.6 (Coordinates on X_c^2)**.** *Put $\mu = 1/x - 1/x' = t - t'$.*

1. *Near the interior of ff_c, (x, μ, y, y') are coordinates on X_c^2, with $\mathrm{ff}_c = \{x = 0\}$, $\mu \in \mathbb{R}$. In them $x' = x/(1 - x\mu)$.*
2. *Near the corner $\mathrm{ff}_c \cap \mathrm{ff}_b$ where $\mu \to +\infty$ (respectively $-\infty$), the functions $\rho_b = 1/|\mu|$ and $\rho_c = x|\mu|$, together with (y, y'), are coordinates; ρ_b defines ff_b and ρ_c defines ff_c.*
3. *The lifted diagonal is $\Delta_c = \{\mu = 0,\ y = y'\}$. It meets ∂X_c^2 only in the interior of ff_c, transversally* [15, Lemma 22].

Proof. The blow-up of the point $(x, s) = (0, 1)$ of the half-plane has the projective coordinates (x, W), $W = (s - 1)/x$, valid where $|s - 1| \leq Cx$ and with $\mathrm{ff}_c = \{x = 0\}$. It also has $(r, w) = (x/(s - 1), s - 1)$ for $s > 1$, with $\mathrm{ff}_b = \{r = 0\}$ and $\mathrm{ff}_c = \{w = 0\}$; the case $s < 1$ is the same with $|w|$. Since $x' = sx$,

$$\mu = \frac{1}{x} - \frac{1}{sx} = \frac{s-1}{sx} = \frac{W}{1 + xW}, \qquad W = \frac{\mu}{1 - x\mu}, \qquad x' = x(1 + xW) = \frac{x}{1 - x\mu}.$$

So $(x, W) \mapsto (x, \mu)$ is a diffeomorphism where $s > 0$, which is (1). In the second chart $x = rw$ and $s = 1 + w$, so $\mu = 1/((1 + w)r)$, $\rho_b = 1/\mu = (1 + w)r$ and $\rho_c = x\mu = w/(1 + w)$. These are r and w up to smooth factors that do not vanish, which is (2). The diagonal is $\{s = 1,\ y = y'\} = \{W = 0,\ y = y'\} = \{\mu = 0,\ y = y'\}$, which gives (3). □

Lemma A.7 (Rapid decay at ff_b is Schwartz decay in μ)**.** *Let f be a smooth function on a neighbourhood of ff_c in X_c^2. Then f vanishes to infinite order at ff_b if and only if, in the coordinates of Lemma A.6(1), $\partial_x^a \partial_\mu^b \partial_{y,y'}^\gamma f = O(\langle \mu \rangle^{-N})$ for all a, b, γ, N, uniformly in x. In other words, f is a smooth function of x up to $x = 0$ with values in $\mathcal{S}(\mathbb{R}_\mu; C^\infty(\partial X^2))$.*

Proof. Only the corners of Lemma A.6(2) matter. Take $\mu > 0$. There $x = \rho_b\rho_c$ and $\mu = 1/\rho_b$, so

$$\partial_{\rho_c} = \rho_b\, \partial_x, \qquad \partial_{\rho_b} = \rho_c\, \partial_x - \mu^2 \partial_\mu.$$

If all $\partial_x^a \partial_\mu^b f$ are $O(\mu^{-\infty})$, then every $\partial_{\rho_b}^j \partial_{\rho_c}^k f$ is a finite sum of such terms multiplied by polynomials in μ, ρ_b and $\rho_c \leq 1$. So it is $O(\rho_b^\infty)$, and f extends smoothly to $\rho_b = 0$ with infinite order vanishing. Conversely $\partial_x = \rho_b^{-1} \partial_{\rho_c}$ and $\partial_\mu = -\rho_b^2 \partial_{\rho_b} + \rho_b\rho_c \partial_{\rho_c}$. If $f = O(\rho_b^\infty)$ with all derivatives, then each $\partial_x^a \partial_\mu^b f$ is $O(\rho_b^\infty) = O(\mu^{-\infty})$. □

A.4 The model: the half-line and the end of a cylinder

We first treat a neighbourhood of the boundary. The reader should keep $H = \mathrm{pt}$ in mind. Then $\mathbb{R} \times H = \mathbb{R}_t$, the end $\{t > T\}$ is the half-line $X = [0, 1/T)_x$ in $x = 1/t$, $\mathfrak{K}^m(\mathrm{pt})$ is the space of distributions on $\mathbb{R}_\mu$ that are classical conormal at 0 and Schwartz at infinity, and every kernel below is a function of (t, t') or of (x, μ). For general H the variables $(y, y') \in H^2$ are carried along inside $\mathfrak{K}^m(H)$. We suppress them from the notation; every formula below holds verbatim.

Fix $T \geq 1$. On $\{t > 0\}$ put $x = 1/t$. In the coordinates (x, μ), the kernel of an operator A with kernel supported in $\{t > 0\}$ is

$$k_A(x, \mu) = \widetilde{K}_A(1/x, \mu).$$

Definition A.8 (The model cusp class)**.** Let $\delta(A) = t^2[\partial_t, A]$. Then $\Psi^m_{c,T}$ is the set of $A \in \Psi^m_{\mathrm{u}}(\mathbb{R} \times H)$ such that

1. $\operatorname{supp} K_A \subset \{t \geq T,\ t' \geq T\}$, and
2. $\delta^j(A) \in \Psi^m_{\mathrm{u}}(\mathbb{R} \times H)$ for every $j \geq 0$.

The commutator $[\partial_t, A]$ has kernel $(\partial_t + \partial_{t'})K_A$, by integration by parts. In the coordinates (t, μ) this is $\partial_t \widetilde{K}_A$ at fixed μ. Since $t^2\partial_t = -\partial_x$, we get

$$k_{\delta(A)} = -\partial_x k_A. \tag{23}$$

Lemma A.9 (Characterization)**.** *Let $A \in \Psi^m_{\mathrm{u}}$ have kernel supported in $\{t \geq T,\ t' \geq T\}$. Then $A \in \Psi^m_{c,T}$ if and only if k_A extends from $(0, 1/T]$ to a smooth map $[0, 1/T] \to \mathfrak{K}^m(H)$. Conversely, every $k \in C^\infty([0, \infty); \mathfrak{K}^m(H))$ with $\operatorname{supp} k \subset \{x \leq 1/T,\ x\mu \leq 1 - xT\}$ is k_A for a unique $A \in \Psi^m_{c,T}$.*

The support condition $x\mu \leq 1 - xT$ says exactly that $t' \geq T$.

Proof. If $A \in \Psi^m_{c,T}$, then by (23) every $\partial_x^j k_A$ is bounded in $\mathfrak{K}^m$ on $(0, 1/T]$. A smooth map $f : (0, b] \to F$ into a Fréchet space, all of whose derivatives are bounded, extends to $C^\infty([0, b]; F)$. Indeed, by the mean value inequality in each seminorm, each $f^{(j)}$ is Lipschitz. So it extends continuously to 0, and $f^{(j)}(x) - f^{(j)}(0) = \int_0^x f^{(j+1)}$ shows that the extension of $f^{(j+1)}$ is the derivative of that of $f^{(j)}$. Conversely, if k is smooth on $[0, 1/T]$, then all $\partial_x^j k$ are bounded. Since $\partial_t = -x^2\partial_x$, the function $\widetilde{K}(t) = k(1/t)$ has bounded t-derivatives, so $A \in \Psi^m_{\mathrm{u}}$. By (23), $\delta^j(A) \in \Psi^m_{\mathrm{u}}$ for all j. □

By Lemmas A.6 and A.7, the condition of Lemma A.9 says that the lifted kernel on X_c^2 is conormal to Δ_c, smooth up to ff_c and vanishing to infinite order at ff_b. This is the condition of [15, Def. 2] near ff_c.

Definition A.10 (Multipliers)**.** $\mathcal{M}_T$ is the set of smooth functions $M(x, \mu)$ on $\Omega_T = \{0 \leq x < 2/T,\ x\mu < 1 - xT/2\}$ (that is, $t > T/2$ and $t' > T/2$) such that $|\partial_x^a \partial_\mu^b M| \leq C_{ab}\langle\mu\rangle^{N_{ab}}$ on Ω_T for all a, b. For $A \in \Psi^m_{c,T}$, MA denotes the operator with kernel Mk_A.

Lemma A.11 (Multipliers)**.** *If $M \in \mathcal{M}_T$ and $A \in \Psi^m_{c,T}$, then $MA \in \Psi^m_{c,T}$. If $M = \mu^l M'$ with $M' \in \mathcal{M}_T$, then $MA \in \Psi^{m-l}_{c,T}$. If M depends holomorphically on a parameter z, with the bounds locally uniform in z, then so does k_{MA}. The following lie in $\mathcal{M}_T$:*

1. *μ, the multiplier of $\mathrm{ad}_t A = [t, A]$;*
2. *$(t/t')^z = (1 - x\mu)^{-z}$, $z \in \mathbb{C}$, the multiplier of $W_z(A) = t^z A t^{-z}$;*
3. *$e^{i\tau\mu}$, $\tau \in \mathbb{R}$, the multiplier of $e^{i\tau t} A e^{-i\tau t}$;*

4. $\chi(t') = \chi(1/x - \mu)$, *for* $\chi \in C^\infty(\mathbb{R})$ *with* χ' *compactly supported.*

Proof. Let $\theta \in C^\infty(\mathbb{R})$ vanish on $(-\infty, 3T/4]$ and equal 1 on $[T, \infty)$. Since $k_A(x,\mu)$ vanishes unless $t' = 1/x - \mu \geq T$, we have $Mk_A = g\,k_A$ with $g(x,\mu) = \theta(1/x-\mu)M(x,\mu)$, extended by 0. The derivative $\partial_x \theta(1/x-\mu) = -x^{-2}\theta'(1/x-\mu)$ is supported where $1/x \leq |\mu| + T$, so $x^{-2} \leq (|\mu|+T)^2$. The same holds for all derivatives. Hence g is smooth on $[0, 1/T] \times \mathbb{R}$ with all derivatives polynomially bounded in μ. By (K1), $x \mapsto g(x,\cdot)k_A(x)$ is a smooth map into $\mathfrak{K}^m$ (into $\mathfrak{K}^{m-l}$ when $M = \mu^l M'$), holomorphic in z when M is. Its support is that of k_A. So $MA \in \Psi^m_{c,T}$ by Lemma A.9.

For the examples: (1) and (3) are clear. For (2), on Ω_T we have $1 - x\mu = t'/t > 0$, and

$$\frac{t}{t'} = 1 + \frac{\mu}{t'} \leq 1 + \frac{2|\mu|}{T}, \qquad \frac{t'}{t} = 1 - \frac{\mu}{t} \leq 1 + \frac{2|\mu|}{T}.$$

Every derivative $\partial_x^a \partial_\mu^b (1-x\mu)^{-z}$ is a polynomial in (x,μ) times $(1-x\mu)^{-z-a-b}$, and is therefore polynomially bounded. The kernel of $t^z A t^{-z}$ is $t^z K_A t'^{-z} = (t/t')^z K_A$. For (4), the derivatives of $\chi(1/x-\mu)$ are handled as those of θ above. □

Lemma A.12 (Derivation identities)**.** *For* $A, B \in \Psi_{c,T}$ *and* $z \in \mathbb{C}$,

$$\delta(AB) = \delta(A)B + W_2(A)\,\delta(B), \qquad \delta \circ \mathrm{ad}_t = \mathrm{ad}_t \circ \delta,$$
$$\delta \circ W_z = W_z \circ \delta - z\,W_{z+1} \circ \mathrm{ad}_t, \qquad \delta(A^*) = W_2\big(\delta(A)^*\big), \qquad W_z(A^*) = (W_{-\bar z}A)^*.$$

Proof. $\delta(AB) = t^2[\partial_t, A]B + t^2 A[\partial_t, B] = \delta(A)B + (t^2 A t^{-2})(t^2[\partial_t, B])$. Next, $[\partial_t, [t, A]] = [t, [\partial_t, A]]$ because $[\partial_t, t] = 1$ is central, and t^2 commutes with t. For the third identity,

$$t^2[\partial_t, t^z A t^{-z}] = z t^{z+1} A t^{-z} + t^{z+2}[\partial_t, A] t^{-z} - z t^{z+2} A t^{-z-1} = W_z(\delta A) - z t^{z+1}[t, A] t^{-z-1}.$$

Finally $[\partial_t, A^*] = [\partial_t, A]^*$, since $\partial_t^* = -\partial_t$. So $\delta(A^*) = t^2(t^{-2}\delta(A))^* = t^2\delta(A)^* t^{-2}$, and $t^z A^* t^{-z} = (t^{-\bar z} A t^{\bar z})^*$. □

Theorem A.13 (Model calculus)**.** $\Psi^{\mathbb{Z}}_{c,T}$ *is a filtered* $*$*-algebra:* $\Psi^m_{c,T}\Psi^{m'}_{c,T} \subset \Psi^{m+m'}_{c,T}$ *and* $(\Psi^m_{c,T})^* = \Psi^m_{c,T}$*. The principal symbol is multiplicative. It is a smooth function of* (x, y, τ, η) *up to* $x = 0$*. The operators* W_z*,* $e^{i\tau t} \cdot e^{-i\tau t}$ *and* ad_t *map* $\Psi^m_{c,T}$ *to* $\Psi^m_{c,T}$*,* $\Psi^m_{c,T}$ *and* $\Psi^{m-1}_{c,T}$ *respectively. Multiplication by* $f \in C^\infty([0,1/T]_x \times H)$ *vanishing near* $x = 1/T$*, and cusp differential operators of order* m *with such coefficients, lie in* $\Psi^m_{c,T}$*.*

Proof. Composition. We show by induction on j that $\delta^j(AB) \in \Psi^{m+m'}_{\mathrm{u}}$ for all $A \in \Psi^m_{c,T}$ and $B \in \Psi^{m'}_{c,T}$. For $j = 0$ this is (K3), which also gives the support condition. By Lemma A.12,

$$\delta^{j+1}(AB) = \delta^j\big(\delta(A)B\big) + \delta^j\big(W_2(A)\delta(B)\big).$$

Here $\delta(A)$ and $W_2(A)$ lie in $\Psi^m_{c,T}$ (Definition A.8 and Lemma A.11), and $\delta(B) \in \Psi^{m'}_{c,T}$. So the induction hypothesis applies to both terms.

Adjoints. We show by induction on j that $\delta^i(A^*) \in \Psi^m_{\mathrm{u}}$ for all $i \leq j$ and all $A \in \Psi^m_{c,T}$. The case $j = 0$ is (K3). By Lemma A.12, $\delta^{j+1}(A^*) = \delta^j W_2(C)$ with $C = \delta(A)^*$. Repeated use of the third identity writes $\delta^j W_2(C)$ as a linear combination of $W_{2+a}\,\mathrm{ad}_t^a\,\delta^{j-a}(C)$, $0 \leq a \leq j$. By the induction hypothesis applied to $\delta(A) \in \Psi^m_{c,T}$, we have $\delta^{j-a}(C) \in \Psi^m_{\mathrm{u}}$. The multipliers of ad_t and W_z preserve Ψ^m_{u} on kernels supported in $\{t, t' \geq T\}$, by the first paragraph of the proof of Lemma A.11. So $\delta^{j+1}(A^*) \in \Psi^m_{\mathrm{u}}$.

Symbols and the rest. By (K3), $\sigma(AB) = \sigma(A)\sigma(B)$. By Lemma A.9 and the continuity of σ_m in (K1), $\sigma_m(A)$ is $\sigma_m(k_A(x))$, a smooth function of x up to 0. The statements about W_z, the conjugation

by $e^{i\tau t}$ and ad_t are Lemma A.11. Multiplication by f has kernel $f(x,y)\delta(\mu)\delta_H(y,y')$, which is smooth in x with values in $\mathfrak{K}^0$. The vector fields $x^2\partial_x = -\partial_t$ and ∂_{y_j} have kernels $-\delta'(\mu)$ and derivatives of δ_H, independent of x. A cusp differential operator is a sum of products of these, so it lies in $\Psi^m_{c,T}$ by the composition result. □

Lemma A.14 (Cut-off translation invariant operators)**.** *Let $\chi \in C^\infty(\mathbb{R})$ vanish on $(-\infty, T]$ and equal 1 on $[2T,\infty)$. Let $\hat{k} \in C^\infty([0,\infty); \mathfrak{K}^m(H))$ vanish for $x \geq 1/(2T)$. Then the operator A with kernel $\chi(t)\chi(t')\hat{k}(1/t, t-t')$ lies in $\Psi^m_{c,T}$, and $k_A - \hat{k}$ vanishes to infinite order at $x = 0$ in $\mathfrak{K}^m$. In particular $\chi\kappa\chi \in \Psi^m_{c,T}$ for $\kappa \in \Psi^m_{\mathrm{sus}}(H)$, with $k_{\chi\kappa\chi} = \kappa + O(x^\infty)$.*

Proof. We have $k_A = \chi(1/x)\chi(1/x-\mu)\hat{k}$. Here $\chi(1/x) = 1$ for $x < 1/(2T)$, and $\chi(1/x-\mu)$ is a multiplier, as in Lemma A.11(4). So $A \in \Psi^m_{c,T}$. The function $1-\chi(1/x-\mu)$ is supported where $\mu \geq 1/x - 2T \geq 1/(2x)$ for small x. There $\langle\mu\rangle^{-N} \leq (2x)^N$. Since $\hat{k}(x)$ is Schwartz away from $\mu = 0$, uniformly with all x-derivatives, $(1-\chi(1/x-\mu))\hat{k}$ is $O(x^N)$ in $\mathfrak{K}^m$ for every N, together with its x-derivatives. □

Example A.15 (The half-line, explicitly)**.** Take $H = \mathrm{pt}$, so that $X = [0, 1/T)_x$. Let $\kappa(\mu) = \frac{1}{2}e^{-|\mu|}$, the kernel of $(D_t^2+1)^{-1} \in \Psi^{-2}_{\mathrm{sus}}$, with $\mathrm{In}(\kappa,\tau) = (1+\tau^2)^{-1}$. By Lemma A.14, $A = \chi(D_t^2+1)^{-1}\chi \in \Psi^{-2}_{c,T}$, with

$$k_A(x,\mu) = \tfrac{1}{2}e^{-|\mu|}\,\chi(1/x)\,\chi(1/x-\mu) = \tfrac{1}{2}e^{-|\mu|} + O(x^\infty).$$

Conjugation by $x = 1/t$ multiplies the kernel by $(1-x\mu)^{-1}$ (Lemma A.11(2) with $z=1$):

$$k_{x^{-1}Ax} = \frac{k_A}{1-x\mu} \sim \sum_{p\geq 0} x^p\mu^p\,\tfrac{1}{2}e^{-|\mu|}, \qquad \text{so} \qquad x^{-1}Ax \sim \sum_{p\geq 0} x^p\ \mathrm{ad}_t^p(A).$$

The leading correction $x\ \mathrm{ad}_t A$ has order -3. By Lemma A.5 its indicial family is $-i\partial_\tau(1+\tau^2)^{-1} = 2i\tau(1+\tau^2)^{-2}$, the derivative in τ predicted by the twist of Lemma 4.4. The operator $x^2D_x + c = -D_t + c$ of Example 4.3 has kernel $i\delta'(\mu) + c\,\delta(\mu)$ on the collar, independent of x. If $c \notin \mathbb{R}$, its inverse on $\mathbb{R}_t$ is convolution with $\kappa_c(\mu) = -i\,e^{ic\mu}\mathbf{1}_{\mu>0}$ when $\mathrm{Im}\,c > 0$, and with $\kappa_c(\mu) = i\,e^{ic\mu}\mathbf{1}_{\mu<0}$ when $\mathrm{Im}\,c < 0$. This kernel decays exponentially, lies in $\mathfrak{K}^{-1}$, and $\mathrm{In}(\kappa_c,\tau) = (\tau+c)^{-1}$.

A.5 The cusp calculus on X

Transplant operators on $\{t > 1/\epsilon\} \times \partial X$ to the collar via $t = 1/x$. Since $\nu_c = dt\,\nu_\partial$ there, kernels correspond. We write $\iota_* B$ for the operator on X defined by $B \in \Psi^m_{c,T}(\mathbb{R}\times\partial X)$ when $T > 1/\epsilon$.

Definition A.16 (Cusp pseudodifferential operators)**.** $\Psi^m_c(X)$ is the space of operators A with

$$K_A = K_{\iota_* B} + K_0 + K_R,$$

where $B \in \Psi^m_{c,T}(\mathbb{R}\times\partial X)$ for some $T > 1/\epsilon$, K_0 is the kernel of a classical pseudodifferential operator of order m with compact support in $X^\circ \times X^\circ$, and $K_R \in \dot{C}^\infty(X^2)$. We put $\Psi_c^{-\infty} = \bigcap_m \Psi^m_c$ and $x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_c(X) = \bigcup_{p,m} x^{-p}\Psi^m_c(X)$.

Proposition A.17 (Agreement with [15])**.** $\Psi^m_c(X)$ *is the space of [15, Def. 2]: the operators whose kernels, lifted to X_c^2, are classical conormal of order m at Δ_c, smooth up to ff_c, and vanishing to infinite order at all other boundary hypersurfaces. Here kernels are normalized with respect to ν_c in the right factor, instead of the half-density normalization of [15, (118)].*

Proof. Away from a neighbourhood of ff_c, the blow-down $X^2_c \to X^2$ is the blow-down of X^2_b away from Φ. There the condition of [15] means: conormal at the interior diagonal, and vanishing to infinite order at lb, rb and ff_b, that is, $\dot{C}^\infty$ on X^2 near the corner. This is also what Definition A.16 gives there. Indeed K_0 is conormal in the interior, K_R is $\dot{C}^\infty$, and on the region $\{x|\mu| \geq c\}$, which is away from ff_c, the kernel k_B is $O(\mu^{-\infty})$ with all derivatives. There x^{-1} and x'^{-1} are bounded by a multiple of $|\mu|$, and $\partial_x|_{x'} = \partial_x|_\mu - x^{-2}\partial_\mu$ and $\partial_{x'} = x'^{-2}\partial_\mu$. So k_B is $O((xx')^\infty)$ with all derivatives. Near ff_c, both definitions say that, in the coordinates (x, μ, y, y'), the kernel is a smooth function of $x \in [0, \delta)$ with values in $\mathfrak{R}^m(\partial X)$. For [15] this follows from Lemmas A.6 and A.7. Conormality at a submanifold transversal to the boundary is defined by smooth extension across ff_c, which is smoothness in x. For us it is Lemma A.9. Given an operator of the type in [15], multiply its kernel by $\chi(t)\chi(t')$ as in Lemma A.14. This gives an element of $\iota_*\Psi^m_{c,T}$. The remainder is supported where x or x' is bounded below, and is the sum of a compactly supported interior pseudodifferential operator and a $\dot{C}^\infty(X^2)$ kernel. □

We use cut-off functions of x alone. For $0 < a < b$, a function $\phi = \phi(x)$ is *of type* (a, b) if $\phi = 1$ for $x \leq a$ and $\phi = 0$ for $x \geq b$.

Lemma A.18 (Localization and mapping)**.** *Let $A \in \Psi^m_c(X)$.*

1. *If $\phi, \psi \in C^\infty(X)$ are functions of x with $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, then $\phi A \psi$ has kernel in $\dot{C}^\infty(X^2)$.*
2. *If $\phi, \psi \in C^\infty(X)$ are functions of x vanishing for $x \geq 1/T$, $T > 1/\epsilon$, then $\phi A\psi \in \iota_*\Psi^m_{c,T} + \dot{C}^\infty(X^2)$. If they have compact support in X°, then $\phi A\psi$ is a compactly supported classical pseudodifferential operator of order m, modulo $\dot{C}^\infty(X^2)$.*
3. *A maps $\dot{C}^\infty(X)$ to $\dot{C}^\infty(X)$ and $C^\infty(X)$ to $C^\infty(X)$ continuously. For $u \in C^\infty(X)$, $(Au)|_{\partial X}$ depends only on $u|_{\partial X}$:*

$$(Au)|_{\partial X} = A_\partial\big(u|_{\partial X}\big), \qquad A_\partial = \mathrm{In}(k_B(0), 0) \text{ has kernel } \int_{\mathbb{R}} k_B(0, \mu)\, d\mu. \tag{24}$$

4. *A^* and A^t, with respect to ν_c, lie in $\Psi^m_c(X)$. If $K_R \in \dot{C}^\infty(X^2)$, then the operators AR and RA have kernels in $\dot{C}^\infty(X^2)$, where R is the operator with kernel K_R.*

Proof. (1) Only the term ι_*B needs proof. If $\phi = 0$ for $x \geq b$ and $\psi = 0$ for $x \leq b'$, with $b < b'$, then on the support $\mu = 1/x - 1/x' \geq 1/x - 1/b' \geq c/x$. So the kernel is in the region of the proof of Proposition A.17, where it is $O((xx')^\infty)$ with all derivatives. The case with the roles of ϕ and ψ exchanged is the same.

(2) $\phi\,\iota_*B\,\psi = \iota_*(\phi B \psi) \in \iota_*\Psi^m_{c,T}$ by Theorem A.13. The term $\phi K_0 \psi$ is a compactly supported interior operator whose kernel is supported in $\{t, t' \geq T\}$. Its k is smooth in x and vanishes near $x = 0$, so it lies in $\iota_*\Psi^m_{c,T}$. The interior case is the same, since on a compact subset of X° the kernel k_B is a classical conormal distribution.

(3) Under $t = 1/x$, $\dot{C}^\infty$ functions supported in the collar correspond to restrictions to $\{t \geq T\}$ of elements of $\mathcal{S}(\mathbb{R} \times \partial X)$. Indeed $\partial_t = -x^2\partial_x$ and $\partial_x = -t^2\partial_t$, so rapid decay of all t-derivatives is equivalent to infinite order vanishing of all x-derivatives. Hence (K4) gives the $\dot{C}^\infty$ statement for ι_*B. The other two terms are clear. For $u \in C^\infty(X)$ supported in the collar,

$$(\iota_*Bu)(x, y) = \iint k_B(x, \mu, y, y')\, u\Big(\frac{x}{1 - x\mu}, y'\Big)\, d\mu\, \nu_\partial(y').$$

The function $u(x/(1 - x\mu), y')$ lies in $\mathcal{M}_T$, because $\partial_x^a\partial_\mu^b(x/(1 - x\mu))$ is polynomially bounded on Ω_T. By (K1), the integrand is smooth in x with values in $\mathfrak{R}^m$. Integration over μ and y' is continuous

from $\mathfrak{K}^m$ to $C^\infty(\partial X)$, by (K2) at $\tau = 0$. So $\iota_* Bu$ is smooth up to $x = 0$. At $x = 0$ the integrand is $k_B(0, \mu, y, y')u(0, y')$, which is (24). The terms K_0 and K_R contribute functions vanishing near ∂X.

(4) For $\iota_* B$ this is Theorem A.13. The transpose is $A^t = \overline{(\bar{A})^*}$, where $\bar{A}$ has kernel $\overline{K_A}$. The other terms are clear. AR has kernel $(A \otimes 1)K_R$, which lies in $\dot{C}^\infty(X^2) = \dot{C}^\infty(X)\widehat{\otimes}\dot{C}^\infty(X)$ [20] by (3). RA has kernel $(1 \otimes A^t)K_R$. □

Lemma A.19. *$x^\infty \Psi_c^{-\infty}(X) = \bigcap_N x^N \Psi_c^{-\infty}(X)$ is the space of operators with kernels in $\dot{C}^\infty(X^2)$. This is the ideal $\mathcal{I}$ of the notes.*

Proof. One inclusion holds by definition. Conversely, let $A \in x^N \Psi_c^{-\infty}$ for every N. Then the Taylor coefficients of k_B at $x = 0$ all vanish. They do not depend on the decomposition, since the other two terms contribute 0 near $x = 0$ or $O(x^\infty)$. So $k_B = O(x^\infty)$ in $\mathfrak{K}^{-\infty} = \mathcal{S}(\mathbb{R}_\mu; C^\infty)$. Since $1/x' = 1/x - \mu \le x^{-1} + |\mu|$, every negative power of x' is bounded by a polynomial in x^{-1} and $|\mu|$. The derivatives $\partial_x|_{x'}$ and $\partial_{x'}$ bring in only such factors. So $K_{\iota_* B} = O((xx')^\infty)$ with all derivatives. □

Theorem A.20 (The cusp calculus is a doubly filtered algebra)**.** *1. $\Psi_c^m(X)\,\Psi_c^{m'}(X) \subset \Psi_c^{m+m'}(X)$, and $\Psi_c^m(X)$ is closed under adjoints and transposes.*

2. For $A \in \Psi_c^m(X)$ and $z \in \mathbb{C}$, $x^{-z} A x^z \in \Psi_c^m(X)$ depends holomorphically on z, and $x^{-z} A x^z - A \in x\Psi_c^{m-1}(X)$.

3. $x^k \Psi_c^m(X) \cdot x^l \Psi_c^{m'}(X) \subset x^{k+l}\Psi_c^{m+m'}(X)$ for $k, l \in \mathbb{Z}$. So $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$ is an algebra filtered by order and by powers of x, and $\mathcal{I} = x^\infty \Psi_c^{-\infty}(X)$ is a two-sided ideal in it.

Proof. (1) Fix $T > 2/\epsilon$ and choose ρ_∂ of type $(1/(8T), 1/(4T))$ and $\tilde{\rho}_\partial$ of type $(1/(2T), 1/T)$, so that $\operatorname{supp}\rho_\partial \cap \operatorname{supp}(1 - \tilde{\rho}_\partial) = \emptyset$. Then $\rho_0 = 1 - \rho_\partial$ vanishes for $x \le 1/(8T)$. Let $\tilde{\rho}_0 = 1 - \phi$ with ϕ of type $(1/(32T), 1/(16T))$; it is equal to 1 near $\operatorname{supp}\rho_0$ and vanishes near ∂X. By Lemma A.18(1), for every $A \in \Psi_c^m$,

$$A \equiv \rho_\partial A \tilde{\rho}_\partial + \rho_0 A \tilde{\rho}_0 \mod \dot{C}^\infty(X^2),$$

and by Lemma A.18(2) the two terms are of collar type ($\iota_* \Psi_{c,T}$) and of interior type, modulo $\dot{C}^\infty$. By Lemma A.18(4) the $\dot{C}^\infty$ terms form an ideal. So it is enough to compose terms of the two types. Two collar terms compose by Theorem A.13, and two interior terms by the interior calculus. For a mixed product $\rho_\partial A \tilde{\rho}_\partial \cdot \rho_0 A' \tilde{\rho}_0$, the function $\tilde{\rho}_\partial \rho_0$ is supported in $\{1/(8T) \le x \le 1/T\}$. Let θ be a function of x, supported in $\{1/(16T) < x < 2/T\}$ and equal to 1 near that set. Then, modulo $\dot{C}^\infty$ (Lemma A.18(1)), $\tilde{\rho}_\partial \rho_0 A' \tilde{\rho}_0 \equiv \tilde{\rho}_\partial \rho_0 A' \theta$, and this lies in $\iota_* \Psi_{c,T/2}^{m'}$ by Lemma A.18(2), since $T/2 > 1/\epsilon$. So the product is a product of collar terms. The other mixed product is treated in the same way. Adjoints and transposes are Lemma A.18(4).

(2) On the collar $x^{-z} = t^z$, so $x^{-z}\iota_* B x^z = \iota_* W_z(B) \in \iota_* \Psi_{c,T}^m$, holomorphic in z (Lemma A.11). Taylor's formula with integral remainder in $w = x\mu$ gives

$$(1 - x\mu)^{-z} = \sum_{p<N} \binom{z+p-1}{p} x^p \mu^p + x^N \mu^N r_N(x, \mu; z), \tag{25}$$

where $r_N = \binom{z+N-1}{N} N \int_0^1 (1-s)^{N-1}(1 - sx\mu)^{-z-N}\, ds$. Here $(1 - sx\mu)^{-1} \le \max(1, (1 - x\mu)^{-1})$, so $r_N \in \mathcal{M}_T$. With $N = 1$ this gives $W_z(B) - B = x \cdot (\mu r_1) B \in x\Psi_{c,T}^{m-1}$. The interior term A_0 changes by $x^{-z}[A_0, x^z]$, which is a compactly supported interior operator of order $m - 1$. The $\dot{C}^\infty$ term is multiplied by $(x'/x)^z$ and stays in $\dot{C}^\infty$.

(3) $x^k A x^l B = x^{k+l}(x^{-l} A x^l) B$, by (1) and (2). The ideal property follows from Lemma A.18(4) and Lemma A.19. □

A.6 The symbol sequence and the quantization map

Let $S^m_{\mathrm{cl}}({}^cT^*X)$ be the classical symbols of order m on the bundle ${}^cT^*X$, smooth up to $x = 0$. Let $\mathcal{G}^m({}^cT^*X)$ be the smooth functions on ${}^cT^*X \setminus 0$ that are homogeneous of degree m. Over the collar, ${}^cT^*X \cong [0,\epsilon) \times T^*\partial X \times \mathbb{R}_\tau$, and over the interior ${}^cT^*X = T^*X^\circ$ with $\tau = x^2\xi$.

Proposition A.21 (The symbol sequence [15, (11)])**.** *There is a principal symbol map $\sigma_m : \Psi^m_c(X) \to \mathcal{G}^m({}^cT^*X)$. It is the usual principal symbol over X°, and the sequence*

$$0 \longrightarrow \Psi^{m-1}_c(X) \hookrightarrow \Psi^m_c(X) \xrightarrow{\sigma_m} \mathcal{G}^m({}^cT^*X) \longrightarrow 0$$

is exact. Moreover $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$, $\sigma_m(A^*) = \overline{\sigma_m(A)}$ and $\sigma_m(x^{-z}Ax^z) = \sigma_m(A)$.

Proof. On the collar, set $\sigma_m(A) = \sigma_m(k_B(x))$. In the interior, take the usual symbol of K_0. These agree over the overlap. Over the interior of the collar, $k_B(x)$ is the kernel of $\iota_* B$ written in the coordinates (t, y), and its symbol is the usual one, written in the covariable $-\tau$ dual to t, which is $x^2\xi$ dual to x. By Lemma A.9 and (K1) the result is smooth up to $x = 0$ on ${}^cT^*X$. Multiplicativity and the adjoint formula hold in the model by Theorem A.13, and in the interior by the standard calculus. By the reduction in the proof of Theorem A.20(1), they hold globally. The conjugation formula follows from Theorem A.20(2). For exactness: if $\sigma_m(A) = 0$, then $\sigma_m(k_B(x)) = 0$ for all x, so k_B is a smooth map into $\mathfrak{K}^m$ with values in the closed subspace $\mathfrak{K}^{m-1}$. Its difference quotients and their limits stay in that subspace, and the topologies agree by (K1). So $k_B \in C^\infty([0, 1/T]; \mathfrak{K}^{m-1})$. The interior term is handled in the same way. Surjectivity follows from Proposition A.22. □

Proposition A.22 (Global quantization [15, Prop. 24])**.** *There is a linear map*

$$q : x^{-\mathbb{Z}} S^{\mathbb{Z}}_{\mathrm{cl}}({}^cT^*X) \longrightarrow x^{-\mathbb{Z}} \Psi^{\mathbb{Z}}_c(X)$$

with $q(x^k S^m_{\mathrm{cl}}) \subset x^k \Psi^m_c$, $q(x^k a) = x^k q(a)$, $\sigma_m(q(a)) = a_m$ (the leading part), and $q(S^{-\infty}_{\mathrm{cl}}) \subset \Psi^{-\infty}_c$. For every $A \in \Psi^m_c$ there is an $a \in S^m_{\mathrm{cl}}$, unique modulo $S^{-\infty}_{\mathrm{cl}}$, with $A - q(a) \in \Psi^{-\infty}_c$. So q induces linear isomorphisms

$$\frac{x^{-\mathbb{Z}} S^{\mathbb{Z}}_{\mathrm{cl}}({}^cT^*X)}{x^{-\mathbb{Z}} S^{-\infty}_{\mathrm{cl}}({}^cT^*X)} \cong \frac{x^{-\mathbb{Z}} \Psi^{\mathbb{Z}}_c(X)}{x^{-\mathbb{Z}} \Psi^{-\infty}_c(X)}, \qquad \frac{x^{\infty} S^{\mathbb{Z}}_{\mathrm{cl}}({}^cT^*X)}{x^{\infty} S^{-\infty}_{\mathrm{cl}}({}^cT^*X)} \cong \frac{x^{\infty} \Psi^{\mathbb{Z}}_c(X)}{x^{\infty} \Psi^{-\infty}_c(X)}.$$

Given $Q \in \Psi^1_c$ with positive principal symbol q, the left-hand sides are $x^{-\mathbb{Z}} C^\infty({}^cS^*X)((\mathsf{q}^{-1}))$ and $\dot{C}^\infty({}^cS^*X)((\mathsf{q}^{-1}))$, which gives the isomorphisms [15, (13)] for $\mathcal{A}_\sigma$ and $\mathcal{I}_\sigma$ as vector spaces.

Proof. Choose $\rho_\partial, \tilde{\rho}_\partial$ as in the proof of Theorem A.20, a finite interior atlas, and a partition of unity $\rho_\partial + \sum_i \rho_i = 1$ with $\tilde{\rho}_i = 1$ near $\operatorname{supp} \rho_i$. Define

$$q(a) = \tilde{\rho}_\partial \, \iota_* \mathrm{Op}_x(\rho_\partial a) \, \tilde{\rho}_\partial + \sum_i \tilde{\rho}_i \, \mathrm{Op}_i(\rho_i a) \, \tilde{\rho}_i.$$

Here Op_i is the left quantization in the chart i, and $\mathrm{Op}_x(b)$ is the operator with kernel $k(x, \mu) = \mathrm{Op}(b(x, \cdot))(\mu)$ from (K1), placed on the cylinder by Lemma A.14. It is smooth in x since Op is continuous and linear. Since x is the left variable, $q(xa) = xq(a)$; we set $q(x^{-k}a) = x^{-k}q(a)$. The symbol identity is (K1).

Given $A \in \Psi^m_c$, let a_m be any element of S^m_{cl} with leading part $\sigma_m(A)$. Then $A - q(a_m) \in \Psi^{m-1}_c$ by Proposition A.21. Repeating, we obtain $a_{m-j} \in S^{m-j}_{\mathrm{cl}}$ with $A - q(\sum_{j<N} a_{m-j}) \in \Psi^{m-N}_c$. Let $a \sim \sum a_{m-j}$ by Borel's lemma, applied to symbols depending smoothly on x. Then $A - q(a) \in$

$\bigcap_N \Psi_c^{m-N}$. Uniqueness: if $q(a) \in \Psi_c^{-\infty}$, then $\sigma(q(a)) = 0$ at every order, so $a \in S_{\mathrm{cl}}^{-\infty}$. If $a = O(x^\infty)$, then $q(a) \in x^\infty \Psi_c$, since its kernel is $O(x^\infty)$ and Lemma A.9 applies. Conversely, if $A \in x^\infty \Psi_c^m$, then for each N, $A = x^N A_N$ and $a \equiv x^N a_N \bmod S_{\mathrm{cl}}^{-\infty}$. So every homogeneous term of a is $O(x^\infty)$, and a Borel sum of these terms gives $a' \in x^\infty S_{\mathrm{cl}}^m$ with $a - a' \in S_{\mathrm{cl}}^{-\infty}$. Then $A - q(a') \in x^\infty \Psi_c^m \cap \Psi_c^{-\infty} = x^\infty \Psi_c^{-\infty}$; the last equality holds because the Taylor coefficients at $x = 0$ of a smooth map into $\mathfrak{K}^{-\infty}$ are the same whether computed in $\mathfrak{K}^{-\infty}$ or in $\mathfrak{K}^m$. So the isomorphism respects the x^∞ subspaces. Finally, a classical symbol modulo $S_{\mathrm{cl}}^{-\infty}$ is the same as its sequence of homogeneous terms $a_{m-j} = \mathsf{q}^{m-j} \cdot (a_{m-j} \mathsf{q}^{j-m})$, with $a_{m-j}\mathsf{q}^{j-m}$ a function on ${}^cS^*X$. Any such sequence is realized, by Borel's lemma. □

Lemma A.23 (Asymptotic summation)**.** *If $A_j \in \Psi_c^{m-j}$, there is $A \in \Psi_c^m$ with $A - \sum_{j<N} A_j \in \Psi_c^{m-N}$ for all N. If $R_j \in x^j \Psi_c^m$, there is $S \in \Psi_c^m$ with $S - \sum_{j<N} R_j \in x^N \Psi_c^m$ for all N.*

Proof. For the first statement, write $A_j = q(a_j)$ modulo $\Psi_c^{-\infty}$, sum the symbols, and set $A = q(a)$. The second statement is Borel's lemma for the Taylor series at $x = 0$ of the kernels $k_{R_j} \in x^j C^\infty([0, 1/T]; \mathfrak{K}^m)$, followed by Lemma A.14. An operator whose k vanishes to order N at $x = 0$ lies in $x^N \Psi_c^m$, by Taylor's formula with remainder. □

A.7 The indicial family, the normal operator and Taylor series

Definition A.24 (Normal operator and indicial family)**.** For $A \in \Psi_c^m(X)$ with collar term $\iota_* B$, the *normal operator* is $N(A) = k_B(0) \in \mathfrak{K}^m(\partial X)$, regarded as an element of $\Psi_{\mathrm{sus}}^m(\partial X)$. The *indicial family* is $\mathrm{In}(A, \tau) = \mathrm{In}(N(A), \tau)$.

$N(A)$ does not depend on the decomposition, since the other terms contribute 0 or $O(x^\infty)$ at $x = 0$. Concretely, $N(A)$ is the restriction of the lifted kernel to ff_c, a convolution kernel in μ by Lemma A.6.

Proposition A.25 (The indicial sequence [15, (12)])**.**

1. (MN's definition.) *For $\tau \in \mathbb{R}$, $\mathrm{In}(A, \tau) = \left(e^{i\tau/x} A e^{-i\tau/x}\right)_\partial$, with $(\cdot)_\partial$ as in* (24)*. This is* (9)*.*
2. *$N(AB) = N(A)N(B)$, $\mathrm{In}(AB, \tau) = \mathrm{In}(A, \tau)\,\mathrm{In}(B, \tau)$, $N(A^*) = N(A)^*$ and $N(x^{-z} A x^z) = N(A)$.*
3. *The sequence $0 \to x\Psi_c^m(X) \hookrightarrow \Psi_c^m(X) \xrightarrow{N} \Psi_{\mathrm{sus}}^m(\partial X) \to 0$ is exact.*

Proof. (1) $e^{i\tau/x} = e^{i\tau t}$, so the conjugated operator has collar kernel $e^{i\tau\mu} k_B$ (Lemma A.11(3)). By (24), its boundary operator has kernel $\int e^{i\tau\mu} k_B(0, \mu)\, d\mu = \mathrm{In}(N(A), \tau)$.

(2) By Lemma A.18(3), $((AB)u)|_\partial = A_\partial((Bu)|_\partial) = A_\partial B_\partial (u|_\partial)$. Every element of $C^\infty(\partial X)$ is a restriction, so $(AB)_\partial = A_\partial B_\partial$. Applying this to the conjugates by $e^{i\tau t}$ gives $\mathrm{In}(AB, \tau) = \mathrm{In}(A, \tau)\,\mathrm{In}(B, \tau)$. Since the Fourier transform (22) is injective, $N(AB) = N(A)N(B)$. For the adjoint: the kernel of B^* is $\overline{K_B(t', t)}$, so $k_{B^*}(x, \mu) = \overline{k_B(x/(1 - x\mu), -\mu)}$ for $x > 0$. Since k_{B^*} is smooth up to $x = 0$ (Theorem A.13), its value at $x = 0$ is the limit, $\overline{k_B(0, -\mu)}$, which is the kernel of $N(A)^*$. The last statement follows from Theorem A.20(2) and part (3), since $x\Psi_c \subset \ker N$.

(3) If $A = xA'$, then $k_B = x k_{B'}$ vanishes at $x = 0$. Conversely, if $N(A) = 0$, then $k_B = x k_1$ with $k_1 = \int_0^1 \partial_x k_B(sx)\, ds \in C^\infty([0, 1/T]; \mathfrak{K}^m)$, which has the same support. So $\iota_* B = x\, \iota_* B_1$, while K_0 and K_R are x times terms of the same type. Surjectivity: $N(\chi\kappa\chi) = \kappa$ by Lemma A.14. □

Proposition A.26 (Conjugation and the twist)**.** *Let $A \in \Psi_c^m(X)$ have collar term $\iota_* B$, and let $z \in \mathbb{C}$, $N \in \mathbb{N}$. Then*

$$x^{-z}(\iota_* B)x^z - \sum_{i<N} \binom{z+i-1}{i} x^i\, \iota_*\big(\mathrm{ad}_t^i B\big) \in x^N \iota_* \Psi_{c,T}^{m-N}.$$

Hence conjugation by x^z preserves orders and x-orders, changes neither σ nor N, and acts on Taylor series at ff_c (Proposition A.27) by $\mathsf{T}(x^{-z}Ax^z) = \sum_i \binom{z+i-1}{i} x^i \,\mathrm{ad}_t^i\, \mathsf{T}(A)$, with ad_t acting on the coefficients. On the collar, $[x, \partial_t] = x^2$, and for $\kappa \in \Psi_{\mathrm{sus}}^m(\partial X)$, $\mathrm{In}([t,\kappa],\tau) = -i\partial_\tau \mathrm{In}(\kappa,\tau)$ and $x^{-1}\kappa x = \sum_{i\geq 0} x^i \,\mathrm{ad}_t^i(\kappa)$ as Taylor series at ff_c.

Here $\binom{z+i-1}{i} = z(z+1)\cdots(z+i-1)/i!$.

Proof. Multiply k_B by (25). The term $x^i\mu^i k_B$ is the kernel of $x^i \,\mathrm{ad}_t^i(B)$, and the remainder $x^N\mu^N r_N k_B$ lies in $x^N\Psi_{c,T}^{m-N}$ by Lemma A.11. The remaining statements about orders, σ and N are Theorem A.20(2) and Propositions A.21 and A.25. For the twist: on the collar $[1/t, \partial_t]u = -\partial_t(1/t)\, u = t^{-2}u = x^2 u$. The indicial family of $[t,\kappa]$ is given by Lemma A.5. The formula for $x^{-1}\kappa x$ is the case $z = 1$, applied to $\chi\kappa\chi$ (Lemma A.14). □

Proposition A.27 (Taylor series at the front face [15, Prop. 25])**.** *For $A \in x^{-k}\Psi_c^m(X)$, let*

$$\mathsf{T}(A) = x^{-k} \sum_{j\geq 0} x^j \,\frac{1}{j!}\, \partial_x^j k_B(0) \in \Psi_{\mathrm{sus}}^m(\partial X)((x)),$$

where $x^k A$ has collar term $\iota_* B$. Write $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x)) = \bigcup_m \Psi_{\mathrm{sus}}^m(\partial X)((x))$, the series whose coefficients have bounded order. Then:

1. *T is well defined, surjective onto $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x))$, respects the order filtration, and has kernel $x^\infty\Psi_c^{\mathbb{Z}}(X)$.*
2. *$\mathsf{T}(AB) = \mathsf{T}(A)\cdot\mathsf{T}(B)$ for the associative product on $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x))$ determined by*

$$(x^j g)(x^l h) = \sum_{i\geq 0} \binom{l+i-1}{i} x^{j+l+i}\, \mathrm{ad}_t^i(g)\, h, \qquad j, l \in \mathbb{Z},$$

with the powers of x written on the left. In particular $x^{-1}\partial_t x = \partial_t - x$, that is, $[x, \partial_t] = x^2$.

Hence $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)/x^\infty\Psi_c^{\mathbb{Z}}(X) \cong \Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)((x))$ as algebras, and the restrictions give [15, (14)].

For $l \leq 0$ only finitely many terms are non-zero. The series in i makes sense in $\Psi_{\mathrm{sus}}((x))$, because x^i raises the x-order by i.

Proof. (1) Well-definedness is as for N. Surjectivity is Lemma A.23 together with Lemma A.14. If $\mathsf{T}(A) = 0$, then $k_B = O(x^\infty)$, so $A \in x^N\Psi_c^m$ for every N, by Proposition A.25(3) applied repeatedly.

(2) Take $k = 0$ for both factors; the general case follows by Theorem A.20(3). Let $\chi = \chi(x)$ be a cut-off of type (a, b) with small b. By Proposition A.25(3) applied repeatedly, and Lemma A.14, $A \equiv \sum_{j<N} x^j \chi g_j \chi$ and $B \equiv \sum_{l<N} x^l \chi h_l \chi$ modulo $x^N\Psi_c$, where $\mathsf{T}(A) = \sum x^j g_j$ and $\mathsf{T}(B) = \sum x^l h_l$. By Proposition A.26,

$$x^j\chi g_j\chi\, x^l\chi h_l\chi \equiv \sum_{i<N} \binom{l+i-1}{i} x^{j+l+i}\, \chi\, \mathrm{ad}_t^i(g_j)\chi\,\chi h_l\chi \mod x^N\Psi_c,$$

using that χ commutes with t. It remains to show that $\mathsf{T}(\chi g\chi\,\chi h\chi) = gh$ for $g, h \in \Psi_{\mathrm{sus}}$. The kernel of $\chi g\chi^2 h\chi - \chi(gh)\chi$ is

$$-\chi(t)\chi(t') \int \kappa_g(t-s)\big(1-\chi(s)^2\big)\kappa_h(s-t')\,ds.$$

The integrand is supported in a bounded set of s, so for large t the factor $\kappa_g(t-s)$ is evaluated far from 0 and is $O(t^{-\infty})$ with all derivatives. If $|\mu|$ is large, then either $t' = t - \mu$ is large, and $\kappa_h(s-t')$ is small, or t is large compared with $|\mu|$. So the kernel, written as a function of (x, μ), is smooth and $O(x^N\langle\mu\rangle^{-N})$ for every N, with all derivatives. That is, its k is $O(x^\infty)$ in $\mathfrak{K}^{-\infty}$. By Lemma A.14, $\mathsf{T}(\chi(gh)\chi) = gh$. The identity $x^{-1}\partial_t x = \partial_t - x$ follows from $\mathrm{ad}_t\,\partial_t = [t, \partial_t] = -1$ and $\mathrm{ad}_t^2\,\partial_t = 0$. Associativity of the product is inherited from Ψ_c. □

Proposition A.28 (Lifting b-operators [15, Prop. 26], first part)**.** *Let $x_b = e^{-1/x}$, and let A be in the small b-calculus $\Psi_b^m(X)$ of [18], defined with respect to x_b. Then $A \in \Psi_c^m(X)$, and $\mathsf{T}(A) = N_b(A)$ is constant in x: the cusp normal operator is the b-normal operator, and $\mathrm{In}(A,\tau) = I(A,\tau)$ for $\tau \in \mathbb{R}$.*

Proof. The b-density dx_b'/x_b' equals dx'/x'^2, so kernels with respect to it are kernels with respect to ν_c. Near ff_b, use the coordinates x_b and $z = \log(x_b/x_b')$. Then $z = -1/x + 1/x' = -\mu$. A small b-kernel is $K_b(x_b, z, y, y')$, smooth in x_b, conormal at $\{z = 0,\ y = y'\}$, and $O(e^{-N|z|})$ for every N together with its derivatives. This is infinite order vanishing at lb and rb, where $x_b'/x_b = e^{-z} \to 0$ or ∞. So $x_b \mapsto K_b(x_b, \cdot)$ is smooth with values in $\mathfrak{K}^m$, and

$$k(x,\mu) = K_b\big(e^{-1/x}, -\mu\big)$$

is smooth in x up to 0, since $e^{-1/x}$ is. It equals $K_b(0, -\mu)$ modulo $O(x^\infty)$, because $e^{-1/x}$ vanishes to infinite order. Away from ff_b, infinite order vanishing in x_b implies it in x, since $x_b^N = e^{-N/x}$. The indicial families agree by (22) and the identification of Section 2.2. □

Remark A.29. The second part of [15, Prop. 26], density of the lifts, is not proved here and is not used in the notes. The cusp structure induced by $x = 1/\log(1/x_b)$ depends on x_b only through its boundary restriction, as discussed in Section 4.6.

A.8 Fredholm theory

Definition A.30. $A \in x^{-p}\Psi_c^m(X;E)$ is *fully elliptic* if $\sigma_m(x^pA)$ is invertible on ${}^cT^*X \setminus 0$ and $\mathrm{In}(x^pA,\tau)$ is invertible on $C^\infty(\partial X; E)$ for every $\tau \in \mathbb{R}$.

By Propositions A.21 and A.25, using x^pA or Ax^p here makes no difference.

Lemma A.31 (Inverses in the suspended algebra)**.** *Let $\kappa \in \Psi_{\mathrm{sus}}^m(H)$ be elliptic, that is, $\sigma_m(\kappa)$ is invertible on $(T^*H \times \mathbb{R}) \setminus 0$. Suppose that $\mathrm{In}(\kappa,\tau)$ is invertible for every $\tau \in \mathbb{R}$. Then there is $\gamma \in \Psi_{\mathrm{sus}}^{-m}(H)$ with $\kappa\gamma = \gamma\kappa = 1$, and $\mathrm{In}(\gamma,\tau) = \mathrm{In}(\kappa,\tau)^{-1}$.*

Proof. Parametrix. Let $b_0 \in S_{\mathrm{cl}}^{-m}$ be equal to $\sigma_m(\kappa)^{-1}$ for $|\zeta| \geq 1$, and put $\beta_0 = \mathrm{Op}(b_0)$. By (K1) and Lemma A.5, $\kappa * \beta_0 = 1 - \rho$ with $\rho \in \Psi_{\mathrm{sus}}^{-1}$. Let β be an asymptotic sum of $\beta_0 * \rho^{*j}$, which exists by (K1). Then $\kappa * \beta - 1 \in \Psi_{\mathrm{sus}}^{-\infty}$. In the same way a left parametrix exists, and the two differ by $\Psi_{\mathrm{sus}}^{-\infty}$. So

$$\mathrm{In}(\kappa)\,\mathrm{In}(\beta) = 1 - S, \qquad \mathrm{In}(\beta)\,\mathrm{In}(\kappa) = 1 - S', \qquad S, S' \in \mathcal{S}\big(\mathbb{R}_\tau; \Psi^{-\infty}(H)\big),$$

by (K2).

The inverse. Let $G(\tau) = \mathrm{In}(\kappa,\tau)^{-1}$. From the two identities, $\mathrm{In}(\beta) = G(1-S)$ and $\mathrm{In}(\beta) = (1-S')G$, so

$$G = \mathrm{In}(\beta) + S'G = \mathrm{In}(\beta) + S'\,\mathrm{In}(\beta) + S'GS.$$

By (K2), $\mathrm{In}(\kappa,\tau)$ is a smooth family of bounded maps $H^{s+m}(H) \to H^s(H)$. So G is smooth, with $\partial_\tau G = -G(\partial_\tau \mathrm{In}\,\kappa)G$. For $|\tau|$ large, $\|S(\tau)\|_{H^s\to H^s} < \frac{1}{2}$. So $G = \mathrm{In}(\beta)(1-S)^{-1}$ is bounded $H^s \to H^{s+m}$ by $C\langle\tau\rangle^{|m|}$, by (K2). On compact sets of τ it is bounded by continuity. (Here $\mathrm{In}(\kappa,\tau)$ is elliptic on H, so its inverse on C^∞ is also its inverse $H^s \to H^{s+m}$.) Since $\partial_\tau \mathrm{In}(\kappa)$ has order $m-1$, the formula for $\partial_\tau G$ and induction show that all $\partial_\tau^j G$ are polynomially bounded as maps $H^s \to H^{s+m}$. Since S and S' are Schwartz in τ as maps $H^{-M} \to H^M$ for every M, the family $S'GS$ is Schwartz in τ with values in $\Psi^{-\infty}(H)$. By (K2) it is $\mathrm{In}(\gamma')$ for some $\gamma' \in \Psi^{-\infty}_{\mathrm{sus}}$. Let $\sigma' \in \Psi^{-\infty}_{\mathrm{sus}}$ be the element with $\mathrm{In}(\sigma') = S'$. Then $\gamma = \beta + \sigma' * \beta + \gamma' \in \Psi^{-m}_{\mathrm{sus}}$, and its indicial family is G. Hence $\kappa\gamma = \gamma\kappa = 1$, by injectivity of the Fourier transform. □

Definition A.32 (Cusp Sobolev spaces)**.** For $s \in \mathbb{R}$, $H^s_c(X)$ is the space of $u \in H^s_{\mathrm{loc}}(X^\circ)$ such that $\rho_0 u \in H^s$ and $\rho_\partial u$, transplanted to the cylinder $\mathbb{R}\times\partial X$, lies in $H^s(\mathbb{R}\times\partial X)$. Here ρ_∂, ρ_0 are as in the proof of Theorem A.20. For $r \in \mathbb{R}$ we also use $x^r H^s_c(X)$. Note that $H^0_c(X) = L^2(X,\nu_c)$.

Lemma A.33. *1. $A \in x^p\Psi^m_c(X)$ is bounded $x^rH^s_c \to x^{r+p}H^{s-m}_c$ for all r,s.*

2. $\bigcap_{r,s} x^rH^s_c(X) = \dot C^\infty(X)$.

3. An operator R with kernel in $\dot C^\infty(X^2)$ maps $\bigcup_{r,s} x^rH^s_c$ to $\dot C^\infty(X)$. It is compact on every $x^rH^s_c$.

Proof. (1) By Theorem A.20(2), $x^{-r-p}Ax^r \in \Psi^m_c$. So we may take $r = p = 0$. Then the collar term is bounded by (K4), and the other two terms are bounded by the interior calculus.

(2) If u lies in every $x^rH^s_c$, then on the cylinder $t^N\rho_\partial u \in H^s$ for all N,s. By Sobolev embedding, all $\partial_t^j\partial_y^\alpha u$ are $O(t^{-\infty})$. Since $\partial_x = -t^2\partial_t$, all x-derivatives are $O(x^\infty)$. The converse is clear.

(3) The space $x^rH^s_c$ is dual to $x^{-r}H^{-s}_c$ under the $L^2(\nu_c)$ pairing. For $v \in x^rH^s_c$, $Rv(z) = \langle K_R(z,\cdot), v\rangle$. So every $\dot C^\infty$ seminorm of Rv is bounded by a seminorm of K_R times $\|v\|$. In particular $\|R\|_{x^rH^s_c\to x^rH^s_c}$ is bounded by a $\dot C^\infty(X^2)$ seminorm of K_R. Finite sums $\sum\phi_i(z)\psi_i(z')$ with $\phi_i,\psi_i \in \dot C^\infty(X)$ are dense in $\dot C^\infty(X^2)$ [20]. So R is a norm limit of finite rank operators, and hence compact. □

Theorem A.34 (Parametrix)**.** *If $A \in \Psi^m_c(X;E)$ is fully elliptic, there is $Q \in \Psi^{-m}_c(X;E)$ with*

$$AQ = 1-R, \qquad QA = 1-R', \qquad R,R' \in x^\infty\Psi^{-\infty}_c(X;E) = \{\text{kernels in } \dot C^\infty(X^2)\}.$$

Conversely, if $A \in x^{-p}\Psi^m_c$ has $B \in x^p\Psi^{-m}_c$ with $AB-1, BA-1 \in \mathcal{I}$, then A is fully elliptic. So full ellipticity is invertibility in $\mathcal{A} = x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_c/\mathcal{I}$, with the orders matched.

Proof. Symbolic step. Let $Q_1 = q(b)$ with b equal to $\sigma_m(A)^{-1}$ away from the zero section. Then $AQ_1 = 1-R_1$ with $R_1 \in \Psi^{-1}_c$ (Proposition A.21). Let $S_1 \sim \sum_{j\ge1} R_1^j$ (Lemma A.23), and put $Q_2 = Q_1(1+S_1)$. As below, $(1-R_1)(1+S_1)-1 \in \Psi^{-N}_c$ for every N, so $AQ_2 = 1-R_2$ with $R_2 \in \Psi^{-\infty}_c$.

Indicial step. By Proposition A.25, $N(A) \in \Psi^m_{\mathrm{sus}}(\partial X)$ satisfies the hypotheses of Lemma A.31. Its symbol is $\sigma_m(A)|_{x=0}$. Let $\gamma = N(A)^{-1} \in \Psi^{-m}_{\mathrm{sus}}$, and put $Q_3 = Q_2 + \chi(\gamma N(R_2))\chi$, with χ as in Lemma A.14. The added term lies in $\Psi^{-\infty}_c$, and

$$N(AQ_3) = 1 - N(R_2) + N(A)\gamma N(R_2) = 1.$$

So $AQ_3 = 1 - R_3$ with $R_3 \in \Psi_c^{-\infty}$ and $N(R_3) = 0$, that is, $R_3 \in x\Psi_c^{-\infty}$ by Proposition A.25(3).

Boundary Neumann series. By Theorem A.20, $R_3^j \in x^j\Psi_c^{-\infty}$. Let $S_3 \sim \sum_{j\geq 1} R_3^j$ in the sense of Lemma A.23, with $S_3 \in \Psi_c^{-\infty}$. Put $Q = Q_3(1 + S_3)$. Since $(1 - R_3)(1 + \sum_{1\leq j\leq N} R_3^j) = 1 - R_3^{N+1}$, we get $AQ - 1 = (1 - R_3)(1 + S_3) - 1 \in x^N\Psi_c^{-\infty}$ for every N. By Lemma A.19 it has kernel in $\dot{C}^\infty(X^2)$. A left parametrix Q' is constructed in the same way, and $Q' - Q = Q'(1 - AQ) - (1 - Q'A)Q \in \mathcal{I}$. So Q is a two-sided parametrix.

Converse. Write $A = x^{-p}A'$ and $B = B'x^p$. Then $A'B' \equiv x^pABx^{-p} \equiv 1 \bmod \mathcal{I}$. Taking σ_0 and N gives $\sigma(A')\sigma(B') = 1$ and $N(A')N(B') = 1$, and similarly in the other order. □

Theorem A.35 (Fredholm theory, (B2))**.** *Let $A \in x^{-p}\Psi_c^m(X; E)$ be fully elliptic. Then:*

1. $\ker A$ *and* $\ker A^*$ *(adjoint with respect to $L^2(X, \nu_c; E)$) are finite dimensional subspaces of $\dot{C}^\infty(X; E)$. For every $r, s \in \mathbb{R}$, $A : x^rH_c^s \to x^{r-p}H_c^{s-m}$ is Fredholm, with kernel* $\ker A$ *and range $\{f : \langle f, v\rangle = 0$ for all $v \in \ker A^*\}$. In particular its index does not depend on r or s.*

2. *$A : \dot{C}^\infty(X; E) \to \dot{C}^\infty(X; E)$ is Fredholm, with the same kernel and the range described in* (1)*.*

3. (Generalized inverse.) *There is $B_0 \in x^p\Psi_c^{-m}(X; E)$ with*

$$AB_0 = 1 - \Pi_1, \qquad B_0A = 1 - \Pi_0.$$

Here Π_0 is the $L^2(\nu_c)$-orthogonal projection onto $\ker A$*, and $\Pi_1 = x^{-p}\Pi_1'x^p$, with Π_1' the orthogonal projection onto* $\ker(x^pA)^*$*. Both are finite rank projections with kernels in $\dot{C}^\infty(X^2)$, that is, in $\mathcal{I}$. Π_1 projects onto a complement of the range of A on $\dot{C}^\infty$.*

Proof. Put $A' = x^pA \in \Psi_c^m$. It is fully elliptic, and $(A')^* = A^*x^p$ is fully elliptic too: its symbol and normal operator are the adjoints (Propositions A.21 and A.25). Let Q be a parametrix for A' as in Theorem A.34.

(1) Conjugating by x^r (Theorem A.20(2)) and using Lemma A.33(1), $A' : x^rH_c^s \to x^rH_c^{s-m}$ and Q in the reverse direction are bounded. Then $A'Q - 1$ and $QA' - 1$ are compact by Lemma A.33(3), so A' is Fredholm. If $A'u = 0$ with $u \in x^rH_c^s$, then $u = R'u \in \dot{C}^\infty$ by Lemma A.33(3). So the kernel is a finite dimensional subspace of $\dot{C}^\infty$, independent of r, s. The same applies to $(A')^*$, and hence to $\ker A^* = x^p\ker(A')^* \subset \dot{C}^\infty$. The range of a Fredholm operator is closed. By duality between $x^rH_c^{s-m}$ and $x^{-r}H_c^{m-s}$, it is the annihilator of the kernel of the adjoint, and that kernel is $\ker(A')^*$. Finally, $A = x^{-p}A'$ and multiplication by x^{-p} is an isomorphism $x^rH_c^{s-m} \to x^{r-p}H_c^{s-m}$.

(2) If $f \in \dot{C}^\infty$ is orthogonal to $\ker A^*$, then by (1) $f = Au$ for some u in some $x^rH_c^s$. Then $u = QA'u + R'u = Qx^pf + R'u \in \dot{C}^\infty$, because Q preserves $\dot{C}^\infty$ (Lemma A.18(3)) and $R'u \in \dot{C}^\infty$. So the range on $\dot{C}^\infty$ is closed, of finite codimension, and A is Fredholm on $\dot{C}^\infty$.

(3) We follow the closed case [14, Ch. 4]. Let Π_0, Π_1' be the orthogonal projections onto $\ker A'$ and $\ker(A')^*$. Their kernels are $\sum_i e_i(z)\overline{e_i(z')}$ for orthonormal bases $e_i \subset \dot{C}^\infty$, so they lie in $\dot{C}^\infty(X^2)$. Let $G : L_c^2 \to H_c^m$ be the inverse of $A' : \ker(A')^\perp \cap H_c^m \to \operatorname{Ran} A'$, extended by 0 on $\ker(A')^*$. Then $GA' = 1 - \Pi_0$ and $A'G = 1 - \Pi_1'$, and

$$G = (QA' + R')G = Q(1 - \Pi_1') + R'G,$$
$$R'G = R'G(A'Q + R) = R'(1 - \Pi_0)Q + R'GR.$$

The terms $Q\Pi_1'$, $R'Q$ and $R'\Pi_0Q$ have kernels in $\dot{C}^\infty(X^2)$ by Lemma A.18(4). The kernel of $R'GR$ is

$$(z, z') \longmapsto \int_X K_{R'}(z, w)\,\big(G\,K_R(\cdot, z')\big)(w)\,\nu_c(w).$$

Here $z' \mapsto K_R(\cdot, z')$ and $z \mapsto K_{R'}(z, \cdot)$ are smooth maps into $L^2(\nu_c)$ all of whose derivatives vanish to infinite order at ∂X, and G is bounded on $L^2(\nu_c)$. So this kernel lies in $\dot{\mathcal{C}}^\infty(X^2)$. Hence $G \in Q + \dot{\mathcal{C}}^\infty(X^2) \subset \Psi_c^{-m}$. Put $B_0 = Gx^p \in x^p \Psi_c^{-m}$. Then $B_0 A = GA' = 1 - \Pi_0$ and $AB_0 = x^{-p} A' G x^p = 1 - x^{-p} \Pi_1' x^p$. The kernel of $x^{-p} \Pi_1' x^p$ is $x^{-p} \Pi_1'(z, z') x'^p$, which still lies in $\dot{\mathcal{C}}^\infty(X^2)$. Its range $x^{-p} \ker(A')^*$ is a complement of $\operatorname{Ran} A = x^{-p} \operatorname{Ran} A'$. □

This is exactly what Lemma 15.2 uses: B_0 represents $[A]^{-1}$ in $\mathcal{A}$, and $\Pi_0, \Pi_1 \in \mathcal{I}$ have finite rank.

A.9 Summary

statement	where proved
(B1) doubly filtered algebra, adjoints	Theorem A.20 (via Theorem A.13)
(B1) symbol sequence [15, (11)]	Proposition A.21
(B1) indicial sequence [15, (12)], $N(A) \in \Psi_{\mathrm{sus}}$	Proposition A.25
(B1) quantization [15, Prop. 24], (13)	Proposition A.22
(B1) Taylor series [15, Prop. 25], (14), twist	Propositions A.27, A.26
(B1) conjugation by x^z	Theorem A.20(2), Proposition A.26
[15, Def. 2] via X_c^2	Proposition A.17
[15, Prop. 26], lift of b-operators	Proposition A.28 (density not proved)
(B2) Fredholm, generalized inverse, $\dot{\mathcal{C}}^\infty$ kernels	Theorems A.34, A.35
quoted: uniform calculus on $\mathbb{R} \times H$	Background fact A.4

B Homological algebra

This appendix proves the homological background facts (B4), (B5) and (B6) of Section 2.5, and it analyses the convergence of the order spectral sequence, which Remark 5.7 only summarises. The arguments are adapted from Wodzicki [23], from [14, Ch. 10] and from the companion notes [7]; we say in each place what is taken from where. The status of each item is as follows.

- (B4), Wodzicki excision: proved in full (Theorem B.7), using three stated facts about nuclear spaces, (F1)–(F3) below.
- (B5), Connes' theorem: proved for $C^\infty(M)$, M compact with corners (Theorem B.17), and for the formal, Laurent and flat coefficient algebras of the notes (Theorem B.18).
- (B6): the Künneth formula is proved in the form used (Proposition B.20); the Eilenberg–Zilber theorem is quoted. Morita invariance for matrices, smoothing-kernel algebras and full corners is proved (Lemma B.22, Proposition B.24).
- Convergence: the algebraic convergence proposition is proved (Proposition B.26), and so is convergence for the closed symbol algebra and for $\mathcal{I}_\sigma$ (Theorem B.30). For $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ the b-calculus argument of [7] does *not* carry over, because of the cusp twist (Example B.31). We prove what survives (Proposition B.33). The remaining step is quoted from [15, p. 12].

B.1 Chains and three facts about nuclear spaces

We use the chain model of Section 5.1. A *filtered nuclear algebra* is an increasing union $\mathfrak{A} = \bigcup_{N\geq 0} \mathfrak{A}_N$ of nuclear Fréchet spaces, each closed in the next, such that multiplication maps $\mathfrak{A}_N \times \mathfrak{A}_M \to \mathfrak{A}_{N+M}$ continuously. Its chain spaces are $C_k(\mathfrak{A}) = \varinjlim_N \mathfrak{A}_N^{\widehat{\otimes}(k+1)}$, and d_i, b, b' are as in Section 5.1. A nuclear Fréchet algebra is the case $\mathfrak{A}_N = \mathfrak{A}$. A closed ideal J and the quotient $\mathfrak{A}/J$ have the pieces $J_N = J \cap \mathfrak{A}_N$ and $\mathfrak{A}_N/J_N$. Homology is always taken algebraically. Filtered colimits of vector spaces are exact, so homology commutes with $\varinjlim_N$. In the proofs below we therefore work with one level N at a time and suppress N, except where a map changes the level. We use the following facts [20, Chs. 43, 50, 51].

(F1) Closed subspaces and Hausdorff quotients of nuclear Fréchet spaces are nuclear Fréchet. If E is nuclear Fréchet and $0 \to F' \to F \to F'' \to 0$ is an exact sequence of Fréchet spaces, then $0 \to E \mathbin{\widehat{\otimes}} F' \to E \mathbin{\widehat{\otimes}} F \to E \mathbin{\widehat{\otimes}} F'' \to 0$ is exact.

(F2) (Kernel theorem.) For compact manifolds with corners $C^\infty(M) \mathbin{\widehat{\otimes}} C^\infty(N) = C^\infty(M \times N)$. The same holds for $\dot{C}^\infty$, and $\mathcal{S}(\mathbb{R}^a) \mathbin{\widehat{\otimes}} \mathcal{S}(\mathbb{R}^b) = \mathcal{S}(\mathbb{R}^{a+b})$.

(F3) If E is nuclear Fréchet, then $E \mathbin{\widehat{\otimes}} \prod_j F_j = \prod_j E \mathbin{\widehat{\otimes}} F_j$ for countable products of Fréchet spaces. For example $C^\infty(N)[[x]]^{\widehat{\otimes}(k+1)} = C^\infty(N^{k+1})[[x_0, \dots, x_k]]$.

Lemma B.1 (Exactness of chain spaces)**.** *Let $J \subset \mathfrak{A}$ be a closed ideal. For $0 \leq s \leq n$ put*

$$H_n^s = \sum_{i\leq s} \mathfrak{A}^{\widehat{\otimes} i} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-i)} \subset \mathfrak{A}^{\widehat{\otimes}(n+1)},$$

the algebraic sum of the images, which are injective by (F1). Put $H_n^s = H_n^n$ for $s > n$. Then:

1. *$\mathfrak{A}^{\widehat{\otimes}(n+1)} \to (\mathfrak{A}/J)^{\widehat{\otimes}(n+1)}$ is onto, with kernel H_n^n;*
2. *$H_n^s/H_n^{s-1} \cong (\mathfrak{A}/J)^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)}$ for $1 \leq s \leq n$;*
3. *$J^{\widehat{\otimes} q} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} r}/J^{\widehat{\otimes}(q+1)} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(r-1)} \cong J^{\widehat{\otimes} q} \mathbin{\widehat{\otimes}} (\mathfrak{A}/J) \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(r-1)}$ for $q, r \geq 1$.*

The same holds for filtered nuclear algebras, with $\varinjlim_N$ applied throughout.

Proof. (3) is (F1) applied to $0 \to J \to \mathfrak{A} \to \mathfrak{A}/J \to 0$ tensored with nuclear factors. For (1) and (2), let $\pi_s : \mathfrak{A}^{\widehat{\otimes}(n+1)} \to (\mathfrak{A}/J)^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n+1-s)}$ project the first s factors. We show by induction that $\ker \pi_s = H_n^{s-1}$. For $s = 1$ this is (F1). Write $\pi_{s+1} = \pi' \circ \pi_s$, where π' projects the factor in slot s. By (F1), $\ker \pi' = (\mathfrak{A}/J)^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)}$, and π_s maps $\mathfrak{A}^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)}$ onto it. Hence $\ker \pi_{s+1} = \ker \pi_s + \mathfrak{A}^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)} = H_n^s$. The case $s = n+1$ is (1), and surjectivity is (F1) again. For (2), π_s restricted to H_n^s has kernel H_n^{s-1} and image $\pi_s(\mathfrak{A}^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)}) = (\mathfrak{A}/J)^{\widehat{\otimes} s} \mathbin{\widehat{\otimes}} J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(n-s)}$. □

In particular $\mathcal{K}_n = H_n^n$ in (13). No continuous linear section of $\mathfrak{A} \to \mathfrak{A}/J$ is needed; the Taylor maps of the notes have none.

Definition B.2. J is *topologically H-unital* if there are continuous maps $E : J_N^{\widehat{\otimes}(k+1)} \to J_N^{\widehat{\otimes}(k+2)}$, for all N and $k \geq 0$, with $b'E + Eb' = 1$ (where $b' = 0$ on C_0). We call E an *extra degeneracy* if moreover $d_0 E = 1$ and $d_{i+1}E = Ed_i$ for $0 \leq i < k$; then $b'E + Eb' = 1$ holds automatically.

A unital algebra has the extra degeneracy $E(c) = 1 \otimes c$. The contractions of Lemmas 5.4 and 6.2 are extra degeneracies. If J is topologically H-unital, then the bar complex is acyclic in degree 0 as well: $b'E = 1$ on $C_0(J)$ says that multiplication $J \mathbin{\widehat{\otimes}} J \to J$ is onto. If E_1, E_2 are extra degeneracies of J_1, J_2, then $E_1 \otimes E_2$ is an extra degeneracy of $J_1 \mathbin{\widehat{\otimes}} J_2$, because the face maps of $J_1 \mathbin{\widehat{\otimes}} J_2$ are the tensor products of the face maps.

B.2 Wodzicki's excision theorem (B4)

Throughout this subsection $J \subset \mathfrak{A}$ is a closed ideal of a filtered nuclear algebra, and J is topologically H-unital with contraction E. We start with the case in which excision fails.

Example B.3 (Failure without H-unitality)**.** Let $\mathfrak{A} = \mathbb{C}[x]$ (with algebraic tensor products) and $J = x\mathbb{C}[x]$. Here J is not H-unital, since $J^2 = x^2\mathbb{C}[x] \neq J$. The algebras are commutative, so $\mathrm{HH}_1(\mathfrak{A}) = \mathbb{C}[x]\,dx$ by HKR (Theorem B.9) and $\mathrm{HH}_1(\mathfrak{A}/J) = \mathrm{HH}_1(\mathbb{C}) = 0$. If excision held, $\mathrm{HH}_1(J) \to \mathrm{HH}_1(\mathfrak{A})$ would be onto. But the HKR form of a chain $\sum a \otimes c$ with $a, c \in J$ is $\sum a\,dc \in x\mathbb{C}[x]\,dx$, so the image misses $[dx]$.

The proof has two steps. Let $\mathcal{K}'_n = J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} n}$ be the chains whose first entry lies in J. It is a subcomplex of $\mathcal{K}$: the faces d_0 and d_n put a_0a_1 or a_na_0 in slot 0, and both lie in J. We show that $C(J) \subset \mathcal{K}'$ and $\mathcal{K}' \subset \mathcal{K}$ are quasi-isomorphisms. The first step uses the bar complex of J directly. The second needs it with coefficients.

Lemma B.4 (Bar complex with coefficients)**.** *Let W be a left $\mathfrak{A}$-module which is a filtered nuclear space, with continuous action $\mathfrak{A}_N \times W_M \to W_{N+M}$, such that $J \cdot W = 0$. Put $P_m = J \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} m} \mathbin{\widehat{\otimes}} W$ for $m \geq 0$, with*

$$b'_W(a_0 \otimes \cdots \otimes a_m \otimes w) = \sum_{i=0}^{m-1} (-1)^i a_0 \otimes \cdots \otimes a_ia_{i+1} \otimes \cdots \otimes w + (-1)^m a_0 \otimes \cdots \otimes a_{m-1} \otimes a_m w.$$

Then $(P_\bullet, b'_W)$ is acyclic, in degree 0 included.

Proof. P is the two-sided bar complex of the right module J and the left module W, so $b'^2_W = 0$. For $0 \leq r \leq m$ let $G^r_m = J^{\widehat{\otimes}(m+1-r)} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} r} \mathbin{\widehat{\otimes}} W$, the chains whose first $m+1-r$ entries lie in J, and put $G^r_m = P_m$ for $r > m$. Then G^r is a subcomplex. A face inside the initial J-block keeps the block, of length $m-r$ in degree $m-1$, so it lands in G^r_{m-1}. The face joining the last block entry to the next entry gives an element of J, and lands in G^{r-1}_{m-1}. A face joining two later entries, or the module face, keeps the block and lands in G^{r-1}_{m-1}. If $r = 0$ the module face is $a_mw = 0$, because $a_m \in J$.

So $G^0 = (C_\bullet(J), b') \mathbin{\widehat{\otimes}} W$ is acyclic, with contraction $E \otimes 1$. For $r \geq 1$, Lemma B.1(3) gives $G^r_m / G^{r-1}_m \cong J^{\widehat{\otimes}(m+1-r)} \mathbin{\widehat{\otimes}} (\mathfrak{A}/J) \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes}(r-1)} \mathbin{\widehat{\otimes}} W$. By the list above, only the faces inside the block survive in the quotient. Hence G^r/G^{r-1} is the bar complex of J (in degrees $m-r \geq 0$) tensored with spectators. Let $\tilde{E}$ apply E to the block. It maps G^{r-1} to G^{r-1}, since it lengthens the block, and $b\tilde{E} + \tilde{E}b \equiv (b'E + Eb') \otimes 1 = 1$ modulo G^{r-1}. So each G^r/G^{r-1} is acyclic. Since $G^{m+1}_j = P_j$ for $j \leq m+1$, we get $H_m(P) = H_m(G^{m+1}) \cong H_m(G^0) = 0$. □

Proposition B.5 (First step)**.** *The inclusion $C(J) \to \mathcal{K}'$ is a quasi-isomorphism.*

Proof. Filter $\mathcal{K}'$ as in the previous proof: $G^r_n = J^{\widehat{\otimes}(n+1-r)} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} r}$ for $r \leq n$, and $G^r_n = \mathcal{K}'_n$ for $r > n$. Then $G^0 = C(J)$. The faces behave as before. The cyclic face d_n puts $a_na_0 \in J$ in front and keeps the block, so it lands in G^{r-1}_{n-1} when $r \geq 1$, and in G^0_{n-1} when $r = 0$. The quotients G^r/G^{r-1}, $r \geq 1$, are bar complexes of J tensored with spectators. They are acyclic by the contraction $\tilde{E}$ of the previous proof. Since $G^r_n = \mathcal{K}'_n$ for $r \geq n$, we get $H_n(C(J)) = H_n(G^0) \cong H_n(G^{n+1}) = H_n(\mathcal{K}')$. □

Proposition B.6 (Second step)**.** *The inclusion $\mathcal{K}' \to \mathcal{K}$ is a quasi-isomorphism.*

Proof. Use the subspaces H^s_n of Lemma B.1: chains with an entry from J in one of the first $s+1$ slots. Then $H^0 = \mathcal{K}'$ and $H^n_n = \mathcal{K}_n$. Each H^s is a subcomplex. Indeed, let $a_i \in J$ with $i \leq s$. A face

d_j with $j < i$ moves it to slot $i-1$; the faces d_{i-1}, d_i and d_n produce an element of J in a slot $\le i$; the other faces leave slot i alone. Fix $s \ge 1$. By Lemma B.1(2),

$$H_n^s / H_n^{s-1} \cong (\mathfrak{A}/J)^{\widehat{\otimes} s} \,\widehat{\otimes}\, J \,\widehat{\otimes}\, \mathfrak{A}^{\widehat{\otimes}(n-s)}.$$

Write an element as $\bar{a}_0 \otimes \cdots \otimes \bar{a}_{s-1} \otimes a_s \otimes \cdots \otimes a_n$ with $a_s \in J$. The faces d_j, $j \le s-1$, produce an element of J in a slot $\le s-1$, so they vanish on the quotient. The faces d_j, $s \le j \le n-1$, multiply neighbours in the segment $(a_s, \dots, a_n)$. The cyclic face puts $\overline{a_n a_0}$ in slot 0, that is, it lets a_n act on the left of $W = (\mathfrak{A}/J)^{\widehat{\otimes} s}$, through the first factor. This action kills J. Moving $(\bar{a}_0, \dots, \bar{a}_{s-1})$ to the end identifies H^s/H^{s-1} with the complex $P(W)$ of Lemma B.4, of degree $n-s$, and the differential with $(-1)^s b'_W$. (For $n = s$ the cyclic face gives $a_s a_0 \in J$ in slot 0, which lies in $H^0 \subset H^{s-1}$.) By Lemma B.4 each quotient is acyclic. The filtration is finite in each degree, so the inclusion $H^0 \subset H^n$ is a quasi-isomorphism near degree n. □

Theorem B.7 (Wodzicki excision, (B4))**.** *Let $J \subset \mathfrak{A}$ be a closed ideal of a filtered nuclear algebra, and suppose that J is topologically H-unital. Then $\mathcal{C}(J) \to \mathcal{K}$ is a quasi-isomorphism, and there is a long exact sequence*

$$\cdots \to \mathrm{HH}_k(J) \to \mathrm{HH}_k(\mathfrak{A}) \to \mathrm{HH}_k(\mathfrak{A}/J) \xrightarrow{\partial} \mathrm{HH}_{k-1}(J) \to \cdots .$$

To compute $\partial[c]$, lift the cycle c to a chain $\tilde{c}$ of $\mathfrak{A}$ (Lemma B.1(1)); then $b\tilde{c} \in \mathcal{K}$ is a cycle, and $\partial[c]$ is its class in $H(\mathcal{K}) \cong \mathrm{HH}(J)$.

Proof. Propositions B.5 and B.6 give the quasi-isomorphism. Lemma B.1(1) gives the short exact sequence (13). Its long exact sequence, with $H(\mathcal{K})$ replaced by $\mathrm{HH}(J)$, is the stated one, and the description of ∂ is the snake lemma. □

This is Background Fact 5.1. Lemma 5.2 on cocycles follows from it as shown there. The theorem uses neither a unit in $\mathfrak{A}$ nor a section of $\mathfrak{A} \to \mathfrak{A}/J$. For algebras with algebraic tensor products the same proof applies, with (F1) replaced by the exactness of $\otimes$ over $\mathbb{C}$.

B.3 Connes' theorem (B5)

We adapt [14, Ch. 10, "Smooth functions"], and add what is needed for boundaries and corners. Let M be a compact manifold with corners of dimension n. Its charts are open subsets of $[0,\infty)^l \times \mathbb{R}^{n-l}$. By (F2), $C_k = C^\infty(M^{k+1})$, and the face maps are restrictions:

$$(d_i u)(z_0, \dots, z_{k-1}) = u(z_0, \dots, z_i, z_i, \dots, z_{k-1}), \qquad (d_k u)(z_0, \dots, z_{k-1}) = u(z_0, \dots, z_{k-1}, z_0).$$

The HKR map is, in local coordinates,

$$\chi(u)(x) = \frac{1}{k!} \sum_{j_1, \dots, j_k} \frac{\partial^k u}{\partial z_1^{j_1} \cdots \partial z_k^{j_k}}(x, \dots, x)\, dx^{j_1} \wedge \cdots \wedge dx^{j_k},$$

with one-sided derivatives at the boundary. On elementary tensors $\chi(a_0 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$. Put $\varphi(a_0; a_1, \dots, a_k) = \sum_{\sigma \in \mathfrak{S}_k} \mathrm{sgn}(\sigma)\, a_0 \otimes a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(k)}$.

Lemma B.8. *$\chi \circ b = 0$, $b\varphi(a_0; a_1, \dots, a_k) = 0$ and $\chi\varphi(a_0; a_1, \dots, a_k) = a_0\, da_1 \wedge \cdots \wedge da_k$. In particular $\chi_* : \mathrm{HH}_k(C^\infty(M)) \to \Omega^k(M)$ is onto.*

Proof. All three identities are algebraic for elementary tensors. Put $T_i = a_0 a_i\, da_1 \wedge \dots \widehat{da_i} \cdots \wedge da_k$. By the Leibniz rule, $(k-1)!\,\chi(b(a_0 \otimes \cdots \otimes a_k)) = T_1 + \sum_{i=1}^{k-1}(-1)^i(T_i + T_{i+1}) + (-1)^k T_k = 0$. In $b\varphi$ the inner faces cancel in pairs (σ against $\sigma \circ (i\ i{+}1)$), and d_0 for σ cancels $(-1)^k d_k$ for the cyclically shifted permutation, whose sign is $(-1)^{k-1}\operatorname{sgn}\sigma$. The third identity is clear. Since $\chi \circ b$ is continuous and the algebraic tensor product is dense, $\chi \circ b = 0$ on all chains. Every k-form is a finite sum $\sum a_0\, df_1 \wedge \cdots \wedge df_k$: use a partition of unity $\sum \rho_\beta = 1$ subordinate to charts and global functions f_i^β equal to the coordinates of the chart on $\operatorname{supp}\rho_\beta$. □

Theorem B.9 (HKR for polynomials [14, 9]). *For $\mathfrak{A} = \mathbb{C}[x_1, \dots, x_n]$, with algebraic tensor products, χ induces $\mathrm{HH}_k(\mathfrak{A}) \cong \mathfrak{A} \otimes \Lambda^k\mathbb{C}^n$, with inverse $a_0\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \mapsto \varphi(a_0; x_{i_1}, \dots, x_{i_k})$. The same holds for $\mathbb{C}[r, r^{-1}]$, with $\mathrm{HH}_0 = \mathbb{C}[r^{\pm1}]$, $\mathrm{HH}_1 = \mathbb{C}[r^{\pm1}]\, dr$, $\mathrm{HH}_k = 0$ for $k \geq 2$.*

Proof. $\mathrm{HH}_k(\mathfrak{A}) = \operatorname{Tor}_k^{\mathfrak{A}^e}(\mathfrak{A}, \mathfrak{A})$, computed by the bar resolution $P_k = \mathfrak{A}^{\otimes(k+2)}$, which is contracted by $1\otimes$. The Koszul complex $K_k = \mathbb{C}[x, y] \otimes \Lambda^k\mathbb{C}^n$, with $\partial e_i = x_i - y_i$, is a second free resolution. In the variables $u = x - y$, $v = y$, ∂ is contraction with the Euler field $\sum u_i \partial_{u_i}$, so the exterior derivative in u divided by the degree in u contracts K in positive u-degree; the augmentation is $f \mapsto f(x, x)$. The map $e_{i_1} \wedge \cdots \wedge e_{i_k} \mapsto \sum_\sigma \operatorname{sgn}\sigma\, 1 \otimes x_{i_{\sigma(1)}} \otimes \cdots \otimes x_{i_{\sigma(k)}} \otimes 1$ is a chain map $K \to P$ over the identity: the inner faces cancel in pairs, and the outer faces give $\sum_l (-1)^{l-1}(x_{i_l} - y_{i_l})$ times the smaller antisymmetrization. By the comparison theorem it induces an isomorphism after $\mathfrak{A} \otimes_{\mathfrak{A}^e}$, where the Koszul side has zero differential. This gives the inverse map φ, and $\chi\varphi = 1$ by Lemma B.8. For $\mathbb{C}[r^{\pm1}]$ the resolution is $0 \to \mathfrak{A}^e \xrightarrow{r-s} \mathfrak{A}^e \to \mathfrak{A} \to 0$. It is exact because $r - s$ is not a zero divisor and $f(r, s) - f(r, r)$ is divisible by $r - s$ (clear the denominator $(rs)^N$ first). □

Normalized chains

The next three results are algebraic and hold verbatim for $C^\infty(M)$, for the polynomial algebras, and for the formal and Laurent algebras of Theorem B.18. Let $D_i = \{z_i = z_{i+1}\} \subset M^{k+1}$ for $0 \leq i < k$ and $D_k = \{z_k = z_0\}$, and

$$N_k = \{u \in C_k :\ u|_{D_i} = 0,\ 0 \leq i < k\}, \qquad M_k = \{u \in N_k :\ u|_{D_k} = 0\} \quad (k \geq 1),$$
$$B_k = d_{k+1}(N_{k+1}), \qquad N_0 = M_0 = C_0.$$

If $u \in N_k$ then $bu = (-1)^k d_k u$, and $d_k u \in N_{k-1}$. So N is a subcomplex. Moreover $B_k \subset M_k$, and $H_k(N) = M_k / B_k$.

Proposition B.10 ([14, Ch. 10]). *$N \subset C$ is a quasi-isomorphism. Hence $\mathrm{HH}_k = M_k / B_k$, and an element of M_k that is a boundary in C lies in B_k.*

Proof. Let $J_m^j = \{u \in C_m :\ u|_{D_i} = 0 \text{ for } i < \min(j, m)\}$; each J^j is a subcomplex, $J^0 = C$, and $J_m^j = N_m$ for $m \leq j$. For $m \geq j + 1$, every $u \in J_m^j$ splits uniquely as $u = u' + u''$, where $u'(z) = u(z_0, \dots, z_j, z_j, z_{j+2}, \dots, z_m)$ does not depend on z_{j+1} and $u'' = u - u' \in J_m^{j+1}$. Let V_m^j be the $u \in J_m^j$ that do not depend on z_{j+1} (and $V_m^j = 0$ for $m \leq j$). Then $J^j = J^{j+1} \oplus V^j$ as graded spaces. On V^j the faces d_i, $i < j$, vanish, and $d_j = d_{j+1}$, so $bu = \sum_{i \geq j+2}(-1)^i d_i u \in V^j$. The map $(Hu)(z_0, \dots, z_{m+1}) = u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_{m+1})$, which inserts a dummy variable in position $j + 2$, satisfies $d_{j+2}H = 1$ and $d_i H = H d_{i-1}$ for $i \geq j + 3$. Hence $bH + Hb = (-1)^j$ on V^j, so V^j is a contractible subcomplex, and $J^{j+1} \subset J^j$ is a quasi-isomorphism. Therefore $H_k(C) \cong H_k(J^{k+1}) = H_k(N)$. □

Lemma B.11. (i) *B_k is a $C^\infty(M^{k+1})$-submodule of M_k.* (ii) *If $f \in C^\infty(M)$ and $g \in M_k$, then $(f(z_j) - f(z_{j+1}))g \in B_k$ for $0 \leq j < k$, and $(f(z_k) - f(z_0))g \in B_k$.*

Proof. (i) If $v \in N_{k+1}$ and $\psi \in C^\infty(M^{k+1})$, then $\tilde{\psi}v \in N_{k+1}$ with $\tilde{\psi}(z_0,\dots,z_{k+1}) = \psi(z_0,\dots,z_k)$, and $d_{k+1}(\tilde{\psi}v) = \psi\, d_{k+1}v$. (ii) For $j < k$ let $w = (-1)^{j+1}(f(z_j) - f(z_{j+1}))\, g(z_0,\dots,z_j,z_{j+2},\dots,z_{k+1}) \in C_{k+1}$. All faces of w vanish except d_{j+1}, because $g \in M_k$ and because of the first factor. So $bw = (f(z_j) - f(z_{j+1}))g$. This lies in M_k and is a boundary, hence in B_k by Proposition B.10. The last case is the sum of the others. □

Example B.12 (The model $k = 1$)**.** Let $M = [0,1]$ and $u \in M_1$, so $u(z_0,z_0) = 0$ and $u = (z_1 - z_0)w$. Then $\chi(u) = w(x,x)\,dx$. If $\chi(u) = 0$, then $u = (z_1 - z_0)^2 g$ with g smooth, and $u = d_2 v$ for $v(z_0,z_1,z_2) = -(z_1 - z_0)(z_2 - z_1)g(z_0,z_1) \in N_2$. So $u \in B_1$, and $\mathrm{HH}_1(C^\infty([0,1])) = \Omega^1([0,1])$. The proof below does the same in general: Taylor expansion at the diagonal, a polynomial lift, and division of a flat remainder.

The local structure of normalized cycles

Near a point $(x_0,\dots,x_0)$ of the diagonal Δ choose a chart around x_0 whose image U is a *convex* subset of $[0,\infty)^l \times \mathbb{R}^{n-l}$, and use the coordinates

$$x = z_0, \qquad y_j = z_j - z_{j-1} \quad (1 \le j \le k)$$

on U^{k+1}. Then $D_{j-1} = \{y_j = 0\}$, $D_k = \{y_1 + \cdots + y_k = 0\}$ and $\Delta = \{y = 0\}$. The set U^{k+1} is convex with non-empty interior, so the Taylor series in y at points of Δ of a smooth function on it is determined by the function. It vanishes on a linear subspace S if the function vanishes on $S \cap U^{k+1}$, because $S \cap U^{k+1}$ is convex with non-empty interior in S and contains $\Delta \cap U^{k+1}$. This replaces the use of Hadamard's lemma in [14], whose integration paths can leave a half-space.

Lemma B.13. *A k-linear form q on $\mathbb{R}^n$ satisfies $q(y_1,\dots,y_{k-1},-(y_1+\cdots+y_{k-1})) \equiv 0$ if and only if it is alternating.*

Proof. If q is alternating the expression is $-\sum_{j<k} q(y_1,\dots,y_{k-1},y_j) = 0$. Conversely, the terms $q(y_1,\dots,y_{k-1},y_j)$ have distinct multidegrees, so each vanishes. Thus q is alternating in the arguments j,k, and these transpositions generate $\mathfrak{S}_k$. □

Lemma B.14. *Let $u \in N_k$. The Taylor series of u at Δ in y has coefficients smooth in $x \in U$. Each of its monomials contains a component of every y_j. Its part of degree k is $u_{(k)} = \sum_I w_I(x)\, y_{1,i_1} \cdots y_{k,i_k}$, and $\chi(u) = \frac{1}{k!}\sum_I w_I\, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$. If $u \in M_k$, then $u_{(k)}(x;\cdot)$ is alternating, and $\chi(u)(x) = 0$ if and only if $u_{(k)}(x;\cdot) = 0$.*

Proof. The Taylor series vanishes on each $\{y_j = 0\}$, which gives the first claim. Since $z_l = x + y_1 + \cdots + y_l$, we have $\partial_{z_l} = \partial_{y_l} - \partial_{y_{l+1}}$ (and $\partial_{z_k} = \partial_{y_k}$). In $\partial_{z_1^{i_1}} \cdots \partial_{z_k^{i_k}} u$ at $y = 0$ only the term in which each y_l is differentiated once survives, and it equals $w_I(x)$. If $u \in M_k$, then $u_{(k)}$ vanishes on $\{y_1 + \cdots + y_k = 0\}$; apply Lemma B.13. □

Lemma B.15 ([14, Ch. 10])**.** *Let $k \ge 1$, $m \ge k+1$. Let $\mathcal{M}_m$ be the homogeneous polynomials of degree m in $(y_1,\dots,y_k)$ vanishing on each $\{y_j = 0\}$ and on $\{y_1 + \cdots + y_k = 0\}$. Let $\mathcal{V}_m$ be the homogeneous polynomials of degree m in $(y_1,\dots,y_{k+1})$ vanishing on each $\{y_j = 0\}$. Then $R : \mathcal{V}_m \to \mathcal{M}_m$, $(RV)(y_1,\dots,y_k) = V(y_1,\dots,y_k,-(y_1+\cdots+y_k))$, is onto.*

Proof. Given $q \in \mathcal{M}_m$, the polynomial chain $u(z) = q(z_1 - z_0,\dots,z_k - z_{k-1})$ lies in M_k and $\chi(u) = 0$. By Theorem B.9 and Proposition B.10 for polynomials, $u = d_{k+1}v$ with $v \in N_{k+1}$. Write v in the coordinates $x, y_1,\dots,y_{k+1}$. Its component of degree 0 in x and m in y lies in $\mathcal{V}_m$, and R maps it to q. □

Lemma B.16. *Let $k \geq 1$ and $u \in M_k$ with $\chi(u) = 0$. Every point $p \in M^{k+1}$ has a neighbourhood on which u agrees with an element of B_k.*

Proof. If $p \notin \Delta$, then $p_j \neq p_{j+1}$ for some $j < k$. Choose $f \in C^\infty(M)$ with $f(p_j) \neq f(p_{j+1})$, and $\psi \in C^\infty(M^{k+1})$ equal to 1 near p and supported where $f(z_j) \neq f(z_{j+1})$. Then $g = \psi u/(f(z_j) - f(z_{j+1})) \in M_k$, and near p, $u = (f(z_j) - f(z_{j+1}))g \in B_k$ by Lemma B.11.

Let $p = (x_0, \dots, x_0)$. By Lemma B.14 and $\chi(u) = 0$, the Taylor series of u at Δ is $\sum_{m \geq k+1} u_{(m)}$ with $u_{(m)} \in C^\infty(U) \otimes \mathcal{M}_m$. Using a linear right inverse of R, choose $V_{(m)} \in C^\infty(U) \otimes \mathcal{V}_m$ with $RV_{(m)} = u_{(m)}$. By Borel's lemma, $V = \sum_m \chi_0(\lambda_m |y|) V_{(m)}$, with $\chi_0 \in C_c^\infty(\mathbb{R})$ equal to 1 near 0 and $\lambda_m \to \infty$ fast enough, is smooth for $x \in U$ up to the boundary and small $y \in (\mathbb{R}^n)^{k+1}$. It has Taylor series $\sum V_{(m)}$ and vanishes on each $\{y_j = 0\}$. Let Θ be a cut-off near $(x_0, \dots, x_0) \in M^{k+2}$, and let $v = \Theta V$, written in z. Then $v \in N_{k+1}$. The chain $s = d_{k+1} v \in B_k$ has the same Taylor series as u at Δ near p, since $z_{k+1} = z_0$ means $y_{k+1} = -(y_1 + \dots + y_k)$.

So $g = u - s \in M_k$ vanishes to infinite order at Δ near p. For $z \in U^{k+1}$ the segment from $(z_0, \dots, z_0)$ to z lies in U^{k+1}, by convexity. Taylor's formula along it gives $|\partial^\alpha g(z)| \leq C_{\alpha,N} |y|^N$ for all N. Hence $g_{j,i} = y_{j,i} g/|y|^2$, extended by 0 on Δ, is smooth up to the boundary, and it vanishes where g does. Let $f_i \in C^\infty(M)$ equal the i-th coordinate near x_0, and let ψ be a cut-off equal to 1 near p. Then near p, $g = \sum_{j,i} y_{j,i} g_{j,i} = \sum_{j,i} (f_i(z_j) - f_i(z_{j-1})) \psi g_{j,i}$, which lies in B_k by Lemma B.11. So $u = s + g$ agrees with an element of B_k near p. □

Theorem B.17 (Connes, (B5))**.** *For a compact manifold with corners M, χ induces isomorphisms $\mathrm{HH}_k(C^\infty(M)) \cong \Omega^k(M)$.*

Proof. For $k = 0$, $b = 0$ on C_1. For $k \geq 1$, χ_* is onto by Lemma B.8. By Proposition B.10 a class is represented by $u \in M_k$, and it vanishes if and only if $u \in B_k$. If $\chi(u) = 0$, Lemma B.16 and compactness give a finite partition of unity $\sum \psi_\beta = 1$ on M^{k+1} and $s_\beta \in B_k$ with $u = s_\beta$ on $\operatorname{supp} \psi_\beta$. Then $u = \sum_\beta \psi_\beta s_\beta \in B_k$ by Lemma B.11(i). □

The coefficient algebras of the notes

Let N be a compact manifold with corners and x a formal variable. Let M be a compact manifold with boundary, with boundary defining function x and a collar $[0, \epsilon) \times \partial M$. We consider

$$C^\infty(N)[[x]], \qquad C^\infty(N)((x)), \qquad x^{-\mathbb{Z}} C^\infty(M), \qquad \dot{C}^\infty(M).$$

The localized algebras are filtered nuclear with pieces $x^{-L} C^\infty(N)[[x]]$ and $x^{-L} C^\infty(M)$. By (F2) and (F3) their chains are $C^\infty(N^{k+1})[[x_0, \dots, x_k]]$, the unions over L of $(x_0 \cdots x_k)^{-L}$ times the previous chains, and $\dot{C}^\infty(M^{k+1})$. The forms are $\Omega(N)[[x]] \oplus \Omega(N)[[x]]\, dx$ (with d the formal differential), the corresponding Laurent forms, $x^{-\mathbb{Z}}\Omega(M)$, and the flat forms $\dot{\Omega}(M)$. The HKR map is given by the same formula.

Theorem B.18 (HKR for the coefficient algebras, (B5))**.** *For each of the four algebras χ induces an isomorphism of HH_k onto the k-forms just listed. The isomorphisms commute with Taylor expansion at ∂M in the collar. The same holds for $\mathcal{S}(\mathbb{R}^d)$, whose forms are the Schwartz forms.*

Proof. Formal series. Run the proof of Theorem B.17 with x as an extra coordinate in which everything is a formal power series. Normalized chains and Lemma B.11 are algebraic. Differences $x_j - x_{j-1}$ are formal variables, and a series in $x_0, \dots, x_k$ is a series in x_0 and the differences. Points of N^{k+1} off the N-diagonal are treated as before. Near the N-diagonal, expand in all differences, lift with

Lemma B.15 (polynomials in the N-coordinates and x), and apply Borel's lemma only in the N-differences. For each fixed monomial in x_0 and the x-differences this is a convergent Borel sum; choose λ_m to control all monomials of degree $\le m$. Every coefficient of the remainder g is then flat at the N-diagonal, so it can be divided by $|y_N|^2$ coefficientwise. If $\dim N = 0$ the remainder is 0.

Laurent coefficients. Let R be $C^\infty(M)$ or $C^\infty(N)[[x]]$ and $R' = R[x^{-1}]$. Proposition B.10 holds for R' as stated. Let $u \in M_k(R')$ with $\chi(u) = 0$, and write $u = fv$ with $f = (x_0 \cdots x_k)^{-L}$ and $v \in C_k(R)$. Multiplication by f is injective, so $v \in M_k(R)$. By Lemma B.14, $v = O(|y|^k)$ at Δ, so the degree-k Taylor part of fv is $f|_\Delta v_{(k)}$ and $\chi(u) = x^{-(k+1)L}\chi(v)$ (as an identity of forms on the interior, hence of Laurent forms). So $\chi(v) = 0$, hence $v = d_{k+1}w$ with $w \in N_{k+1}(R)$, and $u = d_{k+1}(\tilde f w) \in B_k(R')$, where $\tilde f(z_0, \dots, z_{k+1}) = f(z_0, \dots, z_k)$. Surjectivity is Lemma B.8 with $a_0 \in x^{-L}R$.

Flat functions. Taylor expansion in the collar gives the exact sequence $0 \to \dot C^\infty(M) \to C^\infty(M) \to C^\infty(\partial M)[[x]] \to 0$ (Borel's lemma). Its kernel is topologically H-unital by Lemma 6.2, so Theorem B.7 gives a long exact sequence. The HKR maps commute with Taylor expansion, since both sides are computed from derivatives at the diagonal. The sequence of forms $0 \to \dot\Omega \to \Omega(M) \to \Omega(\partial M)[[x]] \oplus \Omega(\partial M)[[x]]\,dx \to 0$ is exact, by Borel's lemma for forms. The HKR maps for the middle and right algebras are isomorphisms, so $\mathrm{HH}_k(C^\infty(M)) \to \mathrm{HH}_k(C^\infty(\partial M)[[x]])$ is onto for every k. Hence the connecting maps vanish, $\mathrm{HH}_k(\dot C^\infty(M))$ is the kernel, and χ identifies it with $\dot\Omega^k(M)$. Finally $\mathcal{S}(\mathbb{R}^d) = \dot C^\infty(\overline{\mathbb{R}^d})$ for the radial compactification, a closed ball, and (F2) identifies the chains. □

Theorem B.18 is the scalar case of [7, Lemma "Continuous HKR for the coefficient kernels"], which proves the flat case directly with a Koszul complex. The version with an adjoined polynomial or Laurent polynomial variable is Corollary B.21 below.

B.4 Künneth and Morita (B6)

The Künneth formula

For unital algebras $\mathfrak{A}, \mathfrak{B}$ the *shuffle map* is

$$\mathrm{sh}\big((a_0 \otimes \cdots \otimes a_p) \otimes (b_0 \otimes \cdots \otimes b_q)\big) \\ = \sum_{\sigma \in \mathrm{Sh}(p,q)} \mathrm{sgn}(\sigma)\,(a_0 \otimes b_0) \otimes \sigma \cdot \big(a_1 \otimes 1, \dots, a_p \otimes 1, 1 \otimes b_1, \dots, 1 \otimes b_q\big),$$

where σ runs over the (p,q)-shuffles: it puts the entries $a_i \otimes 1$ in positions $\sigma(1) < \cdots < \sigma(p)$ and the $1 \otimes b_j$ in order in the remaining positions.

Background fact B.19 (Eilenberg–Zilber (quoted: [9], §4.2))**.** $\mathrm{sh} : \mathrm{Tot}(C(\mathfrak{A}) \otimes C(\mathfrak{B})) \to C(\mathfrak{A} \otimes \mathfrak{B})$ is a chain map and a chain homotopy equivalence. Its homotopy inverse (Alexander–Whitney) and both homotopies are given by universal finite formulas in the face maps and in the degeneracies s_i (insertion of 1 after slot i).

Because the formulas are universal and finite, the same holds for unital filtered nuclear algebras with completed tensor products, using $C_k(\mathfrak{A} \mathbin{\widehat\otimes} \mathfrak{B}) = \mathfrak{A}^{\widehat\otimes(k+1)} \mathbin{\widehat\otimes} \mathfrak{B}^{\widehat\otimes(k+1)}$.

Proposition B.20 (Künneth, (B6))**.** *Let $\mathfrak{A}$ be a unital filtered nuclear algebra and $\mathfrak{B}$ a unital algebra of countable dimension. Put $C_k(\mathfrak{A} \otimes \mathfrak{B}) = \mathfrak{A}^{\widehat\otimes(k+1)} \otimes \mathfrak{B}^{\otimes(k+1)}$. Then*

$$\mathrm{HH}_n(\mathfrak{A} \otimes \mathfrak{B}) \cong \bigoplus_{p+q=n} \mathrm{HH}_p(\mathfrak{A}) \otimes \mathrm{HH}_q(\mathfrak{B}),$$

induced by sh*. If both algebras are commutative, then $\chi(\mathrm{sh}(\alpha \otimes \beta)) = \chi(\alpha) \wedge \chi(\beta)$.*

Proof. By Background Fact B.19, sh is a quasi-isomorphism. The complex $C(\mathfrak{A}) \otimes C(\mathfrak{B})$ is an algebraic tensor product of complexes over $\mathbb{C}$, so its homology is $H(C(\mathfrak{A})) \otimes H(C(\mathfrak{B}))$. For the last statement, each of the $\binom{p+q}{p}$ shuffles contributes $a_0 b_0\, da_1 \wedge \cdots \wedge da_p \wedge db_1 \wedge \cdots \wedge db_q$, since the sign of the reordering cancels $\operatorname{sgn}\sigma$. Multiplying by $1/(p+q)!$ gives $\frac{1}{p!q!}(a_0\, da_1 \wedge \cdots) \wedge (b_0\, db_1 \wedge \cdots)$. □

For two nuclear Fréchet algebras the topological Künneth formula needs a further condition on the boundaries (quoted: [8]); the notes do not need it. What they need is the following.

Corollary B.21 (HKR with a radial variable)**.** *Let R be one of the algebras of Theorems B.17 and B.18, and let $\mathfrak{B} = \mathbb{C}[r, r^{-1}]$ (r of weight 1) or $\mathbb{C}[t]$ (t of weight -1). Grade $C(R \otimes \mathfrak{B})$ by total weight. Then χ induces, in each weight w,*

$$\mathrm{HH}_k(R \otimes \mathfrak{B})_{(w)} \cong \big(\Omega^k(R) \otimes \mathfrak{B} \oplus \Omega^{k-1}(R) \otimes \mathfrak{B}\, d\varrho\big)_{(w)}, \qquad \varrho \in \{r, t\},$$

where $d\varrho$ has the weight of ϱ. In particular, if $V \to Y$ is a vector bundle with a fibre norm and $R = C^\infty(SV)$ (or its Laurent, formal or flat version), the chains of total homogeneity p have homology the homogeneous k-forms of degree p on $V \setminus 0$.

Proof. For unital R combine Proposition B.20 with Theorems B.9, B.17 and B.18. Every map preserves the weight. For $\dot{C}^\infty(M) \otimes \mathfrak{B}$ use the Taylor sequence tensored with $\mathfrak{B}$ and argue as in the flat case of Theorem B.18. The kernel has the extra degeneracy $E \otimes (1 \otimes \cdot)$, the tensor product of extra degeneracies. A homogeneous form of degree p on $V \setminus 0 \cong SV \times (0,\infty)_r$ is $r^p(\alpha + \beta \wedge dr/r)$ with α, β forms on SV, and a homogeneous function of degree p is r^p times a function on SV. □

Morita invariance

Lemma B.22 (Morita for smoothing kernels; adapted from [7, Lemma "Morita invariance for smoothing kernels"])**.** *Let $\mathcal{K} = V \mathbin{\widehat{\otimes}} W$ be as in Lemma 5.4, with $\langle \psi, \tilde{\psi} \rangle = 1$, $e = \tilde{\psi} \otimes \psi$ and $\mathrm{Tr}(v \otimes w) = \langle w, v \rangle$. Let $\mathfrak{A}$ be unital or topologically H-unital. Then*

$$\iota(a_0 \otimes \cdots \otimes a_k) = (e \otimes a_0) \otimes \cdots \otimes (e \otimes a_k),$$
$$\mathrm{Tr}_*\big((\kappa_0 \otimes a_0) \otimes \cdots \otimes (\kappa_k \otimes a_k)\big) = \mathrm{Tr}(\kappa_0 \cdots \kappa_k)\, a_0 \otimes \cdots \otimes a_k$$

are chain maps between $C(\mathfrak{A})$ and $C(\mathcal{K} \mathbin{\widehat{\otimes}} \mathfrak{A})$, with $\mathrm{Tr}_*\, \iota = 1$, and Tr_* is a quasi-isomorphism. For $V = \mathbb{C}^N$, $W = (\mathbb{C}^N)^*$ this gives $\mathrm{HH}(M_N(\mathfrak{A})) \cong \mathrm{HH}(\mathfrak{A})$.

Proof. Write a chain as $(v_0 \otimes w_0 \otimes a_0) \otimes \cdots \otimes (v_k \otimes w_k \otimes a_k)$ and rearrange it as the *letters* $\ell_i = w_i \otimes v_{i+1} \in U = W \mathbin{\widehat{\otimes}} V$ (indices mod $k+1$) together with $a_0 \otimes \cdots \otimes a_k$. Let $\varepsilon(w \otimes v) = \langle w, v \rangle$. Since $(v_i \otimes w_i)(v_{i+1} \otimes w_{i+1}) = \varepsilon(\ell_i)\, v_i \otimes w_{i+1}$, the face d_i applies ε to ℓ_i, deletes it, and multiplies a_i with a_{i+1} (for $i = k$, $a_k a_0$ in slot 0); the other letters are unchanged. Split $U = \mathbb{C}u_0 \oplus U'$ with $u_0 = \psi \otimes \tilde{\psi}$ and $U' = \ker \varepsilon$. The number m of letters in U' is preserved by b, so $C = \bigoplus_m C^{(m)}$.

If $m = 0$, every letter is u_0, every slot is $\tilde{\psi} \otimes \psi = e$, and $C^{(0)} = \iota C(\mathfrak{A})$ with the differential of $\mathfrak{A}$. The projection onto $C^{(0)}$ replaces each letter ℓ by $\varepsilon(\ell)u_0$, that is, it is $\iota\, \mathrm{Tr}_*$, since $\prod_i \varepsilon(\ell_i) = \mathrm{Tr}(\kappa_0 \cdots \kappa_k)$.

If $m \geq 1$, filter by the position w of the last U'-letter. A face d_i with $i < w$ lowers w, d_w vanishes, and d_i with $i > w$ keeps w. On the associated graded complex only the faces $d_{w+1}, \ldots, d_k$ survive. They multiply neighbours in the block $(a_{w+1}, \ldots, a_k, a_0)$, of length $k - w + 1 \geq 1$, with signs $\pm(-1)^{i-w-1}$. So the graded complex is the bar complex of $\mathfrak{A}$ tensored with spectators, and it is contracted by $1 \otimes$ or by E, inserted in front of the block. The filtration is finite in each degree, so $C^{(m)}$ is acyclic for $m \geq 1$. □

For full corners we use a homotopy that we have not found written out in this form; it is the chain-level version of [9, §1.2]. Let T be an algebra containing e, P, Q with $e^2 = e$, $PQ = e$, $eP = P$ and $Qe = Q$. Then $a \mapsto QaP$ is an algebra map $eTe \to T$.

Lemma B.23. *Let $\phi_0, \phi_1 : C(eTe) \to C(T)$ be induced by the inclusion and by $a \mapsto QaP$. For $0 \le j \le k$ put*

$$h_j(a_0 \otimes \cdots \otimes a_k) = a_0 P \otimes Qa_1 P \otimes \cdots \otimes Qa_j P \otimes Q \otimes a_{j+1} \otimes \cdots \otimes a_k,$$

and $H = \sum_j (-1)^j h_j$. Then $bH + Hb = \phi_0 - \phi_1$.

Proof. Use $a_i PQ = a_i$ and $QaPQ = Qa$ for $a, a_i \in eTe$. In h_j the face d_i multiplies two transformed entries if $i < j$, giving $h_{j-1} d_i$. If $i > j+1$ it multiplies two original entries (or a_k with $a_0 P$), giving $h_j d_{i-1}$. The remaining faces satisfy

$$d_0 h_0 = 1, \qquad d_{k+1} h_k = \phi_1, \qquad d_j h_j = d_j h_{j-1} \ (j \ge 1),$$

both sides of the last being $(\dots, Qa_{j-1}P, Qa_j, a_{j+1}, \dots)$. The terms with $i \in \{j, j+1\}$ in bH therefore telescope to $d_0 h_0 - d_{k+1} h_k = \phi_0 - \phi_1$. The terms with $i < j$ or $i > j+1$ add up to $-Hb$. (We also checked the identity by computer algebra for $k \le 4$.) □

Proposition B.24 (Morita for full corners, (B6))**.** *Let S be a unital filtered nuclear algebra and $f \in S$ an idempotent. Suppose there are $u_1, \dots, u_m \in Sf$ and $v_1, \dots, v_m \in fS$ with $\sum_j u_j v_j = 1$. Then the inclusion $fSf \to S$ induces an isomorphism $\mathrm{HH}(fSf) \cong \mathrm{HH}(S)$. Its inverse is induced by $\theta(s_0 \otimes \cdots \otimes s_k) = \sum_{i_0,\dots,i_k} v_{i_0} s_0 u_{i_1} \otimes v_{i_1} s_1 u_{i_2} \otimes \cdots \otimes v_{i_k} s_k u_{i_0}$.*

Proof. In $T = M_{1+m}(S)$, with indices $0, \dots, m$, take $e = E_{00}$, $P = \sum_j u_j E_{0j}$ and $Q = \sum_j v_j E_{j0}$. Then $PQ = e$, $eTe = S$, and $QsP = \sum_{i,j} v_i s u_j E_{ij}$. By Lemma B.23 and the generalized trace $\mathrm{Tr}_* : C(M_{1+m}(S)) \to C(S)$ (a chain map with $\mathrm{Tr}_* \phi_0 = 1$), $1 \simeq \mathrm{Tr}_* \phi_1 = \theta$ on $C(S)$. The image of θ lies in $C(fSf)$. In $T' = M_{1+m}(fSf)$ take $e' = fE_{00}$, $P' = \sum_j f u_j E_{0j}$ and $Q' = \sum_j v_j f E_{j0}$. All entries lie in fSf, $P'Q' = e'$, and $Q'aP' = \sum v_i a u_j E_{ij}$ for $a \in fSf$. The same argument gives $1 \simeq \theta$ on $C(fSf)$. □

Remark B.25 (Bundles). For a subbundle $E \subset X \times \mathbb{C}^N$ of positive rank, with orthogonal projection e, the hypothesis holds in $S = M_N(C^\infty(X)) \subset M_N(\mathfrak{A})$, for every algebra $\mathfrak{A}$ of the notes. Take a partition of unity $\sum \rho_\alpha^2 = 1$ and unit sections s_α of E on $\operatorname{supp} \rho_\alpha$, and put $u_{\alpha,i} = \rho_\alpha e_i s_\alpha^*$ and $v_{\alpha,i} = \rho_\alpha s_\alpha e_i^*$. Then $u_{\alpha,i} e = u_{\alpha,i}$, $e v_{\alpha,i} = v_{\alpha,i}$ and $\sum_{\alpha,i} u_{\alpha,i} v_{\alpha,i} = 1$. With Lemma B.22 this gives $\mathrm{HH}(eM_N(\mathfrak{A})e) \cong \mathrm{HH}(\mathfrak{A})$ for the unital algebras of the notes, as used in Section 13. For the ideals J ($\mathcal{I}$, $\mathcal{I}_\sigma$, $\mathcal{I}_\partial$) the five lemma and Theorem B.7 give the same, provided $eM_N(J)e$ is topologically H-unital. This holds because the contractions of Lemmas 5.4, 6.2 and Propositions 7.2, 9.3 are defined by kernels and can be written with bundle coefficients (sketch).

B.5 Convergence of the order spectral sequence

The algebraic statement

Let $(C_\bullet, b)$ be a complex in which each C_k is a subspace of the space of series $c = \sum_{p \le P} c_p$, $c_p \in C_{p,k}$, containing each $C_{p,k}$, with $b = \sum_{l \ge 0} b_l$ and $b_l : C_{p,k} \to C_{p-l,k-1}$. Put $F_p C = \{c : \ c_q = 0 \text{ for } q > p\}$, $E^1_{p,k} = H_k(C_{p,\bullet}, b_0)$, and let d^1 be induced by b_1. Comparing degrees in $b^2 = 0$ gives $\sum_{i+j=l} b_i b_j = 0$, so d^1 is well defined, $(d^1)^2 = 0$, and $E^2 = H(E^1, d^1)$. The filtration is *complete* if every series $\sum_j e_j$ with $e_j \in F_{p_j} C_k$, $\sup p_j < \infty$ and $p_j \to -\infty$ defines an element of C_k, and b commutes with such sums.

Proposition B.26 (Algebraic convergence [14, Ch. 10])**.** *Suppose that, for an integer n, $E^2_{p,k} = 0$ whenever $p \neq k - n$.*

1. *Every class in $H_k(C, b)$ has a representative in $F_{k-n}C_k$, and $\Theta[c] = [c_{k-n}] \in E^2_{k-n,k}$ is well defined.*
2. *$\ker \Theta = \bigcap_p F_p H_k$, where $F_p H_k$ is the image of $H_k(F_p C)$.*

If the filtration is complete, then $\Theta : H_k(C, b) \to E^2_{k-n,k}$ is an isomorphism, and all differentials d^r, $r \geq 2$, vanish.

Proof. Step (P). Let $c \in F_p C_k$ with $b_0 c_p = 0$ and $b_1 c_p + b_0 c_{p-1} = 0$, and suppose $[c_p] = 0$ in $E^2_{p,k}$. Then $[c_p] = d^1[x]$ for a b_0-cycle $x \in C_{p+1,k+1}$, so $c_p - b_1 x = b_0 y$ with $y \in C_{p,k+1}$. The components of $b(x + y)$ of order $p + 1$ and p are $b_0 x = 0$ and $b_1 x + b_0 y = c_p$. Hence $c - b(x + y) \in F_{p-1}$.

(1). If $c \in F_P C_k$ is a cycle and $P > k - n$, then $E^2_{P,k} = 0$, so (P) lowers P; after finitely many steps $c \in F_{k-n}$. If $c \in F_{k-n}$ is a cycle, then $b_0 c_{k-n} = 0$ and $b_1 c_{k-n} = -b_0 c_{k-n-1}$, so c_{k-n} defines a class in E^2. Let $c = be \in F_{k-n}$ with $e \in F_q$. If $q > k + 1 - n$, then (P) applies to e, since $b_0 e_q = (be)_q = 0$, $b_1 e_q + b_0 e_{q-1} = (be)_{q-1} = 0$ and $E^2_{q,k+1} = 0$. It replaces e by an element of F_{q-1} with the same boundary. Once $q \leq k + 1 - n$, $c_{k-n} = b_1 e_{k+1-n} + b_0 e_{k-n}$, so $[c_{k-n}] = d^1[e_{k+1-n}] = 0$ in E^2.

(2). If $[c] \in F_{k-n-1}H_k$, then $\Theta[c] = 0$ by (1). Conversely, if $c \in F_{k-n}$ and $[c_{k-n}] = 0$, then (P) gives a homologous cycle in F_{k-n-1}. Since $E^2_{p,k} = 0$ for $p < k - n$, repeating (P) finitely often reaches any F_p.

Injectivity. In the situation of (2), the repeated steps give $e_p \in F_{p+1}C_{k+1}$, $p = k - n, k - n - 1, \ldots$, with $c - b\sum_{p \geq P} e_p \in F_{P-1}$. By completeness $c = b\sum_p e_p$.

Surjectivity. Let $x \in C_{k-n,k}$ be a b_0-cycle with $d^1[x] = 0$. Then $b_1 x = -b_0 y$, and $c^{(1)} = x + y$ has $bc^{(1)} \in F_{k-n-2}$. If $z = bc^{(r)} \in F_p$, $p = k - n - r - 1$, then z is a cycle and $p \neq (k - 1) - n$, so (P) gives x', y' of order $\leq p + 1 < k - n$ with $z - b(x' + y') \in F_{p-1}$. Put $c^{(r+1)} = c^{(r)} - x' - y'$. By completeness $c = \lim c^{(r)}$ exists, $bc = 0$ and $\Theta[c] = [x]$.

Degeneration. d^r maps $E^r_{p,k}$ to $E^r_{p-r,k-1}$. Both can be non-zero only if $p = k - n$ and $p - r = k - 1 - n$, that is, $r = 1$. □

This is Proposition 5.6 with its proof completed. Parts (1) and (2) need no completeness and hold in direct limits. The chain model of Section 5.1 is *not* complete. In $C_1(\mathfrak{A}) = \varinjlim_N \mathfrak{A}_N^{\widehat{\otimes}2}$ the series $\sum_{j \geq 1} r^j \otimes r^{-2j}$ has total orders $-j \to -\infty$, but its first entries have unbounded order, so it lies in no $\mathfrak{A}_N^{\widehat{\otimes}2}$. Following [7, §"Convergence and degeneration"], we restore completeness with mixed complexes.

Mixed complexes

Let $\mathfrak{A}$ be one of

- the symbol algebra $\mathcal{A}_M$ of a closed manifold M (Section 3);
- $\mathcal{A}_\sigma$, or $\mathcal{A}_{\partial,\sigma}$;

with n the dimension of the underlying manifold. Let $\mathfrak{B} \subset \mathfrak{A}$ be the subalgebra of symbols of order ≤ 0 without poles (for $\mathcal{A}_{\partial,\sigma}$, the Taylor image of that subalgebra of $\mathcal{A}_\sigma$). Let $\mathfrak{A}_{L,M}$ be the elements with pole order $\leq L$ and order $\leq M$ ($L = 0$ for $\mathcal{A}_M$). The *mixed chains* are

$$C_k(\mathfrak{B}, \mathfrak{A}) = \varinjlim_{L,M} S_{L,M,k}, \qquad S_{L,M,k} = \mathfrak{A}_{L,M} \mathbin{\widehat{\otimes}} \mathfrak{B}^{\widehat{\otimes}k},$$

with the Hochschild differential with coefficients in the $\mathfrak{B}$-bimodule $\mathfrak{A}$: d_0 and d_k put a_0b_1 and b_ka_0 in slot 0, and the inner faces multiply in $\mathfrak{B}$.

Lemma B.27 (Adapted from [7, Lemma "A mixed complex for Laurent complete symbols"])**.** (a) *Each $S_{L,M}$ is a subcomplex. With the order filtration it is complete in the sense of Proposition B.26.* (b) *The inclusion $C(\mathfrak{B},\mathfrak{A}) \to C(\mathfrak{A})$ is a quasi-isomorphism.*

Proof. (a) Let $T \in \mathfrak{B}$ be a symbol of order -1 without poles, elliptic with principal symbol q^{-1} (from (B3)). Its inverse in $\mathfrak{A}$ has order 1 and no poles. Put $u = u_{L,M} = x^{-L}T^{-M}$. By (B1), conjugation by $x^{\pm L}$ preserves orders and smooth coefficients. (In the cusp calculus $[x,P] \in x^2\Psi_c^{m-1}$, even better than in the b-calculus.) Hence $\mathfrak{A}_{L,M} = \mathfrak{B}u = u\mathfrak{B}$: for $a \in \mathfrak{A}_{L,M}$, $au^{-1} = x^{-L}(x^La)T^Mx^L$ is the conjugate by x^{-L} of a pole-free symbol of order ≤ 0. So $\mathfrak{A}_{L,M}$ is a $\mathfrak{B}$-bimodule and $S_{L,M}$ is a subcomplex. A quantization (B1) identifies $\mathfrak{A}_{L,M}$ and $\mathfrak{B}$ with products of their homogeneous components, and the product with the star product $\sum_l P_l$, P_l lowering the order by l. By (F3), $S_{L,M,k}$ is the product over multidegrees $(j_0,\dots,j_k)$, $j_0 \le M$, $j_i \le 0$, of the homogeneous tensor components. At a fixed total order only finitely many multidegrees occur. So $S_{L,M}$ consists of all series in the total order bounded above by M, and b commutes with convergent sums.

(b) We sketch the argument of [7], which uses nothing specific to the b-calculus. Consider the augmented bar complexes of $\mathfrak{A}$-bimodules

$$P_k = \mathfrak{A} \mathbin{\widehat{\otimes}} \mathfrak{B}^{\widehat{\otimes} k} \mathbin{\widehat{\otimes}} \mathfrak{A}, \qquad Q_k = \mathfrak{A} \mathbin{\widehat{\otimes}} \mathfrak{A}^{\widehat{\otimes} k} \mathbin{\widehat{\otimes}} \mathfrak{A} \qquad (k \ge 0),$$

with differential b' and augmentation given by multiplication. On the part of P whose last slot lies in $\mathfrak{B}u$, the maps $h_k(a\otimes\beta\otimes bu) = (-1)^{k+1}a\otimes\beta\otimes b\otimes u$ and $h_{-1}(a) = au^{-1}\otimes u$ form a contraction. With $u = 1$ the same formula contracts Q. So both are resolutions of $\mathfrak{A}$, free on generators $1\otimes\beta\otimes 1$. Comparison maps $P \rightleftarrows Q$ over the identity and the bimodule homotopies are built recursively on generators by lifting cycles with these contractions. The choices are made linear and continuous stage by stage. For this one uses that $\mathfrak{A}_{N,N}$ has a continuous complement in $\mathfrak{A}_{N+1,N+1}$: remove the top-order term, then the lowest boundary coefficient of each remaining homogeneous term, extended into the collar with a cut-off. Finally $a\otimes\beta\otimes a' \mapsto a'a\otimes\beta$ turns P into $C(\mathfrak{B},\mathfrak{A})$ and Q into $C(\mathfrak{A})$, and the comparison maps and homotopies descend. □

Lemma B.28 (Fixed-stage E^1)**.** *Let $q \le M-1$. The b_0-homology of $S_{L,M}$ at total order q is, via χ, the space of homogeneous k-forms of degree q on the cotangent bundle minus its zero section, with coefficients*

- *smooth, for $\mathcal{A}_M$;*
- *in $x^{-L}\Omega$, i.e. x^{-L} times* ordinary *smooth forms on ${}^cT^*X \setminus 0$ (generated by $dx, dy, d\tau, d\eta$), for $\mathcal{A}_\sigma$;*
- *in the formal analogue, for $\mathcal{A}_{\partial,\sigma}$.*

The inclusions of stages induce the obvious inclusions of forms. In the direct limit one gets all smooth, Laurent and formal Laurent homogeneous forms, which is the E^1 of Proposition 7.1.

Proof. The associated graded algebra of $\mathfrak{B}$ is $C^\infty(Y)[t]$, with Y the cosphere bundle, $t = r^{-1}$ of weight -1, and $C^\infty(Y)$ replaced by its formal version for $\mathcal{A}_{\partial,\sigma}$. The graded coefficient module is $x^{-L}t^{-M}C^\infty(Y)[t]$, free of rank one with equal left and right actions. So its Hochschild complex is that of $C^\infty(Y)[t]$ shifted by $x^{-L}t^{-M}$, and Corollary B.21 gives $x^{-L}t^{-M}$ times forms in $\Omega(Y)[t] \oplus \Omega(Y)[t]\,dt$. A homogeneous form of degree q is $t^{-q}(\alpha - \beta\wedge dt/t)$, and lies there if and only if $M-q \ge 0$ and $M-q-1 \ge 0$. The forms on Y are ordinary forms, since $\mathfrak{B}$ has smooth coefficients. □

The stage argument

Proposition B.29 (Adapted from [7, §"Convergence and degeneration"])**.** *Let $D \subset C(\mathfrak{B}, \mathfrak{A})$ be a subcomplex with closed subcomplexes $D_{L,M} = D \cap S_{L,M}$, and suppose:*

(E2) the E^2 term of D vanishes off the line $p = k - n$;

(St) for $L \geq L_0$ and every M, step (P) of Proposition B.26 can be carried out inside $D_{L,M}$ whenever $p \leq M - 2$ and $p \neq k - n$.

Then $\Theta : H_k(D) \to E^2_{k-n,k}(D)$ is an isomorphism, and $d^r = 0$ for $r \geq 2$.

Proof. Parts (1) and (2) of Proposition B.26 hold in the direct limit. For injectivity, let $c \in F_{k-n}$ be a cycle with $[c_{k-n}] = 0$. Perform one step (P) in the direct limit and choose a stage $D_{L,M}$ containing the remainder, with $L \geq L_0$ and M so large that all later orders are $\leq M - 2$. All later steps are off the critical line, so by (St) they stay in $D_{L,M}$, and the infinite sum converges there by Lemma B.27(a). For surjectivity, choose x and y in the direct limit, then a stage containing them, and continue in the same way. Degeneration is as before. □

Theorem B.30 (Convergence for $\mathcal{A}_M$ and $\mathcal{I}_\sigma$)**.** *For the symbol algebra $\mathcal{A}_M$ of a closed manifold, and for $\mathcal{I}_\sigma$, the order spectral sequence converges and degenerates at E^2. Θ is an isomorphism $\mathrm{HH}_k \to E^2_{k-n,k}$, given by the leading symbol of a representative in F_{k-n}.*

Proof. $\mathcal{A}_M$. Take $D = C(\mathfrak{B}, \mathcal{A}_M)$, which computes $\mathrm{HH}(\mathcal{A}_M)$ by Lemma B.27(b). (E2) is Propositions 5.8, 5.9 and 5.10. For (St), let θ be the HKR form of c_p. It is homogeneous of degree p and $\delta\theta = 0$. Then $*\theta$ is closed and homogeneous of degree $w = p + n - k \neq 0$. By Cartan's formula $\eta = \iota_R(*\theta)/w$ satisfies $d\eta = *\theta$, and $\zeta = *^{-1}\eta$ is homogeneous of degree $p + 1 \leq M - 1$ with $\delta\zeta = \pm\theta$. By Lemma B.28, $\pm i\zeta$ is the class of a b_0-cycle x in the same stage at order $p + 1$, with $d^1[x] = [c_p]$ (the proof of Proposition 5.8 applies to mixed chains: it only uses the leading symbols in each slot). Then $c_p - b_1 x$ has zero class in the stage E^1 at order $p \leq M - 1$, so it is $b_0 y$ with y in the stage.

$\mathcal{I}_\sigma$. $\mathcal{I}_\sigma$ is H-unital (Proposition 7.2). By Theorem B.7, $C(\mathcal{I}_\sigma)$ is quasi-isomorphic to the kernel $\mathcal{K}$ of the Taylor map $C(\mathcal{A}_\sigma) \to C(\mathcal{A}_{\partial,\sigma})$. Let D be the kernel of the Taylor map $C(\mathfrak{B}, \mathcal{A}_\sigma) \to C(\mathfrak{B}_\partial, \mathcal{A}_{\partial,\sigma})$. The Taylor maps are onto on every stage, by (F1) and Borel's lemma. So Lemma B.27(b), for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$, and the five lemma show that $D \to \mathcal{K}$ is a quasi-isomorphism. The Taylor map is also onto on the stage E^1 (Borel's lemma for forms). So the stage E^1 of D at orders $q \leq M - 1$ is the space of all flat homogeneous forms of degree q. The operators $*$, $*^{-1}$ and ι_R preserve flat forms: $*$ only introduces the factors $x^{\pm 2}$. So the argument for $\mathcal{A}_M$ applies without change and gives (St). (E2) is the E^2 computation of Proposition 7.2. The maps between the three complexes preserve the order and induce the flat HKR identification on graded pieces. So the resulting isomorphism is the leading-symbol map. □

The Laurent algebras $\mathcal{A}_\sigma$, $\mathcal{A}_{\partial,\sigma}$ and the cusp twist

In [7] condition (St) is verified for the b-calculus analogues of $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ [7, Lemma "Lifts in a fixed coefficient stage"]. The cusp version fails. By Lemma B.28 the stage E^1 of $\mathcal{A}_\sigma$ consists of x^{-L} times ordinary forms. The symplectic star, however, preserves x^{-L} times *cusp* forms.

Example B.31 (The stage obstruction, $n = 1$)**.** Let $n = 1$ and work near one boundary point, with coordinates (x, τ) and $\{a, b\} = \partial_\tau a\, x^2 \partial_x b - x^2 \partial_x a\, \partial_\tau b$. The critical line is $p = k - 1$. Fix L, M and $p \leq M - 2$ with $p \neq -1$. The 0-form $\theta = x^{-L}|\tau|^p$ lies in the stage $E^1_{p,0}$, and $d^1\theta = 0$. The stage

1-forms of degree $p+1$ are $\zeta = x^{-L}(a\,d\tau + c\,dx)$ with a, c smooth and homogeneous of degrees p and $p+1$. For 1-forms $\delta(f\,dg) = \{f, g\}$, so

$$\delta\zeta = \{x^{-L}a, \tau\} + \{x^{-L}c, x\} = -x^2\partial_x(x^{-L}a) + x^{2-L}\partial_\tau c \;\in\; x^{1-L}C^\infty.$$

Hence θ is not in the image, and $[\theta] \neq 0$ in the stage $E^2_{p,0}$, although $[\theta] = 0$ in the direct limit: $\theta = \delta\big(x^{-L-1}|\tau|^p\,d\tau/(L+1)\big)$. So (St) fails in every stage. In the b-calculus the same computation, with $x\partial_{\tilde\xi} \wedge \partial_x$ in place of $x^2\partial_\tau \wedge \partial_x$, gives $-x\partial_x(x^{-L}a) = Lx^{-L}a + O(x^{1-L})$. This reaches the leading coefficient when $L \neq 0$. That is the invertibility of $\mathcal{L}_{x\partial_x} = -L$ on the leading coefficient used in [7]. In the cusp case its replacement $\mathcal{L}_{x^2\partial_x}$ raises the power of x, so it is nilpotent on the leading coefficient.

The same computation in general dimension shows that δ of a stage form never reaches the leading coefficient x^{-L} through the (x,τ)-part of the Poisson structure. On leading coefficients only the boundary part $\partial_\eta \wedge \partial_y$ acts, and for it τ is a Casimir. So the stage E^2 off the critical line contains classes such as $x^{-L}|\tau|^p$ in every stage. We do not need a precise description. Under the transcendental change $\rho = e^{-1/x}$, which turns the cusp structure into a b-structure (Section 4), one has $x^{-L} = (\log 1/\rho)^L$. So the cusp Laurent algebras are b-algebras with *logarithmic* coefficients, on which $\rho\partial_\rho$ acts nilpotently. What survives is the following.

Lemma B.32. *Let $\theta \in x^{-L}\Omega^k$ be homogeneous of degree $p \neq k-n$ with $\delta\theta = 0$. Then $\theta = \delta\zeta$ with $\zeta \in x^{-L-2}\Omega^{k+1}$ homogeneous of degree $p+1$.*

Proof. Let Ω_c be the smooth cusp forms, generated by $dx/x^2, dy, d\tau, d\eta$. Then $\Omega \subset \Omega_c \subset x^{-2}\Omega$. The cusp symplectic form is a smooth non-degenerate cusp 2-form, so $*$ and $*^{-1}$ preserve $x^{-L}\Omega_c$, and so does ι_R. The construction in the proof of Theorem B.30 gives $\zeta \in x^{-L}\Omega_c \subset x^{-L-2}\Omega$. □

Proposition B.33 (Partial convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$)**.** *Let $\mathfrak{A} = \mathcal{A}_\sigma$ or $\mathcal{A}_{\partial,\sigma}$, with the mixed complex $C(\mathfrak{B}, \mathfrak{A})$, which computes $\mathrm{HH}(\mathfrak{A})$.*

1. *Every class in $\mathrm{HH}_k(\mathfrak{A})$ has a representative in F_{k-n}, and $\Theta : \mathrm{HH}_k(\mathfrak{A}) \to E^2_{k-n,k}$ is well defined. Here E^2 is the one computed in Propositions 7.1 and 8.1.*
2. $\ker\Theta = \bigcap_p F_p\,\mathrm{HH}_k(\mathfrak{A})$.
3. *Let $x \in S_{L,M}$ be a b_0-cycle of order $k-n \leq M-2$ with $d^1[x] = 0$. For every $r \geq 1$ there is $c^{(r)} \in S_{L+2r,M}$ with leading term x and $bc^{(r)} \in F_{k-n-r-1}$.*
4. *Θ is an isomorphism if and only if $\mathrm{HH}_k(\mathfrak{A}) \to H_k(\widehat{C})$ is one. Here $\widehat{C}$ is the complex of series $\sum_{p\leq P} c_p$ in which each c_p lies in some stage, with no uniform bound on the poles.*

Proof. (1) and (2) are Proposition B.26(1),(2), which need no completeness. For (3), $[b_1 x] = 0$ already in the stage E^1, by the injectivity in Lemma B.28. Each later step (P) is off the critical line. By Lemma B.32 its primitive lies in $x^{-L'-2}\Omega$, so the corrections lie in the stage $L'+2$. For (4): $\widehat{C}$ is a complex because each b_l preserves the stages. It is complete, and its graded pieces and E^2 agree with those of $C(\mathfrak{B}, \mathfrak{A})$. So $H_k(\widehat{C}) \cong E^2_{k-n,k}$ by Proposition B.26, compatibly with Θ. □

Remark B.34 (What remains quoted). Proposition B.33 reduces the convergence for $\mathcal{A}_\sigma$ and $\mathcal{A}_{\partial,\sigma}$ to one statement. Corrections whose poles grow by up to two per order (part (3)) must be replaceable by corrections with a uniform pole bound. Equivalently, there are no phantom classes $\bigcap_p F_p\,\mathrm{HH}_k \neq 0$, and the towers in (3) can be completed in $C(\mathfrak{B}, \mathfrak{A})$. MN assert this ("as in the boundaryless case",

[15, p. 12]). We have not proved it. The fixed-stage argument of [7] cannot prove it, by Example B.31. In these notes Propositions 7.1 and 8.1, and everything that depends on them, therefore rest on this quoted step. By contrast, Theorem B.30 establishes convergence for $\mathcal{I}_\sigma$ and for the closed case without it. This quoted step is item (B10) of Section 2.5.

C Holomorphic families, residue traces and the suspended eta invariant

This appendix proves the background items (B3), (B7) and (B8) of Section 2.5. We follow the route of [14, Chs. 6 and 15]. Instead of the complex powers Q^{-s} we use *any* holomorphic family $Q(s)$ of order $-s$ with $Q(0) = \mathrm{Id}$. Such families are written down explicitly, and nothing about the spectral theory of Q is needed. The items are proved in the following forms.

(B3′) Proposition C.12 constructs an admissible family $Q(s)$ in the cusp calculus. Its indicial family is translation invariant. Theorem C.14 proves the two-variable meromorphy of $\mathrm{Tr}(A(s,z)x^z)$ for holomorphic families $A(s,z)$ of order $m-s$, with simple poles on the lines $s = m+n-k$, $z = 1-p-j$, and identifies the residues. Proposition C.15 lists which results of Sections 10–15 hold with $Q(s)$ in place of Q^{-s}. It also explains why the final results do not depend on the family.

(B7′) Theorem C.23: $\overline{\mathrm{Tr}}$ is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{sus}}(H)$. Theorem C.24 computes the regularization of $\overline{\mathrm{Tr}}$ by a holomorphic family; this corrects [15, Prop. 8]. Theorem C.29: for operators in APS normal form, $\eta(A_0) = (\pi i)^{-1}\overline{\int}\,\mathrm{Tr}(A_0^{-1}\partial_\tau A_0)\,d\tau$ is the APS eta invariant $\eta_B(0)$. Proposition C.30 relates the functional $\overline{\eta}$ of [15, (89)] to $\eta(A_0)$.

(B8′) Theorem C.7: the residue is a trace. It is given by the local formula and does not depend on the coordinates or on the regularizing family. Proposition C.9 proves uniqueness for $n \geq 2$ (for traces vanishing on smoothing operators).

The quoted inputs are these. From (B1) we use the local structure of the cusp and suspended calculi: composition, symbols, the Taylor series at the front face, and inverses of invertible elliptic suspended families. From [18] we use the small-time heat expansion of $\mathrm{Tr}(Be^{-tB^2})$ and the regularity of $\eta_B(s)$ at $s = 0$. Only Proposition C.30(ii) uses complex powers, and only on the boundary. Every quoted step is marked.

Two statements need correction; we prove the corrected forms, which the main text states.

1. The regular value at $s = 0$ of $\overline{\mathrm{Tr}}(BQ_0^{-s})$ is *not* in general $\overline{\mathrm{Tr}}(B)$, as claimed in [15, Prop. 8]. There is a local correction term (Theorem C.24, Example C.25; corrected statement Proposition 10.12). It propagates to Corollary 10.14(2) and Proposition 15.8; see Correction C.26.

2. If Q^{-s} is replaced by a general family $Q(s)$, then Proposition 15.3 acquires an extra coboundary term, $\frac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}([A,B]\kappa_Q)$ (Remark 15.4). This term vanishes when $[A,B] \in \mathcal{I}$, so the index theorem is unaffected (Proposition C.15).

C.1 Symbols of complex order, traces and Seeley's computation

Let $\overline{\mathbb{R}^N}$ be the radial compactification of $\mathbb{R}^N$. Let $\rho \in C^\infty(\overline{\mathbb{R}^N})$ be a boundary defining function that is positive on $\mathbb{R}^N$ and equal to $|\zeta|^{-1}$ for $|\zeta| \geq 1$.

Definition C.1. Let $U \subset \mathbb{R}^k$ be open and $\mu \in \mathbb{C}$. A *classical symbol of order* μ on $U \times \mathbb{R}^N$ is a function $a = \rho^{-\mu}\beta$ with $\beta \in C^\infty(U \times \overline{\mathbb{R}^N}; M(r, \mathbb{C}))$. The Taylor series of β in $1/|\zeta|$ at the sphere at infinity gives the expansion $a \sim \sum_{j \geq 0} a_{\mu-j}$, where $a_{\mu-j}(u, \zeta) = |\zeta|^{\mu-j}\beta_j(u, \zeta/|\zeta|)$ is homogeneous of degree $\mu - j$ and $|a - \sum_{j<J} a_{\mu-j}| \leq C_J|\zeta|^{\operatorname{Re}\mu - J}$ for $|\zeta| \geq 1$, locally uniformly in u. Let $\Omega \subset \mathbb{C}^l$ be open and $\mu : \Omega \to \mathbb{C}$ holomorphic. A *holomorphic family of order* $\mu(w)$ is $a(w) = \rho^{-\mu(w)}\beta(w)$ with $\beta \in C^\infty(\Omega \times U \times \overline{\mathbb{R}^N})$ holomorphic in w.

A change of ρ multiplies $\beta(w)$ by a holomorphic power of a positive smooth function. So the definition does not depend on ρ.

Lemma C.2 (Seeley's computation [19])**.** *Let $a(w)$ be a holomorphic family of order $\mu(w)$, with compact support in u. Then $I(w; u) = \int_{\mathbb{R}^N} a(w; u, \zeta)\, d\zeta$ is holomorphic on $\{\operatorname{Re}\mu(w) < -N\}$. For every $J \in \mathbb{N}$,*

$$I(w; u) = \sum_{j<J} \frac{1}{j - N - \mu(w)} \int_{S^{N-1}} a_{\mu(w)-j}(w; u, \theta)\, d\theta + I_J(w; u),$$

where I_J is holomorphic on $\{\operatorname{Re}\mu(w) < J - N\}$ and smooth in u. In particular, if $\mu(w) = m - s$ with $w = (s, z')$, then I is meromorphic in w with at most simple poles on the hyperplanes $s = m + N - j$. The residue there is $\int_{S^{N-1}} a_{-N}\, d\theta$, where a_{-N} is the term of homogeneity $-N$ of the symbol $a(w)$ at $s = m + N - j$.

Proof. The integral over $|\zeta| \leq 1$ is holomorphic for all w. For $|\zeta| \geq 1$ write $\zeta = \theta/r$, $r \in (0, 1]$, so that $d\zeta = r^{-N-1} dr\, d\theta$ and $a = r^{-\mu}\beta(w; u, r, \theta)$. Taylor's formula gives $\beta = \sum_{j<J} r^j \beta_j + r^J \beta_{(J)}$, with all terms smooth and holomorphic. For $\operatorname{Re}\mu < -N$, $\int_0^1 r^{j-\mu-N-1} dr = (j - N - \mu)^{-1}$. The remainder $\int_{S^{N-1}} \int_0^1 r^{J-\mu-N-1}\beta_{(J)}\, dr\, d\theta$ is holomorphic for $\operatorname{Re}\mu < J - N$. Since $a_{\mu-j}(\theta) = \beta_j(\theta)$ on $|\theta| = 1$, this proves the formula. The expressions for different J agree where both are defined, so they define the continuation. □

Lemma C.3. *Let M be a closed manifold of dimension N and $E \to M$ a bundle.*

1. *If $A \in \Psi^\mu(M; E)$ with $\operatorname{Re}\mu < -N$, then A is of trace class, its kernel is continuous, and $\operatorname{Tr} A = \int_M \operatorname{tr} K_A|_\Delta$. If $A = \phi \operatorname{Op}(a)\psi$ in a chart, with $\phi, \psi \in C_c^\infty(U)$, then $\operatorname{Tr} A = (2\pi)^{-N} \iint \phi\psi \operatorname{tr} a\, d\zeta\, du$.*
2. *If $A \in \Psi^{\mu_1}$, $B \in \Psi^{\mu_2}$ and $\operatorname{Re}(\mu_1 + \mu_2) < -N$, then $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$.*

Proof. (Adapted from [14, Ch. 15].) (1) With a partition of unity, A is a finite sum of operators $\phi \operatorname{Op}(a)\psi$ plus a smoothing operator. For a smoothing operator the statement is standard. Write $a = a'\langle\zeta\rangle^{\mu/2}$, with a' of order $\mu/2$. In the left quantization $\operatorname{Op}(a')\langle D\rangle^{\mu/2} = \operatorname{Op}(a)$ exactly. The operator $\langle D\rangle^{\mu/2}\psi$ is Hilbert–Schmidt, because $\langle\zeta\rangle^{\operatorname{Re}\mu/2} \in L^2$. The operator $\phi \operatorname{Op}(a')$ is Hilbert–Schmidt, because its kernel has L^2 norm in u' at most $\|a'(u, \cdot)\|_{L^2}$. So A is of trace class. Its trace is the pairing of the two kernels. Since a is integrable in ζ, this pairing is $\int K_A(u, u)\, du$.

(2) Regard the operators as elements of the Hörmander classes $\Psi_{\mathrm{H}}^{\operatorname{Re}\mu}$. The argument of (1) bounds the trace norm of $\phi \operatorname{Op}(a)\psi$ by $C \sup |a|\langle\zeta\rangle^{-\nu}$ for $\nu < -N$. So Tr is continuous on Ψ_{H}^{ν}, $\nu < -N$. Cut off the local symbols of A to $\chi(\epsilon\zeta)a$, keeping the part of A away from the diagonal. This gives smoothing operators $A_\epsilon \to A$ in $\Psi_{\mathrm{H}}^{\operatorname{Re}\mu_1+\delta}$ for every $\delta > 0$. Choose δ with $\operatorname{Re}(\mu_1 + \mu_2) + \delta < -N$. Then $\operatorname{Tr}(AB) = \lim \operatorname{Tr}(A_\epsilon B) = \lim \operatorname{Tr}(BA_\epsilon) = \operatorname{Tr}(BA)$. □

C.2 The Wodzicki residue: item (B8)

Let M be closed, of dimension N. (In (B8), $N = n$.)

Definition C.4. Let $\Omega \subset \mathbb{C}^l$ be open and μ holomorphic on Ω. A family $A(w) \in \Psi^{\mu(w)}(M;E)$ is *holomorphic of order* μ if the following hold.

1. For $\chi \in C^\infty(M^2)$ vanishing near the diagonal, $\chi K_{A(w)}$ is holomorphic with values in $C^\infty(M^2)$.
2. In every chart over which E is trivial, and for $\phi, \psi \in C_c^\infty(U)$, one has $\phi A(w)\psi = \phi \operatorname{Op}(a(w))\psi + R(w)$. Here $a(w)$ is a holomorphic family of classical symbols of order $\mu(w)$, and $R(w)$ is a holomorphic family of smoothing operators.

An *admissible family* is an entire family $Q(s)$ of order $-s$ with $Q(0) = \mathrm{Id}$ whose principal symbol is scalar, $q_s \,\mathrm{Id}_E$.

Lemma C.5. *1. Products of holomorphic families are holomorphic families, and the orders add.*

2. *(Division.) If $F(s)$ is holomorphic of order $\mu(s)$ and $F(s_0) = 0$, then $F(s) = (s - s_0)F_1(s)$, with F_1 holomorphic of order $\mu(s)$. In two variables: if $F(s,z)$ vanishes on $\{s = 0\} \cup \{z = 0\}$, then $F = szF_2$. If $F(0,0) = 0$, then $F = sF_3 + zF_4$. All the F_i are holomorphic of order μ.*
3. *If the principal symbol of a holomorphic family of order μ vanishes identically, it is a holomorphic family of order $\mu - 1$.*

Proof. (1) is the composition theorem with holomorphic parameters. The composition integral and the Borel summation of the symbol expansion can be carried out locally uniformly in w [14, Ch. 15]. We do not repeat the argument.

(2) Away from the diagonal, divide the holomorphic kernel. Near the diagonal, $\phi F(s_0)\psi = 0$ forces $\rho^{-\mu(s_0)}\beta(s_0) \in S^{-\infty}$, so $\beta(s_0)$ vanishes to infinite order at the sphere at infinity. Then $S(s) = \phi \operatorname{Op}(\rho^{-\mu(s)}\beta(s_0))\psi$ is a holomorphic *smoothing* family. Replace $\beta(s)$ by $\beta(s) - \beta(s_0)$ and $R(s)$ by $R(s) + S(s)$. Both now vanish at s_0, and both are divisible by $s - s_0$ by Cauchy's formula. For two variables: if $F(0,z) = 0$, then $F = sF_1$ with z a parameter. Then $F_1(s,0) = 0$ for $s \neq 0$, hence for all s, and $F_1 = zF_2$. If $F(0,0) = 0$, apply the one-variable statement to $F(s,z) - F(0,z)$ and to $F(0,z)$.

(3) We have $\beta|_{\rho=0} = 0$, so $\beta = \rho\beta'$. □

Proposition C.6. *Choose charts U_i trivializing E, a subordinate partition of unity ϕ_i, and $\psi_i \in C_c^\infty(U_i)$ with $\psi_i = 1$ near* $\operatorname{supp}\phi_i$. *Then*

$$Q_M(s) = \sum_i \phi_i \operatorname{Op}\big(\langle\zeta\rangle^{-s}\big)\psi_i, \qquad \langle\zeta\rangle = (1 + |\zeta|^2)^{1/2},$$

is an admissible family. Its principal symbol is $q_s = \sum_i \phi_i |\zeta|_{(i)}^{-s}$, where $|\zeta|_{(i)}$ is the Euclidean length in the coordinates of U_i.

Proof. Each term is an entire family of order $-s$. Since $\operatorname{Op}(1) = \mathrm{Id}$, we get $Q_M(0) = \sum_i \phi_i\psi_i = \sum_i \phi_i = \mathrm{Id}$. □

Complex powers Q^{-s} of a positive elliptic $Q \in \Psi^1$, when they are available, are also admissible. We do not need them.

Theorem C.7 (The residue trace)**.** *Let $A \in \Psi^m(M;E)$, $m \in \mathbb{Z}$. Let $A(s)$ be a holomorphic family of order $m-s$ on a connected open set $\Omega \ni 0$ that meets $\{\operatorname{Re} s > m+N\}$, with $A(0)=A$. For instance $A(s)=AQ(s)$ with Q admissible. Then $\operatorname{Tr} A(s)$ extends from $\{\operatorname{Re} s > m+N\}\cap\Omega$ to a meromorphic function on Ω. It has at most simple poles, at $s \in m+N-\mathbb{N}_0$, and*

$$\operatorname{Res}_{s=0} \operatorname{Tr} A(s) = \operatorname{Tr}_{\mathrm{R}}(A) := (2\pi)^{-N} \sum_i \int_{U_i} \phi_i(u) \int_{|\zeta|=1} \operatorname{tr} a^{(i)}_{-N}(u,\zeta)\, dS(\zeta)\, du, \tag{26}$$

where $a^{(i)}_{-N}$ is the term of homogeneity $-N$ of the full symbol of A in the chart U_i. Consequently:

1. *$\operatorname{Tr}_{\mathrm{R}}(A)$ does not depend on the family $A(s)$, in particular not on Q.*
2. *(Coordinate invariance.) The density $\big(\int_{|\zeta|=1} \operatorname{tr} a_{-N}(u,\zeta)\, dS\big)|du|$ on M does not depend on the chart or on the trivialization of E. On S^*M it is the fibre integral of $\operatorname{tr} a_{-N}\,|\iota_R(\omega^N/N!)|$, where R is the radial vector field. This is the formula (B8) and* (11)*.*
3. *(Trace.) $\operatorname{Tr}_{\mathrm{R}}([A,B]) = 0$ for all $A, B \in \Psi^{\mathbb{Z}}(M;E)$.*
4. *$\operatorname{Tr}_{\mathrm{R}}$ vanishes on Ψ^m for $m < -N$, in particular on $\Psi^{-\infty}$.*
5. *Let Q be admissible. Then $[A,Q(s)] = sM_A(s)$, where M_A is holomorphic of order $m-1-s$. The map $D_Q A := M_A(0) = \frac{d}{ds}[A,Q(s)]|_{s=0}$ is a derivation of order -1, and $\operatorname{Tr}_{\mathrm{R}}(D_Q A) = 0$.*
6. *Let $\widehat{\operatorname{Tr}}_Q(A)$ be the regular value of $\operatorname{Tr}(AQ(s))$ at $s=0$. Then $\widehat{\operatorname{Tr}}_Q([A,B]) = \operatorname{Tr}_{\mathrm{R}}(A\, D_Q B)$. If $Q' = Q + sF$ is another admissible family, then $\widehat{\operatorname{Tr}}_{Q'}(A) = \widehat{\operatorname{Tr}}_Q(A) + \operatorname{Tr}_{\mathrm{R}}(AF(0))$.*

Proof. Write $A(s) = \sum_i \phi_i A(s)\psi_i + \sum_i \phi_i A(s)(1-\psi_i)$. The second sum is a holomorphic smoothing family whose kernel vanishes near the diagonal. So its trace vanishes. By Definition C.4 and Lemma C.3(1), $\operatorname{Tr}(\phi_i A(s)\psi_i) = (2\pi)^{-N} \int \phi_i(u) \int a^{(i)}(s;u,\zeta)\, d\zeta\, du + \operatorname{Tr} R_i(s)$. Here $\operatorname{Tr} R_i(s)$ is entire, and we have used that $\psi_i = 1$ near $\operatorname{supp}\phi_i$. Lemma C.2 gives the continuation and the poles. At $s=0$ the residue is the integral of $a^{(i)}_{-N}(0)$, the term of homogeneity $-N$ of the symbol of $A(0)=A$. This proves (26) and (1).

(2) The left side of (26) is intrinsic. Apply it to ϕA with $\phi \in C^\infty_c(U_i \cap U_j)$ arbitrary. The two local densities then have the same integral against every ϕ, so they agree. In coordinates, $\omega^N/N! = \pm d\zeta \wedge du$. Contracting with $R = \zeta\cdot\partial_\zeta$ and restricting to $|\zeta|=1$ gives $\pm dS\, du$. For h homogeneous of degree $-N$, the form $\iota_R(h\, d\zeta)$ is closed on $\mathbb{R}^N \setminus 0$. So the fibre integral does not depend on the choice of section of $T^*M \setminus 0 \to S^*M$.

(3) Let $A \in \Psi^{m_1}$, $B \in \Psi^{m_2}$ and let Q be admissible. For $\operatorname{Re} s$ large, Lemma C.3(2) gives $\operatorname{Tr}([A,B]Q(s)) = \operatorname{Tr}(ABQ(s)) - \operatorname{Tr}(AQ(s)B) = \operatorname{Tr}(A[B,Q(s)])$. The family $[B,Q(s)]$ vanishes at $s=0$, so it equals $sM_B(s)$ by Lemma C.5. Hence $\operatorname{Tr}([A,B]Q(s)) = s\operatorname{Tr}(AM_B(s))$ has no pole at 0.

(4) There is no pole if $m < -N$.

(5) The principal symbol of $[A,Q(s)]$ of order $m-s$ is $[\sigma_m(A), q_s\mathrm{Id}] = 0$. By Lemma C.5(3),(2), $[A,Q(s)] = sM_A(s)$ with M_A of order $m-1-s$. Differentiating $[AB,Q] = [A,Q]B + A[B,Q]$ shows that D_Q is a derivation. Finally, $\operatorname{Tr}_{\mathrm{R}}(D_Q A) = \operatorname{Res}_{s=0}\operatorname{Tr}(M_A(s))$ by (26). For $\operatorname{Re} s$ large, $\operatorname{Tr}(M_A(s)) = s^{-1}\operatorname{Tr}([A,Q(s)]) = 0$.

(6) By the proof of (3), $\operatorname{Tr}([A,B]Q(s)) = s\operatorname{Tr}(AM_B(s))$. Its value at 0 is $\operatorname{Res}_{s=0}\operatorname{Tr}(AM_B(s)) = \operatorname{Tr}_{\mathrm{R}}(AM_B(0))$. For the last statement, $\operatorname{Tr}(AQ'(s)) = \operatorname{Tr}(AQ(s)) + s\operatorname{Tr}(AF(s))$, and $F(0)$ has order 0. □

Formally $Q(s) = e^{-s\log Q}$, so $\log Q = -Q'(0)$ and $D_Q A = [A, Q'(0)] = [\log Q, A]$. This is the derivation of [15, §6] and of Lemma 3.3. Part (6) is that lemma. The principal symbol of $D_Q A$ is $-i\{\log q, a\}$ with $\log q := -\partial_s q_s|_{s=0}$, at least when the full symbol of Q is scalar in the chosen trivializations. In general a matrix commutator term of lower order may be present [14, Ch. 15].

Remark C.8. Melrose regularizes with families $E(s)$ of order $+s$ and obtains $-\operatorname{Tr}_{\mathrm{R}}$ [14, Ch. 15]. With the convention of order $-s$ used here and in [15], the residue $\operatorname{Tr}_{\mathrm{R}}$ is *plus* the Wodzicki–Guillemin residue [21, 6].

Proposition C.9 (Uniqueness)**.** *Let M be connected and $N \geq 2$. Every trace τ on $\Psi^{\mathbb{Z}}(M)$ that vanishes on $\Psi^{-\infty}(M)$ is a multiple of $\operatorname{Tr}_{\mathrm{R}}$.*

Proof. Work in a coordinate ball U. For a symbol a with u-support in a compact set $K \subset U$, put $P_a = \operatorname{Op}(a)\psi$, with $\psi \in C_c^\infty(U)$ equal to 1 near K. Let $\tilde\psi \in C_c^\infty(U)$ be equal to 1 near $\operatorname{supp}\psi$. In the left quantization, $[u_k, \operatorname{Op}(a)] = i\operatorname{Op}(\partial_{\zeta_k} a)$ and $[D_{u_k}, \operatorname{Op}(a)] = \operatorname{Op}(D_{u_k} a)$ exactly. Hence

$$[\tilde\psi u_k, P_a] = iP_{\partial_{\zeta_k} a}, \qquad [\tilde\psi D_{u_k}, P_a] \equiv P_{D_{u_k} a} \pmod{\Psi^{-\infty}}. \tag{27}$$

The error in the second relation is $\operatorname{Op}(a)(D_{u_k}\psi)$. Its kernel vanishes near the diagonal, so it is smoothing.

Step 1: degrees $j \neq -N$. Let p_j be homogeneous of degree $j \in \mathbb{Z}$ with u-support in K, and let $\chi(\zeta)$ be equal to 0 near 0 and to 1 for $|\zeta| \geq 1$. By Euler's identity, $\sum_k \partial_{\zeta_k}(\zeta_k \chi p_j) = (j+N)\chi p_j + r$, where r has compact support in ζ. By (27), $P_{\chi p_j}$ is a sum of commutators modulo $\Psi^{-\infty}$, so $\tau(P_{\chi p_j}) = 0$. The same applies to symbols $q \in S^{-N-1}$. Write $q \sim \sum_{j \leq -N-1} q_j$ and let b_k be an asymptotic sum of the symbols $-i(j+N)^{-1}\zeta_k \chi q_j$. Then $P_q - \sum_k [\tilde\psi u_k, P_{b_k}] \in \Psi^{-\infty}$, so $\tau(P_q) = 0$.

Step 2: degree $-N$. Let p be homogeneous of degree $-N$ with u-support in K. Put $\varrho(u) = \int_{S^{N-1}} p(u,\theta)\, d\theta$. Since $N \geq 2$, the sphere is connected, and we can solve $\Delta_S w(u,\cdot) = p(u,\cdot) - \varrho(u)/|S^{N-1}|$ on S^{N-1}, smoothly in u and with compact support in u. Extend $\nabla_S w$ to a vector field g that is tangent to the spheres and homogeneous of degree $1-N$. Then $\operatorname{div}_\zeta g = |\zeta|^{-N}\Delta_S w$. So $p - \varrho(u)|\zeta|^{-N}/|S^{N-1}| = \sum_k \partial_{\zeta_k} g_k$. Next fix $\chi_0 \in C_c^\infty(U)$ with $\int \chi_0 = 1$ and put $\nu = \int_U \varrho\, du$. The function $\varrho - \nu\chi_0$ has compact support and integral 0 on the ball U, so it equals $\sum_k \partial_{u_k} b_k$ with $b_k \in C_c^\infty(U)$. By (27),

$$\tau(P_{\chi p}) = \nu\, \tau(P_{\chi g_0}), \qquad g_0 = \chi_0(u)|\zeta|^{-N}/|S^{N-1}|.$$

Since $\operatorname{Tr}_{\mathrm{R}}(P_{\chi p}) = (2\pi)^{-N}\nu$, this gives $\tau(P_{\chi p}) = c_U \operatorname{Tr}_{\mathrm{R}}(P_{\chi p})$ with $c_U = (2\pi)^N \tau(P_{\chi g_0})$.

Step 3: globalization. With a partition of unity and Steps 1–2, $\tau(A) = \sum_i c_{U_i} \operatorname{Tr}_{\mathrm{R}}(\phi_i A \psi_i)$. If $U_i \cap U_j \neq \emptyset$, choose an operator localized in a small ball inside $U_i \cap U_j$ with non-zero residue. This shows $c_{U_i} = c_{U_j}$. Since M is connected, all the constants agree. □

For $N = 1$ the sphere S^0 has two points. A homogeneous function of degree 0 on $\mathbb{R} \setminus 0$ has derivative 0, and there are two independent residues, one for each sign of ξ. This is Correction 10.9. Wodzicki shows that every trace vanishes on $\Psi^{-\infty}$ (quoted: [21]). We do not use this.

C.3 Holomorphic families in the cusp calculus: item (B3)

Fix a collar $[0,\epsilon)_x \times \partial X$ and put $t = 1/x$. We use (B1) in the following form (cf. Sections 4.3 and 4.4). Let $V \subset \mathbb{R}^{n-1}$ be a chart of ∂X over which E is trivial, and let ϕ, ψ be supported in $[0,\epsilon) \times V$, with $\psi = 1$ near $\operatorname{supp}\phi$. If $A \in x^p \Psi_c^m(X; E)$, then

$$\begin{aligned} \phi A \psi &= \phi\ \operatorname{Op}_c(x^p b)\, \psi + R, \\ \operatorname{Op}_c(a)u(t,y) &= (2\pi)^{-n} \int e^{i(t-t')\tau + i(y-y')\cdot\eta} a(1/t, y, \tau, \eta)\, u(t', y')\, dt'\, dy'\, d\tau\, d\eta. \end{aligned} \tag{28}$$

Here b is a classical symbol of order m in $\zeta = (\tau, \eta)$, with coefficients smooth in $(x, y) \in [0, \epsilon) \times V$, and $R \in x^p \Psi_c^{-\infty}(X; E)$. Interior charts have the usual description, and $A - \sum_i \phi_i A \psi_i \in x^p \Psi_c^{-\infty}$. The kernel of an element of $x^p \Psi_c^{-\infty}$ restricted to the diagonal is $x^p g\, x^{-2} |dx\, dy|$ with g smooth up to $x = 0$. The Taylor series (12) of A at the front face is computed from the Taylor series of $x^p b$ in x (powers of x on the left). The coefficient $A_{(j)} \in \Psi_{\mathrm{sus}}^m(\partial X)$ has local symbol equal to the coefficient of x^j in $x^p b$. Finally, if $C \in x^p \Psi_c^\mu$ with $\operatorname{Re} \mu < -n$ and $\operatorname{Re} z > 1 - p$, then Cx^z is of trace class and $\operatorname{Tr}(Cx^z) = \int_X x^z \operatorname{tr} K_C|_\Delta$ (the proof of Lemma C.3(1) applies).

Definition C.10. A family $A(w) \in x^p \Psi_c^{\mu(w)}(X; E)$, $w \in \Omega \subset \mathbb{C}^l$, is *holomorphic* if the following hold. In boundary charts (28) holds with $b = b(w)$ a holomorphic family of classical symbols of order $\mu(w)$, with coefficients smooth up to $x = 0$, and with $R = R(w)$ holomorphic in the Fréchet space $x^p \Psi_c^{-\infty}$. The analogous statement holds in interior charts. And $A(w) - \sum_i \phi_i A(w) \psi_i$ is holomorphic in $x^p \Psi_c^{-\infty}$. An *admissible family* is an entire family $Q(s) \in \Psi_c^{-s}(X; E)$ with $Q(0) = \mathrm{Id}$ and scalar principal symbol.

Lemma C.11. *1. Products of holomorphic families are holomorphic. Orders and weights add.*

2. *The division statements and part (3) of Lemma C.5 hold verbatim.*

3. *If $A(w)$ is holomorphic in $x^p \Psi_c^{\mu(w)}$, then $x^z A(w) x^{-z}$ is holomorphic in (w, z) in $x^p \Psi_c^{\mu(w)}$, and $\partial_z (x^z A x^{-z})|_{z=0} = [\log x, A] \in x^{p+1} \Psi_c^{\mu - 1}$.*

4. *Let $Q_0(w) \in \Psi_{\mathrm{sus}}^{\mu(w)}(\partial X)$ be holomorphic (Definition C.18). Let $\omega, \tilde{\omega} \in C_c^\infty([0, \epsilon))$ be equal to 1 near 0, with $\tilde{\omega} = 1$ on $\operatorname{supp} \omega$. Then $\tilde{\omega} Q_0(w) \omega$ (acting in (t, y)) is holomorphic in $\Psi_c^{\mu(w)}$, and its Taylor series at the front face is $Q_0(w)$.*

5. *All the families act holomorphically on $\dot{C}^\infty(X; E)$.*

Proof (sketch; (B1) with parameters). For (1), (4) and (5), the single-operator statements of (B1) hold with holomorphic dependence on parameters, as in Lemma C.5(1). For (4): $\tilde{\omega}(x') - 1$ and $\omega(x) - 1$ vanish to infinite order at the front face, where $x'/x \to 1$. For (2), the proof of Lemma C.5(2),(3) uses only the form of the symbols. For (3), the kernel of $x^z A x^{-z}$ is $(x/x')^z K_A$. Near the front face, $(x/x')^z = (t'/t)^z = (1 - xT)^z$ with $T = t - t'$. This is smooth and entire in z for $|xT| < 1$, and of polynomial growth in T elsewhere. The kernel is rapidly decreasing in T there. The z-derivative at 0 is $\log(x/x') K_A$, which is $[\log x, A]$. Its leading term is computed in Lemma 4.4(3). □

Proposition C.12 (An explicit admissible family)**.** *Let $\{\chi_i(y)\}$ be a partition of unity on ∂X subordinate to charts V_i over which E is trivial, and let $\tilde{\chi}_i = 1$ near $\operatorname{supp} \chi_i$. Let $\omega, \tilde{\omega}$ be as in Lemma C.11(4). Let $\{\phi_k\}$ be interior cut-offs with $\sum_i \omega \chi_i + \sum_k \phi_k = 1$, and let $\psi_k = 1$ near $\operatorname{supp} \phi_k$. Put*

$$Q(s) = \sum_i \omega \chi_i \operatorname{Op}_c\big(\langle \zeta \rangle^{-s}\big)\, \tilde{\omega} \tilde{\chi}_i + \sum_k \phi_k \operatorname{Op}\big(\langle \xi \rangle^{-s}\big) \psi_k.$$

Then Q is an admissible family in $\Psi_c^{-s}(X; E)$. It is translation invariant to all orders at ∂X: its Taylor series at the front face is

$$Q_0(s) = \sum_i \chi_i \operatorname{Op}_{t,y}\big(\langle (\tau, \eta) \rangle^{-s}\big)\, \tilde{\chi}_i \in \Psi_{\mathrm{sus}}^{-s}(\partial X; E), \qquad Q_0(0) = \mathrm{Id}.$$

Proof. The symbol $\langle \zeta \rangle^{-s} = \rho^s \cdot (\rho \langle \zeta \rangle)^{-s}$ is an entire family of order $-s$ that does not depend on (x, y). So each term is an entire family by (28) and Lemma C.11(4). Its principal symbol is the

positive scalar $\omega\sum_i \chi_i|\zeta|_{(i)}^{-s}+\sum_k \phi_k|\xi|_{(k)}^{-s}$. Since $\mathrm{Op}_c(1)=\mathrm{Id}$, we get $Q(0)=\sum_i \omega\chi_i\tilde\omega\tilde\chi_i+\sum_k \phi_k\psi_k=\mathrm{Id}$. The interior terms have kernels supported away from the front face. The Taylor series of the boundary terms is given by Lemma C.11(4). □

Proposition C.13 (The derivation D_Q)**.** *Let Q be admissible and $A\in x^p\Psi_c^m(X;E)$.*

1. *$[A,Q(s)]=sM_A(s)$, where M_A is holomorphic of order $m-1-s$ and weight x^p. Put*

$$D_QA:=M_A(0)=\frac{d}{ds}[A,Q(s)]\Big|_{s=0}=[\log Q,A],\qquad \log Q:=-Q'(0).$$

   Here $Q'(0)$ is an operator on $\dot C^\infty(X;E)$. It is not in the calculus.

2. *D_Q is a derivation of $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X;E)$. It maps $x^p\Psi_c^m$ into $x^p\Psi_c^{m-1}$ and preserves $\mathcal{I}$, $\mathcal{I}_\sigma$ and $\mathcal{I}_\partial$. For genuine complex powers it is the derivation $\frac{d}{ds}Q^sAQ^{-s}|_{s=0}$ of Section 11.*

3. *If Q has Taylor series $Q_0(s)$ at the front face, then $\mathrm{In}(D_QA)=D_{Q_0}\mathrm{In}(A)$, where $D_{Q_0}B=\frac{d}{ds}[B,Q_0(s)]|_{s=0}$ on $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(\partial X)$.*

4. *If $\tilde Q$ is another admissible family, then $\tilde Q-Q=sF(s)$ with $F(0)\in\Psi_c^0$. Hence*

$$D_{\tilde Q}=D_Q+[L,\cdot],\qquad L=-F(0)=:\log\tilde Q-\log Q\in\Psi_c^0(X;E),$$

   and the two derivations differ by an inner derivation.

5. *$Q(s)Q(u)-Q(s+u)=su\,W(s,u)$, where W is holomorphic of order $-(s+u)$. The operator*

$$\kappa_Q:=-W(0,0)=Q''(0)-Q'(0)^2\in\Psi_c^0(X;E)$$

   is in the calculus. For genuine complex powers $\kappa_Q=0$.

Proof. (1) and (2) are proved as Theorem C.7(5), using Lemma C.11(2). The ideals are preserved because $M_A(s)$ is a family of the same type as A. (3) Take Taylor series at the front face. (4) Use the division lemma. (5) Both $Q(s)Q(u)$ and $Q(s+u)$ are holomorphic families of order $-(s+u)$, and their difference vanishes on $\{s=0\}\cup\{u=0\}$. Apply $\partial_s\partial_u$ at $(0,0)$ as operators on $\dot C^\infty$ (Lemma C.11(5)). □

Theorem C.14 (Two-variable meromorphy)**.** *Let $m,p\in\mathbb{Z}$. Let $\Omega\subset\mathbb{C}^2$ be a connected open set containing points of $\{\mathrm{Re}\,s>m+n,\ \mathrm{Re}\,z>1-p\}$, and let $A(s,z)\in x^p\Psi_c^{m-s}(X;E)$ be holomorphic on Ω. Then $F(s,z)=\mathrm{Tr}(A(s,z)x^z)=\mathrm{Tr}(x^zA(s,z))$ extends meromorphically to Ω. Near every point it has the form $\sum_{\epsilon_1,\epsilon_2\in\{0,1\}}h_{\epsilon_1\epsilon_2}(s,z)\,(s-s_k)^{-\epsilon_1}(z-z_j)^{-\epsilon_2}$ with h holomorphic, $s_k=m+n-k$ and $z_j=1-p-j$, $k,j\in\mathbb{N}_0$. Thus F has at most simple poles on these lines, and szF is holomorphic near $(0,0)$. Moreover:*

1. *(Residue in s.) $\mathrm{Res}_{s=0}F(s,z)=(2\pi)^{-n}\int_{{}^cS^*X}x^z\,\mathrm{tr}\,a_{-n}(0,z)\,{}^c\nu$, where $a_{-n}(0,z)$ is the term of homogeneity $-n$ of the cusp symbol of $A(0,z)$, and ${}^c\nu$ is the density of Lemma 10.5 ($x^{-2}dx\,dy\,dS$ in boundary charts). It depends only on $A(0,\cdot)$. For $A(0,z)\in\mathcal{I}_\sigma$ it is $\mathrm{Tr}_\sigma(A(0,z)x^z)$.*

2. *(Residue in z.) $\mathrm{Res}_{z=0}F(s,z)$ depends only on $A(s,0)$. For $\mathrm{Re}\,s>m+n$ it equals*

$$\mathrm{Res}_{z=0}F(s,z)=\frac{1}{2\pi}\int_{\mathbb{R}}\mathrm{Tr}_{\partial X}\big(A(s,0)_{(1)}(\tau)\big)\,d\tau.$$

3. *(Double residue.) The coefficient of* $(sz)^{-1}$ *is* $\operatorname{Res}_{z=0}\operatorname{Res}_{s=0} F = \operatorname{Res}_{s=0}\operatorname{Res}_{z=0} F = \operatorname{Tr}_{\partial,\sigma}(A(0,0))$*, the functional of Lemma 10.6.*

Proof. For $\operatorname{Re} s > m+n$ and $\operatorname{Re} z > 1-p$, $F = \int_X x^z \operatorname{tr} K_A|_\Delta = \sum \int x^z \operatorname{tr} K_{\phi_i A \psi_i}|_\Delta$, because $A - \sum \phi_i A \psi_i$ has kernel vanishing near the diagonal. The interior terms are entire in z and are treated in s as in Theorem C.7. In a boundary chart, by (28) and $dt = x^{-2}dx$,

$$\int x^z \operatorname{tr} K_{\phi A \psi}|_\Delta = \int_0^\epsilon \int_V x^{z+p-2} D(s,z;x,y)\, dy\, dx,$$
$$D = (2\pi)^{-n} \int \phi \operatorname{tr} b(s,z;x,y,\zeta)\, d\zeta + g(s,z;x,y),$$

with g smooth up to $x=0$ and holomorphic. By Lemma C.2, $D = \sum_{k<K} d_k(s,z;x,y)/(s-s_k) + D^{(K)}$, with all terms smooth in (x,y) up to $x=0$, compactly supported, and holomorphic in (s,z). The term $D^{(K)}$ is holomorphic for $\operatorname{Re} s > m+n-K$. For a smooth compactly supported $h(x)$ depending holomorphically on parameters,

$$\int_0^\epsilon x^{z+p-2} h(x)\, dx = \sum_{j<J} \frac{h^{(j)}(0)}{j!\,(z-z_j)} + (\text{holomorphic for } \operatorname{Re} z > z_J).$$

Applying this to each term gives the local form of F.

(1) The residue at $s=0$ is the contribution of $k = m+n$. It is the integral over (x,y) of

$$x^{z+p-2}(2\pi)^{-n} \int_{S^{n-1}} \phi \operatorname{tr} b_{-n}(0,z)\, dS.$$

The term $b_{-n}(0,z)$ is the homogeneous term of degree $-n$ of the symbol of $A(0,z)$ (Lemma C.2).

(2) The pole at $z=0$ comes from $j = 1-p$. Its residue is the Taylor coefficient of x^{1-p} in $D(s,0)$, that is, the coefficient of x^1 in the diagonal density $x^p D$ of $A(s,0)$. For $\operatorname{Re} s > m+n$ the coefficient is $(2\pi)^{-n}\int (x^p b)_{(1)}\, d\zeta = (2\pi)^{-1}\int [(2\pi)^{1-n}\int (x^p b)_{(1)}\, d\eta]\, d\tau$, plus the corresponding coefficient of the residual part. By (B1) this is the value on the diagonal of the convolution kernel of $A(s,0)_{(1)}(\tau)$, integrated over τ. Integration over y gives the formula.

(3) In the local form, the coefficient of $(sz)^{-1}$ does not depend on the order of the residues. Taking the z-residue of (1) picks the coefficient of x^1 in $a_{-n}(0,0)$. This is the formula of Lemma 10.6. □

Proposition C.15 (Transfer to the functionals of MN)**.** *Let Q be admissible. Define* $Z(A;s,z) = \frac{1}{2}\big(\operatorname{Tr}(Ax^z Q(s)) + \operatorname{Tr}(AQ(s)x^z)\big)$*, the functionals* $\operatorname{Tr}_{\partial,\sigma}$, $\widehat{\operatorname{Tr}}_\partial$, $\widehat{\operatorname{Tr}}_\sigma$ *by* (17), $\widetilde{Z}(A;z) = Z(A;z,z)$, Bf *and* $\mathrm{IF} = \mathrm{IF}_Q$ *as in Sections 14–15, and* $[\log Q, \cdot\,] := D_Q$*. Then:*

1. *Lemmas 10.3, 10.4, 10.5, 10.6, 10.10, 10.11, 10.15, 11.1, 12.1, 15.2, Corollary 10.14(1), Proposition 14.2, and the closed-case Lemma 3.3 and Proposition 3.4 hold as stated. In Lemma 10.15,* $\log(Q'/Q)$ *is replaced by* $\log Q' - \log Q = -F(0)$*, where* $Q' = Q + sF$*.*

2. *Proposition 15.3 holds in the form*

$$\mathrm{IF}_Q(A,B) = \tfrac{1}{2}\big(\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma\big)\big(B\widetilde{D}(A) + \widetilde{D}(A)B\big) + \tfrac{1}{2}\operatorname{Tr}_{\partial,\sigma}\big([A,B]\kappa_Q\big).$$

The last term is the Hochschild coboundary of the 0-cochain $C \mapsto \frac{1}{2}\operatorname{Tr}_{\partial,\sigma}(C\kappa_Q)$ *on* $\mathcal{A}$*. It vanishes if* $[A,B] \in \mathcal{I}_\sigma + \mathcal{I}_\partial$*, in particular if* $[B] = [A]^{-1}$ *in* $\mathcal{A}$*. It vanishes identically for genuine complex powers. Hence Theorem 15.6 and its corollaries hold verbatim.*

3. *(Independence.)* $\operatorname{Tr}_{\partial,\sigma}$ *and* $\widehat{\operatorname{Tr}}_\sigma$ *do not depend on* Q. $\widehat{\operatorname{Tr}}_\partial$ *changes by* $-\operatorname{Tr}_{\partial,\sigma}(\,\cdot\, L)$ *with* $L = \log Q' - \log Q$, D_Q *changes by the inner derivation* $[L, \cdot\,]$, *and* IF_Q *changes by a coboundary. Hence all Hochschild classes and the index are independent of the family.*

The notes use the inverse powers Q^s at five places: Lemmas 10.3, 10.4, 11.1, Proposition 14.2 and Proposition 15.3. The proof replaces each conjugation by $Q^{\pm s}$ with a division identity. We need two lemmas.

Lemma C.16. *Let* $E(z) \in x^q\Psi_c^{k-z}$ *be a holomorphic family near* 0, *and put* $E^\flat = \frac{d}{dz}\big(E(z) - E(0)Q(z)\big)|_{z=0} \in x^q\Psi_c^k$. *Then*

$$z^2 \operatorname{Tr}(E(z)x^z) = \operatorname{Tr}_{\partial,\sigma}(E(0)) + z\Big(\big(\widehat{\operatorname{Tr}}_\partial + \widehat{\operatorname{Tr}}_\sigma\big)(E(0)) + \operatorname{Tr}_{\partial,\sigma}(E^\flat)\Big) + O(z^2).$$

Proof. Write $E(z) = E(0)Q(z) + zE_1(z)$, so that $E_1(0) = E^\flat$. Then $\operatorname{Tr}(E(z)x^z) = \operatorname{Tr}(E(0)Q(z)x^z) + z\operatorname{Tr}(E_1(z)x^z)$. The first term is $Z^R(E(0); z, z)$. By Lemma 10.3 it differs from $Z(E(0); z, z)$ by $\frac{1}{2}z^2\operatorname{Tr}(E(0)P_1(z,z)x^z)$, and z^2 times this is $O(z^2)$. Then use (17) with $s = z$. For the second term, $z^2\operatorname{Tr}(E_1(z)x^z) = \operatorname{Tr}_{\partial,\sigma}(E_1(0)) + O(z)$ by Theorem C.14(3), applied to the two-variable family $(s, z) \mapsto E_1(s)$ and restricted to the diagonal. □

Lemma C.17. *Let* $\mathcal{Q}(z) \in \Psi_c^{-z}$ *be holomorphic with* $\mathcal{Q}(0) = \mathrm{Id}$. *Put* $L(z) = z^{-1}(x^zAx^{-z} - A)$ *and* $N(z) = z^{-1}[A, \mathcal{Q}(z)]$. *Then*

$$\operatorname{Tr}\big([A,B]\mathcal{Q}(z)x^z\big) = z\operatorname{Tr}\big(E(z)x^z\big), \qquad E = B\big(\mathcal{Q}L - N\big), \qquad E(0) = B\widetilde{D}(A),$$

where $\widetilde{D} = [\log x, \cdot\,] - D_\mathcal{Q}$ *and* $D_\mathcal{Q}A = N(0)$.

Proof. For $\operatorname{Re} z$ large, $\operatorname{Tr}([A,B]\mathcal{Q}x^z) = \operatorname{Tr}(B[\mathcal{Q}x^z, A])$ by the trace property. Moreover $[\mathcal{Q}x^z, A] = \mathcal{Q}[x^z, A] + [\mathcal{Q}, A]x^z = z(\mathcal{Q}L - N)x^z$. □

Proof of Proposition C.15. (1) *Lemma 10.3.* The difference is $Z^R - Z^L = \operatorname{Tr}(A[Q(s), x^z])$, and $[Q(s), x^z] = P(s,z)x^z$ with $P = Q(s) - x^zQ(s)x^{-z}$. The family P vanishes for $z = 0$, and also for $s = 0$ because $Q(0) = \mathrm{Id}$. So $P = sz\,P_1$ by Lemma C.11(2), and $Z^R - Z^L = sz\operatorname{Tr}(AP_1x^z)$. The rest of the proof is verbatim.

Lemma 10.4. We have $Z^L([A,B]) = \operatorname{Tr}(A[B, x^zQ(s)])$ and $[B, x^zQ(s)] = x^zK(s,z)$ with $K = (x^{-z}Bx^z)Q(s) - Q(s)B$. Then $K(0,0) = 0$, so $K = sK_1 + zK_2$, and $sz\,Z^L([A,B]) = s\cdot sz\operatorname{Tr}(C_1x^z) + z\cdot sz\operatorname{Tr}(C_2x^z)$ with $C_i = Ax^zK_ix^{-z}$. Both terms vanish at $(0,0)$ by Theorem C.14. For a change of family use $Q' = Q + sF$ in place of $Q'^{-s} = (1 + sC(s))Q^{-s}$. Vanishing on $\mathcal{I}_\sigma + \mathcal{I}_\partial$ follows from Theorem C.14(1),(2).

Lemmas 10.5, 10.6, 10.10, 10.11 and Corollary 10.14(1). These are Theorem C.14(1)–(3) together with the locality of the residues. Independence of Q holds because a residue at $s = 0$ depends only on the value of the family at $s = 0$. Theorem C.7 identifies Tr_σ with Wodzicki's residue on the double $2X$.

Lemma 10.15. We have $sz\,Z(A; Q') = sz\,Z(A; Q) + s\cdot sz\,Z(AF(s))$. Comparing the coefficients of s gives $\widehat{\operatorname{Tr}}_\partial(A; Q') = \widehat{\operatorname{Tr}}_\partial(A; Q) + \operatorname{Tr}_{\partial,\sigma}(AF(0))$.

Lemma 11.1. Write $[B, Q(s)] = sM_B(s)$. Then $M_B(0) = [B, Q'(0)] = [\log Q, B] = D_QB$. By Theorem C.14(3), $\operatorname{Tr}_{\partial,\sigma}(D_QB)$ is the double residue of $\operatorname{Tr}(M_B(s)x^z)$. For $\operatorname{Re} s$ large, $s\operatorname{Tr}(M_B(s)x^z) = \operatorname{Tr}(BQ(s)x^z) - \operatorname{Tr}(Bx^zQ(s)) = \operatorname{Tr}(B[Q(s), x^z]) = sz\operatorname{Tr}(BP_1(s,z)x^z)$. So $sz\operatorname{Tr}(M_Bx^z) = z\cdot sz\operatorname{Tr}(BP_1x^z)$, which vanishes at $(0,0)$. For $[\log x, B]$, write $x^wBx^{-w} = B + wL_B(w)$. Then $w\operatorname{Tr}(L_B(w)x^zQ(s)) = \operatorname{Tr}(Bx^z(x^{-w}Q(s)x^w - Q(s)))$, and $x^{-w}Q(s)x^w - Q(s) = wsP'(w,s)$, because

it vanishes at $w = 0$ and at $s = 0$. So $sz\,\mathrm{Tr}(L_B(0)x^zQ(s)) = s\cdot sz\,\mathrm{Tr}(Bx^zP'(0,s))$, and the double residue of $L_B(0) = [\log x, B]$ vanishes.

Lemma 12.1. In the first line, $\mathrm{Tr}_\partial([A,B]Q(s)) = \mathrm{Tr}_\partial(A[B,Q(s)]) = s\,\mathrm{Tr}_\partial(AM_B(s))$, with M_B as in the previous paragraph. By Theorem C.14(3) its value at $s = 0$ is $\mathrm{Tr}_{\partial,\sigma}(AM_B(0)) = \mathrm{Tr}_{\partial,\sigma}(A[\log Q, B])$. The second line does not involve Q.

Proposition 14.2 and Lemma 15.2. Note that $\mathrm{Tr}([A,B]x^zQ(z)) = \mathrm{Tr}([A,B]\mathcal{Q}_L(z)x^z)$ with $\mathcal{Q}_L = x^zQ(z)x^{-z}$, and that $D_{\mathcal{Q}_L} = D_Q$ because $\mathcal{Q}_L'(0) = Q'(0)$. Apply Lemma C.17 with $\mathcal{Q} = \mathcal{Q}_L$ and with $\mathcal{Q} = Q$, and then Lemma C.16. This gives

$$\begin{aligned}2\widetilde{Z}([A,B];z) &= \sum_{\mathcal{Q}\in\{\mathcal{Q}_L,Q\}} z\,\mathrm{Tr}\big(E_{\mathcal{Q}}(z)x^z\big)\\ &= \frac{2\,\mathrm{Tr}_{\partial,\sigma}(B\widetilde{D}A)}{z} + \sum_{\mathcal{Q}}\Big((\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(B\widetilde{D}A) + \mathrm{Tr}_{\partial,\sigma}(E^\flat_{\mathcal{Q}})\Big) + O(z).\end{aligned}$$

So $\mathrm{Bf}(A,B) = \mathrm{Tr}_{\partial,\sigma}(B\widetilde{D}A)$, as in Proposition 14.2. Lemma 15.2 only uses that $\widetilde{Z}(C;\cdot)$ is entire with value $\mathrm{Tr}\,C$ at 0 for $C \in \mathcal{I}$. This holds because $Q(0) = \mathrm{Id}$.

(2) By the display above, $\mathrm{IF}_Q(A,B) = (\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(B\widetilde{D}A) + \frac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}(E^\flat_{\mathcal{Q}_L} + E^\flat_Q)$. We compute $E^\flat = E'(0) + E(0)\ell$, with $\ell = \log Q = -Q'(0)$, as an identity of operators on $\dot{C}^\infty$. Write $\mathrm{ad}_x = [\log x, \cdot\,]$, $\mathrm{ad}_\ell = [\ell, \cdot\,]$ and $X = \mathrm{ad}_x A$. Then $L(0) = X$, $L'(0) = \frac{1}{2}\,\mathrm{ad}_x^2 A$, $N(0) = \mathrm{ad}_\ell A$ and $N'(0) = \frac{1}{2}[A, \mathcal{Q}''(0)]$. By Proposition C.13(5), $\mathcal{Q}''(0) = \ell^2 + \kappa_Q$ and $\mathcal{Q}_L''(0) = \ell^2 + \kappa_Q - 2[\log x, \ell]$. A direct computation gives

$$E^\flat_Q = B\Big(\tfrac{1}{2}\,\mathrm{ad}_x^2 A - \mathrm{ad}_\ell\,\mathrm{ad}_x A + \tfrac{1}{2}\,\mathrm{ad}_\ell^2 A - \tfrac{1}{2}[A,\kappa_Q]\Big), \qquad E^\flat_{\mathcal{Q}_L} = E^\flat_Q + B\big[A,[\log x,\ell]\big].$$

Here we used $-\frac{1}{2}[A,\ell^2] - (\mathrm{ad}_\ell A)\ell = \frac{1}{2}\,\mathrm{ad}_\ell^2 A$. By the Jacobi identity, $\mathrm{ad}_x\,\mathrm{ad}_\ell - \mathrm{ad}_\ell\,\mathrm{ad}_x = \mathrm{ad}_{[\log x,\ell]}$, and $\widetilde{D} = \mathrm{ad}_x - \mathrm{ad}_\ell$. Hence

$$E^\flat_Q + E^\flat_{\mathcal{Q}_L} = B\big(\widetilde{D}^2 A - [A,\kappa_Q]\big).$$

By Lemma 11.1 and the trace property, $\mathrm{Tr}_{\partial,\sigma}(B\widetilde{D}^2A) = -\,\mathrm{Tr}_{\partial,\sigma}(\widetilde{D}B\,\widetilde{D}A)$. By Lemma 12.1, $\frac{1}{2}(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)([\widetilde{D}A, B]) = -\frac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}(\widetilde{D}A\,\widetilde{D}B)$. Combining,

$$\mathrm{IF}_Q(A,B) = \tfrac{1}{2}(\widehat{\mathrm{Tr}}_\partial + \widehat{\mathrm{Tr}}_\sigma)(B\widetilde{D}A + \widetilde{D}A\,B) - \tfrac{1}{2}\,\mathrm{Tr}_{\partial,\sigma}(B[A,\kappa_Q]),$$

and $-\,\mathrm{Tr}_{\partial,\sigma}(B[A,\kappa_Q]) = \mathrm{Tr}_{\partial,\sigma}([A,B]\kappa_Q)$. The vanishing statements hold because $\mathrm{Tr}_{\partial,\sigma}$ vanishes on $\mathcal{I}_\sigma + \mathcal{I}_\partial \supset \mathcal{I}$.

(3) This follows from (1), (2), Proposition C.13(4) and Lemma 5.3. □

C.4 The regularized trace on the suspended algebra: item (B7)

Let H be a closed manifold of dimension d. In the applications $H = \partial X$ and $d = n-1$.

Definition C.18. For $\mu \in \mathbb{C}$, $\Psi^\mu_{\mathrm{sus}}(H;E)$ is the space of families $\mathbb{R} \ni \tau \mapsto C(\tau) \in \Psi^\mu(H;E)$ of the following form. For a partition of unity ϕ_i subordinate to charts and ψ_i as before,

$$C(\tau) - \sum_i \phi_i\,\mathrm{Op}_y\big(c^{(i)}(\cdot,\tau,\cdot)\big)\psi_i \in \mathcal{S}\big(\mathbb{R};\Psi^{-\infty}(H;E)\big),$$

where each $c^{(i)}(y,\tau,\eta)$ is a classical symbol of order μ in $\zeta = (\tau,\eta) \in \mathbb{R}^{1+d}$. Holomorphic families are defined as in Definition C.4.

By (10) these are the indicial families of translation-invariant operators on $\mathbb{R}_t \times H$ with rapidly decreasing convolution kernels. Composition is pointwise in τ, and $\operatorname{In}([t, C]) = -i\partial_\tau C$. We quote from (B1) and [12]: $\Psi^{\mathbb{Z}}_{\mathrm{sus}}$ is an algebra; ∂_τ maps $\Psi^{\mu}_{\mathrm{sus}}$ to $\Psi^{\mu-1}_{\mathrm{sus}}$; and if $A \in \Psi^{m}_{\mathrm{sus}}$ is elliptic in (τ, η) and $A(\tau)$ is invertible for all τ, then $A^{-1} \in \Psi^{-m}_{\mathrm{sus}}$. For fixed τ, $C(\tau) \in \Psi^{\mu}(H)$.

Lemma C.19. *Let $C(s) \in \Psi^{\mu(s)}_{\mathrm{sus}}(H; E)$ be holomorphic on Ω with $\operatorname{Re}\mu < -d$. Then $f(\tau; s) = \operatorname{Tr} C(s)(\tau)$ is a holomorphic family of classical symbols on $\mathbb{R}$ of order $\mu(s) + d$:*

$$f(\tau; s) \sim \sum_{j \geq 0} f_j^{\pm}(s) |\tau|^{\mu(s)+d-j}, \quad \tau \to \pm\infty,$$

$$f_j^{\pm}(s) = (2\pi)^{-d} \sum_i \iint \phi_i \operatorname{tr} c^{(i)}_{\mu(s)-j}(s; y, \pm 1, \omega)\, d\omega\, dy.$$

The remainders are $O(|\tau|^{\operatorname{Re}\mu+d-J})$, locally uniformly in s.

Proof. By Lemma C.3, $f = (2\pi)^{-d} \sum_i \iint \phi_i \operatorname{tr} c^{(i)}(s; y, \tau, \eta)\, d\eta\, dy$ plus a Schwartz function of τ. Split $c = \sum_{j<J}(1 - \chi(\zeta)) c_{\mu-j} + r_J$, where χ is supported in $|\zeta| < \frac{1}{2}$. For $|\tau| \geq 1$ substitute $\eta = |\tau|\omega$. By homogeneity, $\int c_{\mu-j}(y, \tau, \eta)\, d\eta = |\tau|^{\mu-j+d} \int c_{\mu-j}(y, \pm 1, \omega)\, d\omega$. This integral converges since $\operatorname{Re}(\mu - j) < -d$. Finally $\int |r_J|\, d\eta \leq C \int \langle (\tau, \eta) \rangle^{\operatorname{Re}\mu - J} d\eta \leq C' \langle \tau \rangle^{\operatorname{Re}\mu - J + d}$. □

Regularized integrals. Let $\mathcal{P}$ be the space of $u \in C^\infty(\mathbb{R})$ such that, for every N, $u(\tau) = \sum c^{\pm}_{\gamma,l} |\tau|^{\gamma} (\log|\tau|)^l + O(|\tau|^{-N})$ for $\pm\tau \geq 1$. The sum is finite, over $\operatorname{Re}\gamma > -N$ and $l \in \mathbb{N}_0$. Using $\int_1^T \tau^\gamma (\log\tau)^l\, d\tau$, one sees that for $u \in \mathcal{P}$,

$$\int_{-T}^{T} u\, d\tau = \sum d_{\gamma,l} T^\gamma (\log T)^l + c + o(1), \qquad T \to \infty,$$

with a finite sum over $\operatorname{Re}\gamma \geq 0$, $(\gamma, l) \neq (0, 0)$. We put $\overline{\int} u\, d\tau := c$.

Lemma C.20. *1. $\overline{\int}$ is linear. It equals $\int$ on integrable elements of $\mathcal{P}$. It vanishes on polynomials and on odd functions.*

2. $\mathcal{P}$ is closed under multiplication by τ and under $I : u \mapsto \int_0^\tau u$. Classical symbols of any order lie in $\mathcal{P}$. The negative powers in the expansion of Iu are obtained by integrating the expansion of u term by term.

3. Let $u \in C^\infty(\mathbb{R})$ with $u^{(k)} \in \mathcal{P}$ for $0 \leq k \leq p$. Let $e_k^{\pm}$ be the coefficient of $|\tau|^{-1-k}$ (without logarithm) in $u^{(k)}$ at $\pm\infty$. Then

$$\overline{\int} u\, d\tau = \frac{(-1)^p}{p!} \overline{\int} \tau^p u^{(p)}\, d\tau + \sum_{k=0}^{p-1} \frac{(-1)^k}{(k+1)!} \left(e_k^+ + (-1)^k e_k^-\right).$$

Proof. (1) and (2) are immediate from the definitions. For the last part of (2), $\int_0^\tau = \int_0^1 + \int_1^\tau$, and a constant of integration affects only the power τ^0. For (3), integrate by parts p times on $[-T, T]$:

$$\int_{-T}^{T} u = \sum_{k<p} \frac{(-1)^k}{(k+1)!} \left[\tau^{k+1} u^{(k)}\right]_{-T}^{T} + \frac{(-1)^p}{p!} \int_{-T}^{T} \tau^p u^{(p)}.$$

Now take constant terms. The constant term of $T^{k+1} u^{(k)}(T)$ is e_k^+. That of $(-T)^{k+1} u^{(k)}(-T)$ is $(-1)^{k+1} e_k^-$. □

Lemma C.21 (Finite parts of holomorphic families)**.** *Let $f(\cdot;s)$ be a holomorphic family of classical symbols on $\mathbb{R}$ of order $\mu_0 - s$, $\mu_0 \in \mathbb{Z}$, on a connected open set $\Omega \ni 0$ meeting $\{\operatorname{Re} s > \mu_0 + 1\}$. Then $\int f(\tau;s)\,d\tau$ extends meromorphically to Ω, with at most simple poles at $s \in \mu_0 + 1 - \mathbb{N}_0$. Let $j_0 = \mu_0 + 1$ and $e(s) = f_{j_0}^+(s) + f_{j_0}^-(s)$, with $e = 0$ if $j_0 < 0$. Then*

$$\operatorname{Res}_{s=0} \int f\,d\tau = e(0), \qquad \operatorname{FP}_{s=0} \int f\,d\tau = \overline{\int} f(\tau;0)\,d\tau + e'(0).$$

Proof. Write $f = \sum_{j<J} \chi_\pm f_j^\pm(s)|\tau|^{\mu_0 - s - j} + r_J$, where $\chi_\pm$ is the indicator of $\{\pm\tau \geq 1\}$ and r_J is holomorphic with values in L^1 for $J > \operatorname{Re}\mu_0 + 2$. Since $\int_1^\infty \tau^{\mu_0 - s - j}\,d\tau = (s - (\mu_0 + 1 - j))^{-1}$, the continuation is

$$\int_{-1}^1 f + \sum_{j<J,\pm} \frac{f_j^\pm(s)}{s - (\mu_0 + 1 - j)} + \int \Big(r_J - \chi_{[-1,1]} \sum_{j<J} (\dots) \Big).$$

Here the last integral is holomorphic. At $s = 0$ the term j_0 contributes $e(s)/s = e(0)/s + e'(0) + O(s)$. The other terms are holomorphic near 0. On the other hand, $\overline{\int} f(\cdot;0)$ is given by the same expression at $s = 0$, with the j_0 term replaced by 0, because $\int_1^T \tau^{-1} d\tau = \log T$ has constant term 0. For $j \neq j_0$, $\int_1^T \tau^{\mu_0 - j}$ has constant term $-(\mu_0 - j + 1)^{-1}$, which is the value at $s = 0$ of the continued term. □

Definition C.22 (Melrose's regularized trace)**.** For $C \in \Psi_{\mathrm{sus}}^m(H;E)$, $m \in \mathbb{Z}$, choose $p \in \mathbb{N}_0$ with $p \geq m + d + 1$ and put $h_p(\tau) = \operatorname{Tr}\big(\partial_\tau^p C(\tau)\big)$. This is a classical symbol of order $m + d - p < 0$ by Lemma C.19. Define

$$\overline{\operatorname{Tr}}(C) = \frac{1}{2\pi} \overline{\int} I^p h_p\,d\tau.$$

This is the definition [15, (58)–(59)]: $I^p h_p$ is the p-fold integral from 0, and $\overline{\int}$ takes the constant term of $\int_{-T}^T$.

Theorem C.23 (The trace property, (B7))**.** *1. $\overline{\operatorname{Tr}}(C)$ does not depend on p. If $m < -d - 1$, then $\overline{\operatorname{Tr}}(C) = (2\pi)^{-1} \int \operatorname{Tr} C(\tau)\,d\tau$. In particular $\overline{\operatorname{Tr}}$ is the functional of Lemma 10.11 on $\Psi_{\mathrm{sus}}^{-\infty}$.*

2. If $C(-\tau) = -C(\tau)$, then $\overline{\operatorname{Tr}}(C) = 0$.

3. $\overline{\operatorname{Tr}}([A,B]) = 0$ for all $A, B \in \Psi_{\mathrm{sus}}^{\mathbb{Z}}(H;E)$. Thus $\overline{\operatorname{Tr}}$ is a trace on $\Psi_{\mathrm{sus}}^{\mathbb{Z}}(H;E)$.

Proof. (1) We have $h_{p+1} = h_p'$, so $I^{p+1} h_{p+1} = I^p(h_p - h_p(0)) = I^p h_p - h_p(0)\tau^p/p!$. A polynomial has $\overline{\int} = 0$. If $m < -d - 1$, then $\operatorname{Tr} C(\tau)$ is integrable and differs from $I^p h_p$ by its Taylor polynomial of degree $p - 1$ at 0.

(2) Then h_p has parity $(-1)^{p+1}$. Each integration from 0 reverses the parity, so $I^p h_p$ is odd, and $\int_{-T}^T$ of it vanishes.

(3) Let $A \in \Psi_{\mathrm{sus}}^{m_1}$ and $B \in \Psi_{\mathrm{sus}}^{m_2}$, and take $p > m_1 + m_2 + d$. Since composition is pointwise in τ,

$$\partial_\tau^p [A,B](\tau) = \sum_{l=0}^p \binom{p}{l} \big[\partial_\tau^l A(\tau), \partial_\tau^{p-l} B(\tau)\big].$$

Each commutator is a commutator of operators on H of orders adding to $m_1 + m_2 - p < -d$. By Lemma C.3(2) its trace vanishes. So $h_p \equiv 0$ and $\overline{\operatorname{Tr}}([A,B]) = 0$. □

The *suspended residue* of $C \in \Psi^m_{\mathrm{sus}}(H;E)$ is the Wodzicki residue of C, regarded as a translation-invariant operator on $\mathbb{R}\times H$, per unit length in t:

$$\widetilde{\mathrm{Tr}}(C) = (2\pi)^{-(d+1)} \sum_i \int \phi_i \int_{S^d} \operatorname{tr} c^{(i)}_{-1-d}(y,\zeta)\, dS(\zeta)\, dy.$$

Theorem C.24 (Holomorphic regularization of $\overline{\mathrm{Tr}}$)**.** *Let $C(s) \in \Psi^{m-s}_{\mathrm{sus}}(H;E)$, $m \in \mathbb{Z}$, be holomorphic for $\operatorname{Re} s > -\frac{1}{2}$. For instance $C(s) = CQ_0(s)$ with Q_0 a holomorphic suspended family of order $-s$ and $Q_0(0) = \mathrm{Id}$. Put*

$$c^{\pm}(s) = (2\pi)^{-d} \sum_i \iint \phi_i \operatorname{tr} c^{(i)}_{-1-d-s}(s;y,\pm 1,\omega)\, d\omega\, dy, \qquad \gamma(s) = c^+(s) + c^-(s),$$

where $c_{-1-d-s}(s)$ is the homogeneous term of degree $-1-d-s$ of the symbol of $C(s)$. Then $G(s) = (2\pi)^{-1}\int \operatorname{Tr} C(s)(\tau)\, d\tau$, defined for $\operatorname{Re} s > m+d+1$, extends meromorphically to $\operatorname{Re} s > -\frac{1}{2}$. Moreover

$$\operatorname{Res}_{s=0} G = \frac{\gamma(0)}{2\pi} = \widetilde{\mathrm{Tr}}(C(0)), \qquad \mathrm{FP}_{s=0} G = \overline{\mathrm{Tr}}(C(0)) + \frac{\gamma'(0)}{2\pi}.$$

The correction $\gamma'(0)$ is a local functional of the symbols of $C(s)$ near $s=0$.

Proof. Fix $p \geq m+d+2$. On $\operatorname{Re} s > -\frac{1}{2}$ the family $\partial_\tau^p C(s)$ has order with real part $< -d$. By Lemma C.19, $h_p(\cdot;s) = \operatorname{Tr} \partial_\tau^p C(s)$ is a holomorphic family of classical symbols of order $m-s+d-p$. Its homogeneous term of degree $-1-d-s-p$ is $\partial_\tau^p c_{-1-d-s}$. Differentiation under the integral sign is legitimate since $\operatorname{Re} s > -1$. Hence the coefficient $a^{\pm}(s)$ of $|\tau|^{-1-p-s}$ in h_p is the coefficient produced by $\partial_\tau^p(c^{\pm}(s)|\tau|^{-1-s})$:

$$a^+(s) = (-1)^p (1+s)_p c^+(s), \qquad a^-(s) = (1+s)_p c^-(s), \qquad (1+s)_p = \prod_{k=1}^{p}(k+s).$$

For $\operatorname{Re} s > m+d+1$ and s outside a discrete set, the function $f_s = \operatorname{Tr} C(s)$ is integrable and $f_s - I^p h_p(\cdot;s)$ is a polynomial. By Lemma C.20(3) (with no boundary constants, since no exponent equals $-1-k$),

$$G(s) = \frac{1}{2\pi} \frac{(-1)^p}{p!} \overline{\int} \tau^p h_p(\tau;s)\, d\tau.$$

The family $\tau^p h_p(\cdot;s)$ is a holomorphic family of classical symbols of order $m+d-s$. The coefficient of $|\tau|^{-1-s}$ in it is $\epsilon(s) = a^+(s) + (-1)^p a^-(s) = (-1)^p (1+s)_p \gamma(s)$. By Lemma C.21,

$$\operatorname{Res}_{s=0} G = \frac{(1+s)_p \gamma(s)}{2\pi\, p!}\Big|_{s=0} = \frac{\gamma(0)}{2\pi},$$

$$\mathrm{FP}_{s=0} G = \frac{1}{2\pi}\Big(\frac{(-1)^p}{p!} \overline{\int} \tau^p h_p(\tau;0)\, d\tau + H_p \gamma(0) + \gamma'(0)\Big),$$

where $H_p = \sum_{k=1}^{p} \frac{1}{k}$ is the derivative of $(1+s)_p/p!$ at $s=0$. Now evaluate $\overline{\mathrm{Tr}}(C(0)) = (2\pi)^{-1}\overline{\int} I^p h_p(\cdot;0)$ with Lemma C.20(3), taking $u = I^p h_p$. The coefficient of $|\tau|^{-1}$ in u is $b^+ = (-1)^p a^+(0)/p!$ at $+\infty$ and $b^- = a^-(0)/p!$ at $-\infty$. Then $e_k^+ = (-1)^k k!\, b^+$ and $e_k^- = k!\, b^-$. The boundary sum is

$$H_p(b^+ + b^-) = H_p \tfrac{(-1)^p}{p!} \epsilon(0) = H_p \gamma(0).$$

So $\mathrm{FP}_{s=0} G - \overline{\mathrm{Tr}}(C(0)) = \gamma'(0)/2\pi$.

Finally, let g be homogeneous of degree $-(d+1)$ on $\mathbb{R}^{d+1}\setminus 0$. The substitution $\zeta = (\tau, \tau\omega)$ in $\int_{1\leq|\zeta|\leq e} g\, d\zeta = \int_{S^d} g\, dS$ gives $\int_{S^d} g\, dS = \int g(1,\omega)\, d\omega + \int g(-1,\omega)\, d\omega$. Hence $\gamma(0) = 2\pi\, \widetilde{\mathrm{Tr}}(C(0))$. □

Example C.25 (FP $\neq \overline{\mathrm{Tr}}$)**.** Let H be a point, $C = \langle\tau\rangle$ (the indicial family of $\langle D_t\rangle = Q_0$), and $Q_0(s) = \langle\tau\rangle^{-s}$. Then $\int_{\mathbb{R}}\langle\tau\rangle^{1-s}\,d\tau = \sqrt{\pi}\,\Gamma(\frac{s-2}{2})/\Gamma(\frac{s-1}{2}) = \frac{1}{s} + \log 2 - \frac{1}{2} + O(s)$. On the other hand, $\int_{-T}^{T}\langle\tau\rangle\,d\tau = T^2 + \log T + \log 2 + \frac{1}{2} + o(1)$. So

$$\mathrm{FP}_{s=0}\,\frac{1}{2\pi}\int\langle\tau\rangle^{1-s}d\tau = \frac{\log 2 - \frac{1}{2}}{2\pi} \neq \frac{\log 2 + \frac{1}{2}}{2\pi} = \overline{\mathrm{Tr}}(\langle\tau\rangle).$$

This agrees with Theorem C.24. The coefficient of $|\tau|^{-1-s}$ in $\langle\tau\rangle^{1-s}$ is $\frac{1-s}{2}$ at each end, so $\gamma(s) = 1-s$, $\gamma(0)/2\pi = 1/2\pi$ is the residue, and the correction is $\gamma'(0)/2\pi = -1/2\pi$.

Correction to MN C.26 (MN Proposition 8, Corollary 2 and Proposition 17)*.* The residue statement of [15, Prop. 8] (Proposition 10.12) is correct. The claim that the regular value of $\overline{\mathrm{Tr}}(BQ_0^{-s})$ at $s = 0$ is $\overline{\mathrm{Tr}}(B)$ is false in general (Example C.25). The correct statement is Theorem C.24 with $C(s) = BQ_0(s)$. Correspondingly, Corollary 10.14(2) reads

$$\widehat{\mathrm{Tr}}_\partial(A) = \overline{\mathrm{Tr}}\big(A_{(1)}\big) + \frac{\gamma'_{A_{(1)}}(0)}{2\pi},$$

where $\gamma_{A_{(1)}}$ is computed from $A_{(1)}Q_0(s)$. Here Q is translation invariant to all orders at ∂X, with Taylor series $Q_0(s)$. Indeed, by Lemma 10.3 and Theorem C.14(2), $\widehat{\mathrm{Tr}}_\partial(A)$ is the regular value at $s = 0$ of $(2\pi)^{-1}\int \mathrm{Tr}\big(A_{(1)}Q_0(s)\big)\,d\tau$, because $(AQ(s))_{(1)} = A_{(1)}Q_0(s)$. Proposition 15.8 then holds with a local correction; see Proposition C.30.

Remark C.27 (Parity (sketch))*.* Say that a symbol of order μ has *parity type* ε if $c_{\mu-k}(y,-\zeta) = \varepsilon(-1)^k c_{\mu-k}(y,\zeta)$ for all k. Differential operators of order m, their parametrices, ∂_τ of them, and the family $\langle\zeta\rangle^{-s}$ have this property. The types are $(-1)^m$, $(-1)^m$, $(-1)^{m-1}$ and $+1$. The property is preserved by composition and by changes of coordinates on H, since the terms of the composition and coordinate-change formulas are ∂_ζ^α of symbols multiplied by polynomials in ζ of matching parity. For $C = A_0^{-1}[t, A_0]$ with A_0 the indicial family of a differential operator, $C\,Q_0(s)$ has type -1. Its term of degree $-1-d-s$ has $k = d$, so $c^+(s) + c^-(s) = (1 + (-1)^{d+1})\int c(1,\omega)\,d\omega$. This vanishes when d is even, that is, when $n = \dim X$ is *odd*. In that case $\gamma \equiv 0$. In the APS case, n even, parity alone does not kill γ.

C.5 The eta invariant and Dirac operators

Definition C.28. Let $A \in \Psi^m_{\mathrm{sus}}(H; E, F)$ be elliptic and invertible. Then $A^{-1}\partial_\tau A \in \Psi^{-1}_{\mathrm{sus}}(H; E)$, and Melrose's eta invariant [12] is

$$\eta(A) = \frac{1}{\pi i}\overline{\int_{\mathbb{R}}}\,\mathrm{Tr}\big(A^{-1}\partial_\tau A\big)\,d\tau := \frac{2\pi}{\pi i}\,\overline{\mathrm{Tr}}\big(A^{-1}\partial_\tau A\big) = 2\,\overline{\mathrm{Tr}}\big(A^{-1}[t, A]\big).$$

The last equality uses $\mathrm{In}([t, A]) = -i\partial_\tau A$ from (10). The notation $\overline{\int}\,\mathrm{Tr}(\cdot)\,d\tau$ always means $2\pi\overline{\mathrm{Tr}}$. The pointwise trace exists only for $d = 0$.

Theorem C.29 (Dirac operators, (B7))**.** *Let B be a self-adjoint, invertible, first-order elliptic differential operator on H. Let $\sigma : E \to F$ be a unitary bundle isomorphism, and let $A_0(\tau) = \sigma(i\tau + B)$. This is the indicial family of $\sigma(-\partial_t + B)$, the APS normal form $\sigma(\partial_u + B)$ in $u = -t$ of Corollary 15.10. Then*

$$\eta(A_0) = \eta_B(0), \qquad \eta_B(s) = \sum_{\lambda\in\mathrm{spec}\,B}\mathrm{sgn}\,\lambda\,|\lambda|^{-s}.$$

Proof. We have $[t, A_0] = -i\partial_\tau A_0 = \sigma$, so $A_0^{-1}[t, A_0] = (B + i\tau)^{-1} =: C \in \Psi^{-1}_{\mathrm{sus}}(H; E)$. Split $C = C_e + C_o$ with $C_e(\tau) = B(B^2 + \tau^2)^{-1}$ and $C_o(\tau) = -i\tau(B^2 + \tau^2)^{-1}$. Both lie in Ψ^{-1}_{sus}. By Theorem C.23(2), $\overline{\mathrm{Tr}}(C) = \overline{\mathrm{Tr}}(C_e)$. We must show $\overline{\mathrm{Tr}}(C_e) = \frac{1}{2}\eta_B(0)$.

Let $\theta(t) = \mathrm{Tr}(Be^{-tB^2}) = \sum_\lambda \lambda e^{-t\lambda^2}$. It decays exponentially as $t \to \infty$. From [18, Section "The eta invariant"] we quote the small-time expansion $\theta(t) \sim \sum_{k \geq 0} a_k t^{(k-d-1)/2}$ as $t \downarrow 0$. We also quote the Mellin representation $\Gamma(\frac{s+1}{2})\eta_B(s) = \int_0^\infty t^{(s-1)/2}\theta(t)\, dt$ and the regularity of η_B at 0, which is equivalent to $a_d = 0$. Split $\theta = \sum_{k<K} a_k t^{(k-d-1)/2} + r_K$ on $(0, 1]$, with $K > d + 1$. Then

$$\sqrt{\pi}\,\eta_B(0) = \int_1^\infty \frac{\theta}{\sqrt{t}}\, dt + \int_0^1 \frac{r_K}{\sqrt{t}}\, dt + \sum_{k<K,\, k\neq d} \frac{2a_k}{k-d}. \tag{29}$$

Fix $q > (d + 1)/2$, put $p = 2q$, and let $E_q(x) = e^{-x} - \sum_{k<q}(-x)^k/k!$. Define $F(\tau) = \int_0^\infty \theta(t) E_q(t\tau^2)\, dt$. The integral converges: near $t = 0$, $E_q = O(t^q)$ and $\theta = O(t^{-(d+1)/2})$. The polynomial part of $E_q(t\tau^2)$ has degree $< p$ in τ. Differentiation under the integral and Fubini over the eigenvalues give

$$F^{(p)}(\tau) = \int_0^\infty \theta(t)\, \partial_\tau^p e^{-t\tau^2}\, dt = \sum_\lambda \partial_\tau^p \frac{\lambda}{\lambda^2 + \tau^2} = \mathrm{Tr}\big(\partial_\tau^p C_e(\tau)\big) = h_p.$$

The series converges absolutely because $\sum |\lambda|^{-p-1} < \infty$. So $F - I^p h_p$ is a polynomial, and $\overline{\mathrm{Tr}}(C_e) = (2\pi)^{-1}\overline{\int} F\, d\tau$.

By Fubini and the substitution $u = \sqrt{t}\,\tau$,

$$\int_{-T}^{T} F\, d\tau = \int_0^\infty \theta(t)\, t^{-1/2} W(T\sqrt{t})\, dt, \qquad W(R) = \int_{-R}^{R} E_q(u^2)\, du = \sqrt{\pi} - \sum_{l<q} c_l R^{2l+1} + e(R),$$

with $c_l = 2(-1)^l/(l!(2l + 1))$ and $e(R) = -2\int_R^\infty e^{-u^2} du$. The function e is bounded and decays faster than any power. We find the constant term as $T \to \infty$ piece by piece.

- On $[1, \infty)$: $\int_1^\infty \theta\, t^{-1/2} W(T\sqrt{t}) = \sqrt{\pi}\int_1^\infty \theta t^{-1/2} - \sum_l c_l T^{2l+1}\int_1^\infty \theta t^l + o(1)$. The constant term is $\sqrt{\pi}\int_1^\infty \theta t^{-1/2}$.

- The remainder r_K on $(0, 1]$ gives $\sqrt{\pi}\int_0^1 r_K t^{-1/2}$ by the same argument. The e-term tends to 0 by dominated convergence.

- For $a_k t^\beta$, $\beta = (k - d - 1)/2$, substitute $R = T\sqrt{t}$:

$$\int_0^1 t^{\beta - 1/2} W(T\sqrt{t})\, dt = 2T^{-2\beta-1}\int_0^T R^{2\beta} W(R)\, dR.$$

  Insert $W = \sqrt{\pi} - \sum_l c_l R^{2l+1} + e$ on $[1, T]$. If $k \neq d$, the only constant term is $2\sqrt{\pi}/(2\beta + 1) = 2\sqrt{\pi}/(k - d)$. The other terms are non-zero powers of T, possibly times $\log T$, or $o(1)$. The term $k = d$ is absent because $a_d = 0$.

Comparing with (29), $\overline{\int} F\, d\tau = \sqrt{\pi}\cdot\sqrt{\pi}\,\eta_B(0) = \pi\eta_B(0)$. Hence $\overline{\mathrm{Tr}}(C) = \frac{1}{2}\eta_B(0)$ and $\eta(A_0) = 2\overline{\mathrm{Tr}}(C) = \eta_B(0)$. □

The proof also shows the formal identity $\pi^{-1}\int(\lambda + i\tau)^{-1} d\tau = \operatorname{sgn}\lambda$ of Corollary 15.10, now with the regularization made precise. As a numerical check, take $H = S^1$ and $B = D_\theta + a$, $0 < a < 1$. The partial integrals $\int_{-T}^{T} F$ computed from the spectrum $\mathbb{Z} + a$ have constant term $\pi(1 - 2a) = \pi\eta_B(0)$.

Proposition C.30 (The functional $\overline{\eta}$ of MN (89))**.** *Let $A \in \Psi_c^m(X; E, F)$ be fully elliptic, with $A_0 = \mathrm{In}(A)$, and let Q be admissible and translation invariant to all orders at ∂X, with Taylor series $Q_0(s)$ (on E and on F).*

(i) $\overline{\eta}(A) = \eta(A_0) + \delta_Q(A)$, where $\delta_Q(A) = (2\pi)^{-1}\gamma'(0)$ and γ is computed in Theorem C.24 from $A_0^{-1}[t, A_0]Q_0(s)$ and $[t, A_0]A_0^{-1}Q_0(s)$. The correction $\delta_Q(A)$ is a local functional of the boundary symbols. By Remark C.27 it vanishes when n is odd and A is a differential operator (sketch).

(ii) In the situation of Theorem C.29, suppose that $Q_0(s) = (D_t^2 + B^2)^{-s/2}$ on E and $\sigma(D_t^2 + B^2)^{-s/2}\sigma^{-1}$ on F. Then $\overline{\eta}(A) = \eta(A_0) = \eta_B(0)$. Such Q exist if one quotes the fact that $(D_t^2 + B^2)^{-s/2}$ is an entire family in $\Psi_{\mathrm{sus}}^{-s}(H)$ (Seeley's complex powers [19] with the parameter τ, quoted: [15, App. B]). Then take $Q(s) = \tilde{\omega}Q_0(s)\omega + Q_X(s)(1-\omega)$, with Q_X from Proposition C.12.

Proof. (i) As in the proof of Proposition 15.8, $(A^{-1}[\log x, A])_{(1)} = -A_0^{-1}[t, A_0] =: -C_1$ and $([\log x, A]A^{-1})_{(1)} = -[t, A_0]A_0^{-1} =: -C_2$. By Correction C.26, $\overline{\eta}(A) = -\widehat{\mathrm{Tr}}_\partial(A^{-1}[\log x, A] + [\log x, A]A^{-1}) = \overline{\mathrm{Tr}}(C_1) + \overline{\mathrm{Tr}}(C_2) + (2\pi)^{-1}\gamma'_{C_1+C_2}(0)$. By Theorem C.23(3), $\overline{\mathrm{Tr}}(C_2) = \overline{\mathrm{Tr}}(C_1)$, and $2\overline{\mathrm{Tr}}(C_1) = \eta(A_0)$.

(ii) We have $C_1 = (B + i\tau)^{-1}$ and $C_2 = \sigma C_1 \sigma^{-1}$, so $\mathrm{Tr}(C_2 Q_0^F(s)) = \mathrm{Tr}(C_1 Q_0^E(s))$. For $\mathrm{Re}\, s > d$, the eigenvalue sum and the τ-integral converge absolutely. The odd part integrates to 0, and

$$\frac{1}{2\pi}\int \mathrm{Tr}\big(C_1(B^2+\tau^2)^{-s/2}\big)d\tau = \frac{1}{2\pi}\sum_\lambda \lambda \int_{\mathbb{R}} (\lambda^2+\tau^2)^{-1-s/2}d\tau = \frac{\Gamma(\frac{1+s}{2})}{2\sqrt{\pi}\,\Gamma(1+\frac{s}{2})}\,\eta_B(s).$$

This is regular at $s = 0$ with value $\frac{1}{2}\eta_B(0)$. By Correction C.26 (before the comparison with $\overline{\mathrm{Tr}}$), $\widehat{\mathrm{Tr}}_\partial(A^{-1}[\log x, A]) = \widehat{\mathrm{Tr}}_\partial([\log x, A]A^{-1}) = -\frac{1}{2}\eta_B(0)$. So $\overline{\eta}(A) = \eta_B(0)$, which is $\eta(A_0)$ by Theorem C.29. For the construction of Q: the family is admissible since $\tilde{\omega}\omega = \omega$, and the term $Q_X(s)(1-\omega)$ has kernel supported away from the front face. □

Remark C.31 (The regularizer $(A^*A+1)^{1/2}$ (sketch)). Theorem 15.13 uses $Q = (A^*A+1)^{1/2m}$, whose indicial family is $(D_t^2 + B^2 + 1)^{1/2}$ for A as in Theorem C.29. Expanding $(1 + (\tau^2 + B^2)^{-1})^{-s/2}$ binomially in the computation above (for $|\lambda| > 1$) gives

$$\overline{\eta}(A) = \eta_B(0) + \frac{1}{\sqrt{\pi}} \sum_{1 \le k \le d/2} \frac{(-1)^k}{k \cdot k!}\, a_{d-2k},$$

where the a_j are the heat coefficients of $\theta(t)$. For Dirac operators compatible with a Clifford structure, $\theta(t) = O(t^{1/2})$ as $t \downarrow 0$ (local cancellation of Bismut–Freed [2], quoted). So all a_j with $j \le d$ vanish, and $\overline{\eta}(A) = \eta_B(0)$ also for this Q. For general first-order B the extra terms need not vanish. They are local and are compensated in the index formula by β_Q (Proposition C.15(3)).

Remark C.32 (Signs). In [18] the APS operator is $P = c(du)(\partial_u + A^+)$ near the boundary, with $u = \log x_b \to -\infty$ on the end, and $\mathrm{ind}\, P = \int \hat{A}\,\mathrm{ch} - \frac{1}{2}(\eta(A^+) + h)$. The cusp picture has $x_b = e^{-1/x}$, so $u = -t$ and $\partial_u = -\partial_t$. Hence $A = \sigma(-\partial_t + B)$ with $\sigma = c(du)$ and $B = A^+$, and Theorem C.29 gives $\eta(A_0) = \eta(A^+)$. Thus the corrected formula $\mathrm{Ind}\, A = \mathrm{AS}(A) - \frac{1}{2}\eta(A_0)$ of Theorem 15.13, in which $\delta_Q = 0$ here (Remark C.31), reproduces APS (with $h = 0$). The sign of the eta term is fixed by the indicial convention (9), $\mathrm{In}(A, \tau) = e^{i\tau t}Ae^{-i\tau t}$; with $e^{-i\tau t}Ae^{i\tau t}$ instead, $\eta(A_0)$ would change sign. This confirms the sign of the eta term in Correction 15.7.

Acknowledgement of sources

The mathematics in these notes is that of Melrose and Nistor [15]. The spectral sequence and HKR arguments follow [14, Ch. 10]. The Hochschild tools, the contraction lemmas and the comparison with the b-calculus follow the companion notes [7]. The appendices adapt [14], [23] and [7], as indicated there. Corrections and the worked examples are our own and are marked as such.

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